

The topology of algebraic manifolds
<https://github.com/tbrazel/math231br>

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0.1 About

These notes are from a MATH213BR, a second-semester graduate algebraic topology topics class taught at Harvard in spring of 2026. This particular course is modeled off of Hirzebruch's amazing 1962 text *Neue topologische Methoden in der algebraischen Geometrie*. Theorem counters and page references refer to the 1995 reprinting in English [Hir78].

0.1.1 Overview

Part 1: We begin with the theory of sheaves of sets and discrete groups on topological spaces, developing their cohomology and comparing it with singular and de Rham cohomology. We continue with fiber bundles and develop their basic theory. We then discuss vector bundle theory and Chern and Pontryagin classes.

Part 2: We discuss oriented and complex cobordism, genera, the L -genus and Todd genus, and the index of a $4n$ -manifold. We prove Hirzebruch's signature theorem and sketch the proof of Riemann-Roch for algebraic manifolds.

0.1.2 References

Aside from the main text, which we deviate from at various points, some other references which have informed the presentation here include:

- ▷ *sheaf theory*: Godement's original work [God58], and of course Serre's [Ser55]
- ▷ *fiber bundles*: Steenrod's 1951 printing [Ste51]
- ▷ *Kähler manifolds*: Michael Wong's 2013 [course notes](#)
- ▷ *cobordism*: Dan Freed's notes [Fre12] and Haynes Miller's notes [Mil]
- ▷ *genera*: Hirzebruch's book [HBJ94]
- ▷ *complex manifolds*: Griffiths and Harris [GH78]

0.1.3 Acknowledgements

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Contents

0.1	About	2
1	Course notes	5
1.1	Sheaves	5
1.2	Sheaf cohomology	18
1.3	Comparison with singular cohomology	36
1.4	Vector bundles and sheaves of sections	40
1.5	Fiber bundles	47
1.6	Principal G -bundles	55
1.7	Vector bundles revisited	62
1.8	Chern classes	67
1.9	Pontryagin classes	72
1.10	Cobordism	74
1.11	Genera	78
1.12	Hirzebruch's signature theorem	85
1.13	Complex genera	91
1.14	The holomorphic Euler characteristic	98
1.15	Hirzebruch-Riemann-Roch	102
A	Appendices	107
A.1	Category theory	107
A.2	Point-set topology	109
A.3	Normalized versus unnormalized cochains	113
B	Student appendices	117
B.1	Thom's computation of MSO_* (Joaquim Bocrésion)	118
B.2	Connecting Elliptic Integrals, Curves, and Functions through Elliptic Genera (Grayson Kemplin)	126
B.3	The Hodge-de Rham spectral sequence (Brian Siew)	139
B.4	The Calabi-Eckmann Manifolds and their $\bar{\partial}$ -Cohomology (Carl Scandellius)	150
B.5	Leray Spectral Sequence (David Ettel)	159

Chapter 1

Course notes

1.1 Sheaves

The definition of a *sheaf* in Hirzebruch is perhaps a bit different from what we're used to. He wants to think of sheaves as *living over* a space X , whereas we might be comfortable thinking about sheaves as local data across a space. We'll see that these perspectives are equivalent.

Definition 1.1.1. A *sheaf of abelian groups* over a topological space X is a topological space S with the data of a surjective continuous map $\pi: S \rightarrow X$, for which

- ▷ π is a *local homeomorphism*, meaning for every point $s \in S$ there is an open neighborhood $U \ni s$ for which

$$\pi|_U: U \rightarrow \pi(U)$$

is a homeomorphism

- ▷ for every $x \in X$, the *stalk* $S_x := \pi^{-1}(x)$ has the structure of an *abelian group*
- ▷ the abelian group structure on S_x is *continuous in x* . Precisely, forming inverses in each fiber defines a continuous function:

$$\begin{aligned} S &\rightarrow S \\ s &\mapsto -s, \end{aligned}$$

and if we denote by $S \times_X S$ the subspace

$$S \times_X S := \{(s_1, s_2) \in S \times S: \pi(s_1) = \pi(s_2)\} \subseteq S \times S,$$

then the operation of addition is a continuous function:

$$\begin{aligned} S \times_X S &\rightarrow S \\ ((\alpha, x), (\beta, x)) &\mapsto (\alpha + \beta, x). \end{aligned}$$

We should think about a sheaf of abelian groups as being a space where we can do abelian group operations (addition, subtraction), but where the abelian group in which we're working is *changing* according to the topology of X .

Example 1.1.2 (Zero sheaf). The identity map $X \rightarrow X$ is a sheaf where the abelian group structure on each stalk is just the trivial group with one element.

Example 1.1.3 (Constant sheaves). For any abelian group A and any space X , we get a *constant sheaf*

$$X \times A \rightarrow X,$$

where π is the projection onto X . When $A = \{0\}$, this recovers the zero sheaf above.

Example 1.1.4 (Skyscraper sheaves). If we want build a sheaf with some *prescribed* fiber A over a point $x \in X$ (we want a name for the map of the inclusion of a point, so let's call it $i: \{x\} \rightarrow X$), we have an nice way to do this – namely, we can take the discrete set underlying the abelian group A , and build the space which we call i_*A , defined as

$$i_*A = \frac{X \times A}{(y, a_1) \sim (y, a_2) \text{ for } y \neq x}.$$

That is, we identify all the points of A together over every point in X *except* our specified point x . We call it a *skyscraper sheaf* because it has this big tall stalk at x and is flat everywhere else. This space comes equipped with a projection back to X which is a continuous local homeomorphism, and the abelian group structure on the stalks over points other than x are just the structure on the one-element group. Note that this resulting space i_*A is very much not Hausdorff.

Example 1.1.5 (c.f. [Bre97, 1.3]). The *line with doubled origin* is the skyscraper sheaf over $\mathbb{R}$ defined by $i_*(\mathbb{Z}/2)$, where $i: \{0\} \hookrightarrow \mathbb{R}$ is the inclusion of the origin.

Notation 1.1.6. For any sheaf of abelian groups $\pi: S \rightarrow X$, we have the so-called *zero section* $z: X \rightarrow S$, sending each element $x \in X$ to the zero element in S_x . This map is continuous and has the property that

$$\pi \circ z = \text{id}_X.$$

Exercise 1.1.7. Let $\pi: X \rightarrow S$ be any sheaf and $z: X \rightarrow S$ the zero section. Then $\text{im}(z) \subseteq S$ is open.

Example 1.1.8. The projection $\mathbb{R} \rightarrow S^1$ cannot be given the structure of a sheaf of abelian groups.

Proof. We're tempted to say it looks like a sheaf, since each fiber looks like $\mathbb{Z}$, however it turns out there is no way to endow these fibers simultaneously with the additive structure of the integers. Suppose towards a contradiction there was. Then composing with the zero section, we get a composite which is the identity:

$$S^1 \xrightarrow{z} \mathbb{R} \xrightarrow{\pi} S^1.$$

Applying π_1 or H_1 for instance, we have that the identity on $\mathbb{Z}$ factors through zero, which is a contradiction. $\square$

Remark 1.1.9. Just as we have a sheaf of abelian groups, we can have a sheaf with other structures as well — we could have a sheaf of R -modules for instance. All that's important is that:

1. our structure is a set with extra data

2. we have that data in every stalk
3. the data varies continuously in the stalks

Definition 1.1.10. A *morphism of sheaves* between $\pi: S \rightarrow X$ and $\tilde{\pi}: \tilde{S} \rightarrow X$ is a continuous map $f: S \rightarrow \tilde{S}$ satisfying the following properties:

▷ $\tilde{\pi} \circ f = \pi$, that is, we have a commutative diagram

$$\begin{array}{ccc} S & \xrightarrow{f} & \tilde{S} \\ & \searrow \pi & \swarrow \tilde{\pi} \\ & X & \end{array}$$

Note this implies that f restricts to a map on each stalk, that is, we get maps $f_x: S_x \rightarrow \tilde{S}_x$ for each $x \in X$

▷ for each $x \in X$, the induced map on stalks

$$f_x: S_x \rightarrow \tilde{S}_x$$

is an abelian group homomorphism.

Exercise 1.1.11. Show that every morphism between sheaves of abelian groups is a local homeomorphism.

Definition 1.1.12. We say a morphism of sheaves $f: S \rightarrow \tilde{S}$ is:

1. *injective* if each f_x is injective
2. *surjective* if each f_x is surjective.

We say a sequence of maps of sheaves

$$S \xrightarrow{f} T \xrightarrow{g} U$$

is *short exact* if $S_x \xrightarrow{f_x} T_x \xrightarrow{g_x} U_x$ is a short exact sequence of abelian groups for every $x \in X$.

1.1.1 Sheaves and presheaves

Given any sheaf $S \rightarrow X$, we can ask when we have *sections*, which are maps $X \rightarrow S$ that are right inverses to the projection map π .

Definition 1.1.13. We define the set of *sections* by

$$\Gamma(X, S) := \{s: X \rightarrow S \text{ continuous} \mid \pi \circ s = \text{id}_X\}.$$

Note that $\Gamma(X, S)$ is an abelian group by adding sections pointwise. The zero element of $\Gamma(X, S)$ is the zero section.

Note that we don't need sections to be defined on all of X — we could look at sections defined on subspaces. For instance if $U \subseteq X$ is an open subspace, we can define

$$\Gamma(U, S) := \{s: U \rightarrow S \mid \pi \circ s = \text{id}_U\}.$$

If $V \subseteq U$ is an open subspace, there is a natural *restriction* map

$$\begin{aligned} \text{res}_V^U: \Gamma(U, S) &\rightarrow \Gamma(V, S) \\ s &\mapsto s|_V. \end{aligned}$$

Notation 1.1.14. For X a topological space, we denote by $\text{Open}(X)$ the category whose objects are open subspaces of X and whose morphisms are inclusions. Then the constructions above tell us that $\Gamma(-, S)$ defines a functor

$$\Gamma(-, S): \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}.$$

This is called the *presheaf* associated to S .

Terminology 1.1.15. If $S \rightarrow X$ is a sheaf, an element of $\Gamma(X, S)$ will be called a *global section*. This is to indicate it is defined globally, i.e. everywhere on X , in contrast with sections only defined locally on some open subspace U .

Let's think about what structures of S we can recover from its presheaf.

Question 1.1.16. Can we recover the stalk S_x from the presheaf $\Gamma(-, S)$?

We can't just take Γ and evaluate it on the one-point set $\{x\}$ since this won't be open in most reasonable topological spaces. Instead, we can take elements in $\Gamma(U, S)$ for any $U \ni x$, and glue sections together along restriction of open sets containing x . This is called a *colimit* (in classical terminology, a *direct limit*):

$$\text{colim}_{U \ni x} \Gamma(U, S).$$

Proposition 1.1.17. For every sheaf of abelian groups $S \rightarrow X$ and every $x \in X$, the canonical “evaluation at x ” map

$$\text{ev}_x: \text{colim}_{U \ni x} \Gamma(U, S) \rightarrow S_x$$

is an isomorphism of abelian groups.

Proof. If $s_1(x) = s_2(x) = a$, then since π is a local homeomorphism, there is some small neighborhood $V \ni a$ on which $\pi|_V: V \rightarrow \pi(V)$ is a local homeomorphism. In particular since $s_1|_{\pi(V)}$ and $s_2|_{\pi(V)}$ are both sections (right inverses) of $\pi|_V$ they must agree. Since they agree on a sufficiently small open neighborhood around x , they agree in the colimit. This establishes injectivity.

To see surjectivity, if $a \in S_x$, we want to create a section s defined in a neighborhood U of x for which $s(x) = a$. We again leverage the fact that π is a local homeomorphism to find some neighborhood $V \ni a$ for which $\pi: V \xrightarrow{\sim} \pi(V)$ is a homeomorphism. Then the inverse to this map satisfies $\pi^{-1}(x) = a$. $\square$

This gives us an alternative way to look at stalks. Now we know we can recover stalks from the underlying presheaf.

Question 1.1.18. Can we recover the sheaf S from the presheaf $\Gamma(-, S)$?

That is, suppose someone hands you a functor

$$F: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab},$$

and tells you it is of the form $\Gamma(-, S)$ for some sheaf S , but they don't tell you what S is. Can you rebuild S ? Let's denote by

$$F_x := \text{colim}_{U \ni x} F(U)$$

the stalk at a point $x \in X$. Then we know how to construct the sheaf as a *set*, namely it is

$$\coprod_{x \in X} F_x \rightarrow X.$$

The question then becomes how to topologize the thing on the left. Each F_x has the discrete topology as an abelian group, and the disjoint union has the discrete topology a priori – but if we accept this topology, then in general we won't have the property that the projection map is a local homeomorphism. So we need to hunt for a different topological structure. The trick to finding it is that we know how to write down sections, and we want to *force sections to be continuous*.

We denote by

$$\text{germ}_x: F(U) \rightarrow F_x$$

the structure map (part of the data of the colimit) to the stalk (c.f. [Proposition 1.1.17](#); we called this an evaluation map earlier when we knew $F(U)$ was comprised of actual sections. We don't know that a priori now). For any $s \in F(U)$, we then get a composite

$$\begin{aligned} U &\rightarrow \coprod_x F_x \\ x &\mapsto \text{germ}_x(s). \end{aligned}$$

This should be like a section of the sheaf, so we want it to be continuous. We want the preimage of opens to be open, and since U is indeed open, an easy way to try to force continuity is to ask for the image of this map above to be open.

Definition 1.1.19. We define a *basis*¹ for a topology on $\coprod_x F_x$ by

$$[(s, U)] := \{\text{germ}_x(s) \mid x \in U\}$$

for any $s \in F(U)$.

It now suffices to verify that this satisfies the axioms for a basis:

- B1** *The basis elements cover our space:* Take some $x \in X$ and $a \in F_x$. We want to argue there is a basis element containing it. By the construction of F_x as a colimit of abelian groups, there is some $V \in \text{Open}(X)$ and some $s \in F(V)$ for which $\text{germ}_x(s) = a$. That is, by definition, $a \in [(s, V)]$.

¹See [\[Mun00, II§13\]](#) if this is an unfamiliar term.

B2 Common refinement: Suppose we have $[(s_1, U_1)]$ and $[(s_2, U_2)]$ and $a \in [(s_1, U_1)] \cap [(s_2, U_2)]$. Then we want to find another basis element contained in the intersection of these two and containing a . Let $x = \pi(a)$, then $x \in U_1 \cap U_2$ by definition. So we can look in $F(U_1 \cap U_2)$. By the construction of the colimit, since we have a commutative diagram

$$\begin{array}{ccc}
 F(U_1) & & \\
 & \searrow & \\
 & F(U_1 \cap U_2) & \longrightarrow F_x \\
 & \nearrow & \\
 F(U_2) & &
 \end{array}$$

since $\text{germ}_x(s_1) = \text{germ}_x(s_2)$, their images in $F(U_1 \cap U_2)$ agree. Call that image s . Then we have that

$$a \in [(s, U_1 \cap U_2)] \subseteq [(s_1, U_1)] \cap [(s_2, U_2)] \subseteq \coprod_{x \in X} F_x.$$

So we've checked that we get a basis for a topology.

Remark 1.1.20. In general, the sheaf $\coprod_{x \in X} F_x$ constructed in this way will be very far from being Hausdorff ([Bre97, p. 3], [Ser55]).

It turns out we have in fact built a sheaf! That is, the projection map will be a local homeomorphism.

Lemma 1.1.21. Let $F: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$ be a presheaf, and topologize $\coprod_{x \in X} F_x$ by giving it the basis of Definition 1.1.19. Then the natural projection

$$\pi: \coprod_{x \in X} F_x \rightarrow X$$

is a local homeomorphism.

Proof. Pick any $a \in F_x$. Then by construction, there exists some open set U and some $s \in F(U)$ so that $\text{germ}_x(s) = a$. In particular, we claim the map

$$\pi|_{[(s, U)]}: [(s, U)] \rightarrow U$$

is a local homeomorphism. It is clearly continuous, and it admits a continuous inverse given by $s: U \rightarrow F(U) \rightarrow [(s, U)]$ by construction. $\square$

We now have a way to build a sheaf out of a presheaf! So we have a way to go from presheaves to sheaves. I think we've answered Question 1.1.18, which was if someone hands you a presheaf $F: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$, and tells you it is of the form $\Gamma(-, S)$ for some sheaf S , then we can reconstruct S .

Let's now give a slightly different question pandering to a more cynical worldview. Suppose someone hands you a presheaf $F: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$ and tells you it is the presheaf of sections attached to some sheaf, but we don't believe them. How might we argue that this cannot come from any sheaf?

Here's an idea of how we might find a way to disagree:

▷ if F was of the form $\Gamma(-, S)$ for some S , then $F(U)$ would be equal to $\Gamma(U, S)$

▷ we now know how we would reconstruct S : it is $\coprod_{x \in X} F_x$ equipped with the topology defined in [Definition 1.1.19](#)

▷ so we can check: is the canonical map $F(U) \xrightarrow{\theta_U} \Gamma(U, \coprod_x F_x)$ a bijection?

Let's recall how this map θ_U worked — each $s \in F(U)$ determines a function $U \rightarrow \coprod_{x \in U} F_x$ given by

$$\begin{aligned} \theta_U: F(U) &\rightarrow \Gamma(U, \coprod_x F_x) \\ s &\mapsto [x \mapsto \text{germ}_x(s)]. \end{aligned} \tag{1.1.22}$$

If θ_U is not a bijection for some open U , then F cannot have come from a sheaf. So let's see when θ_U could fail to be a bijection — that is, when is it injective and when is it surjective? We phrase the following as lemmas, although there's no mathematical content, it's just unwinding definitions.

Lemma 1.1.23 (When is θ_U a monomorphism?). Let U, θ_U as above. Let's first see what it means for θ_U to send two elements in $F(U)$ to the same thing in $\Gamma(U, \coprod_{x \in X} F_x)$.

1. We have that $\theta_U(s) = \theta_U(t)$ if and only if s and t agree locally — that is, for every $x \in U$ there is some open $V \ni x$ for which $s|_V = t|_V$.

This tells us the content of being a monomorphism is that “being equal locally implies you are equal.” In other words:

2. θ_U is a monomorphism if and only if, for every $x \in X$, if there is a neighborhood $V \ni x$ with $s|_V = t|_V \in F(V)$, then $s = t$ in $F(U)$.

As the V 's chosen this way form an open cover of U , we can reword this as:

3. θ_U is a monomorphism if and only if, for every open cover $\{U_i\}$ of U , if $s, t \in F(U)$ satisfies $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$.

Proof. For the first point, we have that $\theta_U(s) = \theta_U(t)$ if and only if the functions $u \mapsto \text{germ}_u(s)$ and $u \mapsto \text{germ}_u(t)$ agree, that is, only if $\text{germ}_u(s) = \text{germ}_u(t)$ in F_u for all $u \in U$. By construction of F_u as a colimit, we have that there must exist some neighborhood $V \ni u$ for which $s|_V = t|_V$ in $F(V)$. The second point is a rephrasing of the first, and (3) is a rephrasing of the second point. $\square$

Lemma 1.1.24 (When is θ_U an epimorphism?). Let U, θ_U be as above, and suppose [Lemma 1.1.23\(3\)](#) holds. We want to know when θ_U hits some element $a \in \Gamma(U, \coprod_{x \in X} F_x)$

1. Let $a \in \Gamma(U, \coprod_{x \in X} F_x)$. Then for each $x \in U$, we have that $a(x) \in F_x$. By the construction of F_x as a colimit, this means there is some open neighborhood $U_x \ni x$, and some element $s_x \in F(U_x)$ so that, under the structure map $F(U_x) \rightarrow F_x$, we have that $s_x \mapsto a(x)$. Another way to say that is that the section $\theta_{U_x}(s_x)$ agrees with a at the point x . Since π is a local homeomorphism, we have that a and $\theta_{U_x}(s_x)$ must agree on some open neighborhood $V_x \ni x$ which, without loss of generality, we may assume to be a subspace of U_x . So we have that $\theta_{V_x}(s_x) = a|_{V_x}$ and $\theta_{V_y}(s_y) = a|_{V_y}$, so under restriction we get

$$\theta_{V_x \cap V_y}(s_x) = a|_{V_x \cap V_y} = \theta_{V_x \cap V_y}(s_y).$$

If we assumed that the θ 's were injective, [Lemma 1.1.23\(3\)](#), this says that

$$s_x|_{V_x \cap V_y} = s_y|_{V_x \cap V_y}.$$

And we have that a is in the image of θ_U if and only if there is some $s \in F(U)$ so that $s|_{V_x} = s_x$ for each $x \in U$. In other words, θ_U is surjective if, any time locally defined sections agree on overlaps, there is a section defined on all of U restricting to them.

2. With the hypotheses above, we have that θ_U is surjective if and only if, for every cover $\{U_i\}_{i \in I}$ of U , and elements $s_i \in F(U_i)$ satisfying $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, there exists an element $s \in F(U)$ so that $s|_{U_i} = s_i$ for each i .

So what conditions were essential to showing that θ_U was a bijection for every U ? It was **Lemma 1.1.23(3)**, which tells us if sections agree *locally* then they agree *globally*, and **Lemma 1.1.24(2)**, which told us that sections defined locally and agreeing on overlaps can be *glued* together to get a global section. These two conditions are called *locality* and *gluing*. Let's summarize our observations:

Theorem 1.1.25. A presheaf $F: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$ is the presheaf of sections attached to a sheaf if and only if the two conditions hold:

- Sh1 (Locality)** For any open set $U \subseteq X$ and any open cover $\{U_i\}_{i \in I}$ of U , if $s, t \in F(U)$ satisfy $s|_{U_i} = t|_{U_i}$ for each $i \in I$, then $s = t$. Phrased differently, the map

$$F(U) \rightarrow \prod F(U_i)$$

is injective.

- Sh2 (Gluing)** If $\{U_i\}_{i \in I}$ is an open cover of U , and we have elements $s_i \in F(U_i)$ for each i so that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j , then there exists some $s \in F(U)$ so that $s|_{U_i} = s_i$ for each i . Phrased differently, $F(U)$ surjects onto the equalizer of the parallel restriction maps

$$\prod F(U_i) \rightrightarrows \prod F(U_i \cap U_j).$$

Remark 1.1.26. Let E denote the equalizer $E = \text{eq}(\prod F(U_i) \rightrightarrows \prod F(U_i \cap U_j))$. By functoriality, the map $F(U) \rightarrow \prod F(U_i)$ factors through E , and $E \rightarrow \prod F(U_i)$ is always a monomorphism. So for presheaves of abelian groups, locality reduces to the statement that $F(U) \rightarrow E$ is injective, while gluing restricts to the statement that $F(U) \rightarrow E$ is surjective. Hence altogether the sheaf condition can be compressed to the assertion that, for any $U \subseteq X$ and any open set $\{U_i\}$ of U , the sequence is an equalizer:

$$F(U) \rightarrow \prod F(U_i) \rightrightarrows \prod F(U_i \cap U_j).$$

Remark 1.1.27. In an abelian category, an equalizer of two maps $f, g: A \rightrightarrows B$ is the same as the kernel of their difference $A \xrightarrow{f-g} B$. So the locality and gluing conditions can be condensed into the statement that, for every open subspace $U \subseteq X$ and every open cover $\{U_i\}$ of U , the sequence

$$0 \rightarrow F(U) \rightarrow \prod F(U_i) \rightarrow \prod F(U_{ij})$$

is exact, where that rightmost map is the difference of the two restrictions. If we ever want to talk about sheaves of *sets* we can't do this, because subtracting two functions doesn't make sense unless they are group homomorphisms.

Corollary 1.1.28. There is an isomorphism of categories between

$$\{\text{sheaves on } X\} \simeq \left\{ \begin{array}{c} \text{the full subcategory of presheaves on } X \\ \text{satisfying locality and gluing} \end{array} \right\}.$$

Proof. We have shown a bijection on objects, so it suffices to verify that there is also a bijection on morphisms of each type.

Let S and S' be sheaves. Then a map of sheaves $f: S \rightarrow S'$ clearly induces a map of presheaves, since any section $U \rightarrow S$ can be postcomposed with f to get a section $U \rightarrow S'$, hence we get an induced map $\Gamma(U, S) \rightarrow \Gamma(U, S')$ which is easily seen to be compatible with restriction.

Conversely, suppose we have two presheaves $F, G \in \text{Fun}(\text{Open}(X)^{\text{op}}, \text{Ab})$ which satisfy locality and gluing, and a morphism of presheaves $F \rightarrow G$. We want to see how this induces a morphism of sheaves $\Pi_x F_x \rightarrow \Pi_x G_x$. We first observe that, by taking colimits, there is a natural induced map on stalks $F_x \rightarrow G_x$ for every X , so we get a commutative diagram of *sets*

$$\begin{array}{ccc} \Pi_{x \in X} F_x & \xrightarrow{f} & \Pi_{x \in X} G_x \\ & \searrow & \swarrow \\ & X & \end{array}$$

We now want to check that this induced map f is actually continuous. Since we can check continuity on a basis of the target space (reference needed), we can take some $[(t, U)] \subseteq \Pi_{x \in X} G_x$ and try to see its preimage is open in $\Pi_{x \in X} F_x$. We can check that

$$f^{-1}([(t, U)]) = \{\text{germ}_x(s) \mid x \in U \text{ and } \text{germ}_x(f \circ s) = \text{germ}_x(t)\}.$$

Pick one such point $\text{germ}_x(s) \in f^{-1}([(t, U)])$. Since $\text{germ}_x(f \circ s) = \text{germ}_x(t)$, there is some open neighborhood $V \ni x$ for which $(f \circ s)|_V = t|_V$. This means every germ of s over V lies in the preimage of f , that is,

$$\text{germ}_x(s) \in [(s, V)] \subseteq f^{-1}([(t, U)]).$$

In particular this implies that $f^{-1}([(t, U)])$ is open and therefore that f is continuous.

Verifying these two assignments are inverse is immediate by construction in one direction, and is the content of θ_U being bijective in the other direction. $\square$

Remark 1.1.29 (Important). What we have been calling sheaves (following Hirzebruch, Bredon, Serre) are now more commonly called the *espace étalé* of a sheaf, while the associated presheaf, satisfying locality and gluing, is what is known contemporarily as a sheaf. The equivalence of categories above tells us it doesn't quite matter which one we talk about, but it's important to keep in mind our definition is technically different than the modern one.

Remark 1.1.30. The big advantage now is that we can describe a sheaf *in terms of its sections*! We don't even need the sections to form a sheaf, we can always construct a sheaf out of a presheaf if need be.

Example 1.1.31 (Orientation sheaf). Let M be an n -dimensional manifold, and define a sheaf by the presheaf

$$\begin{aligned} o_M: \text{Open}(M)^{\text{op}} &\rightarrow \text{Ab} \\ U &\mapsto H_n(M, M - U; \mathbb{Z}). \end{aligned}$$

We call this the *orientation sheaf* of M (something to check: relative homology classes satisfy locality and gluing). Then an *orientation* of M is exactly a global section $\omega \in \Gamma(M, o_M)$ with the property that the stalk of ω at each $x \in M$ is a generator of the abelian group $o_{M,x} = H_n(M, M - \{x\}; \mathbb{Z}) \cong \mathbb{Z}$. This is just a rephrasing of [Hat01, pp. 233–236].

Remark 1.1.32. The *espace étalé* of the orientation sheaf is *not* the double cover of M , since this will not be a sheaf of abelian groups, only a sheaf of $\mathbb{Z}/2$ -torsors. The *espace étalé* of $\mathcal{o}_M$ will have fiber $\mathbb{Z}$ (if we are defining $\mathcal{o}_M$ by homology with integral coefficients) and will be the infinite-sheeted cover corresponding to the orientation character $\pi_1(M) \rightarrow \text{Aut}(\mathbb{Z}) = \mathbb{Z}/2$ given by the first Stiefel-Whitney class. We'll learn more about this later on.

1.1.2 Digression: big and small sites

To briefly tie what we've been doing into the language of Grothendieck topologies, we have the slice category $\text{Top}/_X$ of spaces over X , which comes equipped with a Grothendieck topology given by open subsets. We could call $\text{Top}/_X$ the *big site*, and in this terminology $\text{Open}(X) \subseteq \text{Top}/_X$ is the *small site*, which is the full subcategory on open subsets of X .

1.1.3 Sheaves of functions

The following is a space-level analogue of the idea that representable presheaves are sheaves.

Example 1.1.33 (Representable sheaves). Let X and Y be any spaces. Then the assignment

$$\begin{aligned} C^0(-, Y): \text{Open}(X)^{\text{op}} &\rightarrow \text{Set} \\ U &\mapsto \text{Hom}_{\text{Top}}(U, Y) \end{aligned}$$

is a sheaf of sets. We call this the *representable sheaf* attached to Y , although this is maybe a slight abuse of terminology?²

Proof. We just have to verify that the presheaf of functions valued in Y satisfies locality and gluing, which is immediate. $\square$

Question 1.1.34. When would this be a sheaf of groups or of abelian groups? That is, when can we endow $\text{Hom}_{\text{Top}}(U, Y)$ with a group structure compatibly for all U . One way to insist on this is to consider $\text{Hom}_{\text{Top}}(-, Y)$ by extending its domain to all of Top , and then ask for it to be valued in groups rather than sets. This is the same data (by the Yoneda lemma) as asking for Y to come equipped with continuous multiplication and inversion maps which turn Y into a group compatibly with its topology. This is the data of being a *topological group*, which we'll define in [Definition 1.5.1](#).

Example 1.1.35. Let X be any space, and denote by

$$\mathbb{C}_c: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$$

the sheaf sending $U \subseteq X$ to the abelian group of complex-valued continuous functions $U \rightarrow \mathbb{C}$. This is the “representable sheaf” attached to $\mathbb{C}$, in the sense of [Example 1.1.33](#). We call $\mathbb{C}_c$ the *sheaf of germs of local complex-valued continuous functions*, following [\[Hir78, p. 23\]](#).

In general for spaces, we can take a sheaf of continuous functions valued in $\mathbb{R}$, in $\mathbb{C}$, in some other space, whatever we like. If X has more structure, we can ask for these functions to respect the additional structure!

Example 1.1.36. Let X be a differentiable manifold. We denote by $\mathbb{C}_b$ the sheaf of local complex-valued differentiable functions.³

²Technically speaking, Y should be viewed as a representable sheaf on $\text{Top} = \text{Top}/_*$, which is the big site over the one-point space. We can pull this back along the projection $X \rightarrow *$ and we get a sheaf $\text{Hom}_{\text{Top}/_X}(-, X \times Y)$, which, when restricted to the small site $\text{Open}(X)$, agrees with $\text{Hom}_{\text{Top}}(-, Y)$ by adjunction.

³Recall a complex-valued function is differentiable if and only if both its real and imaginary parts are.

Example 1.1.37. Let X be a complex manifold. We let $\mathbb{C}_\omega$ denote the sheaf of local holomorphic functions. We will prefer the notation $\mathcal{O}_X$ in this class.

Proposition 1.1.38. If X is any complex manifold and $p \in X$ is any point, there is an isomorphism between $\mathcal{O}_{X,p}$ and the ring of convergent power series in n variables.

Proof. An element of $\mathcal{O}_{X,p}$ is an equivalence class of tuple (f, U) where $U \ni p$ is open and f is holomorphic on U . By cofinality we can assume that (U, p) is homeomorphic $(B, 0)$ where B is some open ball around the origin in Euclidean space. In particular f can be identified with its Taylor series expansion on the open ball given by its radius of convergence. We note that $(f_1, U_1) \sim (f_2, U_2)$ if the Taylor series expansions of f_1 and f_2 agree in a neighborhood of p . $\square$

Example 1.1.39. Similarly the stalks of the sheaf $\mathcal{M}_X$ of germs of meromorphic functions are given by the ring of convergent Laurent series with finitely many negative variables (finite principal parts) [For81, p. 42].

Exercise 1.1.40. If X is a compact connected complex manifold, show that $\Gamma(X, \mathbb{C}_\omega) = \mathbb{C}$.

1.1.4 Sheafification

Let $\text{Ab}(X)$ denote the category of sheaves of abelian groups. It is clearly a full subcategory of $\text{Fun}(\text{Open}(X)^{\text{op}}, \text{Ab})$. Our goal is to show the following:

Theorem 1.1.41. The category $\text{Ab}(X)$ is a localization of $\text{Fun}(\text{Open}(X)^{\text{op}}, \text{Ab})$, that is, the inclusion admits a left adjoint called *sheafification*. Moreover sheafification preserves stalks.

Our goal is to define the sheafification of a presheaf F over X . We claim we've already seen this construction, it's simply the space $\coprod_{x \in X} F_x \rightarrow X$ we built previously.

Definition 1.1.42. Let $F: \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$ be a presheaf. We will define the *sheafification* of F to be the sheaf $F^\sharp$ defined as $\coprod_{x \in X} F_x$, topologized as in Definition 1.1.19.

Remark 1.1.43. We can interpret $F^\sharp(U) = \Gamma(U, \coprod_{x \in X} F_x)$ as the set of those tuples $(f_x)_{x \in U} \in \prod_{x \in U} F_x$ for which, for any $x \in U$ there exists an open neighborhood $V \ni x$ and a section $s \in F(V)$ satisfying $s_v = f_v$ for all $v \in V$ (see e.g. [Aut, 007X]).

There is clearly a natural map of presheaves $F \rightarrow F^\sharp$, and we've already seen it! We called it θ_U in Equation (1.1.22). It induces an isomorphism on stalks by definition. The thing to check is that the sheafification construction is functorial (omitted) and defines a left adjoint to the inclusion of sheaves in presheaves. Adjunction follows from the following universal property:

Proposition 1.1.44. Let $F \in \text{Fun}(\text{Open}(X)^{\text{op}}, \text{Ab})$ and $G \in \text{Ab}(X)$. Then any map $F \rightarrow G$ factors uniquely through $F^\sharp$.

Proof. By construction there is a commutative diagram

$$\begin{array}{ccc} F & \longrightarrow & F^\sharp \\ \downarrow & & \downarrow \\ G & \longrightarrow & G^\sharp, \end{array}$$

so it suffices to check that $G \rightarrow G^\sharp$ is an equivalence of sheaves, but this is true because it is a stalkwise isomorphism.⁴ $\square$

⁴This wouldn't work if G was just a presheaf, as although $G \rightarrow G^\sharp$ is a stalkwise isomorphism that does not imply it is an isomorphism of presheaves, only of sheaves.

Exercise 1.1.45 (Constant sheaf). Let $A \in \mathbf{Ab}$. Then there is a silly presheaf on X which sends every open subset $U \subseteq X$ to A , and every restriction map to the identity on A . We call this the *constant presheaf* valued at A . We will denote by $\underline{A}$ the sheafification of this presheaf, and call this the *constant sheaf* valued at A . By construction.

1. Verify that, under the isomorphism of categories [Corollary 1.1.28](#), the constant sheaf $\underline{A}$ corresponds to the sheaf [Example 1.1.3](#) defined as $X \times A \rightarrow X$, where A has the discrete topology.
2. Show that $\Gamma(U, \underline{A})$ can be described as locally constant functions valued in A .

1.1.5 Subsheaves, kernels and cokernels

Notation 1.1.46. For a topological space X , we denote by $\mathbf{Ab}(X)$ the category of *sheaves of abelian groups* on X , and morphisms between them. Again following the isomorphism of categories of [Corollary 1.1.28](#) we're allowed to work with local homeomorphisms of abelian groups over X or presheaves on X satisfying the sheaf condition. So every definition and statement we make here will have two equivalent formulations in each model of sheaves. We will provide both at the start but will begin dropping one for convenience as we continue.

Definition 1.1.47 (Subsheaf). Let $F: \mathbf{Open}(X)^{\text{op}} \rightarrow \mathbf{Ab}$ be a sheaf. We say that a sheaf G is a *subsheaf* of F if $G(U) \leq F(U)$ is a subgroup for each $U \in \mathbf{Open}(X)$, and for any $V \subseteq U$, the restriction map $\text{res}_U^V: F(U) \rightarrow F(V)$ is equal to the restriction map $G(U) \rightarrow G(V)$ for G .

Definition 1.1.48 (Subsheaf, espace étalé model). We say that a sheaf $\pi_1: S_1 \rightarrow X$ is a *subsheaf* of $\pi_2: S_2 \rightarrow X$ if

1. $S_1 \subseteq S_2$ is an open subspace
2. $\pi_1 = \pi_2|_{S_1}$
3. The stalk $(S_1)_x$ is a subgroup of the stalk $(S_2)_x$ for all $x \in X$.

Example 1.1.49. Let X be a complex manifold.

1. We can think of subsheaves

$$\mathcal{O}_X \subseteq \mathbb{C}_b \subseteq \mathbb{C}_c$$

as the subsheaves of complex valued continuous functions of differentiable, and then holomorphic sections.

2. Let $\mathcal{M}_X$ denote the sheaf of local meromorphic sections. Then $\mathcal{O}_X \subseteq \mathcal{M}_X$ is a subsheaf, since every holomorphic function is meromorphic.
3. Let $\mathcal{O}_X^*$ denote the sheaf of germs of nowhere-vanishing holomorphic functions:

$$\mathcal{O}_X^*(U) := \{f: U \rightarrow \mathbb{C} \text{ holomorphic, and } f(p) \neq 0 \text{ for all } p \in U\}.$$

Note that, while we have a subset inclusion $\mathcal{O}_X^*(U) \subseteq \mathcal{O}_X(U)$ for every U , this is only a subsheaf of *sets*. It is not a subsheaf of groups because the group operation in $\mathcal{O}_X^*(U)$ is multiplicative, while in $\mathcal{O}_X(U)$ it is additive.

Example 1.1.50. For X a complex manifold and $p \in X$, we have that the stalk $\mathcal{O}_{X,p}^*$ can be identified with the multiplicative group of convergent power series around p whose constant term is nonzero (c.f. [Proposition 1.1.38](#)).

Just as subgroups of abelian groups are kernels of group homomorphisms, we can construct subsheaves as kernels of abelian sheaf homomorphisms.

Proposition 1.1.51. If $f: A \rightarrow B$ is a morphism of sheaves over X , we denote by $\ker(f)$ the *kernel*, defined to be the sheaf of A consisting of those points $a \in A$ so that $f_{\pi(a)}(a) = 0$ in $B_{\pi(a)}$.

Proof. We have to verify this is a subsheaf. Let K denote the kernel of f , and let's write out the diagram so we can picture it:

$$\begin{array}{ccccc} K & \longrightarrow & A & \xrightarrow{f} & B \\ & \searrow \pi_K & \downarrow \pi_A & \swarrow \pi_B & \\ & & X & & \end{array}$$

It is clear that π_K is just π_A restricted to K by construction, and moreover the stalks K_x are subgroups of A_x , again by construction. So the only thing to verify is that $K \subseteq A$ is an open subspace. To see this, we note that the kernel of f is the preimage of zero in B , which is the image of the zero section $z_B: X \rightarrow B$. This is open in B by [Exercise 1.1.7](#), and $K = f^{-1}(\text{im}(z_B))$, which will also be open since f is continuous. $\square$

Exercise 1.1.52.

1. Let $f: S_1 \rightarrow S_2$ be a morphism of sheaves over X . Give a reasonable definition of an *image* subsheaf $\text{im}(f) \subseteq S_2$.
2. Give a reasonable definition of a *cokernel* sheaf $\text{coker}(f)$ over f .
3. Prove an analogue of the first isomorphism theorem for sheaves of abelian groups over spaces.
4. How does your first iso theorem look on stalks? (c.f. [Definition 1.1.12](#))

Now that we have a notion of *subsheaf* and *cokernel*, we can build short exact sequences of sheaves out of subsheaves (recall the definition of exactness from [Definition 1.1.12](#)).

Example 1.1.53. Let $\mathbb{C}_c^*$ be the sheaf of germs of locally never zero complex valued continuous functions. There is a morphism of sheaves $\exp$, defined on sections by

$$\begin{aligned} \mathbb{C}_c(U) &\rightarrow \mathbb{C}_c^*(U) \\ f &\mapsto \exp(2\pi i \cdot f). \end{aligned}$$

The kernel of this is isomorphic to the constant sheaf $\mathbb{Z}$, because f and $f + n$ map to the same thing for any $n \in \mathbb{Z}$. We claim that we get a short exact sequence of sheaves

$$0 \rightarrow \underline{\mathbb{Z}} \rightarrow \mathbb{C}_c \xrightarrow{\exp} \mathbb{C}_c^* \rightarrow 0,$$

where as a reminder $\underline{\mathbb{Z}}$ denotes the constant sheaf at the integers ([Exercise 1.1.45](#)). The fact that exponentiation is an epimorphism of sheaves is a stalkwise condition, saying that for every value $x \in X$, we can find a sufficiently small neighborhood in which we can take a branch and define $\log(z)$.

Example 1.1.54. Let X be any complex manifold. Let $\mathcal{O}_X \subseteq \mathcal{M}_X$ be the subsheaf of holomorphic sections inside meromorphic sections, as in [Example 1.1.49](#). Then we can *define* the *sheaf of principal parts* (see e.g. [\[Gun18, \(4.2\)\]](#)) as

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{M}_X \rightarrow \mathcal{P}_X \rightarrow 0.$$

The terminology comes from the *principal part* of a meromorphic function around a pole [\[SS03, p. 75\]](#).

Proposition 1.1.55. If X is a Riemann surface, and $p \in X$ is any point, we have that $\mathcal{P}_{X,p}$ is given by the ring of finite negative Laurent expansions in a local coordinate around p .

Proof. As taking stalks is exact, we can describe $\mathcal{P}_{X,p}$ as the cokernel of the natural map $\mathcal{O}_{X,p} \rightarrow \mathcal{M}_{X,p}$. This inclusion sits a Taylor series inside the world of convergent Laurent series, so the quotient is exactly those terms which do not come from Taylor series, i.e. the principal part of a convergent Laurent expansion. $\square$

Example 1.1.56. For X a complex manifold we define the *sheaf of divisors* Div_X to be the cokernel

$$0 \rightarrow \mathcal{O}_X^* \rightarrow \mathcal{M}_X^* \rightarrow \text{Div}_X \rightarrow 0.$$

By our understanding of the stalks of the first two sheaves we can understand the stalks of the latter.

Exercise 1.1.57. If X is a Riemann surface, show that $\text{Div}_{X,p} \cong \mathbb{Z}$ for every point $p \in X$.

1.2 Sheaf cohomology

A big result here that we won't prove is the following:

Theorem 1.2.1. We have that $\text{Ab}(X)$ is an abelian category.

Proof. This is a standard example of an abelian category. See [\[God58, II§2\]](#) or [\[Gro57\]](#). $\square$

An abelian category is a setting in which we can do *homological algebra*. In particular, we will be allowed to resolve sheaves by nicer sheaves, or ones which are easier to deal with in some respect.

Example 1.2.2 (Representability of sections). Let $U \subseteq X$ be an open set. Then the functor

$$\begin{aligned} \Gamma(U, -) : \text{Ab}(X) &\rightarrow \text{Ab} \\ S &\mapsto \Gamma(U, S) \end{aligned}$$

is corepresentable, meaning there exists a sheaf we will denote by $\mathbb{Z}_U$ for which $\text{Hom}_{\text{Ab}(X)}(\mathbb{Z}_U, -) \cong \Gamma(U, -)$ are naturally isomorphic as abelian groups.⁵

Proof. Denote by $h_U = \text{Hom}_{\text{Open}(X)}(-, U)$ the presheaf of sets⁶

$$h_U(V) = \begin{cases} * & V \subseteq U \\ \emptyset & V \not\subseteq U. \end{cases}$$

⁵This is poor notation. Better notation is to let $j: U \hookrightarrow X$ denote the inclusion, and write $j_*\mathbb{Z}_U$.

⁶Note this is *different* than the “representable” presheaf we defined in [Example 1.1.33](#). There we looked at all continuous maps into U , whereas here we only want the natural inclusion of V in U , if it exists.

By the Yoneda lemma, we have a natural bijection

$$\mathrm{Hom}_{\mathrm{Fun}(\mathrm{Open}(X)^{\mathrm{op}}, \mathrm{Set})}(h_U, F) \cong F(U) \in \mathrm{Set}$$

for any sheaf of sets F .

Let's now bootstrap to abelian groups. Let $\mathbb{Z}[h_U]$ be the free presheaf of abelian groups on the sheaf of sets h_U — by this we mean it is defined as

$$\mathbb{Z}[h_U](V) = \begin{cases} \mathbb{Z} & V \subseteq U \\ \{0\} & V \not\subseteq U. \end{cases}$$

It is clear then that we have a natural group isomorphism

$$\mathrm{Hom}_{\mathrm{Fun}(\mathrm{Open}(X)^{\mathrm{op}}, \mathrm{Ab})}(\mathbb{Z}[h_U], F) \cong F(U) \in \mathrm{Ab}$$

for any presheaf of abelian groups F . Finally if F was a sheaf, then any morphism of presheaves $\mathbb{Z}[h_U] \rightarrow F$ factors uniquely through the sheafification of $\mathbb{Z}[h_U]$ by [Proposition 1.1.44](#). We denote this by

$$\mathbb{Z}_U := (\mathbb{Z}[h_U])^\#.$$

It is clear now that we have a natural isomorphism $\mathbb{Z}_U \cong \Gamma(U, -)$ (c.f. [\[Aut, 03CP, 03CQ\]](#)). $\square$

Remark 1.2.3. We can describe $\mathbb{Z}_U(V)$ explicitly via our understanding of sheafification in [Remark 1.1.43](#) — it is the abelian group of locally constant functions $f: U \cap V \rightarrow \mathbb{Z}$ which extend to a locally constant function on all of V , but where the function satisfies $f(v) = 0$ if $v \in V \setminus (U \cap V)$. This latter condition is equivalent to the statement that the support of f is closed in V , which is often how you'll see $\mathbb{Z}_U(V)$ described:

$$\mathbb{Z}_U(V) = \left\{ \begin{array}{l} \text{locally constant functions } U \cap V \rightarrow \mathbb{Z} \\ \text{with support closed in } V \end{array} \right\}$$

Observe that if $U_1 \subseteq U_2$ then for any sheaf F we have a restriction map $F(U_2) \rightarrow F(U_1)$, which is of the form $\mathrm{Hom}_{\mathrm{Ab}(X)}(\mathbb{Z}_{U_2}, F) \rightarrow \mathrm{Hom}_{\mathrm{Ab}(X)}(\mathbb{Z}_{U_1}, F)$ — by Yoneda this is precomposition with a morphism of sheaves $\mathbb{Z}_{U_1} \rightarrow \mathbb{Z}_{U_2}$. Therefore the assignment $U \mapsto \mathbb{Z}_U$ extends to a functor

$$\begin{aligned} \mathrm{Open}(X) &\rightarrow \mathrm{Ab}(X) \\ U &\mapsto \mathbb{Z}_U \end{aligned} \tag{1.2.4}$$

We'll use this later on.

Exercise 1.2.5. Verify the claim in [\[Gro57, 1.9.2\]](#) that the collection $\{\mathbb{Z}_U\}_{U \in \mathrm{Open}(X)}$ forms a system of *generators* for the category $\mathrm{Ab}(X)$.

1.2.1 Čech cohomology of a presheaf

Definition 1.2.6. Let $\mathcal{U} = \{U_i\}_{i \in I}$ be a cover of X , and let F be a presheaf of abelian groups.

1. For every $q \geq 0$ we define the *group of q -cochains* to be the abelian group:

$$C^q(\mathcal{U}, F) = \prod_{i_0, \dots, i_q \in I} F(U_{i_0} \cap \dots \cap U_{i_q}). \tag{1.2.7}$$

2. For each $q \geq 0$ we define the *coboundary map* to be the group homomorphism

$$\delta^q: C^q(\mathcal{U}, F) \rightarrow C^{q+1}(\mathcal{U}, F), \quad (1.2.8)$$

defined by the formula

$$(\delta^q f)(i_0, \dots, i_{q+1}) = \sum_{k=0}^{q+1} (-1)^k f(i_0, \dots, \widehat{i_k}, \dots, i_{q+1}).$$

The element $f(i_0, \dots, \widehat{i_k}, \dots, i_{q+1})$ is naturally valued in $F(U_{i_0} \cap \dots \cap U_{i_{k-1}} \cap U_{i_{k+1}} \cap \dots \cap U_{i_{q+1}})$, but we want to think about it as being valued in $F(\bigcap_{j=0}^{q+1} U_{i_j})$ under the restriction map in order to form the sum above. We've chosen to suppress this restriction from the notation, but Hirzebruch includes it (c.f. [Hir78, p. 26]).

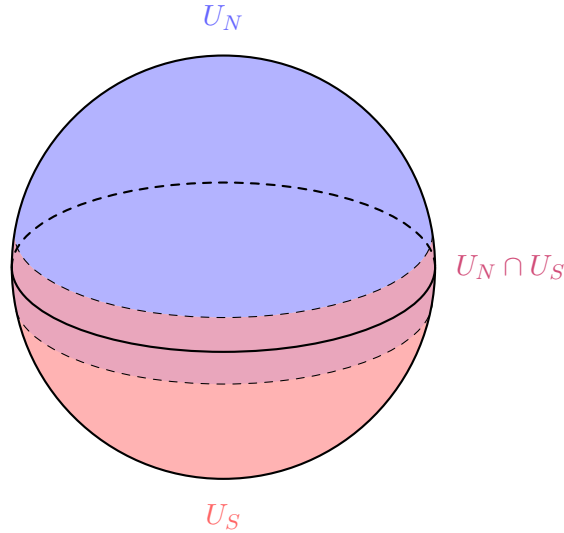
Exercise 1.2.9. Show that $\delta^{q+1} \circ \delta^q = 0$, so that we get a cochain complex

$$\dots \rightarrow C^{q-1}(\mathcal{U}, F) \xrightarrow{\delta^{q-1}} C^q(\mathcal{U}, F) \xrightarrow{\delta^q} C^{q+1}(\mathcal{U}, F) \xrightarrow{\delta^{q+1}} \dots \quad (1.2.10)$$

Definition 1.2.11. We define the *Cech cohomology* of the presheaf F over the cover $\mathcal{U}$ to be

$$\hat{H}^q(\mathcal{U}, F) = \ker(\delta^q) / \text{im}(\delta^{q-1}).$$

Example 1.2.12. Consider the sphere S^2 , equipped with an open cover $\mathcal{U} = \{U_n, U_s\}$, where U_n and U_s denote the northern and southern hemispheres, respectively, plus a little overlap on the equator. Let's compute the sheaf cohomology in the constant sheaf $\underline{\mathbb{Z}}$ which we defined in [Exercise 1.1.45](#):



We have that:

- ▷ $\check{C}^0(\mathcal{U}, \underline{\mathbb{Z}}) = \underline{\mathbb{Z}}(U_n) \times \underline{\mathbb{Z}}(U_s)$. Since each of these open sets is connected, a locally constant function $U_n \rightarrow \mathbb{Z}$ is just constant – so we get that $\underline{\mathbb{Z}}(U_n) \cong \mathbb{Z}$ by evaluation at the north pole, say, and similarly for U_s . Hence $\check{C}^0(\mathcal{U}, \underline{\mathbb{Z}}) \cong \mathbb{Z} \times \mathbb{Z}$.
- ▷ $\check{C}^1(\mathcal{U}, \underline{\mathbb{Z}}) = \underline{\mathbb{Z}}(U_n \cap U_s)$, since there is only one double overlap to consider. Again since it is connected, we get that $\check{C}^1(\mathcal{U}, \underline{\mathbb{Z}}) \cong \mathbb{Z}$.

▷ $\check{C}^q(\mathcal{U}, \mathbb{Z}) = 0$ for $q \geq 2$ since there are no triple overlaps in our cover.

So our chain complex looks like

$$0 \rightarrow \mathbb{Z} \oplus \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0.$$

We want to understand the differential δ^0 . Given $f \in \check{C}^0(\mathcal{U}, \mathbb{Z})$, it consists of two functions which we'll call $f_n: U_n \rightarrow \mathbb{Z}$ and $f_s: U_s \rightarrow \mathbb{Z}$. They are both locally constant, hence constant. The differential is then of the form

$$\begin{aligned} \delta^0(f): U_n \cap U_s &\rightarrow \mathbb{Z} \\ u &\mapsto f_n(u) - f_s(u). \end{aligned}$$

Altogether we get the chain complex

$$0 \rightarrow \mathbb{Z} \oplus \mathbb{Z} \xrightarrow{\begin{pmatrix} 1 \\ -1 \end{pmatrix}} \mathbb{Z} \rightarrow 0.$$

So we get

$$\check{H}^q(\mathcal{U}, \mathbb{Z}) = \begin{cases} \mathbb{Z} & q = 0 \\ 0 & q = 1 \\ 0 & q \geq 2 \end{cases}$$

Definition 1.2.13. Let X be a space, and let $\mathcal{U} = \{U_i\}_{i \in I}$ and $\mathcal{V} = \{V_j\}_{j \in J}$ be two open covers of X . We say that $\mathcal{V}$ is a *refinement* of $\mathcal{U}$ if for every $V \in \mathcal{V}$ there exists some $U \in \mathcal{U}$ so that $V \subseteq U$.

Exercise 1.2.14. Recall the cover $\mathcal{U}$ of the 2-sphere we constructed in [Example 1.2.12](#). Find a refinement $\mathcal{V}$ of $\mathcal{U}$ for which the Čech cohomology is exactly

$$\check{H}^q(\mathcal{V}, \mathbb{Z}) = \begin{cases} \mathbb{Z} & q = 0 \\ 0 & q = 1 \\ \mathbb{Z} & q = 2 \\ 0 & q > 2. \end{cases}$$

Definition 1.2.15. Let X be a topological space. We denote by $\text{Cov}(X)$ the poset (category) of open covers of X under refinement. That is, an object of $\text{Cov}(X)$ is an open cover $\mathcal{U}$ of X , and for any two covers $\mathcal{U}, \mathcal{V} \in \text{Cov}(X)$, we have exactly one morphism $\mathcal{V} \rightarrow \mathcal{U}$ if $\mathcal{V}$ refines $\mathcal{U}$, otherwise there are no morphisms from $\mathcal{V}$ to $\mathcal{U}$.

Exercise 1.2.16. If X is a space, F is a presheaf on X , and $\mathcal{U}, \mathcal{V}$ are open covers of X for which $\mathcal{V}$ is a refinement of $\mathcal{U}$, define a natural restriction morphism

$$\check{H}^q(\mathcal{U}, F) \rightarrow \check{H}^q(\mathcal{V}, F).$$

Hint: You should need to make choices in your construction – argue that the map ultimately doesn't depend on the choices you make.

In particular, the exercise above indicates that Čech cohomology is functorial in the cover. This motivates the following definition:

Definition 1.2.17. We define the *Čech cohomology* of a (pre)sheaf F on X by

$$\check{H}^q(X, F) := \operatorname{colim}_{\mathcal{U} \in \operatorname{Cov}(X)} \check{H}^q(\mathcal{U}, F).$$

That is, it is the colimit of the Čech cohomology along all possible covers, filtered by refinement.

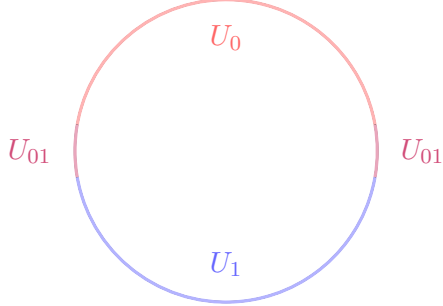
We have very little control over these refinement maps, as the following result illustrates:

Proposition 1.2.18. Let $\mathcal{V}$ be a refinement of the cover $\mathcal{U}$ on a space X . Then:

- ▷ the refinement map $\check{H}^0(\mathcal{U}, F) \rightarrow \check{H}^0(\mathcal{V}, F)$ is an isomorphism
- ▷ the refinement map $\check{H}^1(\mathcal{U}, F) \rightarrow \check{H}^1(\mathcal{V}, F)$ is injective
- ▷ higher refinement maps $\check{H}^q(\mathcal{U}, F) \rightarrow \check{H}^q(\mathcal{V}, F)$ need not be injective for $q \geq 2$. An explicit example is on Math Overflow here: [MO462472](#).

Proof. The statement $q = 0$ will follow from [Proposition 1.2.20](#). For $q = 1$ a nice proof is in [For81, 12.4]. $\square$

Example 1.2.19. On S^1 , we can take a simple cover consisting of only all of S^1 , letting $F = \mathbb{Z}$ be the constant sheaf $\mathbb{Z}$. Then the Čech cochains complex for this is the chain complex $\mathbb{Z}[0]$ concentrated in degree zero. This gives $\check{H}^0(\mathcal{U}, F) = \mathbb{Z}$ and vanishing cohomology elsewhere. On the other hand, we can take a cover $S^1 = U_0 \cup U_1$, where U_0 and U_1 are the upper and lower semicircle, plus a little extra so that they overlap:



Note that $\check{C}^1(\mathcal{U}, F) = \mathbb{Z}^{\times 2}$ in this instance, since $U_0 \cap U_1$ has two connected components. The chain complex here then looks like

$$\check{C}^\bullet(\mathcal{U}, F) : \quad 0 \rightarrow \mathbb{Z} \oplus \mathbb{Z} \xrightarrow{\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}} \mathbb{Z} \oplus \mathbb{Z} \rightarrow 0 \rightarrow \dots$$

so we get cohomology in degrees 0 and 1, as expected. This is an example of how *refining* the cover can detect more cohomological features.

Proposition 1.2.20. For any space X and any sheaf $F \in \operatorname{Ab}(X)$, we have a canonical isomorphism of abelian groups

$$\check{H}^0(X, F) \cong \Gamma(X, F).$$

Proof. Given any open cover $\mathcal{U} = \{U_i\}$, an element $f \in \check{H}^0(\mathcal{U}, F)$ is an element $f_i \in F(U_i)$ for each i (this is the condition of lying in $C^0(\mathcal{U}, F)$), which lies in the kernel of $\delta^0: C^0 \rightarrow C^1$. This means that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for each i, j . Since F is a sheaf, by gluing, the f_i 's glue together to give a well-defined global section $f \in \Gamma(X, F)$. This is easily seen to be an isomorphism. $\square$

Exercise 1.2.21. Verify that Čech cohomology is functorial in the sheaf, that is for any X and any $q \geq 0$ we have a functor

$$\check{H}^q(X, -): \text{Ab}(X) \rightarrow \text{Ab}.$$

Before we get to **Proposition 1.2.24** it's worth saying some words about exactness of sheaves versus presheaves.

Remark 1.2.22 (On exactness). By definition, a sequence of presheaves $F_1 \rightarrow F_2 \rightarrow F_3$ on X is exact if

$$0 \rightarrow F_1(U) \rightarrow F_2(U) \rightarrow F_3(U) \rightarrow 0$$

is an exact sequence of abelian groups for all $U \in \text{Open}(X)$. If these are all sheaves, then we say the sequence is exact if

$$0 \rightarrow (F_1)_x \rightarrow (F_2)_x \rightarrow (F_3)_x \rightarrow 0$$

is exact for every point $x \in X$. If a sequence of sheaves is exact (when considered as presheaves) then it is still exact when considered as sheaves, e.g. by **Corollary A.1.7** or by the phrase “sheafification is exact.” The converse *does not hold*! Consider the short exact sequence of sheaves

$$0 \rightarrow \underline{\mathbb{Z}} \rightarrow C^0(-, \mathbb{R}) \rightarrow C^0(-, S^1) \rightarrow 0.$$

This is exact (over any space X) since it is exact in each fiber. However over an arbitrary space it won't be exact as a sequence of presheaves, because not every circle-valued function lifts to a real-valued function (e.g. if it has non-trivial winding number).⁷

Exercise 1.2.23. If $F \rightarrow G$ is a map of presheaves which is a monomorphism of sheaves (it is injective on every stalk) then it is a monomorphism of presheaves (it is injective on every section).

Proposition 1.2.24. If

$$0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$$

is a short exact sequence of presheaves, it induces a long exact sequence

$$\begin{aligned} 0 \rightarrow \check{H}^0(X, F_1) &\rightarrow \check{H}^0(X, F_2) \rightarrow \check{H}^0(X, F_3) \\ &\rightarrow \check{H}^1(X, F_1) \rightarrow \check{H}^1(X, F_2) \rightarrow \check{H}^1(X, F_3) \\ &\rightarrow \check{H}^2(X, F_1) \rightarrow \cdots \end{aligned}$$

Sketch. The first step is to show that, for any cover $\mathcal{U}$, the functor

$$\check{C}^q(\mathcal{U}, -): \text{Fun}(\text{Open}(X)^{\text{op}}, \text{Ab}) \rightarrow \text{Ab}$$

is exact. A diagram chase then implies we get a long exact sequence on $\check{H}^*(\mathcal{U}, -)$. Then we take a colimit over all $\mathcal{U}$ in the filtered category of covers $\text{Cov}(X)$, and use that filtered colimits preserve exactness to conclude (**Example A.1.4** and **Corollary A.1.7**). The details are spelled out in [Hir78, pp. 28–29]. $\square$

⁷Indeed we can think of the winding number as measuring the failure of lifting, giving us a class in $H^1(X, \underline{\mathbb{Z}}) = [X, S^1]$.

This begs a very interesting question — when do the Čech cohomology of a presheaf F and of its sheafification $F^\sharp$ agree? We need two preliminary results to lead up to a partial answer to this question.

Proposition 1.2.25. Let F be a presheaf over a space, and suppose $F^\sharp = 0$. For any point $x \in X$, any open $U \ni x$ and any $f \in F(U)$, there exists some smaller open neighborhood $V \subseteq U$ containing x for which $f|_V = 0$.

Proof. This is because the stalk $F_x = F_x^\sharp$ must be zero. □

Lemma 1.2.26. Let X be a paracompact space, and F a presheaf on X . Then if $F^\sharp$ is the zero sheaf, we claim that $\check{H}^n(X, F) = 0$ for all $n \geq 0$.

Proof. We show that no cochains survive refinement. That is, if $f \in \check{C}^q(\mathcal{U}, F)$ for some cover $\mathcal{U} = \{U_i\}_{i \in I}$, we will find a refinement of the cover which kills f . Without loss of generality assume that $\mathcal{U}$ is locally finite. By the Dieudonné shrinking theorem ([Theorem A.2.11](#)), we may find a cover $\mathcal{W} = \{W_i\}_{i \in I}$ for which $\overline{W_i} \subseteq U_i$.

Let $J = X$ as a set — we will produce a refinement of the cover $\mathcal{U}$, now indexed over J . For each $x \in X$, we pick an open neighborhood $V_x \ni x$ for which:

1. if $x \in U_i$ then $V_x \subseteq U_i$ and if $x \in W_i$ then $V_x \subseteq W_i$
2. if $V_x \cap W_i \neq \emptyset$ then $V_x \subseteq U_i$
3. if $x \in U_{i_0} \cap \cdots \cap U_{i_q}$ then $f(i_0, \dots, i_q)$ restricts to zero in $F(V_x)$.

Since $\mathcal{U}$ and $\mathcal{W}$ are locally finite, there are only finitely many U_i and W_i containing x to worry about, so the first two conditions can be fulfilled. Why are we allowed to guarantee the last condition can hold? By [Proposition 1.2.25](#) some sufficiently small neighborhood of x in $U_{i_0} \cap \cdots \cap U_{i_q}$ will have the property that $f(i_0, \dots, i_q)$ restricts to zero in it. We can pick V_x to be that neighborhood, perhaps intersecting it further finitely many times to make it satisfy the first two conditions.

We now claim that f refines to zero in this new cover $\{V_x\}_{x \in X}$. Pick any tuple $(x_0, \dots, x_q)$, and let i_j be such that $x_j \in W_{i_j} \subseteq U_{i_j}$ for each j . By (1), this tells us that $V_{x_j} \subseteq W_{i_j} \subseteq U_{i_j}$. We want to argue that the image of $f(i_0, \dots, i_q)$ under the restriction map

$$F(U_{i_0} \cap \cdots \cap U_{i_q}) \rightarrow F(V_{x_0} \cap \cdots \cap V_{x_q})$$

is zero. If the intersection of the V_{x_i} 's is empty, there is nothing to show, so suppose it is nonempty. Then $V_{x_0} \cap V_{x_k}$ is nonempty for each k . Since $V_{x_k} \subseteq W_{i_k}$, we have that $V_{x_0} \cap W_{i_k} \neq \emptyset$, and by point (2) this implies that $V_{x_0} \subseteq U_{i_k}$. Since this is true for all k , we have that $V_{x_0} \subseteq U_{i_0} \cap \cdots \cap U_{i_q}$. However by (3), since $x_0 \in V_{x_0} \subseteq U_{i_0} \cap \cdots \cap U_{i_q}$, we have that $f(i_0, \dots, i_q)$ restricts to zero on V_{x_0} and therefore on the smaller subset $V_{x_0} \cap \cdots \cap V_{x_q}$. □

Theorem 1.2.27. If X is paracompact, and F is a presheaf with corresponding sheaf $F^\sharp$, then the natural map $\check{H}^q(X, F) \rightarrow \check{H}^q(X, F^\sharp)$ is an isomorphism for all $q \geq 0$.

Proof. Consider the sheafification map $F \rightarrow F^\sharp$ and take its kernel and cokernel in the category of presheaves

$$0 \rightarrow K \rightarrow F \rightarrow F^\sharp \rightarrow C \rightarrow 0.$$

As $F \rightarrow F^\sharp$ sheafifies to an isomorphism, and since sheafification is an exact functor, we have that K and C sheafify to zero. Hence their higher cohomology vanishes by [Lemma 1.2.26](#), and using the long exact sequence on cohomology attached to short exact sequences of presheaves, we obtain that the induced maps $\check{H}^q(X, F) \rightarrow \check{H}^q(X, F^\sharp)$ are all isomorphisms. □

So over paracompact spaces, there is no difference between the cohomology of a presheaf and of its resulting sheafification. A corollary of this we will use often is the following:

Proposition 1.2.28. If X is a paracompact space, and $0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$ is an exact sequence of *sheaves* on X , then it induces a long exact sequence

$$0 \rightarrow \check{H}^0(X, F_1) \rightarrow \check{H}^0(X, F_2) \rightarrow \check{H}^0(X, F_3) \rightarrow \check{H}^1(X, F_1) \rightarrow \dots$$

Proof. Let $F_1 \rightarrow F_2 \rightarrow F_3$ be a short exact sequence of *sheaves*. Define a *presheaf* F_2/F_1 to be the cokernel of the map $F_1 \rightarrow F_2$ in the category of sheaves. By [Proposition 1.2.24](#) we get a long exact sequence on Čech cohomology of presheaves

$$0 \rightarrow \check{H}^0(X, F_1) \rightarrow \check{H}^0(X, F_2) \rightarrow \check{H}^0(X, F_2/F_1) \rightarrow \check{H}^1(X, F_1) \rightarrow \dots$$

Note that F_1 and F_2 are already sheaves, and we claim that the natural map $(F_2/F_1)^\# \rightarrow F_3$ is an isomorphism. This is clear since sheafification is exact. Therefore by [Theorem 1.2.27](#) we have canonical isomorphisms $\check{H}^q(X, F_2/F_1) \xrightarrow{\sim} \check{H}^q(X, F_3)$ for any $q \geq 0$. Plugging these into the sequence above we get our desired result. $\square$

1.2.2 Acyclic resolutions

In classical topology, spaces were called *acyclic* if every cycle on the space was a boundary — in other words the space had the homology type of a point or any contractible space. Similarly a connective cochain complex can be considered acyclic if its cohomology is concentrated in degree zero. For Čech cohomology, we say a sheaf is acyclic if its Čech cochains complex is an acyclic complex. In other words:

Definition 1.2.29. A sheaf $F \in \text{Ab}(X)$ is *acyclic* if $\check{H}^p(X, F) = 0$ for $p > 0$.

Definition 1.2.30. Take $F \in \text{Ab}(X)$. An *acyclic resolution* of F is a (possibly infinite) exact sequence of sheaves

$$0 \rightarrow F \rightarrow A^0 \rightarrow A^1 \rightarrow \dots$$

where each A^i is acyclic for $i \geq 0$.

Exercise 1.2.31. Argue that the global sections functor

$$\Gamma(X, -): \text{Ab}(X) \rightarrow \text{Ab}$$

is left exact, and then argue that it sends an acyclic resolution of F to a chain complex in Ab .

It turns out we can use acyclic resolutions to compute cohomology!

Theorem 1.2.32 (Čech=sheaf cohomology). Let $F \in \text{Ab}(X)$ be a sheaf over a paracompact space, and consider any acyclic resolution $F \rightarrow A^\bullet$. Then $\check{H}^q(X, F)$ is canonically isomorphic to the cohomology of the chain complex obtained by applying $\Gamma(X, -)$ to the acyclic resolution.

Proof. We write out the sequence we're looking at:

$$\Gamma(X, A^0) \xrightarrow{d^0} \Gamma(X, A^1) \xrightarrow{d^1} \dots$$

We will proceed by induction on q .

Base case $q = 0$: Since $\Gamma(X, -F)$ is a left exact functor ([Exercise 1.2.31](#)), we get that

$$0 \rightarrow \Gamma(X, F) \rightarrow \Gamma(X, A^0) \xrightarrow{d_*^0} \Gamma(X, A^1)$$

is left exact. In particular, we get that the kernel of d_*^0 is exactly $\Gamma(X, F)$, which agrees with $\check{H}^0(X, F)$ by [Proposition 1.2.20](#). Hence $\check{H}^0(X, F)$ is the 0th cohomology of $\Gamma(X, -)$ applied to the acyclic resolution, establishing our base case.

Inductive step: For any $p \geq 0$, we let $K^p := \ker(A^p \rightarrow A^{p+1})$. Note by definition of our resolution, we have that $K^0 \cong F$. By exactness and the first isomorphism theorem, $A^p/K^p \cong K^{p+1}$, so we get short exact sequences of sheaves for each $p \geq 0$:

$$0 \rightarrow K^p \rightarrow A^p \rightarrow K^{p+1} \rightarrow 0.$$

Applying the long exact sequence on cohomology ([Proposition 1.2.28](#)), we get

$$\cdots \rightarrow \check{H}^q(X, K^p) \rightarrow \check{H}^q(X, A^p) \rightarrow \check{H}^q(X, K^{p+1}) \xrightarrow{\partial} \check{H}^{q+1}(X, K^p) \rightarrow \check{H}^{q+1}(X, K^{p+1}) \rightarrow \cdots$$

Since the A^p 's are acyclic, the boundary maps ∂ are isomorphisms $\check{H}^q(X, K^{p+1}) \cong \check{H}^{q+1}(X, K^p)$ for $q \geq 1$. Note though that $K^0 = F$, so we get that

$$\check{H}^1(X, K^{q-1}) \cong \check{H}^q(X, F) \quad \text{for } q \geq 1.$$

We claim $\check{H}^1(K^{q-1})$ is identified canonically with the q th cohomology of the sequence $\{\Gamma(X, A^i)\}$. Indeed this follows from looking at the long exact sequence attached to $0 \rightarrow K^{q-1} \rightarrow A^{q-1} \rightarrow K^q \rightarrow 0$, from which we get

$$H^0(X, A^{q-1}) \xrightarrow{d_*^{q-1}} H^0(X, K^q) \rightarrow H^1(X, K^{q-1}) \rightarrow 0.$$

Since $H^0(X, K^q) = \ker(d_*^q)$, we have that $H^1(X, K^{q-1})$ is $\ker(d_*^q)$ modulo the image of d_*^{q-1} , hence it is the cohomology of the complex. $\square$

This provides us a new way to compute Čech cohomology over a paracompact space!

Definition 1.2.33. Recall an *injective sheaf* is an injective object in $\text{Ab}(X)$, meaning that the functor

$$\text{Hom}_{\text{Ab}(X)}(-, I): \text{Ab}(X)^{\text{op}} \rightarrow \text{Ab}$$

is exact.

Example 1.2.34. Let I be an injective abelian group, and $i: \{x\} \rightarrow X$ the inclusion of a point. Then the skyscraper sheaf $i_*I \in \text{Ab}(X)$ is injective.

Proof. The data of a map into a skyscraper sheaf is completely determined by what happens on the stalk over x (this is a pullback-pushforward adjunction) – this gives us a natural isomorphism of functors $\text{Ab}(X) \rightarrow \text{Ab}$:

$$\text{Hom}_{\text{Ab}(X)}(-, i_*I) \cong \text{Hom}_{\text{Ab}}((-)_x, I).$$

the right hand side is exact since taking stalks is an exact functor, hence i_*I is injective. $\square$

Definition 1.2.35. Let $F \in \text{Ab}(X)$ be a sheaf. An *injective resolution* of F is an exact sequence

$$0 \rightarrow F \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

for which each sheaf I^j is injective.

Remark 1.2.36. In general, one deals with *injective resolutions* rather than acyclic ones. This allows us to reword the above in the language of derived categories, and define cohomology as the right derived functors of the global sections functor. In order to do this, we should show that injective sheaves are in fact acyclic.

First we need a lemma.

Lemma 1.2.37 ([Aut, 03AS, 03AU]). Let X be any space, and $\mathcal{U} = \{U_i\}$ any open cover. Define a chain complex of abelian sheaves $\mathbb{Z}_{\mathcal{U},\bullet}$ by

$$\mathbb{Z}_{\mathcal{U},p} = \bigoplus_{i_0, \dots, i_p} \mathbb{Z}_{U_{i_0} \cap \dots \cap U_{i_p}},$$

where recall $\mathbb{Z}_U$ is the sheaf corepresenting $\Gamma(U, -)$ as in [Example 1.2.2](#). The differentials on the chain complex $\mathbb{Z}_{\mathcal{U},\bullet}$ are the sum over $(-1)^k$ times the canonical map

$$\mathbb{Z}_{U_{i_0} \cap \dots \cap U_{i_p}} \rightarrow \mathbb{Z}_{U_{i_0} \cap \dots \cap \widehat{U_{i_k}} \cap U_{i_p}}.$$

Then we claim that $\text{Hom}_{\text{Ab}(X)}(\mathbb{Z}_{\mathcal{U},p}, F) \cong \check{C}^p(\mathcal{U}, F)$ for any $F \in \text{Ab}(X)$, and that the differentials $\mathbb{Z}_{\mathcal{U},p+1} \rightarrow \mathbb{Z}_{\mathcal{U},p}$ induce the Čech cochain differentials $\delta^p: \check{C}^p(\mathcal{U}, F) \rightarrow \check{C}^{p+1}(\mathcal{U}, F)$ after applying $\text{Hom}_{\text{Ab}(X)}(-, F)$.

Corollary 1.2.38. Every injective sheaf is acyclic.

Proof. If $I \in \text{Ab}(X)$ is injective, then $\text{Hom}_{\text{Ab}(X)}(-, I)$ is an exact functor. In particular it sends $\mathbb{Z}_{\mathcal{U},\bullet}$ to a cochain complex which is exact except for in the first term. Since this is exactly the Čech cochains complex, this implies that $\check{H}^q(\mathcal{U}, I) = 0$ for $q \geq 1$. As this is true for every $\mathcal{U}$, we conclude that $\check{H}^q(X, I) = 0$ for all $q \geq 1$. $\square$

The converse does not hold:

Example 1.2.39. Let A be a non-injective abelian group, and $i: \{x\} \rightarrow X$ the inclusion of a point. Then we claim the skyscraper sheaf i_*A is acyclic but need not be injective.

Proof. Take any cover of X and refine it so that $U_0 \ni x$ and $U_i \not\ni x$ for $i > 0$. In particular there are no sections of i_*A over overlaps, so $\check{H}^q(\mathcal{U}, i_*A) = 0$ for all $q > 0$. This remains true under refinement, so this vanishing passes to the colimit. Failure of injectivity of the sheaf i_*A reduces to failure of injectivity of A as an abelian group by looking at maps on stalks as in the proof of [Example 1.2.34](#). $\square$

So [Corollary 1.2.38](#) lets us conclude what we may have already expected:

Corollary 1.2.40. Over a paracompact Hausdorff space X , we have that Čech cohomology is canonically isomorphic to the right derived functors of global sections.

Remark 1.2.41. The contemporary perspective is to *first* define cohomology as these right derived functors, *then* to compare with Čech cohomology. The classical perspective of Serre, Godement and others, and what we presented above, is the exact opposite.

Exercise 1.2.42. Note that [Theorem 1.2.32](#) has already told us that cohomology can be computed using an injective resolution, and that this doesn't depend on the choice of injective resolution. The yoga of derived functors lets us conclude this also. If we wanted to try to prove it by hand, we could do it as follows: Prove that if $F \rightarrow I^\bullet$ is any injective resolution and $F \rightarrow J^\bullet$ is any other resolution, there exists a map of chain complexes $f^\bullet$ making the diagram commute:

$$\begin{array}{ccc} F & \longrightarrow & I^\bullet \\ & \searrow & \downarrow f^\bullet \\ & & J^\bullet \end{array}$$

Show that $f^\bullet$ is a quasi-isomorphism and moreover is well-defined up to chain homotopy.

1.2.3 Existence of injectives

We now know a new way to take cohomology, which is leveraging an acyclic resolution, or if we want to use some category theory, take the derived functors of global sections. This perspective is only valuable if we can actually find acyclic/injective resolutions. How do we know they even exist? Let's argue that they do.

Proposition 1.2.43. For every $F \in \text{Ab}(X)$, there exists an injective sheaf I and a monomorphism $F \rightarrow I$.

Proof. We can embed the abelian group F_x into an injective abelian group I_x . Define a presheaf by

$$\begin{aligned} \mathcal{I}: \text{Open}(X)^{\text{op}} &\rightarrow \text{Ab} \\ U &\mapsto \prod_{x \in U} I_x. \end{aligned}$$

Note that $\mathcal{I}$ is actually a sheaf, since it is a product of skyscraper sheaves, one for each point of X . Observe that the stalk of the sheaf $\mathcal{I}_x$ need not be equal to I_x , however it will contain I_x as a retract. We claim that

$$\text{Hom}_{\text{Ab}(X)}(F, \mathcal{I}) \cong \prod_{x \in X} \text{Hom}_{\text{Ab}}(F_x, I_x).$$

There is a map from the left to the right by taking $g: \mathcal{F}(U) \rightarrow \mathcal{I}(U)$ for any open $U \ni x$, then noting that $\mathcal{F}(U) \rightarrow \mathcal{I}(U)$ maps to I_x along the projection, then taking a colimit over U , inducing $\mathcal{F}_x \rightarrow \mathcal{I}_x$. For the reverse direction, if we have maps $f_x: F_x \rightarrow I_x$ for each x , we can define $\mathcal{F}(U) \rightarrow \mathcal{I}(U)$ by the composite

$$\mathcal{F}(U) \rightarrow \prod_{x \in U} F_x \xrightarrow{\prod f_x} \prod_{x \in U} I_x = \mathcal{I}(U),$$

and we have these maps on each section, which give a map of sheaves. We claim these are inverse operations.

We have to check two things: that $\mathcal{I}$ is injective, and that $F \rightarrow \mathcal{I}$ is a monomorphism. That it is a monomorphism can be seen sectionwise, which induces monomorphisms on stalks since filtered colimits preserve monomorphisms.

To see $\mathcal{I}$ is injective, it suffices to show observe each skyscraper sheaf is injective by [Example 1.2.34](#) and then use that injective objects are closed under arbitrary products. $\square$

Corollary 1.2.44. For any space X , we have that every $F \in \text{Ab}(X)$ admits an injective resolution.

Proof. By [Proposition 1.2.43](#), we start with

$$0 \rightarrow F \rightarrow I^0,$$

and then take the quotient I^0/F and embed it in some injective I^1 . The composite $F \rightarrow I^0 \rightarrow I^1$ is obviously zero since it factors through the quotient by construction, so we get

$$0 \rightarrow F \rightarrow I^0 \rightarrow I^1.$$

We then embed I^1/I^0 into some injective I^2 , and so on. □

1.2.4 Godement resolutions

We can expand the results of the previous section by establishing the so-called *Godement resolution*, which is a functorial way of producing an acyclic resolution. We first need some preliminary definitions:

Definition 1.2.45. Let $f: X \rightarrow Y$ be a continuous map of spaces.

- ▷ Given a sheaf $F \in \text{Ab}(X)$, we define the *pushforward sheaf* f_*F on Y by the rule

$$\begin{aligned} f_*F: \text{Open}(Y) &\rightarrow \text{Ab} \\ U &\mapsto F(f^{-1}(U)). \end{aligned}$$

We claim that f_*F satisfies locality and gluing (exercise) and therefore f_* induces a functor which we call *pushforward*:

$$f_*: \text{Ab}(X) \rightarrow \text{Ab}(Y)$$

- ▷ Given a sheaf $G \in \text{Ab}(Y)$, we define a presheaf on X by

$$\begin{aligned} f_{\text{pre}}^{-1}G: \text{Open}(X) &\rightarrow \text{Ab} \\ U &\mapsto \text{colim}_{\substack{V \in \text{Open}(Y) \\ V \supseteq f(U)}} G(V). \end{aligned}$$

We define the *inverse image sheaf* to be the sheafification of this, that is $f^{-1}G := (f_{\text{pre}}^{-1}G)^\sharp$. Again we claim this is functorial:

$$f^{-1}: \text{Ab}(Y) \rightarrow \text{Ab}(X).$$

Example 1.2.46. In the particular case when our map is of the form $i: \{x\} \rightarrow X$, that is, the inclusion of a point, we have that a sheaf on the one-point space is just a single abelian group A . The pushforward sheaf i_*A is *exactly* the skyscraper sheaf of A at the point $x \in X$. This is the reason for the notation for skyscraper sheaves in [Example 1.1.4](#).

Proposition 1.2.47. Let $f: X \rightarrow Y$ be any map of spaces. Then pullback and pushforward form an adjunction:

$$f^{-1}: \text{Ab}(Y) \rightleftarrows \text{Ab}(X): f_*.$$

Proof. For a proof (in slightly more generality), see the arguments in [Aut, 008P] and the previous sections. $\square$

Remark 1.2.48. Right adjoints preserve injective objects, so the pushforward of an injective sheaf is still injective. In particular pushforward of an injective abelian group (viewed as an injective sheaf over a point) is still injective. This is alternative way to see that skyscraper sheaves of injective abelian groups are injective sheaves (Example 1.2.34). And indeed, we essentially used the adjunction Proposition 1.2.47 to prove it.

An interesting case comes from *forgetting* the topology on X , and considering pullback-pushforward along the resulting natural map.

Notation 1.2.49. For any space X , we denote by X^δ the topological space with the exact same underlying set as X , but with the discrete topology (every subset is open). There is a natural map $\epsilon: X^\delta \rightarrow X$.⁸

We define a functor

$$\begin{aligned} C^0: \text{Ab}(X) &\rightarrow \text{Ab}(X) \\ F &\mapsto \epsilon_* \epsilon^{-1} F. \end{aligned}$$

As a presheaf, $\epsilon_* \epsilon^{-1} F$ sends U to $\prod_{x \in U} F_x$, that is, it is the sheaf of germs at all points. It is also clear that it comes equipped with a natural map of sheaves $j: F \rightarrow C^0(F)$ (which is the unit of the pullback-pushforward adjunction).

Proposition 1.2.50. Let X be an arbitrary space, and $F \in \text{Ab}(X)$ any sheaf of abelian groups.

1. The map $j: F \rightarrow C^0(F)$ is a monomorphism.
2. The sheaf $C^0(F)$ is acyclic

Proof. The first point is immediate – if $f, g \in F(U)$ have the property that $j(f) = j(g) \in C^0(F)(U) = \prod_{x \in U} F_x$, this means they have the same value on stalks of all points in U , which by the sheaf condition means they must agree. The difficult part is the second point, arguing that $C^0(F)$ is acyclic.

Fix any open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of X . We have that the Čech p -cochains of $C^0(F)$ can be rewritten (by swapping products around) as:

$$\begin{aligned} \check{C}^p(\mathcal{U}, C^0(F)) &= \prod_{i_0, \dots, i_p \in I} C^0(F)(U_{i_0} \cap \dots \cap U_{i_p}) = \prod_{i_0, \dots, i_p \in I} \prod_{x \in U_{i_0} \cap \dots \cap U_{i_p}} F_x \\ &= \prod_{x \in X} \prod_{\substack{i_0, \dots, i_p \in I \\ U_{i_0} \cap \dots \cap U_{i_p} \ni x}} F_x. \end{aligned}$$

So the Čech differential acts pointwise on each x . In other words if we define

$$\check{C}^p(\mathcal{U}, x, F) := \prod_{\substack{i_0, \dots, i_p \in I \\ U_{i_0} \cap \dots \cap U_{i_p} \ni x}} F_x,$$

⁸There is a “discrete-forgetful” adjunction $(-)^{\delta}: \text{Set} \rightleftarrows \text{Top}: U$, where U forgets the topology on a space and returns its underlying set, and $(-)^{\delta}$ endows a set with the discrete topology. This map $X^{\delta} \xrightarrow{\epsilon} X$ is simply the counit of this adjunction, hence is natural in X .

then it is clear that $\check{C}^\bullet(\mathcal{U}, C^0(F))$ is a product over the individual complexes $\check{C}^\bullet(\mathcal{U}, x, F)$. It therefore reduces to showing that $\check{C}^\bullet(\mathcal{U}, x, F)$ is an acyclic complex.

Fix an arbitrary point $x \in X$, and find one index $i' \in I$ for which $x \in U_{i'}$. Then (if we use *unnormalized Čech cochains*⁹) we can define a chain homotopy

$$h: \check{C}^p(\mathcal{U}, x, F) \rightarrow \check{C}^{p-1}(\mathcal{U}, x, F),$$

by defining $(hc)(i_0, \dots, i_{p-1}) = c(i', i_0, \dots, i_{p-1})$. We can check, for $p \geq 1$, this is a null-homotopy for the identity map on the cochain complex. Therefore $\check{C}^p(\mathcal{U}, x, F)$ has no cohomology above H^0 , and therefore neither does $\check{C}^p(\mathcal{U}, C^0(F))$. $\square$

Since C^0 is functorial, we can use it to (functorially!) build an acyclic resolution.

Notation 1.2.51. We define $C^1: \text{Ab}(X) \rightarrow \text{Ab}(X)$ to be the functor sending F to $C^0(\text{coker}(F \xrightarrow{j} C^0(F)))$, that is, we look at $j: F \rightarrow C^0(F)$, take its cokernel, and then hit it with C^0 . We get a composite map which we call d^0 :

$$C^0(F) \rightarrow \text{coker}(j) \rightarrow C^0(\text{coker}(j)) =: C^1(F),$$

and it is clear that the composite $F \rightarrow C^0(F) \rightarrow C^1(F)$ is zero, since it factors through the cokernel of F .

Iteratively, we define

$$C^{n+1}(F) = C^0(\text{coker}(C^{n-1}(F) \xrightarrow{d^{n+1}} C^n(F))),$$

which receives an analogous map d^n from $C^n(F)$.

Definition 1.2.52. We define the *Godement resolution* to be the acyclic resolution of F obtained as

$$0 \rightarrow F \rightarrow C^0(F) \xrightarrow{d^0} C^1(F) \xrightarrow{d^1} C^2(F) \rightarrow \dots$$

as above.

Remark 1.2.53. A few things about this are worth noting:

- ▷ each of these constructions C^n are *functorial*
- ▷ the functors $C^n(-): \text{Ab}(X) \rightarrow \text{Ab}(X)$ are exact for each n .

Exactness of $C^0(-)$ is immediate, since it is a stalkwise construction, and exactness of $C^n(-)$ can be proved inductively.

What is the benefit of all this nonsense? It allows us to give an analogue and a repro of **Proposition 1.2.28** (on a paracompact space, a short exact sequences of sheaves induces a long exact sequence on cohomology) for sheaf cohomology, and *without paracompactness*.

Proposition 1.2.54. Let X be any topological space, and let $0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$ be a short exact sequence of sheaves on X . This induces a long exact sequence on sheaf cohomology

$$0 \rightarrow H^0(X, F_1) \rightarrow H^0(X, F_2) \rightarrow H^0(X, F_3) \rightarrow H^1(X, F_1) \rightarrow \dots$$

⁹There are two notions of Čech cochains, one is *normalized*, where each of $i_0, \dots, i_p$ needs to be distinct, and the other is *unnormalized*, where we're allowed to have repetitions. They end up giving completely the same answer, but some operations like this one are much easier to state in the language of unnormalized cochains.

Proof. We build the Godement resolutions $F_i \rightarrow C^\bullet(X, F_i)$ functorially for each $i = 1, 2, 3$, and exactness of the C^n functors implies that $C^\bullet(X, F_1) \rightarrow C^\bullet(X, F_2) \rightarrow C^\bullet(X, F_3)$ is a short exact sequence of chain complexes. A standard diagram chase argument gives the desired long exact sequence on cohomology. $\square$

When X is paracompact, sheaf cohomology agrees with Čech cohomology, so the previous result lets us reprove [Proposition 1.2.28](#).

Remark 1.2.55. While $C^n(F)$ is acyclic, there is no reason to expect it to be injective. A modified construction works, however, to produce functorial injective resolutions:

1. begin with a functorial injective embedding of abelian groups into injective ones
2. modify the construction of $C^0(F)$ by applying this functor to each stalk
3. take the inclusion, cokernel, iterate.

Details are in [\[Aut, 01DG\]](#).

1.2.5 Fine sheaves

Another source of acyclic sheaves comes from so-called *fine sheaves* on paracompact spaces.

Definition 1.2.56. Let F be a sheaf over a paracompact space X . We say that F is *fine* if, for every locally finite open covering $\{U_i\}$ of X , we have a system of homomorphisms

$$h_i: F \rightarrow F,$$

so that

1. For every i , the support of h_i is closed in X and contained in U_i
2. $\sum_i h_i$ is the identity (the sum makes sense because the cover is locally finite).

Example 1.2.57. The sheaf $C^0(-, \mathbb{C})$ of germs of complex-valued continuous functions over a paracompact space is fine.

Proof. Fix a locally finite open cover $\{U_i\}$, and pick a partition of unity $\{\phi_i\}$. Then we can define

$$\begin{aligned} h_i: C^0(-, \mathbb{C}) &\rightarrow C^0(-, \mathbb{C}) \\ f &\mapsto f \cdot \phi_i. \end{aligned}$$

It is clear the h_i 's sum to the identity, and the closed subsets in the definition of fineness are given by their support. $\square$

Example 1.2.58. If X is a differentiable manifold, then the sheaf of germs of differentiable functions $\mathbb{C}_b$ is fine.

Proof. The same proof works, aside from one subtlety – in order for h to be a map of sheaves, we can't just multiply by arbitrary ϕ_i , since this might break differentiability— we need the ϕ_i 's themselves to be differentiable. Hence this hinges on the existence of a differentiable partition of unity, which was proven by de Rham (at least in the smooth setting? see [\[de 84, I.2\]](#)). $\square$

Remark 1.2.59. If X is a complex manifold and $\mathcal{O}_X$ is the sheaf of holomorphic functions, it is not necessarily the case that $\mathcal{O}_X$ is fine. An analogous argument to the above doesn't work because there are no holomorphic partitions of unity (this would violate the maximum modulus principle, for instance). Additionally, if $\mathcal{O}_X$ was fine then $H^1(X, \mathcal{O}_X)$ would vanish by [Theorem 1.2.61](#), but this isn't true when X is a Riemann surface of positive genus, for instance.

Proposition 1.2.60. Let F be fine over a paracompact Hausdorff space X , and let $Z \subseteq X$ any closed subspace. Define $\Gamma(Z, F)$ to be the colimit of sections defined in open neighborhoods around Z . Then the natural restriction map

$$\Gamma(X, F) \rightarrow \Gamma(Z, F)$$

is surjective, that is, every section defined around Z can be extended to a section on X .¹⁰

Proof. Pick any section $s \in \Gamma(Z, F)$, which by construction of the colimit is defined in some neighborhood $U \supseteq Z$. Pick an open cover $\{U_i\}$ of X , let $V_i = U \cap U_i$, and let s_i be the restriction of s to $U \cap U_i$. Finally let $V = X - Z$, so altogether we have an open cover $\{V\} \cup \{V_i\}_i$ of X . We may assume it is locally finite without loss of generality. Since F is fine, we may pick h_i 's which vanish outside $V_i \cap Z$ for all i , and sum to the identity. We also have an h which vanishes outside some closed subset of V , we don't have any control over this. However the sum of all these h 's and h_i 's is the identity everywhere. Then we can take

$$\sigma = \sum h_i s_i.$$

Away from Z this is zero, and this is equal to s on Z . Hence $\sigma \mapsto s$ under the restriction $\Gamma(X, F) \rightarrow \Gamma(Z, F)$ and we are done. $\square$

Theorem 1.2.61. Let F be a fine sheaf on a paracompact space X . Then F is acyclic.

Proof. It suffices to show that $\check{H}^q(\mathcal{U}, F) = 0$ for all $q \geq 1$ and any locally finite open cover $\mathcal{U}$. We will define a *chain homotopy*

$$k^q: \check{C}^q(\mathcal{U}, F) \rightarrow \check{C}^{q-1}(\mathcal{U}, F)$$

using our functions h_i provided by the fineness of the sheaf F . Fix some $f \in \check{C}^q(\mathcal{U}, F)$, then we want to define $(k^q f)(i_0, \dots, i_{q-1})$. For an arbitrary $i \in I$, we define a section $t_f(i, i_0, \dots, i_{q-1}) \in F(U_{i_0} \cap \dots \cap U_{i_{q-1}})$ by

$$t_f(i, i_0, \dots, i_{q-1})(x) = \begin{cases} h_i \cdot f(i, i_0, \dots, i_{q-1}) & x \in U_i \cap U_{i_0} \cap \dots \cap U_{i_{q-1}} \\ 0 & x \notin U_i \cap U_{i_0} \cap \dots \cap U_{i_{q-1}}. \end{cases}$$

We then define

$$(k^q f)(i_0, \dots, i_{q-1}) = \sum_{i \in I} t_f(i, i_0, \dots, i_{q-1}).$$

This sum makes sense since $\mathcal{U}$ is locally finite. If $\delta^q: \check{C}^q(\mathcal{U}, F) \rightarrow \check{C}^{q+1}(\mathcal{U}, F)$ denotes the boundary operator, then we can verify (exercise) that, for $q \geq 1$ we get:

$$k^{q+1} \circ \delta^q + \delta^{q-1} \circ k^q = \text{id} = \text{id} - 0.$$

This gives us a chain homotopy between the identity and the zero map, implying that $\check{H}^q(\mathcal{U}, F) = 0$ for all $q \geq 1$. $\square$

Example 1.2.62. On any paracompact space X , the sheaves $C^0(-, \mathbb{R})$ and $C^0(-, \mathbb{C})$ are acyclic.

¹⁰The terminology for sheaves satisfying this condition is that they are *soft* — in that language, this proposition states “fine sheaves are soft.”

1.2.6 Leray covers

When is a short exact sequence of sheaves actually a short exact sequence of presheaves? This occurs when it is a *sectionwise* short exact sequence. We claim that a stalkwise monomorphism induces a sectionwise monomorphism, so if $F_1 \rightarrow F_2 \rightarrow F_3$ is a short exact sequence of sheaves, then by looking at the long exact sequence on *sheaf cohomology*¹¹, we obtain

$$0 \rightarrow F_1(U) \rightarrow F_2(U) \rightarrow F_3(U) \rightarrow H^1(U, F_1) \rightarrow \dots$$

hence failure of exactness as a sequence of presheaves over an open set U is measured by the vanishing of the sheaf cohomology group $H^1(U, F_1)$.

Now what about long exact sequences? Suppose for instance, we had an injective resolution of a sheaf F given by

$$0 \rightarrow F \rightarrow I^0 \xrightarrow{d^0} I^1 \rightarrow \dots$$

For an open set U , can we see whether

$$0 \rightarrow F(U) \rightarrow I^0(U) \rightarrow I^1(U) \rightarrow I^2(U) \rightarrow \dots \quad (1.2.63)$$

is long exact? Here is where we squint at [Equation \(1.2.63\)](#) and say that it looks an awful lot like the sequence used to compute the derived functors of $\Gamma(-, F)$, but now over the open subspace U instead of over X . Indeed it is, but only if we know that I^j is still injective (or at least acyclic) when considered as a sheaf on U . This wouldn't necessarily be true if we had used an acyclic resolution (a restriction of an acyclic resolution to an open subspace need not remain an acyclic resolution), but because we used an *injective* resolution, when we restrict to an open subspace it remains an injective resolution (exercise). In summary if $F \rightarrow I^\bullet$ is an injective resolution, then [Equation \(1.2.63\)](#) will be exact exactly if $F|_U$ is acyclic.

An interesting case emerges when $U = U_{i_0} \cap \dots \cap U_{i_q}$ is the overlap of some open spaces coming from an open cover $\mathcal{U}$ of X :

Proposition 1.2.64. Let X be a space, $\mathcal{U}$ an open cover, and F a sheaf. Suppose that the sheaf cohomology satisfies $H^p(U_{i_0 \dots i_q}, F|_{U_{i_0 \dots i_q}}) = 0$ for all $p \geq 1$ and all finite intersections of elements in our cover. Then for any injective resolution of sheaves

$$F \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

the sequence of Čech cochains is long exact for every q :

$$0 \rightarrow \check{C}^q(\mathcal{U}, F) \rightarrow \check{C}^q(\mathcal{U}, I^0) \rightarrow \check{C}^q(\mathcal{U}, I^1) \rightarrow \check{C}^q(\mathcal{U}, I^2) \rightarrow \dots \quad (1.2.65)$$

Proof. This follows from the formula $\check{C}^q(\mathcal{U}, G) = \prod_{i_0, \dots, i_q} G(U_{i_0 \dots i_q})$ for any sheaf G , together with the above discussion and the observation that an arbitrary product of long exact sequences of abelian groups is still long exact.¹² $\square$

The sheaves F satisfying this nice property deserve a special name for being so very special.

¹¹We don't know we get a long exact sequence on Čech cohomology for a short exact sequence of sheaves, only of presheaves.

¹²This is axiom AB4* in [Gro57]. Arbitrary products are always left exact, and axiom AB4 says arbitrary coproducts preserve monomorphisms. The dual axioms AB4* which Grothendieck leaves as an exercise to us is that arbitrary products preserve epimorphisms, hence are right exact. We can also just see this by hand in the category of abelian groups.

Definition 1.2.66. Let X be a space and $\mathcal{U}$ an open cover. We say F is *Leray* for $\mathcal{U}$ if

$$H^p(U_{i_0 \dots i_q}, F|_{U_{i_0 \dots i_q}}) = 0$$

for all $p \geq 1$ and all $q \geq 0$.

We'll now see why this key property of Leray covers is so important – it tells us we can compute Čech cohomology along a Leray cover!

Theorem 1.2.67. Let $\mathcal{U}$ be an open cover of a space X , and suppose F is a Leray cover with respect to $\mathcal{U}$. Then we obtain an isomorphism

$$\check{H}^q(\mathcal{U}, F) \xrightarrow{\sim} H^q(X, F)$$

for all $q \geq 0$.

Proof. Take any injective resolution $F \rightarrow A^\bullet$. By functoriality of Čech cochains and naturality of differentials we get a commutative infinite grid (see [Bre97, p. 193]):

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \Gamma(X, F) & \longrightarrow & \Gamma(X, I^0) & \longrightarrow & \Gamma(X, I^1) & \longrightarrow & \Gamma(X, I^2) & \longrightarrow & \dots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \check{C}^0(\mathcal{U}, F) & \longrightarrow & \check{C}^0(\mathcal{U}, I^0) & \longrightarrow & \check{C}^0(\mathcal{U}, I^1) & \longrightarrow & \check{C}^0(\mathcal{U}, I^2) & \longrightarrow & \dots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \check{C}^1(\mathcal{U}, F) & \longrightarrow & \check{C}^1(\mathcal{U}, I^0) & \longrightarrow & \check{C}^1(\mathcal{U}, I^1) & \longrightarrow & \check{C}^1(\mathcal{U}, I^2) & \longrightarrow & \dots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \check{C}^2(\mathcal{U}, F) & \longrightarrow & \check{C}^2(\mathcal{U}, I^0) & \longrightarrow & \check{C}^2(\mathcal{U}, I^1) & \longrightarrow & \check{C}^2(\mathcal{U}, I^2) & \longrightarrow & \dots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & \vdots & & \vdots & & \vdots & & \vdots
 \end{array}$$

The first row is not necessarily exact (it computes $H^*(X, F)$) nor is the first column necessarily exact (it computes $\check{H}^*(\mathcal{U}, F)$). However we claim all other rows and columns are exact. The columns are exact because they Čech cohomology in an injective sheaf, and we have seen this vanishes in [Corollary 1.2.38](#). The rows are exact by the Leray condition – this was the reason for [Equation \(1.2.65\)](#).

We claim now that the cohomology of the first row is isomorphic to the cohomology of the first column! This can be done by taking an element in a kernel/image and chasing it diagonally through the grid (TODO add details). $\square$

The procedure of computing Čech cohomology is a bit hairy. For a fixed cover $\mathcal{U}$ it is a fairly easy process to compute $\check{H}^*(\mathcal{U}, F)$, however the process of refining the cover to produce a colimit is a bit unclear. A massive upshot of [Theorem 1.2.67](#) is that, in the process of computing Čech cohomology over nice spaces, it gives you a way to tell when you've refined your cover *enough* to stop refining and call it a day!

Corollary 1.2.68. Let X be a paracompact Hausdorff space, and $F \in \text{Ab}(X)$. If $\mathcal{U}$ is any open cover for which F is Leray, then the natural colimiting map

$$\check{H}^q(\mathcal{U}, F) \rightarrow \check{H}^q(X, F)$$

is an isomorphism for all q .

Proof. This is a combination of [Theorem 1.2.67](#) and the comparison between Čech and sheaf cohomology on paracompact Hausdorff spaces we saw in [Corollary 1.2.40](#). $\square$

1.3 Comparison with singular cohomology

Theorem 1.3.1. If X is paracompact, Hausdorff, and locally contractible, and A is any abelian group, there is a natural isomorphism between singular cohomology and sheaf cohomology with constant coefficients

$$H_{\text{sing}}^*(X, A) \xrightarrow{\sim} H^*(X, \underline{A}).$$

In particular as X is paracompact, these are both isomorphic to Čech cohomology $\check{H}^*(X, \underline{A})$ by [Theorem 1.2.32](#).

Proof (sheafy version). Here we're loosely following [[Bre97](#), III.1] and [this note of Mustață](#).

Step 1: Singular cochain sheaves resolve the constant sheaf: For any $q \geq 0$, define a presheaf as

$$\begin{aligned} C_{\text{sing}}^q : \text{Open}(X)^{\text{op}} &\rightarrow \text{Ab} \\ U &\mapsto C_{\text{sing}}^q(U, A), \end{aligned}$$

where we recall $C_q(U) = \bigoplus_{\Delta^q \xrightarrow{\sigma} U} \mathbb{Z}$ is the free abelian group on the set of singular q -chains, and by definition $C_{\text{sing}}^q(U, A) = \text{Hom}_{\text{Ab}}(C_q(U), A) = \prod_{\sigma : \Delta^q \rightarrow U} A$ is the set of *singular cochains* on U . Since cochains form a chain complex, it is clear that we get a chain complex of presheaves of abelian groups for any F . Moreover there is a natural map of presheaves

$$\underline{A} \rightarrow C_{\text{sing}}^0$$

sending any locally constant function $s : U \rightarrow A$ to the singular cochain which associates, to the point $u \in U$, the value $s(u) \in A$. So we get a sequence of presheaves

$$\underline{A} \rightarrow C_{\text{sing}}^0 \rightarrow C_{\text{sing}}^1 \rightarrow \cdots$$

Observe that C_{sing}^q is not a sheaf — this is because we can have simplices landing in U_i 's which agree on overlaps, but do not glue to one big simplex (exercise: come up with an explicit example illustrating this). So we now sheafify everything in sight, and we get

$$\underline{A} \rightarrow (C_{\text{sing}}^0)^\sharp \rightarrow (C_{\text{sing}}^1)^\sharp \rightarrow \cdots \quad (1.3.2)$$

We first claim this is an exact sequence of sheaves (a resolution of the constant sheaf), so we have to verify exactness on stalks. Since X is locally contractible, there is a cofinal system of open neighborhoods $U \ni x$ which are contractible ([Definition A.2.22](#)). On such a contractible neighborhood, we observe that $\underline{A}(U) = A$ (since U is connected), and we get that all the singular cohomology vanishes, i.e. $H_{\text{sing}}^i(U, A) = 0$ for $i > 0$. In other words, the sequence

$$0 \rightarrow \underline{A}(U) \rightarrow C_{\text{sing}}^0(U) \rightarrow C_{\text{sing}}^1(U) \rightarrow \cdots$$

is exact. Hence the stalks are exact and we have that [Equation \(1.3.2\)](#) is a resolution of sheaves.

Step 2: Singular cochain sheaves are acyclic: The next claim to verify is that these $(C_{\text{sing}}^q)^\sharp$ sheaves are acyclic. We'll do this by reference to [Theorem 1.2.27](#) — since our base space is paracompact it suffices to show the *presheaves* C_{sing}^q are acyclic in Čech cohomology. Observe that the Čech cochains for this presheaf on an open cover $\mathcal{U} = \{U_i\}$ look like:

$$\check{C}^q(\mathcal{U}, C_{\text{sing}}^q) = \prod_{i_0, \dots, i_q} C_{\text{sing}}^q(U_{i_0 \dots i_q}) = \prod_{i_0 \dots i_q} \prod_{\Delta^p \xrightarrow{\sigma} U_{i_0 \dots i_q}} A$$

The trick is to rearrange this – if $\sigma: \Delta^p \rightarrow X$ is any simplex, we'll denote by $I_\sigma = \{i \in I \mid \sigma(\Delta^p) \cap U_i \neq \emptyset\}$, this is the indexing set of the collection of open sets in our cover that a given simplex σ hits. This is because, unless $i_0, \dots, i_q \in I_\sigma$, then σ does not contribute a copy of A to $C_{\text{sing}}^q(U_{i_0 \dots i_q})$. Letting $\check{C}^q(I_\sigma; A)$ denote the group $\prod_{i_0, \dots, i_q \in I_\sigma} A$, we can rearrange the above to equal

$$\check{C}^q(\mathcal{U}, C_{\text{sing}}^q) = \prod_{\sigma} \check{C}^q(I_\sigma, A).$$

So acyclicity of C_{sing}^q reduces to checking that the sequence

$$\dots \rightarrow \check{C}^{q-1}(I_\sigma, A) \rightarrow \check{C}^q(I_\sigma, A) \rightarrow \check{C}^{q+1}(I_\sigma, A) \rightarrow \dots \quad (1.3.3)$$

is exact (again, because products of long exact sequences of abelian groups are long exact). We do this by constructing an explicit chain homotopy — we've fixed our σ , and let's fix a preferred element which we'll call $j \in I_\sigma$ (it doesn't matter what it is). Then we define a map

$$h: \check{C}^q(I_\sigma, A) \rightarrow \check{C}^{q-1}(I_\sigma, A)$$

by defining $h(c)(i_0, \dots, i_{q-1}) := c(j, i_0, \dots, i_{q-1})$. We claim that $d^{q-1} \circ h + h \circ d^q = \text{id}$, that is, h exhibits the identity map on [Equation \(1.3.3\)](#) as null-homotopic. This is because, for any $c \in \check{C}^{q-1}(I_\sigma, A)$, we have

$$(d^{q-1}h)(c)(i_0, \dots, i_q) = \sum_{k=0}^q (-1)^k (hc)(i_0, \dots, \hat{i}_k, \dots, i_q) = \sum_{k=0}^q (-1)^k c(j, i_0, \dots, \hat{i}_k),$$

and

$$(hd^q)(c)(i_0, \dots, i_q) = (d^q c)(j, i_0, \dots, i_q) = c(i_0, \dots, i_q) + \sum_{k=1}^q (-1)^{k+1} c(j, i_0, \dots, \hat{i}_k, \dots, i_q).$$

Summing these two terms, everything cancels except $c(i_0, \dots, i_q)$, so $d^{q-1} \circ h + h \circ d^q = \text{id}$, hence [Equation \(1.3.3\)](#) is exact, and therefore the C_{sing}^q presheaves are acyclic. No aspect of this used paracompactness of X ! However we use paracompactness to conclude that the *sheaves* $(C_{\text{sing}}^q)^\sharp$ are acyclic, as desired.

Step 2.5: Pause, what do we have so far: The previous two steps tell us that the constant sheaf cohomology $H^q(X, \underline{A})$ can be computed as derived functors of global sections along the acyclic resolution given by the sheafification of the singular cochain sheaves — explicitly, as the cohomology of the following chain complex:

$$0 \rightarrow \Gamma(X, \underline{A}) \rightarrow (C_{\text{sing}}^0)^\sharp(X) \rightarrow (C_{\text{sing}}^1)^\sharp(X) \rightarrow \dots$$

Note that we have a natural morphism of complexes

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \Gamma(X, \underline{A}) & \longrightarrow & (C_{\text{sing}}^0)(X) & \longrightarrow & C_{\text{sing}}^1(X) \longrightarrow \dots \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \Gamma(X, \underline{A}) & \longrightarrow & (C_{\text{sing}}^0)^\#(X) & \longrightarrow & (C_{\text{sing}}^1)^\#(X) \longrightarrow \dots
 \end{array}$$

This induces a map between their cohomology, and by what we've seen above, this gives us a map $H_{\text{sing}}^p(X, A) \rightarrow H^p(X, \underline{A})$. We're aiming to argue it is an isomorphism by showing that the vertical maps induce a quasi-isomorphism.

Step 3: Cochains to sheafified cochains is levelwise surjective: Suppose we have some $q \geq 0$ and an element $c \in (C_{\text{sing}}^q)^\#(X)$. We want to lift it to $C_{\text{sing}}^q(X)$. Fix an open cover $\mathcal{U}$. Since we have sheafified, we can repackage the data of c as the data of, for every $U_i \in \mathcal{U}$, a cochain $c_i \in C_{\text{sing}}^q(U_i, A)$, so that the cochains c_i and c_j agree on the overlap. Surjectivity then reduces to checking the existence of a global element $\chi \in C_{\text{sing}}^q(X, A)$ which restricts to c_i on each U_i . Recall what a q -cochain is – it *tells you* a particular value in A for every q -simplex in its domain. The c_i 's give us a rule of how to give a value of A for every q -simplex landing in U_i for each i , and these rules agree for simplices on the overlaps, so now we have to assign a value for every simplex landing in X . The trick is to use barycentric subdivision to chop up an arbitrary simplex in X into much smaller simplices, each of which has the property that it either lands completely in a U_i or is disjoint from it. This can't be accomplished on arbitrary spaces, but it can on paracompact Hausdorff ones, since they admit locally finite covers.

Step 4: Cochains to sheafified cochains has null-homologous kernel: We now have a map of chain complexes $C_{\text{sing}}^\bullet(X) \rightarrow (C_{\text{sing}}^\bullet)^\#(X)$ which is levelwise surjective. We want to show that its kernel is null-homologous. We claim that this also follows from barycentric subdivision. In particular we can describe, at each level q , the kernel of $C_{\text{sing}}^q(X) \rightarrow (C_{\text{sing}}^q)^\#(X)$ as those q -cochains c satisfying the following condition: there exists some open cover $\mathcal{U}$ so that, if σ has image inside any one (maybe more) of the U_i 's, then $c(\sigma) = 0$. However it is a standard result in barycentric subdivision to see that the complex of such things is null-homologous (see [Hat01, 2.21]). $\square$

If we limit our attention to sufficiently nice spaces, e.g. differentiable manifolds, we have an alternative proof of [Theorem 1.3.1](#) via homotopy theory.

Proof sketch (nerve lemma). Suppose X is paracompact and it admits a *good cover* $\mathcal{U}$ (when X is a Riemannian manifold, we proved this existence in [Proposition A.1.12](#)). The nerve lemma [Wei52, §5] (see also [Hat01, 4G.3]) says that $|N(\mathcal{U})| \simeq X$, that is, the geometric realization of the nerve of this cover recovers the homotopy type of the space X . Since singular cohomology is a homotopy invariant, we can replace $H_{\text{sing}}^*(X, A)$ with $H_{\text{sing}}^*(|N(\mathcal{U})|, A)$ without loss of generality. One can prove that the simplicial cochains complex on $N(\mathcal{U})$ is canonically identified with the Čech cochains complex for $\mathcal{U}$. We can compose this isomorphism with the natural map comparing simplicial and singular cochains:

$$\check{C}^*(\mathcal{U}, \underline{A}) \cong C_\Delta^\bullet(N(\mathcal{U}), A) \rightarrow C_{\text{sing}}^\bullet(|N(\mathcal{U})|, A).$$

The latter map is a quasi-isomorphism by the standard comparison theorem between simplicial and singular cohomology (see for instance [Hat01, §2.1]).

Thus for any good cover $\mathcal{U}$, we have that Čech cohomology of the cover agrees with singular cohomology. We use cofinality of good covers ([Proposition A.1.10](#)) to conclude. $\square$

Remark 1.3.4. It turns out by a recent theorem of Sella you can weaken this and drop the paracompact+Hausdorff condition, and only assume X is semi-locally contractible [Sel16].

Remark 1.3.5. When X is a manifold and $A = \mathbb{R}$, we'll provide a different proof of this fact, via the *de Rham theorem* (Theorems 1.4.33 and 1.4.35).

1.4 Vector bundles and sheaves of sections

Definition 1.4.1. A *real vector bundle* $\pi: E \rightarrow X$ is a map of spaces so that

- ▷ π is a surjection
- ▷ $\pi^{-1}(x)$ has the structure of a real vector space for every $x \in X$
- ▷ for every $x \in X$ there exists an open neighborhood $U \ni x$, an integer k , and a homeomorphism ϕ fitting into a commutative diagram

$$\begin{array}{ccc} U \times \mathbb{R}^k & \xrightarrow{\phi} & \pi^{-1}(U) \\ & \searrow & \swarrow \pi \\ & U, & \end{array}$$

so that $\phi(x, -): \mathbb{R}^k \xrightarrow{\sim} \pi^{-1}(x)$ is an isomorphism of real vector spaces for each $x \in U$.

A *complex* vector space is the same definition, but we replace $\mathbb{R}$ with $\mathbb{C}$ above.

Remark 1.4.2. A vector bundle (unless it is a rank zero vector bundle) is *not* a sheaf. Even though it is a continuous surjection $\pi: E \rightarrow X$, and every fiber has the structure of an abelian group, that fiber has a non-discrete topology, which prevents π from being a local homeomorphism.

Notation 1.4.3. Given a vector bundle $\pi: E \rightarrow X$ and a point $x \in X$, it is common notation to write E_x to mean the fiber $\pi^{-1}(x)$. This is supposed to remind us of stalks.

Example 1.4.4. Given any space X and any rank $k \geq 0$, we have the *trivial vector bundle*

$$\pi: X \times \mathbb{R}^k \rightarrow X.$$

Definition 1.4.5. Let $\pi_1: E_1 \rightarrow X$ and $\pi_2: E_2 \rightarrow X$ be two vector bundles over the same base space. A *morphism* of vector bundles is a continuous map $f: E_1 \rightarrow E_2$ so that:

1. $\pi_2 \circ f = \pi_1$, that is, the diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{f} & E_2 \\ & \searrow \pi_1 & \swarrow \pi_2 \\ & X & \end{array}$$

2. for every point $x \in X$, the restriction $f_x := f|_{(E_1)_x}: (E_1)_x \rightarrow (E_2)_x$ is a linear transformation.

We say f is an *isomorphism* if f is a homeomorphism (equivalently, f_x is an invertible linear transformation for each $x \in X$).

Terminology 1.4.6. We say a vector bundle is *trivial* if it is isomorphic to the trivial vector bundle.

Example 1.4.7. The Möbius strip can be thought of as a vector bundle over the sphere S^1 . It is not isomorphic to the trivial vector bundle on S^1 .

Example 1.4.8. On any manifold, we have a *tangent bundle*

$$TM \rightarrow M,$$

which is a vector bundle of rank equal to the dimension of M . A section of the bundle is a vector field on M .

Example 1.4.9. Let M be a real n -dimensional manifold. If M has trivial tangent bundle (such manifolds are called *parallelizable*) then we can pick any nonzero vector $\vec{v} \in \mathbb{R}^n$, and create a section σ which is the composite

$$\begin{aligned} M &\rightarrow M \times \mathbb{R}^n \cong TM \\ m &\mapsto (m, \vec{v}). \end{aligned}$$

Since the vector bundle isomorphism $M \times \mathbb{R}^n \cong TM$ is an invertible linear transformation in each fiber, it sends $\vec{v}$ to some nonzero tangent vector over each point in M . Thus the composite map $\sigma: M \rightarrow TM$ is a *nonvanishing* vector field across all of M .

Example 1.4.10. The hairy ball theorem follows from the statement that the tangent bundle on the 2-sphere is nontrivial.

Theorem 1.4.11. The only spheres whose tangent bundles are trivial are S^0 , S^1 , S^3 , and S^7 .

Example 1.4.12. On any manifold we have a *cotangent bundle*

$$T^*M \rightarrow M,$$

where the fiber over $p \in M$ is the dual vector space $(T_p M)^*$. That is. A section of the cotangent bundle can be thought of as a way to coherently “measure” tangent vectors along the manifold.

Notation 1.4.13. If $E \rightarrow X$ is a vector bundle and $U \subseteq X$ is some open subset, we denote by $\text{Sec}(U, E)$ the set of continuous sections:

$$\text{Sec}(U, E) := \{s: U \rightarrow E|_U : \pi \circ s = \text{id}_U\}.$$

Note that $\text{Sec}(U, E)$ is an $\mathbb{R}$ -module. In fact, it is a $C(U, \mathbb{R})$ -module.

Proposition 1.4.14. Let X be a space, and $E \rightarrow X$ a vector bundle. Then the functor

$$\text{Sec}(-, E): \text{Open}(X)^{\text{op}} \rightarrow \text{Ab}$$

is a sheaf.

Proof. Sections satisfy locality and gluing. □

Question 1.4.15. If you were handed the sheaf $\text{Sec}(-, E)$ but not told what E is, could you reconstruct the vector bundle E ? If so, demonstrate how to do it, and if not, explain what extra data you might need in order to build E from its sheaf of sections.

Remark 1.4.16. We can form new vector bundles out of old ones by doing linear algebra operations “fiberwise.” That is, we can make sense of direct sums, tensor products, duals, exterior powers, symmetric powers, etc. of existing vector bundles.

Definition 1.4.17. Let M be a smooth n -manifold. Then we denote by

$$\Omega_{\text{dR}}^k = \text{Sec}_{\text{smooth}}(-, \wedge^k T^*M)$$

the sheaf of smooth sections of the k th exterior power of the cotangent bundle.

Example 1.4.18. We have that Ω_{dR}^0 is the sheaf of real-valued smooth functions on M , and Ω_{dR}^1 is the sheaf of smooth sections of the cotangent bundle.

Remark 1.4.19 (Key properties of wedges). If you haven't seen exterior/wedge powers before, here are the key properties to keep in mind:

1. we have that

$$\alpha \wedge (\beta + \gamma) = \alpha \wedge \beta + \alpha \wedge \gamma$$

2. we have

$$\alpha \wedge \beta = -\beta \wedge \alpha$$

3. and

$$\alpha \wedge \alpha = 0.$$

Formally, the exterior algebra of a vector space is the tensor algebra $V^{\otimes \bullet}$ modded out by the two-sided ideal $\langle v \otimes v \mid v \in V \rangle$.

Differential forms are expressible in local coordinates, where they admit a nicer shape. Let's see some examples:

Example 1.4.20. On $M = \mathbb{R}^3$, an element of $\Gamma(\mathbb{R}^3, \Omega_{\text{dR}}^1)$ is expressible as

$$f_1(x, y, z)dx + f_2(x, y, z)dy + f_3(x, y, z)dz,$$

where the f_i 's are differentiable real-valued functions, dx is the section $\mathbb{R}^3 \rightarrow T^*\mathbb{R}^3$ which projects a tangent vector to its x -component, and similarly for dy and dz . In multivariable calculus we often rewrite these as *vector fields*

$$F_1(x, y, z)\hat{i} + F_2(x, y, z)\hat{j} + F_3(x, y, z)\hat{j}.$$

Example 1.4.21. Again on $M = \mathbb{R}^3$, an element of $\Gamma(\mathbb{R}^3, \Omega_{\text{dR}}^2)$ is something of the form

$$f_1(x, y, z)dx \wedge dy + f_2(x, y, z)dz \wedge dx + f_3(x, y, z)dy \wedge dx,$$

where again the f_i 's are differentiable and real-valued, and $dx \wedge dy \in \Gamma(\mathbb{R}^3, \wedge^2 T^*M)$ is the section which takes an ordered pair of tangent vectors, projects them onto the xy -plane, and spits out the signed volume of the parallelogram they span. In general we make a choice (this is secretly the Hodge star operator which we'll learn later) to identify

$$dx \wedge dy \leftrightarrow \hat{k}$$

$$dy \wedge dz \leftrightarrow \hat{i}$$

$$dz \wedge dx \leftrightarrow \hat{j},$$

and rewrite a differential 2-form as a vector field.

Definition 1.4.22. We define the *exterior derivative* of a differential 0-form f to be

$$df = \sum_{i=1}^n \frac{df}{dx_i}(x_1, \dots, x_n) dx_i.$$

More generally, we define the exterior derivative of a differential k -form

$$\omega = \sum_{i_1 < \dots < i_k} f_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}$$

as

$$d\omega = \sum_{i_1 < \dots < i_k} df_{i_1, \dots, i_k} \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Remark 1.4.23. Given any $f, g \in \Omega_{\text{dR}}^0(M)$, we have that

$$d(f \cdot g) = f dg + g df.$$

This is just the product rule for derivatives.

Exercise 1.4.24. Let M be a smooth manifold. Show that the exterior derivative in [Definition 1.4.22](#) defines a morphism of abelian sheaves.

Terminology 1.4.25.

- ▷ A form ω is said to be *closed* if $d\omega = 0$.
- ▷ A form ω is said to be *exact* if it is of the form $\omega = d\alpha$ for some α . In this case, α is said to be a *primitive* of ω .

1.4.1 The Poincaré lemma

Lemma 1.4.26 (Poincaré Lemma). On a smooth manifold M , the sequence

$$\underline{\mathbb{R}} \xrightarrow{d} \Omega_{\text{dR}}^0 \xrightarrow{d} \Omega_{\text{dR}}^1 \xrightarrow{d} \dots \quad (1.4.27)$$

is a long exact sequence of sheaves in $\text{Ab}(M)$.

Proof. Exactness in $\text{Ab}(M)$ is detected on stalks. This is not the same as being exact on each open set (we might call the latter *sectionwise exactness*), however sectionwise exactness does imply exactness on stalks by [Corollary A.1.7](#). For a given stalk over a point x , to argue exactness we take a colimit over the opposite category of the poset of open sets of M which contain x . This admits a cofinal system given by those open sets $U \ni x$ for which U is homeomorphic to (an open ball in) $\mathbb{R}^n$ ([Proposition A.1.10](#)). In fact since we will be making arguments which leverage differentiability of sections of our sheaves on open neighborhoods, we need to know a bit more - that there is a cofinal system of open sets of M containing x which are *diffeomorphic* to an open ball in $\mathbb{R}^n$. This is satisfied since M is a smooth manifold. Hence we can reduce to proving exactness of [Equation \(1.4.27\)](#) when taking sections over the open ball B of radius 1 around the origin in $\mathbb{R}^n$.

We leave it as an exercise to show that $d^2 = 0$ (i.e. that exact forms are closed), and we will prove the hard part, namely that closed forms are exact. Let $\omega \in \Omega^k(B)$ and suppose that $d\omega = 0$.

We will show that ω is exact by explicitly constructing a primitive. Since the exterior derivatives are sectionwise abelian group homomorphisms, we may reduce to the case where ω is of the form

$$\omega = f(x_1, \dots, x_n) dx_I$$

for some differentiable function f and some multiindex $I = \{i_1 < i_2 < \dots < i_k\} \subseteq \{1, \dots, n\}$. What does $d\omega = 0$ mean here? It means precisely that $\frac{\partial f}{\partial x_j} = 0$ for every $j \notin I$.

We claim that the following $(k-1)$ -form α is a primitive for ω :

$$\alpha := \sum_{j=1}^k (-1)^{j-1} x_{i_j} \left(\int_0^1 t^{k-1} f(tx_1, \dots, tx_n) dt \right) dx_{I \setminus \{i_j\}}.$$

This form doesn't come out of thin air – it is obtained explicitly by pulling back along the scaling map $h(x, t) = tx$ for $t \in [0, 1]$ and then applying a homotopy operator which “integrates out” the t . For more details about how you could have come up with this primitive, see for instance [BT82, I§4].

We check that $d\alpha = \omega$. To simplify notation, let's let

$$P(x_1, \dots, x_n) := \int_0^1 t^{k-1} f(tx_1, \dots, tx_n) dt.$$

Then we have that

$$\begin{aligned} d\alpha &= \sum_{j=1}^k (-1)^{j-1} (P dx_{i_j} + x_{i_j} dP) \wedge dx_{I \setminus \{i_j\}} \\ &= kP dx_I + \sum_{j=1}^k (-1)^{j-1} x_{i_j} dP \wedge dx_{I \setminus \{i_j\}}. \end{aligned} \tag{1.4.28}$$

We have a decent handle on that first term, so we want to better understand the second term. We compute dP , using that we can pass partial derivatives inside the integral, to get

$$dP = \sum_{i=1}^n \left(\int_0^1 t^k \frac{df}{dx_i}(tx_1, \dots, tx_n) dt \right) dx_i,$$

where we notice the power on t increased because of the chain rule. Since dP has a dx_i in it for every i , when we wedge with $dx_{I \setminus \{i_j\}}$ we will get zero if $i \in I \setminus \{i_j\}$. Similarly we will get zero if $i \notin I$, because $\frac{df}{dx_i} = 0$ for any such i by ω being closed. So the only nonvanishing term in the summand, after wedging with $dx_{I \setminus \{i_j\}}$, is when $i = i_j$. That is,

$$\begin{aligned} dP \wedge dx_{I \setminus \{i_j\}} &= \left(\int_0^1 t^k \frac{df}{dx_{i_j}}(tx_1, \dots, tx_n) dt \right) dx_{i_j} \wedge dx_{I \setminus \{i_j\}} \\ &= \left(\int_0^1 t^k \frac{df}{dx_{i_j}}(tx_1, \dots, tx_n) dt \right) (-1)^{j-1} dx_I. \end{aligned}$$

Plugging this back in to the second term in the expression for $d\alpha$ in Equation (1.4.28), we get

$$\sum_{j=1}^k (-1)^{j-1} x_{i_j} dP \wedge dx_{I \setminus \{i_j\}} = \sum_{j=1}^k x_{i_j} \left(\int_0^1 t^k \frac{df}{dx_{i_j}}(tx_1, \dots, tx_n) dt \right) dx_I.$$

Putting everything together we get that

$$d\alpha = \left(k \left(\int_0^1 t^{k-1} f(tx_1, \dots, tx_n) dt \right) + \sum_{j=1}^k \int_0^1 t^k x_{i_j} \frac{df}{dx_{i_j}}(tx_1, \dots, tx_n) dt \right) dx_I$$

So it suffices to check that the smooth function in front of dx_I in the expression above is actually equal to $f(x_1, \dots, x_n)$. This comes from recognizing that the product rule has been applied! First note that we can change our sum to be indexed over all x_i 's without a problem, since all the extra terms it adds are zero. Thus we can rewrite the above as

$$\begin{aligned} & \int_0^1 \left(kt^{k-1} f(tx_1, \dots, tx_n) + \sum_{i=1}^n t^k x_i \frac{df}{dx_i}(tx_1, \dots, tx_n) \right) dt \\ &= \int_0^1 \left(f(tx_1, \dots, tx_n) \cdot \frac{d}{dt}(t^k) + t^k \cdot \frac{d}{dt} f(tx_1, \dots, tx_n) \right) dt \\ &= \int_0^1 \frac{d}{dt} \left(t^k f(tx_1, \dots, tx_n) \right) dt \\ &= t^k f(tx_1, \dots, tx_n) \Big|_{t=0}^{t=1} \\ &= f(x_1, \dots, x_n). \end{aligned}$$

This completes the proof. $\square$

In general, as we know, applying $\Gamma(M, -)$ to the sequence in [Equation \(1.4.27\)](#) will turn it into a chain complex, but will fail to preserve exactness. Here's an easy case where it does preserve exactness, which we know from multivariable calculus:

Proposition 1.4.29. On $\mathbb{R}^3$, taking global sections of the resolution from the Poincaré lemma gives an *exact sequence*

$$0 \rightarrow C^0(\mathbb{R}^3, \mathbb{R}) \xrightarrow{d} \Omega_{\text{dR}}^1(\mathbb{R}^3) \xrightarrow{d} \Omega_{\text{dR}}^2(\mathbb{R}^3) \xrightarrow{d} \Omega_{\text{dR}}^3(\mathbb{R}^3) \rightarrow 0. \quad (1.4.30)$$

Proof. Identifying Ω^1 and Ω^2 with vector fields as in [Examples 1.4.20](#) and [1.4.21](#), these differentials take a more familiar shape:

$$0 \rightarrow C^0(\mathbb{R}^3, \mathbb{R}) \xrightarrow{\text{grad}} \Omega_{\text{dR}}^1(\mathbb{R}^3) \xrightarrow{\text{curl}} \Omega_{\text{dR}}^2(\mathbb{R}^3) \xrightarrow{\text{div}} \Omega_{\text{dR}}^3(\mathbb{R}^3) \rightarrow 0.$$

In this case exactness at each stage has a nice interpretation:

exactness at $\Omega_{\text{dR}}^1(\mathbb{R}^3)$	the curl of a vector field is zero if and only if the vector field is a gradient
exactness at $\Omega_{\text{dR}}^2(\mathbb{R}^3)$	the divergence of a vector field is zero if and only if the vector field is a curl
exactness at $\Omega_{\text{dR}}^3(\mathbb{R}^3)$	every differentiable function on $\mathbb{R}^3$ is the divergence of some vector field

$\square$

In particular let's compute the cohomology of the sequence [Equation \(1.4.30\)](#). It's exact everywhere except at $C^0(\mathbb{R}^3, \mathbb{R})$. The homology at this point is the kernel of the gradient, and we

see that this is isomorphic to $\mathbb{R}$, since any two functions with the same gradient differ by a scalar. In particular the cohomology of the cochain complex above is

$$H^* = \begin{cases} \mathbb{R} & * = 0 \\ 0 & * > 0 \end{cases}$$

This looks an awful lot like the singular cohomology of $\mathbb{R}^3$. We know that sheaf cohomology with constant coefficients computes singular cohomology, so we'd love to say that what we just did is compute the sheaf cohomology of $\mathbb{R}^3$ with constant coefficients $\mathbb{R}$, and the only thing preventing us from saying that is just verifying that the de Rham sheaves are acyclic. We actually know one of these already — we've shown that the sheaf of differentiable real-valued functions is fine (and therefore acyclic) on a differentiable manifold, since we have access to differentiable partitions of unity, so this is the case Ω_{dR}^0 . We show this works for the higher sheaves as well:

Proposition 1.4.31. If M is a differentiable manifold, then Ω_{dR}^k is fine.

Proof. Pick any *differentiable* partition of unity ϕ_i subordinate to an open cover, and define homeomorphisms by multiplication by h_i on each open cover. This is fine by a similar proof to [Example 1.2.57](#). More generally, a sheaf of modules over a fine sheaf of commutative rings is fine. $\square$

Definition 1.4.32. We define *de Rham* cohomology as the cohomology of the *de Rham complex*:

$$\Omega_{\text{dR}}^0(M) \xrightarrow{d} \Omega_{\text{dR}}^1(M) \xrightarrow{d} \Omega_{\text{dR}}^2(M) \xrightarrow{d} \dots$$

Theorem 1.4.33 (Sheafy de Rham theorem). Let M be a smooth manifold. Then there is an isomorphism

$$H_{\text{dR}}^p(M) \cong H^p(M, \mathbb{R}).$$

Proof. We have shown that the sheaves of differential forms are fine [Proposition 1.4.31](#), and therefore acyclic by [Theorem 1.2.61](#). $\square$

We now have some equalities of cohomology theories along a manifold M :

$$\underbrace{H_{\text{dR}}^*(M)}_{\text{de Rham}} \xrightarrow{1.4.33} \underbrace{H^*(M, \mathbb{R})}_{\text{sheaf}} \xrightarrow{1.2.32} \underbrace{\check{H}^*(M, \mathbb{R})}_{\check{\text{Cech}}} \xrightarrow{1.3.1} \underbrace{H_{\text{sing}}^*(X, \mathbb{R})}_{\text{singular}}$$

The comparison between Čech and singular cohomology was very precise, that is, we cooked up an explicit map sending singular cochains to Čech cochains and showed it is an isomorphism on cohomology. The other isomorphisms are more abstract — Čech and de Rham cohomology agree with sheaf cohomology in this context because they arise as the cohomology of a resolution of a constant sheaf $\mathbb{R}$ by an acyclic resolution of sheaves (being the Čech cochains sheaves and the de Rham sheaves, respectively).

So our takeaway from all of this is that de Rham and singular cohomology agree along a manifold, but right now we only really know they agree abstractly. It would be nice to find an explicit map exhibiting this isomorphism — this is the content of the *de Rham theorem*.

Our isomorphism in question is exhibited by integration. At the level of presheaves, given an open neighborhood $U \subseteq M$ and an element $\omega \in \Omega_{\text{dR}}^k(U)$, we obtain a singular cochain by integration along simplices in U :

$$\begin{aligned} \Omega_{\text{dR}}^k(U) &\rightarrow C_{\text{sing}}^k(U; \mathbb{R}) \\ \omega &\mapsto \left[Z \mapsto \int_Z \omega \right]. \end{aligned}$$

Exercise 1.4.34. Show that this induces a map of chain complexes between the de Rham complex and the (smooth) singular chains complex (hint: Stokes' theorem!)

The de Rham theorem states that integration exhibits our desired isomorphism between de Rham and singular cohomology:

Theorem 1.4.35 (de Rham theorem). If M is a smooth manifold, then the map given by integration of differential forms induces an isomorphism

$$H_{\text{dR}}^*(M) \xrightarrow{\sim} H_{\text{sing}}^*(M, \mathbb{R}).$$

We don't prove this - to see it spelled out, we refer to [Lee12, Chapter 16]. Specifically [Lee12, 16.12] is the result, while naturality is in [Lee12, 16.11] and the comparison between smooth and singular chains is [Lee12, 16.6].

Remark 1.4.36. If we delete a point from $\mathbb{R}^2$, we know that this introduces non-trivial cohomology, since $\mathbb{R}^2 \setminus \{0\}$ deformation retracts onto the unit vectors, which form the sphere S^1 . Following the de Rham theorem, this tells us we should expect the existence of a global 1-form on $\mathbb{R}^2 \setminus \{0\}$ which has zero derivative but is *not* the gradient of a smooth function on $\mathbb{R}^2 \setminus \{0\}$. Such a thing exists precisely because we have deleted the origin, meaning we should be allowed to find something new which was *not allowed* when the origin was present. An example is, for instance:

$$\omega = \frac{x \, dx - y \, dy}{x^2 + y^2}. \quad (1.4.37)$$

Such a form doesn't exist on $\mathbb{R}^2$, but *does* on $\mathbb{R}^2 \setminus \{0\}$. This is the way in which differential forms see the creation of cohomology classes by deleting regions from a contractible Euclidean space.

Exercise 1.4.38. Prove that $H_{\text{dR}}^1(\mathbb{R}^2 \setminus \{0\})$ is a 1-dimensional real vector space spanned by the form ω from Equation (1.4.37).

1.5 Fiber bundles

Previously we talked about sheaves of abelian groups. Much of what we talked about goes through if we have sheaves of groups, or just sheaves of sets. The important thing was that the sheaf was a *local homeomorphism*, in other words the fibers of the projection came with a discrete topology.

There are many interesting groups which come equipped with a natural topological structure, for instance S^1 or $\text{GL}_n(\mathbb{R})$. However our notion of sheaf above won't be able to accurately encompass the idea of having these sorts of groups as fibers unless we forget their topologies, which lands us back in the world of sheaves as above. To that end, we want to discuss a new family of objects in which stalks are allowed to have a non-discrete topology. These are *fiber bundles*.

Definition 1.5.1. A *topological group* is a group object in the category of spaces. Explicitly, this means it is a *space* G , equipped with an identity element $e \in G$ and continuous multiplication and inversion maps

$$\begin{aligned} m: G \times G &\rightarrow G \\ i: G &\rightarrow G \end{aligned}$$

which satisfy the expected identities.

Proposition 1.5.2. Let G be a topological group. If the one-point subgroup $\{e\} \leq G$ is a closed subspace, then G is Hausdorff (and clearly the backwards direction holds as well because Hausdorff implies T_1).

Proof. Being Hausdorff is equivalent to the image of the diagonal $\Delta: G \rightarrow G \times G$ being closed. Consider the map

$$\begin{aligned} G \times G &\rightarrow G \\ (g_1, g_2) &\mapsto g_1 g_2^{-1}. \end{aligned}$$

By the axioms of topological groups, this is continuous. We have that $\text{im}(\Delta)$ is the preimage of $\{e\}$ under this map, hence closed if the identity subgroup is closed. $\square$

Remark 1.5.3. Some authors, e.g. [Mun00], assume that topological groups are T_1 , which is the separation axiom guaranteeing one-point subspaces are automatically closed. Hence all topological groups are Hausdorff from this perspective.

Exercise 1.5.4 ([Mun00, p. 146]). If G is a topological group and $H \leq G$ is a subgroup, then the projection map $G \rightarrow G/H$ is open.

Example 1.5.5. A *Lie group* is a further strengthening of a topological group – it is a group object in the subcategory of smooth manifolds.

Notation 1.5.6 ([Hir78, p. 39]). Let G be a topological group, and X a space. Let G_c denote the presheaf defined by

$$\begin{aligned} \text{Open}(X)^{\text{op}} &\rightarrow \text{Grp} \\ U &\mapsto \text{Hom}_{\text{Top}}(U, G). \end{aligned}$$

Note that G_c is a sheaf by [Example 1.1.33](#).

If G has more structure (e.g. it is a *Lie group*) and X is a manifold, we could take a sheaf of differentiable functions valued in the group, for instance — denote this by $G_{\mathfrak{b}}$. Or if G is a complex Lie group and X is a complex manifold, we could send U to holomorphic functions $U \rightarrow G$ — we denote this by G_{ω} .

Definition 1.5.7. Let G be a topological group, fix a space F equipped with a faithful¹³ continuous action of G , and let X be a space. Let We define a *fiber bundle over X with fiber F and structure group G* to be a continuous map $\pi: W \rightarrow X$ if there exists additional data:

- ▷ an open cover $\{U_i\}$ of X
- ▷ homeomorphisms $h_i: \pi^{-1}(U_i) \rightarrow U_i \times F$ over U_i
- ▷ elements $g_{ij} \in \Gamma(U_i \cap U_j, G_c)$

so that

$$(h_i h_j^{-1})(u, f) = (u, g_{ij}(u)f) \in (U_i \cap U_j) \times F$$

for all $u \in U_i \cap U_j$ and $f \in F$.

¹³Hirzebruch uses the word *effective* for the group action, meaning if $g \cdot f = f$ for all f then $g = e$. We use the word *faithful* for this same notion.

Terminology 1.5.8. If we want to refer to the cover $\mathcal{U}$, we will say the fiber bundle is *trivialized over $\mathcal{U}$* . The idea being that, via the homeomorphisms π_i , over each $U_i \in \mathcal{U}$ our fiber bundle does indeed look trivial.

Example 1.5.9. For any F and G , we have the *trivial fiber bundle* defined by the projection $\pi: X \times F \rightarrow X$, whose structure group is the trivial group.

Remark 1.5.10. In the previous example it's not really necessary that the structure group is the trivial group, all we want is that the transition functions g_{ij} to be the identity in whatever the structure group G is. This is part of a more general idea that the structure group is a choice, not necessarily an intrinsic part of the data of a bundle. The structure group could be enlarged by just sitting the g_{ij} 's inside a bigger group (provided the action on F is still *faithful*!), however it is not true that the structure group for a fiber bundle can always be reduced. The question of when we have “reduction of structure group” for a bundle is an interesting line of thought we won't get into here.

Exercise 1.5.11. Repackage the definition of a vector bundle into the definition of a certain type of fiber bundle. That is, if $E \rightarrow X$ is a vector bundle, how can we view it as a fiber bundle - what is the fiber and what is the structure group?

Example 1.5.12. The Möbius strip

$$M = \frac{[0, 2\pi] \times [0, 1]}{(0, t) \sim (2\pi, 1 - t)}$$

can be viewed as a fiber bundle over S^1 via the projection map

$$\begin{aligned} \pi: M &\rightarrow S^1 \\ (\theta, t) &\mapsto e^{i\theta}. \end{aligned}$$

Here $F = [0, 1]$ is the interval, and we can think of $G = \mathbb{Z}/2$ acting via $t \mapsto 1 - t$ on F .

Example 1.5.13. In the special case where $F = G$, and the action is G acting on itself by left translation, we call our fiber bundle a *principal fiber bundle*. More on this later.

1.5.1 Čech 1-cocycles

Given a topological group G , we can define the sheaf of groups G_c as in [Notation 1.5.6](#), and therefore construct its Čech cohomology following [Definition 1.2.11](#). Note that in the previous definition we had assumed our sheaf to be a sheaf of abelian groups, so we have to be a bit careful about products versus sums.

Exercise 1.5.14 (Suggested). Let G be non-abelian, and define $\check{C}^i(\mathcal{U}, G)$ as in [Equation \(1.2.7\)](#). Try to write down a formula for the differential δ^q (but remember you can't take *sums* any more).

1. If you wrote down a solid definition for δ^1 , great! Check if it is a group homomorphism.
2. Oops
3. Try to redefine it to make it a group homomorphism? Try to turn $\check{C}^\bullet(\mathcal{U}, G)$ into a complex.
4. Get really frustrated and give up

The takeaway is that *we can't define higher nonabelian Čech cohomology*. Nevertheless we can still actually make sense of $\check{H}^1(\mathcal{U}, G)$ when G is nonabelian, and it turns out to be a crucially important object in the theory of fiber bundles.

Definition 1.5.15. Given a topological group G and a space X with an open cover $\mathcal{U} = \{U_i\}$, we define the set of *Čech 1-cocycles* as the “kernel” of δ^1 . Even though δ^1 isn't a group homomorphism, it is still a map of pointed sets

$$\begin{aligned}\delta^1: \check{C}^1(\mathcal{U}, G) &\rightarrow \check{C}^2(\mathcal{U}, G) \\ (g_{ij}) &\mapsto g_{ij}g_{ik}^{-1}g_{jk}.\end{aligned}$$

We can take the fiber over the identity element to be our “kernel” and we obtain

$$\check{Z}^1(\mathcal{U}, G) = \left\{ (g_{ij}) \in \prod G(U_i \cap U_j) \mid g_{ik} = g_{jk}g_{ij} \text{ on } G(U_i \cap U_j \cap U_k) \text{ for all } i, j, k \right\}.$$

Exercise 1.5.16. Verify, if $(g_{ij}) \in \check{Z}^1(\mathcal{U}, G)$, that $g_{ii} = 1$ for every i and that $g_{ij} = g_{ji}^{-1}$.

Two Čech cocycles define the same element of $\check{H}^1(\mathcal{U}, G)$ if they differ by an element in

$$\text{im}(\delta^0: \check{C}^0(\mathcal{U}, G) \rightarrow \check{C}^1(\mathcal{U}, G)) \leq \check{Z}^1(\mathcal{U}, G),$$

where δ^0 sends a 0-cochain $\{\phi_i: U_i \rightarrow G\}$ to $g_{ij} = \phi_i\phi_j^{-1}$.

Unwinding this definition, we have:

Definition 1.5.17.

1. Two Čech cocycles $(g_{ij}), (\gamma_{ij}) \in \check{Z}^1(\mathcal{U}, G_c)$ are *equivalent* if there exist elements $f_i \in \Gamma(U_i, G_c)$ for which

$$\gamma_{ij} = f_i^{-1}g_{ij}f_j$$

in $\Gamma(U_i \cap U_j, G)$ for all i, j .

2. We define the *non-abelian Čech cohomology group* $\check{H}^1(\mathcal{U}, G)$ to be the pointed set $\check{Z}^1(\mathcal{U}, G)$ modulo the equivalence relation defined above.

Exercise 1.5.18. Check carefully that $\check{H}^1(\mathcal{U}, G)$ defined above is indeed a group.

1.5.2 Comparing cocycles to fiber bundles

The point of all this is we can construct a fiber bundle out of a cocycle and vice versa.

Construction 1.5.19 (Fiber bundle from cocycle). Fix G acting faithfully on F as above, and an open cover $\mathcal{U}$ of X . Given a cocycle $(g_{ij}) \in \check{Z}^1(\mathcal{U}, G)$, we can assemble a fiber bundle by taking $U_i \times F$ for each i , and gluing them together along the overlaps using the g_{ij} 's:

$$\begin{aligned}U_i|_{U_i \cap U_j} \times F &\xrightarrow{\sim} U_j|_{U_i \cap U_j} \times F \\ (u, f) &\mapsto (u, g_{ij}(u) \cdot f).\end{aligned}$$

Since this data is a cocycle, all the gluings agree across triple overlaps (exercise: check this), so we obtain a space $W = \cup_i (U_i \times F) / \sim$ by modding out by this gluing. This can easily be seen to define a fiber bundle with fiber F .

Construction 1.5.20 (Cocycle from fiber bundle). Given a fiber bundle $\pi: W \rightarrow F$ with fiber F and structure group G , we claim that the elements g_{ij} from [Definition 1.5.7](#) define a Čech 1-cocycle. (Exercise: check this is well-defined)

Theorem 1.5.21. The constructions in [Constructions 1.5.19](#) and [1.5.20](#) are inverse, and define a bijection

$$\{G\text{-fiber bundles with fiber } F \text{ trivialized over } \mathcal{U}\} \leftrightarrow \check{Z}^1(\mathcal{U}, G). \quad (1.5.22)$$

Remark 1.5.23 (Important). There is no F on the right hand side of the bijection above, and nor should there be. This tells us that $\check{Z}^1(\mathcal{U}, G)$ is allowed to classify G -fiber bundles with *any* fiber F , so long as we fix the fiber and the action ahead of time. It doesn't classify bundles with different fibers simultaneously.

Example 1.5.24 (Identity to identity). The identity cocycle corresponds to the trivial fiber bundle along the bijection in [Equation \(1.5.22\)](#).

Question 1.5.25. If we mod out by coboundaries on the right hand side of the bijection of [Equation \(1.5.22\)](#), what does this correspond to as an equivalence class on fiber bundles?

This leads us to the idea of (iso)morphisms of fiber bundles.

1.5.3 Morphisms of fiber bundles

Given two fiber bundles with the same fiber and structure group over a base space B , we want to come up with a notion of a *morphism* between fiber bundles. Here is a very reasonable definition that we might expect:

Definition 1.5.26. Fix a topological group G acting faithfully on a space F , and let $\pi_1: E_1 \rightarrow B$ and $\pi_2: E_2 \rightarrow B$ be two fiber bundles with fiber F and structure group G . We define a *morphism of fiber bundles* between them to be any continuous map $h: E_1 \rightarrow E_2$ making the diagram commute:

$$\begin{array}{ccc} E_1 & \xrightarrow{h} & E_2 \\ & \searrow \pi_1 & \swarrow \pi_2 \\ & B & \end{array}$$

It is clear that the identity map on E_1 gives a fiber bundle isomorphism from E_1 to itself, and it is moreover clear how to compose fiber bundle isomorphisms.

From this it seems obvious what we should call an *isomorphism* of fiber bundles: it should be any morphism h as in [Definition 1.5.26](#) that has a two-sided inverse, which is equivalent to h being a homeomorphism. Surely this is the definition right?

Wrong. Kinda.

We can certainly find places in the literature where isomorphisms of fiber bundles are described in this way, but to my knowledge this is ahistorical. Originally, in order to say that two fiber bundles are *equivalent* or *isomorphic*, we insist that in each fiber the homeomorphism $F \xrightarrow{\sim} F$ induced by h is not just any homeomorphism of spaces, it is in fact induced by an element of the structure group. If we wanted to be pedantic we might call this latter notion *isomorphism in the strict sense*, following [\[Ste51, §2.4\]](#).

What do we mean by “induced by the structure group”? If $b \in B$, we could look at the induced map $h_b: \pi_1^{-1}(b) \rightarrow \pi_2^{-1}(b)$, and we want to say that this arises via the action of some element $g \in G$. In order to do this, we have to identify these fibers with F , and there is a canonical way to do this, coming from the trivializations of each fiber bundle. To that end, we make the following definition:

Definition 1.5.27. Fix a topological group G acting faithfully on a space F , and let $\pi_1: E_1 \rightarrow B$ and $\pi_2: E_2 \rightarrow B$ be two fiber bundles with fiber F and structure group G . Let ϕ denote the trivializations for E_1 and ψ the trivializations for E_2 .

Let $h: E_1 \rightarrow E_2$ be a continuous map. We say it is an *isomorphism of fiber bundles* if:

1. the diagram commutes (i.e. h is a morphism of fiber bundles):

$$\begin{array}{ccc} E_1 & \xrightarrow{h} & E_2 \\ & \searrow \pi_1 & \swarrow \pi_2 \\ & B & \end{array}$$

2. for every $b \in B$ there exists some open neighborhood $U \ni b$ trivializing both bundles, so that if we consider the composite of morphisms over U :

$$U \times F \xrightarrow{\phi^{-1}} \pi_1^{-1}(U) \xrightarrow{h} \pi_2^{-1}(U) \xrightarrow{\psi} U \times F, \quad (1.5.28)$$

there exists a continuous assignment we call $h_U: U \rightarrow G$ for which the map above sends $(u, f) \mapsto (u, h_U(u) \cdot f)$.

That is, the action on the fiber over u occurs via some element in G , but that element can change continuously in U . In nice cases, we can pick the U 's to come from an open cover — let's make this a precise definition:

Definition 1.5.29. Continuing the notation of **Definition 1.5.27**, let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of B . We say that h is a *fiber bundle isomorphism witnessed along $\mathcal{U}$* if, in **Equation (1.5.28)**, the open neighborhood $U \ni b$ can always be chosen to be an element in $\mathcal{U}$.

The following proposition is straightforward:

Proposition 1.5.30. If h is a vector bundle isomorphism as in **Definition 1.5.27**, then h is a homeomorphism.

Proof. Since G acts by automorphisms of the fiber, the entire composite in **Equation (1.5.28)** has a continuous inverse, hence it is a homeomorphism. Since ϕ and ψ are homeomorphisms by definition, we see that h is a homeomorphism when restricted to $\pi_1^{-1}(U) \rightarrow \pi_2^{-1}(U)$. As this is true for any U , it is clear that the total map h is a homeomorphism. $\square$

We see that h determines these maps $h_U: U \rightarrow G$, but could we have gone the other way and assembled h out of the h_U 's? The following theorem tells us when this is possible:

Theorem 1.5.31. Let $\pi_i: E_i \rightarrow B$ be two fiber bundles with the same structure group G and fiber F . Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of B which trivializes both bundles, and for any $i \in I$ let $\phi_i: \pi_1^{-1}(U_i) \xrightarrow{\sim} U_i \times F$ and $\psi_i: \pi_2^{-1}(U_i) \xrightarrow{\sim} U_i \times F$ denote these trivializations. Let $g_{ij}, \gamma_{ij}: U_i \cap U_j \rightarrow G$ be the cocycles attached to E_1 and E_2 , respectively. Then the following sets are in canonical bijection:

1. the set of vector bundle isomorphisms $h: E_1 \xrightarrow{\sim} E_2$ witnessed along $\mathcal{U}$
2. the set whose elements are tuples of continuous maps $\left\{ U_i \xrightarrow{h_i} G \right\}_{i \in I}$ so that, for all $i, j \in I$, we have that $\gamma_{ij} \cdot h_i = h_j \cdot g_{ij}$ are equal as maps $U_i \cap U_j \rightarrow G$, where the multiplication is occurring pointwise in G .

Proof. Given a vector bundle isomorphism h , by definition, the h_U 's are determined by the definition of vector bundle isomorphism. We argue that they satisfy the desired property. For any i, j , consider the commutative diagram of maps over $U_i \cap U_j$

$$\begin{array}{ccccc}
 U_i \cap U_j \times F & \xrightarrow{(u,f) \mapsto (u, h_i(u)f)} & & \xrightarrow{\psi_i} & U_i \cap U_j \times F \\
 \downarrow (u,f) \mapsto (u, g_{ij}(u)f) & \swarrow \phi_i & \pi_1^{-1}(U_i \cap U_j) \xrightarrow{h} \pi_2^{-1}(U_i \cap U_j) & \searrow \psi_j & \downarrow (u,f) \mapsto (u, \gamma_{ij}(u)f) \\
 U_i \cap U_j \times F & \swarrow \phi_j & & \searrow \psi_j & U_i \cap U_j \times F \\
 & \xrightarrow{(u,f) \mapsto (u, h_j(u)f)} & & &
 \end{array}$$

The outer triangles commute by definition of the cocycles, and the upper and lower trapezoids commute by definition of the h_i 's. Tracing an element along the outermost rectangle, we see that we must have $\gamma_{ij}(u) \cdot h_i(u) \cdot f = h_j(u) \cdot g_{ij}(u) \cdot f$ for all $u \in U_i \cap U_j$ and all $f \in F$. Since G acts faithfully, this is the same as saying that $\gamma_{ij} \cdot h_i = h_j \cdot g_{ij}$ are equal as maps $U_i \cap U_j \rightarrow G$.

Conversely, suppose we are given some elements $\{U_i \xrightarrow{h_i} G\}$ satisfying this condition. For any $i \in I$, we can define a map $h_{U_i}: \pi_1^{-1}(U_i) \rightarrow \pi_2^{-1}(U_i)$ as the unique homeomorphism filling in the dashed arrow in the diagram

$$\begin{array}{ccc}
 \pi_1^{-1}(U_i) & \xrightarrow{\quad h_{U_i} \quad} & \pi_2^{-1}(U_i) \\
 \phi_i \downarrow \cong & & \cong \downarrow \psi_i \\
 U_i \times F & \xrightarrow[\quad (u,f) \mapsto (u, h_i(u)f) \quad]{\cong} & U_i \times F
 \end{array}$$

We want to check that the h_{U_i} 's glue together to give a well-defined map $E_1 \rightarrow E_2$, that is, we must check they agree on the overlaps. Restricting the square above to $U_i \cap U_j$, and flipping it over and turning it into a trapezoid suggestively, we get a commutative diagram

$$\begin{array}{ccc}
 U_i \cap U_j \times F & \xrightarrow{(u,f) \mapsto (u, h_i(u)f)} & U_i \cap U_j \times F \\
 \swarrow \phi_i & \pi_1^{-1}(U_i \cap U_j) \xrightarrow{h_{U_i|U_i \cap U_j}} \pi_2^{-1}(U_i \cap U_j) & \searrow \psi_i
 \end{array}$$

On the analogous diagram for h_{U_j} , we similarly get a commutative diagram:

$$\begin{array}{ccc}
 & \pi_1^{-1}(U_i \cap U_j) \xrightarrow{h_{U_j|U_i \cap U_j}} \pi_2^{-1}(U_i \cap U_j) & \\
 \swarrow \phi_j & & \searrow \psi_j \\
 U_i \cap U_j \times F & \xrightarrow{(u,f) \mapsto (u, h_j(u)f)} & U_i \cap U_j \times F
 \end{array}$$

Patching these diagrams together with the one defining the 1-cocycles for each bundle, we get a big

diagram (we don't know it all commutes yet):

$$\begin{array}{ccc}
 U_i \cap U_j \times F & \xrightarrow{(u,f) \mapsto (u, h_i(u)f)} & U_i \cap U_j \times F \\
 \downarrow (u,f) \mapsto (u, g_{ij}(u)f) & \swarrow \phi_i \quad \pi_1^{-1}(U_i \cap U_j) \xrightarrow{h_{U_i|U_i \cap U_j}} \pi_2^{-1}(U_i \cap U_j) \searrow \psi_i & \downarrow (u,f) \mapsto (u, \gamma_{ij}(u)f) \\
 & \parallel \quad \pi_1^{-1}(U_i \cap U_j) \xrightarrow{h_{U_j|U_i \cap U_j}} \pi_2^{-1}(U_i \cap U_j) \parallel & \\
 U_i \cap U_j \times F & \xleftarrow{\phi_j} \quad \pi_1^{-1}(U_i \cap U_j) \xrightarrow{h_{U_j|U_i \cap U_j}} \pi_2^{-1}(U_i \cap U_j) \searrow \psi_j & U_i \cap U_j \times F \\
 & \searrow \psi_j & \\
 & \xrightarrow{(u,f) \mapsto (u, h_j(u)f)} &
 \end{array}$$

All the outer trapezoids commute by construction, but we don't know the inner square commutes a priori. If we had chosen the h_i 's at random, there is absolutely no reason to expect it to commute. However, since $\gamma_{ij} \cdot h_i = h_j \cdot g_{ij}$, the outermost square commutes, and therefore the innermost square must commute as well. But commutativity of this inner square (where the vertical arrows are equalities) is exactly saying that $h_{U_i|U_i \cap U_j} = h_{U_j|U_i \cap U_j}$. Therefore all the h_{U_i} 's glue together to give a fiber bundle isomorphism $h: E_1 \rightarrow E_2$. $\square$

The elegance of this theorem is it allows us to reframe fiber bundle isomorphisms in the language of non-abelian Čech cohomology:

Theorem 1.5.32. Fix a topological group G acting faithfully on a space F . Let B be any space and $\mathcal{U}$ any open cover of B . Then there is a natural bijection:

$$\frac{\{G\text{-fiber bundles with fiber } F \text{ trivialized over } \mathcal{U}\}}{\text{isomorphism witnessed along } \mathcal{U}} \leftrightarrow \check{H}^1(\mathcal{U}, G). \quad (1.5.33)$$

Proof. The bijection between fiber bundles and 1-cocycles is in [Theorem 1.5.21](#), and it is clear by [Theorem 1.5.31](#) that fiber bundles being isomorphic is *exactly the same* as the associated 1-cocycles being equivalent, in the sense of [Definition 1.5.17](#).¹⁴ $\square$

As every isomorphism of fiber bundles is necessarily witnessed along some cover, we're tempted to take colimits along all cover $\mathcal{U}$, and say that $\check{H}^1(X, G)$ is in bijection with *all* principal G -bundles on X . However this is wrong! There's a subtlety in that we'll actually only be able to detect the *numerable* bundles.

Remark 1.5.34. Suppose we have a principal G -bundle P over X , and we've picked two covers $\mathcal{U}$ and $\mathcal{V}$ which trivialize the bundle, so we get classes $u \in \check{H}^1(\mathcal{U}, G)$ and $v \in \check{H}^1(\mathcal{V}, G)$ which each correspond to P . We want to be able to argue that u and v agree in the colimit. In order to do this, we need to take a common refinement of $\mathcal{U}$ and $\mathcal{V}$ and show they differ by a 0-cochain along this refined cover. Unless the cover is locally finite, there's no reason to expect this is possible – we may need to make arguments about modifying infinitely many overlaps at once, and there's no reasonable way to do this continuously unless there is extra structure.

Over paracompact spaces, we can always take locally finite refinements, so we don't have to worry about this. We therefore get the following:

¹⁴In fact more is true: not only are the equivalence relations the same, the *data* which exhibits the equivalence is that same. That is, the elements $h_i: U_i \rightarrow G$ we use to exhibit an isomorphism between two bundles are the same data as the 0-cocycles we use to exhibit cohomology of the 1-cocycles.

Theorem 1.5.35. Fix a topological group G and a faithful action on some space F . Then for any *paracompact* space X , we have that $\check{H}^1(X, G_c)$ is in bijection with the set of isomorphism classes of fiber bundles over X with structure group G and fiber F .

Proof. Take colimits over $\mathcal{U} \in \text{Cov}(X)^{\text{op}}$ in [Theorem 1.5.32](#) and exploit that we have locally finite refinements. $\square$

Example 1.5.36. In the theorem above, the trivial fiber bundle corresponds to the identity element of the group $H^1(X, G_c)$.

1.6 Principal G -bundles

Any faithful G -action on a fiber F yields a study of fiber bundles with structure group G and fiber F . There is a distinguished example of such an F , namely G itself, where the action is via left translation. These are called *principal fiber bundles*. Here's a precise definition.

Definition 1.6.1. A fiber bundle $\pi: Y \rightarrow X$ with structure group G and fiber F is said to be *principal* if G acts simply transitively on Y , if G is homeomorphic to F along the action by left translation.

Example 1.6.2. The trivial bundle $G \times X \rightarrow X$ is a principal G -bundle, corresponding to the identity element in $\check{H}^1(X, G)$ by [Example 1.5.24](#).

Remark 1.6.3. There's a pretty horrible conflation between “a trivial principal bundle” and “the trivial principal bundle” in the literature. Writing a trivial principal G -bundle as $G \times X \rightarrow X$ is a very bad practice, since it assumes we know where the identity is in each fiber. A priori we don't know this, that's why Steenrod warned us to be careful about this [[Ste51](#), pp.37-38]. To illustrate by example, consider the fold map

$$S^1 \amalg S^1 \rightarrow S^1.$$

Let the nonidentity element of C_2 swap the two copies of S^1 upstairs, and let the identity element fix them. Then we'd be correct to say the projection above is a trivial principal C_2 -bundle. But it is *not* the same data as $S^1 \times C_2 \rightarrow S^1$, since we don't have a preferred copy of S^1 in the fiber corresponding to the identity element. We can *pick* one, and define a fiberwise homeomorphism $S^1 \amalg S^1 \cong S^1 \times C_2$, but there are two ways to do this. This is a shade of a more common general conflation between a group G and a *torsor* for it, which is by definition any G -set with faithful transitive G -action.

We can always *pick* an identity continuously in each fiber. That is, if $Y \rightarrow X$ is a trivial G -bundle, we can take a section $\sigma: X \rightarrow Y$ and define a trivialization of the fiber by allowing $\sigma(x)$ to be the identity in Y_x . We claim this defines an isomorphism of fiber bundles between Y and $G \times X$. Steenrod calls this the *cross-section theorem* [[Ste51](#), I.8.3].

Theorem 1.6.4 (Cross-section theorem). Let $\pi: Y \rightarrow X$ be a principal G -bundle. Then it is trivial if and only if it admits a global section.

Sketch. Verify the data of a global section $\sigma: X \rightarrow Y$ is the same data as a homeomorphism between π and the product bundle $G \times X \rightarrow X$. $\square$

So we see something interesting, namely that a principal G -bundle defines a class in $H^1(X, G)$, and that the failure of this class to be the identity element is measuring the failure of the existence of a global section. Phrased differently, the cohomology class in $H^1(X, G)$ is *obstructing* the existence of a global section. We'll see this theme again in the theory of vector bundles.

1.6.1 Factor spaces

A crucial example of principal bundles comes from factor groups.

Definition 1.6.5. If G is a topological group and $H \leq G$ is a closed subgroup, we define the *factor space* to be the set of left cosets G/H , equipped with a topology defined by $U \subseteq G/H$ being open if and only if $\pi^{-1}(U) \subseteq G$ is open, where π is the projection map

$$\pi: G \rightarrow G/H.$$

Exercise 1.6.6. If G is a topological group and $H \leq G$ is a closed subgroup then G/H is Hausdorff. We have that G/H is Hausdorff.

Here is an important theorem, which can be found in [Ste51, p. 30] in slightly different language.

Theorem 1.6.7. Let G be any topological group and $H \leq G$ a closed subgroup. Let $\pi: G \rightarrow G/H$ be the surjective open map corresponding to projection onto cosets. Let $\text{Sec}(-, \pi)$ be the sheaf of sets on G/H given by local continuous sections of π . Then the following are equivalent:

1. $\pi: G \rightarrow G/H$ is a fiber bundle with fiber H and structure group H acting on itself via left translation
2. the stalk $\text{Sec}(-, \pi)_{eH}$ at the identity coset is nonempty.

In the classical literature, an element of this stalk is called a *local cross-section*, and demonstrating existence of local cross-sections was the general strategy for proving various projections were fiber bundles.

Proof. The forwards direction is immediate, since fiber bundles always admit locally defined sections, so we need to argue the backwards direction. Suppose the stalk is nonempty, then by definition there is some open neighborhood $U \ni eH$ and a continuous section $s: U \rightarrow \pi^{-1}(U)$. We see that $s(U) \subseteq G$ is just some subspace, and we can use the H -action to permute it around in G . We claim this doesn't create any overlaps, that is, $s(U) \cap h \cdot s(U) = \emptyset$ for all $h \neq e$. Indeed if it did, we would have some $u_1, u_2 \in U$ and $h \in H$ for which

$$s(u_1) = h \cdot s(u_2).$$

Applying π and leveraging that π is H -equivariant, we get that $u_1 = u_2$, hence $s(u_1) = s(u_2)$ in G , and $h = e$.

We therefore have an injective continuous map

$$\begin{aligned} U \times H &\xrightarrow{\sim} \pi^{-1}(U) \\ (u, h) &\mapsto h \cdot s(u). \end{aligned} \tag{1.6.8}$$

We claim this is a homeomorphism. We see that for any $y \in \pi^{-1}(U)$, there is a unique $h \in H$ for which $hs(u) \ni y$, which defines for us an H -equivariant function

$$\pi^{-1}(U) \rightarrow H.$$

We want to argue that this is continuous. By H -equivariance, it suffices to check continuity by looking at a neighborhood basis of the origin in H . Let $K \subseteq H$ be any such basis, then the preimage of K under the map above is the elements $y \in \pi^{-1}(U)$ so that $ks(u) \ni y$ for some $k \in K$. That is,

it is exactly $K \cdot s(U)$. Since $s(U)$ is open in $\pi^{-1}(U)$ so is $K \cdot s(U) = \cup_{k \in K} s(U)$, since each k acts via a homeomorphism, and we just get a union of opens.

Hence we get an continuous map

$$\phi: \pi^{-1}(U) \rightarrow U \times H$$

which is the product of π and the continuous map $\pi^{-1}(U) \rightarrow H$ defined above, and it is easy to check it is inverse to Equation (1.6.8), hence a homeomorphism.

We can now prove that $\pi: G \rightarrow G/H$ is a fiber bundle. Consider the open cover $\mathcal{U} = \{g \cdot U\}_{g \in G}$ of G/H , where U is as above. By H -equivariance of all the constructions above, we have homeomorphisms

$$g\phi: \pi^{-1}(gU) \xrightarrow{\sim} gU \times H$$

for every homeomorphism in the open cover. Given an overlap $g_1U \cap g_2U \neq \emptyset$, we have that $g_2^{-1}g_1U = U$, hence $g_2g^{-1} \in H$. Therefore the composite

$$(g_1U \cap g_2U) \times H \xrightarrow{(g_2\phi)^{-1}} \pi^{-1}(g_1U \cap g_2U) \xrightarrow{g_1\phi} (g_1U \cap g_2U) \times H$$

is constant on that second factor H , and is just multiplication by g_2g^{-1} on the first factor. This is clearly a section of the structure group for our bundle, which is H , therefore we are done. $\square$

Theorem 1.6.9 ([Gle50, 4.1]). If G is an arbitrary topological group, and $H \leq G$ is a closed subgroup which is a compact Lie group, then $G \rightarrow G/H$ is a fiber bundle with fiber H and structure group H and action given by left translation.

Remark 1.6.10. It's entirely possible to not admit a local cross-section. Steenrod refers to an unpublished result of Hanner for an example [Ste51, p. 33]. Karube gives us another example – let $G = \prod_{i=1}^{\infty} S^1$, let $\pi_n: G \rightarrow S^1$ be the projection onto the n th circle. Take $H \leq G$ to be any closed subgroup for which $\pi_n(H) \leq S^1$ is finite and nonzero. Then $G \rightarrow G/H$ admits no cross-section, hence is not a fiber bundle [Kar58].

1.6.2 Identifying factor spaces

We now know that we can build a principal H -bundle $G \rightarrow G/H$ for any G, H as before. This doesn't necessarily help us understand G/H as a topological space though. A nice strategy to assemble explicit fiber bundles is to start with a transitive G -action on a space, and let H be the stabilizer of any point of that space.

Proposition 1.6.11. Suppose G acts transitively on a Hausdorff space Y , let $y \in Y$ be any arbitrary basepoint, and let $p: G \rightarrow Y$ be the continuous map sending g to gy . Define the *stabilizer subgroup*

$$G_y := \{g \in G \mid gy = y\}.$$

Then we have that

1. $G_y \leq G$ is closed,
2. p factors through the quotient by G_y , inducing a map

$$\begin{aligned} G/G_y &\rightarrow Y \\ g &\mapsto g \cdot y \end{aligned} \tag{1.6.12}$$

which is continuous and bijective

3. if furthermore G is compact, then [Equation \(1.6.12\)](#) is a homeomorphism.

Proof. The closed condition is immediate (exercise!). We define a map

$$\begin{aligned} G/G_y &\rightarrow Y \\ g &\mapsto g \cdot y. \end{aligned}$$

This is continuous since the action of G on Y is continuous, it is injective by definition of the isotropy subgroup, and surjective by transitivity. If G is compact, then so is G/G_y (image of compact is compact). Therefore [Equation \(1.6.12\)](#) is a homeomorphism since every bijective continuous maps from a compact space to a Hausdorff space is automatically a homeomorphism. $\square$

Notation 1.6.13. Given a fiber bundle $f: Y \rightarrow X$ with fiber F , we often write it as what's called a *fiber sequence*

$$F \hookrightarrow Y \rightarrow X.$$

This should in some sense remind us of a short exact sequence.

Example 1.6.14 (Orthogonal group fiber sequence). Recall a real matrix is *orthogonal* if its transpose and inverse coincide. Define the *orthogonal group* $O(n)$ as

$$O(n) := \{M \in \mathrm{GL}_n(\mathbb{R}) : M^t = M^{-1}\}.$$

We claim $O(n)$ is compact (exercise). There is a natural inclusion of $O(n)$ into $O(n+1)$ by sticking a 1 in the bottom right corner and a 0 elsewhere:

$$\begin{aligned} O(n) &\rightarrow O(n+1) \\ M &\mapsto \begin{pmatrix} M & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Note that $O(n)$ acts transitively on the (Hausdorff) unit sphere $S^{n-1} \subseteq \mathbb{R}^n$. Fixing any unit vector $v \in S^{n-1}$, we can verify that its isotropy group is isomorphic to $O(n-1)$. Hence by [Proposition 1.6.11](#) we get a homeomorphism

$$O(n)/O(n-1) \cong S^{n-1}.$$

Hence we have a fiber sequence

$$O(n-1) \hookrightarrow O(n) \rightarrow S^{n-1}.$$

Example 1.6.15. Completely analogously, the *unitary group* $U(n)$ is defined to be those complex matrices whose conjugate transpose are equal to their inverse:

$$U(n) := \{M \in \mathrm{GL}_n(\mathbb{C}) : M^* M = I\}.$$

This group is also compact, acting on $\mathbb{C}^n$ and also on the unit sphere $S^{2n-1} \subseteq \mathbb{C}^n$. By a similar argument to above, we have a homeomorphism $U(n)/U(n-1) \cong S^{2n-1}$, and a fiber sequence

$$U(n-1) \hookrightarrow U(n) \rightarrow S^{2n-1}.$$

Example 1.6.16. The *compact symplectic group* $\mathrm{Sp}(n)$ is the subgroup of $\mathrm{GL}_n(\mathbb{H})$ preserving the standard Hermitian inner product on $\mathbb{H}^n$. We claim that $\mathrm{Sp}(n)$ acts transitively on the unit sphere S^{4n-1} with stabilizer $\mathrm{Sp}(n-1)$, giving a fiber sequence

$$\mathrm{Sp}(n-1) \hookrightarrow \mathrm{Sp}(n) \rightarrow S^{4n-1}.$$

Example 1.6.17 (Stiefel manifolds). Let

$$\begin{aligned} V_k(\mathbb{R}^n) &:= \{M \in M_{n \times k}(\mathbb{R}) : M^t M = I\} \\ V_k(\mathbb{C}^n) &:= \{M \in M_{n \times k}(\mathbb{C}) : M^* M = I\} \\ V_k(\mathbb{H}^n) &:= \{M \in M_{n \times k}(\mathbb{C}) : M^* M = I\} \end{aligned}$$

denote the *Stiefel manifolds* of k -frames in $\mathbb{R}^n$, $\mathbb{C}^n$, and $\mathbb{H}^n$, respectively. With the subspace topology from the Euclidean space of matrices, these are manifolds. Since orthogonal matrices preserve frames, we have that $O(n)$ acts transitively on $V_k(\mathbb{R}^n)$. The isotropy subgroup of a given frame is $O(n-k)$. Hence we have

$$\begin{aligned} O(n)/O(n-k) &\cong V_k(\mathbb{R}^n) \\ U(n)/U(n-k) &\cong V_k(\mathbb{C}^n) \\ \mathrm{Sp}(n)/\mathrm{Sp}(n-k) &\cong V_k(\mathbb{H}^n). \end{aligned}$$

An important space to define is the *Grassmannian* $\mathrm{Gr}(k, n)$, which is the moduli space of k -dimensional subspaces of an n -dimensional vector space. Even before defining it, we note that it should receive a map from the Stiefel manifold, given by taking the *span* of a given k -frame. Since the $O(k)$ -action on a given frame doesn't affect its span, it makes sense we should mod out by this.

Definition 1.6.18. We define the *Grassmannian* to be

$$\begin{aligned} \mathrm{Gr}_{\mathbb{R}}(k, n) &:= O(k) \backslash V_k(\mathbb{R}^n) \cong O(k) \backslash O(n) / O(n-k) \\ \mathrm{Gr}_{\mathbb{C}}(k, n) &:= U(k) \backslash V_k(\mathbb{C}^n) \cong U(k) \backslash U(n) / U(n-k) \\ \mathrm{Gr}_{\mathbb{H}}(k, n) &:= \mathrm{Sp}(k) \backslash V_k(\mathbb{H}^n) \cong \mathrm{Sp}(k) \backslash \mathrm{Sp}(n) / \mathrm{Sp}(n-k). \end{aligned}$$

Then, by definition, we have principal fiber bundles

$$\begin{aligned} O(k) &\hookrightarrow V_k(\mathbb{R}^n) \rightarrow \mathrm{Gr}_{\mathbb{R}}(k, n) \\ U(k) &\hookrightarrow V_k(\mathbb{C}^n) \rightarrow \mathrm{Gr}_{\mathbb{C}}(k, n) \\ \mathrm{Sp}(k) &\hookrightarrow V_k(\mathbb{H}^n) \rightarrow \mathrm{Gr}_{\mathbb{H}}(k, n). \end{aligned}$$

1.6.3 Dold's classification theorem

Recall by [Theorem 1.5.35](#), given a topological group G and a space X , then isomorphism classes of principal G -bundles over X are classified by the set $H^1(X, G_c)$.

Notation 1.6.19. Recall, given two spaces X, Y , that $[X, Y]$ denotes the set of homotopy classes of maps between them.

Definition 1.6.20 ([Dol63, 7.2]). Let G be a topological group. We say a space B is *classifying* for G if there exists a natural isomorphism

$$T: [-, B] \Rightarrow \check{H}^1(-, G_c) \tag{1.6.21}$$

of functors $\mathrm{Top}^{\mathrm{op}} \rightarrow \mathrm{Set}$.

Remark 1.6.22 (Yoneda and the homotopy category). We are tempted to make a Yoneda-flavored argument about classifying spaces for principal G -bundles, since [Equation \(1.6.21\)](#) looks a lot like a natural transformation from a representable functor to a non-representable one. However for any space X , we note that $[X, B]$ is not a hom-set in the category Top of topological spaces. We can fix this, however, by defining the *homotopy category* of topological spaces $\text{Ho}(\text{Top})$.

Proposition 1.6.23. Let G be a topological group. There is a natural bijection

$$\text{Nat}([-, B], \check{H}^1(-, G_c)) \cong \check{H}^1(B, G_c).$$

In particular if B is *classifying*, the choice of natural isomorphism corresponds to a preferred element in $\check{H}^1(B, G_c)$, whose associated principal G -bundle we denote by EG , and call it the *universal principal G -bundle*. This has the property that any other principal G -bundle $E \rightarrow X$ is of the form $E \cong f^*EG$, where $f: X \rightarrow B$ is some (homotopy class of) map.

Proof sketch. Once the category theory is set up correctly, the remainder of this proposition is just the Yoneda lemma. Define $\text{Ho}(\text{Top})$ to be the *homotopy category* of spaces — objects are the same as those in Top , and hom-sets are homotopy classes of maps between two spaces:

$$\text{Hom}_{\text{Ho}(\text{Top})}(X, Y) := [X, Y].$$

Then $[-, B]$ is indeed a representable functor out of $\text{Ho}(\text{Top})$. In order to apply Yoneda, however, we need to know that $\check{H}^1(-, G_c)$ descends to a functor out of $\text{Ho}(\text{Top})$, that is, non-abelian Čech cohomology is homotopy invariant. This is left as an exercise. $\square$

Example 1.6.24 (Universal C_2 -bundles). Let C_2 be the cyclic group of order two. For any n , the sphere double covers real projective space:

$$S^n \rightarrow \mathbb{RP}^n,$$

and this is a principal C_2 -bundle. None of these are universal, but if we let $n \rightarrow \infty$, we get a principal C_2 -bundle:

$$S^\infty \rightarrow \mathbb{RP}^\infty.$$

We will very soon see that this is universal! What this means is that if X is any space, and $E \rightarrow X$ is any principal C_2 -bundle, then there exists some map $f: X \rightarrow \mathbb{RP}^\infty$ so that E is the pullback of S^∞ , that is, we have a pullback diagram of spaces:

$$\begin{array}{ccc} E & \longrightarrow & S^\infty \\ \downarrow & \lrcorner & \downarrow \\ X & \xrightarrow{f} & \mathbb{RP}^\infty. \end{array}$$

We now want to figure out how to argue whether a given space is *classifying* for principal G -bundles. The *classification theorem* of Dold gives us a lovely way to do this:

Definition 1.6.25. We say that a space is *contractible* if it has the homotopy type of a point. If the space is a CW complex, then this is equivalent to the statement that all its homotopy groups vanish.

Theorem 1.6.26 (Classification theorem, [Dol63, 7.5]). Let $\eta: E \rightarrow B$ be a numerable ([Definition A.2.13](#)) principal G -bundle. Then it is classifying if and only if E is contractible.

Sketch. First suppose $\eta: E \rightarrow B$ is contractible, and take some other principal G -bundle $\zeta: V \rightarrow X$. We want to find a map $f: X \rightarrow B$ for which $V \cong f^*E$.

We define a new bundle

$$(\zeta, \eta) \rightarrow X,$$

where the fiber over x is the set of all possible bundle maps

$$\begin{array}{ccc} V_x & \longrightarrow & E \\ \downarrow & & \downarrow \\ x & \longrightarrow & B. \end{array}$$

It is clear to see this bundle has fibers $(\zeta, \eta)_x \cong E$, since the G -action on the fibers implies that the bundle map is determined by the image of one point. Moreover, numerability of ζ implies numerability of (ζ, η) .

A section of this bundle would be the data we're looking for — it would be a bundle map

$$\begin{array}{ccc} V & \longrightarrow & E \\ \downarrow & & \downarrow \\ X & \longrightarrow & B, \end{array}$$

and this will necessarily be a pullback, since every principal G -bundle map is an isomorphism. So it suffices to argue that (ζ, η) admits a global section. We can define a section locally on one of our covers over which E is trivialize, then use contractability of E to extend it to a global section (see [Dol63, 2.6, 2.7, 2.8]).

For the other direction, suppose $\eta: E \rightarrow B$ is universal. If we can prove the existence of a contractible numerable principal bundle ([Dol63, §8]), call it $C \rightarrow X$, then it is also universal by the previous direction, so we have a composite

$$\begin{array}{ccccc} E & \longrightarrow & C & \longrightarrow & E \\ \downarrow & & \downarrow & & \downarrow \\ B & \longrightarrow & X & \longrightarrow & B. \end{array}$$

The map on bottom is homotopic to the identity, so the composite map on top is a homeomorphism $E \rightarrow E$. However this factors through a contractible space, hence E must be contractible as well. $\square$

Remark 1.6.27. We now know how to prove that $S^\infty \rightarrow \mathbb{RP}^\infty$ is the universal C_2 -bundle as we claimed in Example 1.6.24. By Dold's classification theorem Theorem 1.6.26, it suffices to argue that the infinite sphere S^∞ is contractible, which is a basic exercise in topology (see for instance [Hat01, p. 19]).

Proposition 1.6.28. Let G be a topological group, and suppose that B_1 and B_2 are two classifying spaces for principal G -bundles. Then B_1 and B_2 are homotopy equivalent.

Proof. Both B_1 and B_2 represent the functor

$$\check{H}^1(-, G): \text{Ho}(\text{Top})^{\text{op}} \rightarrow \text{Set},$$

therefore they must be isomorphic in the domain category $\text{Ho}(\text{Top})$. One checks that B_1 and B_2 being isomorphic in $\text{Ho}(\text{Top})$ is the same as them being homotopy equivalent as spaces.¹⁵ $\square$

¹⁵An isomorphism class of object in $\text{Ho}(\text{Top})$ is often referred to as a *homotopy type*. This is potentially overloaded terminology, but at least among CW complexes there is no confusion using this terminology.

Notation 1.6.29. Since the classifying space for principal G -bundles (assuming it exists, we haven't argued this yet) is well-defined up to homotopy equivalence, one usually calls it “ BG ” and the universal principal G -bundle over it is denoted $EG \rightarrow BG$. In casual conversation you'll often see topologists write something like

$$\mathbb{RP}^\infty = BC_2,$$

and this is intended to mean that $\mathbb{RP}^\infty$ is (the homotopy type of) the classifying space for C_2 .

Theorem 1.6.30 (Existence of universal bundles). Let G be a topological group. Then a universal G -bundle exists.

Proof. Omitted, see [AGP02, §12.2] or [May, §16.5]. $\square$

1.6.4 Universal bundles for the general linear group

Notation 1.6.31. Denote by $V_k(\mathbb{R}^\infty)$ the *infinite Stiefel manifold*, defined as the colimit

$$V_k(\mathbb{R}^\infty) := \operatorname{colim}_{n \rightarrow \infty} V_k(\mathbb{R}^n)$$

along the natural inclusions. Similarly we define

$$\operatorname{Gr}_{\mathbb{R}}(k, \infty) := \operatorname{colim}_{n \rightarrow \infty} \operatorname{Gr}_{\mathbb{R}}(k, n).$$

Proposition 1.6.32. The induced map

$$V_k(\mathbb{R}^\infty) \rightarrow \operatorname{Gr}_{\mathbb{R}}(k, \infty)$$

is still a principal $O(k)$ -bundle.

Proof sketch. We can commute the quotient by $O(k)$ with the colimit defining the infinite Stiefel manifold (todo: add more details). $\square$

Theorem 1.6.33. The space $V_k(\mathbb{R}^\infty)$ is contractible, and therefore

$$V_k(\mathbb{R}^\infty) \rightarrow \operatorname{Gr}_{\mathbb{R}}(k, \infty)$$

is a universal principal $O(k)$ -bundle. Similar statements for $U(k)$ and $\operatorname{Sp}(k)$.

Proof sketch. This will follow by Dold's classification theorem (Theorem 1.6.26). We have to check that $\operatorname{Gr}(k, \infty)$ is paracompact so that the infinite Stiefel manifold gives a numerable cover. The proof the the infinite Stiefel manifold is contractible is essentially a generalization of the proof that S^∞ is contractible. $\square$

1.7 Vector bundles revisited

Exercise 1.7.1. There is a natural inclusion $O(k) \subseteq \operatorname{GL}_k(\mathbb{R})$. Show that, if X is paracompact, then the map

$$\check{H}^1(X, O(k)) \rightarrow \check{H}^1(X, \operatorname{GL}_k(\mathbb{R}))$$

is a bijection. That is, we can always reduce the structure group to the orthogonal group $O(k)$ if we like.

A similar thing is true for $U(k) \subseteq \mathrm{GL}_k(\mathbb{C})$ and complex vector bundles. So we can redefine a vector bundle in the language of fiber bundles.

Definition 1.7.2. A *real/complex/symplectic* vector bundle of rank k is a fiber bundle with structure group $O(k)/U(k)/\mathrm{Sp}(k)$ and fiber $\mathbb{R}^k/\mathbb{C}^k/\mathbb{H}^k$.

Definition 1.7.3. If $E \rightarrow X$ is a real vector bundle, we denote by $\mathrm{Fr}(E)$ the *frame bundle*, whose fibers $\mathrm{Fr}(E)_x$ consist of all the orthonormal frames in E_x . We claim that $\mathrm{Fr}(E)$ is a principal $O(k)$ -bundle over X .

Exercise 1.7.4. For a paracompact space X , under the sequence of natural bijections

$$\mathrm{Prin}_{O(k)}(X) \leftrightarrow \check{H}^1(X, O(k)) \leftrightarrow \mathrm{Vect}_{\mathbb{R}}^k(X),$$

we have that $\mathrm{Fr}(E)$ corresponds to the vector bundle $E \rightarrow X$.

This tells us that:

- ▷ a vector bundle can be recovered by its frame bundle
- ▷ every principal $O(k)$ -bundle is the frame bundle of some vector space, uniquely determined up to isomorphism
- ▷ if a space B is classifying for $O(k)$ -bundles then it is classifying for vector bundles!

We know which space is classifying for $O(k)$ – it is the infinite Stiefel manifold $V_k(\mathbb{R}^\infty)$. So we just want to understand, under the bijection between principal $O(k)$ -bundles and real rank k vector bundles, what does $V_k(\mathbb{R}^\infty)$ correspond to?

The answer is the so-called *tautological bundle*. Let's see a small-dimensional example:

Notation 1.7.5. We denote by

$$E_k(\mathbb{R}^n) := \{(v, P) \in \mathbb{R}^n \times \mathrm{Gr}_{\mathbb{R}}(k, n) \mid v \in P\}.$$

This is, by construction, a vector bundle over $\mathrm{Gr}_{\mathbb{R}}(k, n)$. We call this the *tautological vector bundle*.

Proposition 1.7.6. The tautological bundle $E_k(\mathbb{R}^n)$ is pulled back from $E_k(\mathbb{R}^{n+1})$ along the inclusion

$$\mathrm{Gr}_{\mathbb{R}}(k, n) \subseteq \mathrm{Gr}_{\mathbb{R}}(k, n+1),$$

Proposition 1.7.7. There is a tautological bundle

$$E_k(\mathbb{R}^\infty) \rightarrow \mathrm{Gr}_{\mathbb{R}}(k, \infty)$$

which pulls back to the tautological bundle on $\mathrm{Gr}_{\mathbb{R}}(k, n)$ for any n .

Remark 1.7.8. Similar statements are true for $\mathrm{Gr}_{\mathbb{C}}(k, n)$ and $\mathrm{Gr}_{\mathbb{C}}(k, \infty)$, except the tautological bundle is now a complex (in fact *holomorphic*) vector bundle.

Example 1.7.9. Over $\mathbb{CP}^n = \mathrm{Gr}_{\mathbb{C}}(1, n+1)$, the tautological line bundle has transition functions

$$g_{ij} = z_j/z_i$$

over the standard open cover $U_i = \{[z_0 : \dots : z_n] \mid z_i \neq 0\}$.

Remark 1.7.10. Along the isomorphism $\mathrm{Gr}_{\mathbb{C}}(1, n) \cong \mathbb{CP}^{n-1}$, we have that $E_1(\mathbb{C}^n)$ is $\mathcal{O}_{\mathbb{P}^{n-1}}(-1)$.

Theorem 1.7.11. If X is paracompact, the maps

$$\begin{aligned} [X, \mathrm{Gr}_{\mathbb{R}}(k, \infty)] &\rightarrow \mathrm{Vect}_{\mathbb{R}}^k(X) \\ [X, \mathrm{Gr}_{\mathbb{C}}(k, \infty)] &\rightarrow \mathrm{Vect}_{\mathbb{C}}^k(X). \end{aligned}$$

is a bijection, natural in X , defined by pulling back the tautological vector bundle.

Proof sketch. Combining Dold's classification theorem [Theorem 1.6.26](#), and universality of the Stiefel manifolds [Theorem 1.6.33](#), the only thing remaining to check is that the tautological vector bundle is an associated bundle to the Stiefel manifolds, which is an immediate application of [Exercise 1.7.4](#). A self-contained proof can alternatively be found in [\[Hat, 1.16\]](#). $\square$

Remark 1.7.12. If X is compact (say, a finite-dimensional CW complex), then any map $X \rightarrow \mathrm{Gr}(k, \infty)$ lands in some $\mathrm{Gr}(k, N)$ for a finite N . In particular there is a more refined version of the classification theorem above for such spaces – see for instance [\[Ste51, 19.4\]](#).

Remark 1.7.13. If X is not paracompact, we still get a well-defined map

$$[X, \mathrm{Gr}_{\mathbb{R}}(k, \infty)] \rightarrow \mathrm{Vect}_{\mathbb{R}}^k(X),$$

whose image consists of all the numerable vector bundles on X .

1.7.1 Vector bundles with more structure

A real vector bundle over a paracompact space X is equivalent to an element in $\check{H}^1(X, \mathrm{GL}_r(\mathbb{R}))$, hence it is defined by some 1-cocycle (g_{ij}) of continuous maps

$$g_{ij}: U_i \cap U_j \rightarrow \mathrm{GL}_r(\mathbb{R}).$$

Note that each of the coordinates of the matrices g_{ij} outputs are continuous in the input, so we can think about each of the coordinates as a continuous map $U_i \cap U_j \rightarrow \mathbb{R}$, that is, a section in $C^0(U_i \cap U_j, \mathbb{R})$. We could therefore define a sheaf $\mathrm{GL}_r(C^0(-, \mathbb{R}))$ of $r \times r$ matrices of continuous sections which have determinant nonzero everywhere, and it is clear to see that this sheaf of groups is isomorphic to the one above.

From this perspective, it becomes clear how to ask for our cocycles to have more structure. We can take GL_r of the sheaf of differentiable, smooth, holomorphic, etc. maps, and we get the following definition.

Definition 1.7.14. Let X be a manifold which is smooth or differentiable, and let τ be some differentiability property of continuous real-valued functions of X (i.e. differentiable functions, C^p -functions, smooth functions). Then we define a τ -vector bundle on X is a vector bundle assembled from a 1-cocycle in $\check{H}^1(X, \mathrm{GL}_r(C_{\tau}^0(-, \mathbb{R})))$. Equivalently, a τ -vector bundle is a 1-cocycle

$$g_{ij}: U_i \cap U_j \rightarrow \mathrm{GL}_r(\mathbb{R})$$

so that each of the g_{ij} 's has property τ .¹⁶

¹⁶Note: there is some math here, namely that a map to $\mathrm{GL}_r(\mathbb{R})$ has a differentiability property if and only if all the coordinate projections to the individual entries of the matrix have that property.

Example 1.7.15. If M is a differentiable (resp. smooth) manifold, then the tangent bundle TM is differentiable (resp. smooth).

In general there is no obstruction to giving a topological vector bundle a smooth structure.

Proposition 1.7.16. If X is a smooth manifold, then every real (or complex) topological vector bundle on X is isomorphic to a smooth vector bundle.

Proof sketch. An elegant resolution to this problem is to endow $\mathrm{Gr}_{\mathbb{R}}(r, \infty)$ with a smooth structure, and then argue that a vector bundle classified by a map $X \rightarrow \mathrm{Gr}_{\mathbb{R}}(r, \infty)$ is smooth if and only if the map is smooth. It then suffices to show that any continuous map can be homotoped into a smooth one, which is Whitney’s approximation theorem (see for instance [Lee12, 10.16]). $\square$

A similar definition makes sense over complex manifolds:

Definition 1.7.17. Let X be a complex manifold, and $\xi: E \rightarrow X$ a complex topological vector bundle defined by transition maps g_{ij} valued in $\mathrm{GL}_r(\mathbb{C})$. We say ξ is *holomorphic* if the transition maps are holomorphic.

It is *not* true that every complex topological bundle on a complex manifold admits a holomorphic structure. We’ll see some explicit examples later, but first let’s see a sheafy perspective on where the obstruction arises.

The sheaf on X given by $\mathrm{GL}_r(\mathbb{C})$ can be thought of GL_r of the sheaf $C^0(-, \mathbb{C})$, that is, invertible $r \times r$ matrices of locally-defined continuous $\mathbb{C}$ -valued functions. Inside of $C^0(U, \mathbb{C})$ is the subsheaf $\mathcal{O}_X(U)$ of *holomorphic* $\mathbb{C}$ -valued functions. Hence there is a natural subsheaf inclusion

$$\mathrm{GL}_r(\mathcal{O}_X) \rightarrow \mathrm{GL}_r(\mathbb{C}).$$

This map, coming from “forgetting the holomorphic structure” induces a map

$$H^1(X, \mathrm{GL}_r(\mathcal{O}_X)) \rightarrow H^1(X, \mathrm{GL}_r(\mathbb{C})).$$

The former object is holomorphic bundles on X , and the latter is topological bundles.

Idea: Can we detect the failure of a complex topological bundle to admit a holomorphic structure by taking the cokernel of the inclusion $\mathrm{GL}_r(\mathcal{O}_X) \rightarrow \mathrm{GL}_r(\mathbb{C})$ and looking at the long exact sequence on H^1 ?

Why this won’t work: Unfortunately, the inclusion $\mathrm{GL}_r(\mathcal{O}_X) \subseteq \mathrm{GL}_r(\mathbb{C})$ is not a normal subgroup (conjugating holomorphic things by continuous things breaks holomorphic-ness), so there’s no reason the quotient will be a sheaf of groups. Even if it was a sheaf of groups, we want to say something about H^2 so we’d need it to be a sheaf of *abelian* groups.

However if both $\mathrm{GL}_r(\mathcal{O}_X)$ and $\mathrm{GL}_r(\mathbb{C})$ happened to be abelian, this wouldn’t be a problem. And indeed there is a nice case where this works, i.e. when $r = 1$. In this case, we get a natural map of sheaves $\mathcal{O}_X^* \rightarrow C^0(-, \mathbb{C}^*)$, taking a holomorphic nowhere zero section and forgetting that it was holomorphic. We can take its cokernel, and conclude confidently that the failure of a complex line bundle to admit a holomorphic structure is measured by this cokernel. A clever way to describe this failure is to compare the exponential sequences for holomorphic and continuous maps. We get a commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathcal{O}_X & \xrightarrow{\exp} & \mathcal{O}_X^* \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & C^0(-, \mathbb{C}) & \xrightarrow{\exp} & C^0(-, \mathbb{C}^*) \longrightarrow 0 \end{array}$$

Looking at the long exact sequence on cohomology, we get

$$\begin{array}{ccccccccccc}
 H^1(X, \mathbb{Z}) & \longrightarrow & H^1(X, \mathcal{O}_X) & \longrightarrow & H^1(X, \mathcal{O}_X^*) & \longrightarrow & H^2(X, \mathbb{Z}) & \longrightarrow & H^2(X, \mathcal{O}) & \longrightarrow & \dots \\
 \parallel & & \downarrow & & \downarrow & & \parallel & & \downarrow & & \\
 H^1(X, \mathbb{Z}) & \longrightarrow & 0 & \longrightarrow & H^1(X, \mathbb{C}^*) & \xrightarrow{\sim} & H^2(X, \mathbb{Z}) & \longrightarrow & 0 & \xrightarrow{\sim} & \dots
 \end{array}$$

Here the vanishing on bottom comes from $C^0(-, \mathbb{C})$ being fine, as we've seen (Example 1.2.57). So what is the obstruction to lifting? We've just shown the following by a nice diagram chase:

Proposition 1.7.18. On a complex manifold X , a complex line bundle admits a holomorphic structure if and only if the image of its class in $H^2(X, \mathcal{O}_X)$ vanishes.

If X is compact and Kähler we will see this is equivalent to being a Hodge class of type $(1, 1)$, but this sheafy obstruction argument works for any complex manifold.

1.7.2 Operations with vector bundles

Various operations we can do with vector bundles are reflected by operations on Grassmannians that satisfy associativity up to homotopy (up to what's called *higher coherence*). We can define operations on vector bundles either directly or by universal property combined with Yoneda. One advantage of the Yoneda perspective is we immediately obtain naturality of all the following constructions:

Proposition 1.7.19 (Operations with vector bundles). Given vector bundles $V_i \rightarrow X$, we can construct new vector bundles via:

1. direct sum (called *Whitney sum* in the older literature)
2. tensor product
3. hom bundles
4. duals
5. exterior powers
6. symmetric powers.

Moreover if the V_i 's are differentiable or complex analytic so are the resulting bundles

Proof. See [Hir78, 3.6.1] and the following pages for how these operations work in terms of cocycle data. Hirzebruch refers us to Bourbaki for a proof. $\square$

As a special case of obtaining sheaves from sections of fiber bundles, we obtain sheaves from sections of vector bundles. These have some special notations:

Proposition 1.7.20 ([Hir78, p. 46]). Let $V \rightarrow X$ be a vector bundle. We can define sheaves in the presence of more or less structure as follows:

- ▷ $\mathfrak{C}(V)$, the sheaf of germs of local continuous sections of a vector bundle over a space X
- ▷ $\mathfrak{A}(V)$, the sheaf of germs of local *differentiable* sections of a differentiable vector bundle over a differentiable manifold.

- ▷ $\Omega(V)$ the sheaf of germs of local *holomorphic* sections of a complex analytic vector bundle over a complex manifold X .

If X is paracompact, then $\mathfrak{C}(V)$ and $\mathfrak{A}(V)$ are fine. This follows by the existence of continuous (resp. differentiable) partitions of unity.

1.8 Chern classes

1.8.1 An axiomatic description

The general way to discuss Chern classes is to (a) lay out their axioms, (b) show the axioms uniquely describe them, and (c) construct them.

Definition 1.8.1. Given a complex vector bundle $E \rightarrow X$, we define the Chern classes $c_k(E) \in H^{2k}(X, \mathbb{Z})$ and the total Chern class $c(E) = \sum_{k \geq 0} c_k(E) \in H^*(X, \mathbb{Z})$ by the following axioms:

1. $c_0(E) = 1$ for all vector bundles
2. if $f: Y \rightarrow X$ is any map, then $f^*c(E) = c(f^*E)$ in $H^*(Y, \mathbb{Z})$
3. for any short exact sequence of vector bundles

$$E_1 \rightarrow E_2 \rightarrow E_3$$

on X , we have that $c(E_2) = c(E_1)c(E_3)$ in $H^*(X, \mathbb{Z})$.

4. the tautological line bundle $E_1(\mathbb{C}^n)$ on $\mathbb{C}P^n$ has total Chern class

$$c(E_1(\mathbb{C}^n)) = 1 - h \in H^*(\mathbb{C}P^n; \mathbb{Z})$$

where $h \in H^2(\mathbb{C}P^n; \mathbb{Z})$ is the hyperplane class.

Proposition 1.8.2. The Chern classes are uniquely determined by their axioms (that is, if we have elements $d_k(E) \in H^{2k}(X, \mathbb{Z})$ for any bundle E satisfying the axioms above, then $c_k(E) = d_k(E)$).

1.8.2 Constructing characteristic classes

A *characteristic class* is a way to assign, to every vector bundle over X a cohomology class over X .¹⁷ We want this assignment to be natural in X , that is, natural under pullbacks. We have an elegant way to talk about these following the Pontryagin–Steenrod theorem, combined with the Yoneda embedding.

Definition 1.8.3. A *complex characteristic class* for rank r vector bundles is a natural transformation

$$\text{Vect}_{\mathbb{C}}^r(-) \Rightarrow H^k(-, \mathbb{Z})$$

of functors $\text{Top}^{\text{op}} \rightarrow \text{Set}$ (or perhaps from $\text{Ho}(\text{Top})^{\text{op}} \rightarrow \text{Set}$). Here k can be whatever we want.

When we restrict our attention to paracompact spaces, we have that $\text{Vect}_{\mathbb{C}}^r(-)$ is *representable*, when viewed as a functor out of $\text{Ho}(\text{Top})$. So a characteristic class is a natural transformation of the form

$$[-, \text{Gr}_{\mathbb{C}}(r, \infty)] \Rightarrow H^k(-, \mathbb{Z}).$$

But by the Yoneda lemma, the set of natural transformations of the above is in bijection with the cohomology class $H^k(\text{Gr}_{\mathbb{C}}(r, \infty))$. So we get the following:

¹⁷More generally, to every principal G -bundle a cohomology class.

Slogan 1.8.4. A complex characteristic class for rank r vector bundles is an integral cohomology class of the complex Grassmannian $\mathrm{Gr}_{\mathbb{C}}(r, \infty)$.

Theorem 1.8.5 (Borel). We have an isomorphism

$$H^*(\mathrm{Gr}_{\mathbb{C}}(r, \infty); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_r],$$

where c_i lives in degree $2i$.

We can apply this to answer the question of when characteristic classes exist:

- ▷ when k is odd, there is one and only characteristic class $\mathrm{Vect}_{\mathbb{C}}^r(-) \Rightarrow H^k(-, \mathbb{Z})$, which is the zero map
- ▷ for $k = 2$, all characteristic classes are integer multiples of c_1
- ▷ for $k = 4$, all characteristic classes are $\mathbb{Z}$ -linear combinations of c_1^2 and c_2

Constructing Chern classes: Given any complex vector bundle $E \rightarrow X$ of rank r over a paracompact space X , it is pulled back from the tautological bundle via some map

$$f: X \rightarrow \mathrm{Gr}_{\mathbb{C}}(r, \infty).$$

We define $c_i(E)$ as the pullback of $c_i \in H^{2i}(\mathrm{Gr}_{\mathbb{C}}(r, \infty), \mathbb{Z})$ under the map f .

Constructing the first Chern class via cocycles on a complex manifold: If X is paracompact, then by the exponential sequence, we can define c_1 to be the composite bijection

$$c_1: \mathrm{Vect}_{\mathbb{C}}^1(X) \xrightarrow{\sim} H^1(X, \mathbb{C}^\times) \xrightarrow{\sim} H^2(X, \mathbb{Z}).$$

Suppose X is a complex manifold, then there is a very nice explicit construction of c_1 . Given a complex line bundle on X , we pick an open cover $\{U_i\}$ of X trivializing the line bundle. After potentially shrinking the U_i 's if needed, we may assume they are simply connected and therefore admit well-defined logarithms. Given a cohomology class in $H^1(X, \mathbb{C}^\times)$, we can express it in terms of the transition functions g_{ij} . Pick a logarithm $e^{2\pi i f_{ij}} = g_{ij}$ for each. Then the triple overlap condition says that

$$g_{ij}g_{jk}g_{ki} = 1.$$

Taking a logarithm on both sides, we have a well-defined map

$$f_{ij} + f_{ik} + f_{jk}: U_{ijk} \rightarrow \mathbb{Z}.$$

We see that this is a Čech 2-cocycle, and define its class in $\check{H}^2(\mathcal{U}, \mathbb{Z})$ to be c_1 of the line bundle.

1.8.3 Flag bundles

We had this data of a *frame bundle*, which was the principal $U(k)$ -bundle associated to complex vector bundle $E \rightarrow X$ of rank k . The fiber over a point $x \in X$ was an ordered sequence of vectors $(v_1, \dots, v_k)$ which were unit vectors and pairwise orthogonal. We're going to define a similar construction, where we don't care about the precise vectors, but only their *spans*. This is called the *flag bundle*.

Definition 1.8.6. Let $\text{Fl}(\mathbb{C}^r)$ denote the topological space whose points are: *complete flags*, that is, a sequence of linear subspaces

$$\{0\} \subsetneq F_1 \subsetneq F_2 \subsetneq \cdots \subsetneq F_r = \mathbb{C}^r.$$

We claim that there is a homeomorphism

$$\text{Fl}(\mathbb{C}^r) \cong U(r)/U(1)^{\times r},$$

where $U(1)^{\times r} \subseteq U(r)$ is the subgroup of diagonal unitary matrices.

Remark 1.8.7. We have that the flag bundle is obtained from the frame bundle by taking spans. Explicitly, we take the frame bundle $\text{Fr}(E)$, and mod out by the action of $U(1)^{\times r}$ in each fiber.¹⁸

Example 1.8.8. If $L \rightarrow X$ is a line bundle, then $\text{Fl}(L) \cong X$.

Example 1.8.9. If $E \rightarrow X$ is rank two, then $\text{Fl}(E) \rightarrow X$ has fibers over x given by $\mathbb{P}(E_x)$, since the data of a flag is the same as the data of a line through the origin in E_x .

Definition 1.8.10. We can generalize this construction of the flag bundle of a rank two bundle as a projectivization. Given any vector bundle $E \rightarrow X$ of rank r , we denote by $\mathbb{P}(E)$ the *projectivization* of the vector bundle, whose fiber over $x \in X$ is $\mathbb{P}(E_x)$. The projectivized bundle $\mathbb{P}(E) \rightarrow X$ is a fiber bundle with structure group $\text{PGL}_r(\mathbb{C})$ and fiber $\mathbb{CP}^{r-1}$.¹⁹

While the flag bundle and projectivization agree for rank two bundles, they don't for rank three bundles, since a line in $\mathbb{C}^3$ is not the same data as a complete flag. Let's let $E \rightarrow X$ be rank three, and consider the map

$$\pi: \mathbb{P}(E) \rightarrow X.$$

Just as projective space has a tautological line bundle over it, so does the projectivization of a vector bundle. We define

$$\mathcal{S}_E = \{\ell \in \mathbb{P}(E) \mid \ell \subseteq E_x, \pi(\ell) = x\}.$$

We have that $\mathcal{S}_E$ can be viewed as a subbundle of $\pi^{-1}(E)$ over $\mathbb{P}(E)$, so we get a short exact sequence of vector bundles

$$\mathcal{S}_E \rightarrow \pi^{-1}(E) \rightarrow Q_E$$

over $\mathbb{P}(E)$. This quotient bundle Q_E has rank two, and we can now define

$$\pi': \mathbb{P}(Q_E) \rightarrow \mathbb{P}(E).$$

Claim 1.8.11. In the explanation above, we have that $\mathbb{P}(Q_E) \cong \text{Fl}(E)$.

To justify this claim, let's work through the definition of a point in $\mathbb{P}(Q_E)$. Over $\mathbb{P}(E)$, the short exact sequence splits (every short exact sequence of topological or smooth bundles splits) as:

$$\pi^{-1}(E) \cong \mathcal{S}_E \oplus Q_E.$$

So a point $q \in \mathbb{P}(Q_E)$ has a projection to $\ell \in \mathbb{P}(E)$, which represents a line in $\mathcal{S}_E$. The point q , via the tautological bundle on $\mathbb{P}(Q_E)$, represents a line in Q_E , which is fiberwise identified with the orthogonal complement of $\mathcal{S}_E$ in $\pi^{-1}(E)$. So the data is an ordered tuple of two linearly independent vectors in E_x for each $x \in X$.

This is a rough sketch, but it details a general construction of the flag bundle — projectivize, pull back, split, iterate.

¹⁸There's a sense in which they are associated, but $U(r)$ doesn't act faithfully on $\text{Fl}(\mathbb{C}^r)$ so we have to modify our earlier definitions to make this make any sense. This is omitted.

¹⁹We can't keep the structure group as $\text{GL}_r(\mathbb{C})$ because it won't act faithfully on the fiber, so we need to mod out by the scaling action by diagonal matrices with the same entry along the diagonal. This yields $\text{PGL}_r(\mathbb{C}) = \text{GL}_r(\mathbb{C})/\mathbb{C}^\times$.

1.8.4 The splitting principle

Before proving the splitting principle, we need the following:

Lemma 1.8.12. We have that $H^*(\mathbb{P}(E), \mathbb{Z})$ is a free module over $H^*(X, \mathbb{Z})$ generated by $x = c_1(\mathcal{S}^*)$, subject to the relation

$$x^n + c_1(E)x^{n-1} + \dots + c_n(E),$$

where n is the rank of E .

This is a consequence of the Leray-Hirsch theorem.

Theorem 1.8.13 (Splitting principle for projectivizations). Let $E \rightarrow X$ be a complex vector bundle. Then if $\pi: \text{Fl}(E) \rightarrow X$ denotes the flag bundle, we have that:

1. The induced map $\pi^*: H^*(X, \mathbb{Z}) \rightarrow H^*(\text{Fl}(E), \mathbb{Z})$ is injective
2. The pullback bundle $\pi^*E \rightarrow \text{Fl}(E)$ decomposes as a sum of line bundles:

$$\pi^*(E) \cong L_1 \oplus \dots \oplus L_r.$$

Proof. It suffices to prove this inductively. If E has rank 1, this is clear, since $\text{Fl}(E) \cong X$. Suppose we know it up to rank $r - 1$, and let $E \rightarrow X$ be any bundle of rank r . Then we have that

$$\pi: \mathbb{P}(E) \rightarrow X,$$

and pulling back we get that $\pi^{-1}(E) \cong \mathcal{S}_E \oplus Q_E$, where Q_E has rank $n - 1$. We can apply the lemma to π^* to get injectivity, then we apply the inductive step to conclude. $\square$

Slogan 1.8.14. Any statement we want to prove about vector bundles, we can assume the vector bundle is split into a sum of line bundles.

1.8.5 More about Chern classes

Exercise 1.8.15. For any space X , the group structure on $H^1(X, \mathbb{C}^*)$ corresponds, under the isomorphism $H^1(X, \mathbb{C}^*) \cong \text{Vect}_{\mathbb{C}}^1(X)$, to tensor products of line bundles. The unit is the trivial line bundle, and the inverse is the dual bundle.

Once we've proven this exercise, the following proposition is immediate (since we defined c_1 as a connecting homomorphism in the long exact sequence of sheaf cohomology coming from the exponential sequence, it is clear that it is a group homomorphism).

Proposition 1.8.16. The map

$$c_1: \text{Vect}_{\mathbb{C}}^1(X) \rightarrow H^2(X, \mathbb{Z})$$

is a group homomorphism, so $c_1(L^*) = -c_1(L)$ for any line bundle L .

Corollary 1.8.17. Let $E \rightarrow X$ be a bundle and E^* the dual vector bundle. Then $c_k(E^*) = (-1)^k c_k(E)$.

Proof. Let $\pi: \text{Fl}(E) \rightarrow X$ be the flag bundle map. By the splitting principle, $\pi^*: H^*(X, \mathbb{Z}) \rightarrow H^*(\text{Fl}(E), \mathbb{Z})$ is injective on cohomology, so it suffices to prove that $c_k(\pi^*E^*) = (-1)^k c_k(\pi^*E)$ in $H^*(\text{Fl}(E), \mathbb{Z})$. Also by the splitting principle, we have that

$$\pi^*E \cong L_1 \oplus \dots \oplus L_k.$$

Since dualizing a bundle is a fiberwise construction, it is clearly natural under pullback (i.e. we can dualize then pull back, or pull back and then dualize. Moreover, dualization commutes with direct sums, so we get that

$$\pi^*E^* \cong L_1^* \oplus \dots \oplus L_k^*.$$

We compute the Chern polynomial of π^*E^* as

$$c(\pi^*E^*) = \prod_i c(L_i^*) = \prod_i (1 - c_1(L_i)).$$

Since $c(\pi^*E) = \prod_i (1 + c_1(L_i))$, the result is clear. $\square$

Proposition 1.8.18. Let $E \rightarrow X$ be a trivial complex vector bundle of rank r . Then its total Chern class is $c(E) = 1$, that is, $c_k(E) = 0$ for $k \geq 1$.

Proof. The map $X \rightarrow \text{Gr}_{\mathbb{C}}(r, \infty)$ classifying a trivial bundle is homotopic to a constant map, hence factors through a point. Therefore the induced map on cohomology factors through the integral cohomology of a point, which is just $\mathbb{Z}$ in degree zero. $\square$

Proposition 1.8.19. We have that

$$c(T\mathbb{CP}^n) = (1 + h)^{n+1}.$$

Proof. Letting ϵ^1 denote the trivial line bundle on projective space, and E_1 the tautological bundle, there is a short exact sequence

$$0 \rightarrow \epsilon^1 \rightarrow (E_1^*)^{\oplus(n+1)} \rightarrow T\mathbb{CP}^n \rightarrow 0.$$

Therefore we have that

$$c(T\mathbb{CP}^n) = \prod_{i=1}^{n+1} c(E_1^*) = \prod_{i=1}^{n+1} (1 + h).$$

$\square$

1.8.6 Gysin integration

Let X be a smooth n -dimensional manifold, let $E \rightarrow X$ be a smooth rank r real vector bundle, and let $\pi: \text{Fl}(E) \rightarrow X$ be the projection map from the total flag space. We recall that π^* is injective on cohomology by the splitting principle. There is a map back the other way, called the *Gysin map* or *integration in fibers*:

$$\pi_*: H^*(\text{Fl}(E), \mathbb{Z}) \rightarrow H^{*-r(r-1)/2}(X, \mathbb{Z}).$$

We won't define it formally in this class, but the most elegant definitions come from the world of de Rham cohomology. Since $\text{Fl}(E) \rightarrow X$ is a principal $O(r)$ -bundle, locally $\text{Fl}(E)$ looks like a

neighborhood in X crossed with $O(r)$. A differential form on this neighborhood looks like a form on $O(r)$ crossed with a form on a neighborhood of X , so we can integrate out the part of the form coming from $O(r)$. This drops the dimension of the form by $\dim O(r) = r(r-1)/2$. For a formal construction, see [BT82, §6] for instance.

While we stated this for smooth bundles over smooth manifolds, this holds in the topological context just as well (we just need slightly different machinery to define it). We state some properties of Gysin integration without proof:

Proposition 1.8.20 (Properties of Gysin maps). Let X be a paracompact space, let $E \rightarrow X$ be a vector bundle of rank r , and let $\pi: \text{Fl}(E) \rightarrow X$ be the associated principal $O(r)$ -bundle.

1. *Projection*: Given any cohomology class $\omega \in H^*(\text{Fl}(E), \mathbb{Z})$, and any cohomology class $\alpha \in H^*(X, \mathbb{Z})$, we have that

$$\pi_*(\pi^*\alpha \cdot \omega) = \alpha \cdot \pi_*\omega.$$

2. *Interchange/base change* Let $f: Y \rightarrow X$ be any continuous map, and consider the pullback bundle:

$$\begin{array}{ccc} f^*\text{Fl}(E) & \xrightarrow{i} & \text{Fl}(E) \\ p \downarrow & \lrcorner & \downarrow \pi \\ Y & \xrightarrow{f} & X. \end{array}$$

Then we have that

$$f^*\pi_* = p_*i^*$$

as maps

$$H^*(\text{Fl}(E), \mathbb{Z}) \rightarrow H^{*-r(r-1)/2}(Y, \mathbb{Z}).$$

1.9 Pontryagin classes

As discussed, Chern classes are characteristic classes of complex vector bundles over spaces. We define some characteristic classes of real vector bundles, called *Pontryagin classes*, by turning a real vector bundle into a complex vector bundle then taking its Chern classes.

Definition 1.9.1. Let $V \in \text{Vect}_{\mathbb{R}}$ be a real vector space. We define its *complexification* to be the tensor product $V \otimes_{\mathbb{R}} \mathbb{C} \in \text{Vect}_{\mathbb{C}}$. If $E \rightarrow X$ is a real vector bundle, we define $E \otimes_{\mathbb{R}} \mathbb{C} \rightarrow X$ to be the complexification of the real vector bundle, obtained by complexifying in every fiber. Note that complexification doubles the real dimension of any vector space, hence the real rank of any vector bundle.

Definition 1.9.2. Let $E \rightarrow X$ be any real vector bundle. We define its *Pontryagin classes* by

$$p_k(E) := (-1)^k c_{2k}(E \otimes_{\mathbb{R}} \mathbb{C}) \in H^{4k}(X, \mathbb{Z}).$$

We define the *total Pontryagin class* to be

$$p(E) = \sum_{k \geq 0} p_k(E) \in H^*(X, \mathbb{Z}).$$

Remark 1.9.3. We can alternatively consider the inclusion $O(r) \subseteq U(r)$ of $r \times r$ orthogonal matrices as living inside of the group of $r \times r$ unitary matrices. This induces a map

$$\psi: \check{H}^1(X, O(r)) \rightarrow \check{H}^1(X, U(r)),$$

which is the same as complexification – that is, it sends a real rank r vector bundle to a complex rank r vector bundle. We can define Pontryagin classes of a bundle E as the Chern classes of $\psi(E)$ [Hir78, 4.5]. This has a subtlety in that this technically allows us to define odd Chern classes of the complexification, however all of these are 2-torsion.

Proposition 1.9.4. Let E be a real vector bundle. Then $c_{2k+1}(E \otimes_{\mathbb{R}} \mathbb{C})$ is 2-torsion.

Proof sketch. For any complex vector bundle V , we can define a conjugate bundle $\bar{V}$ by taking the complex conjugate of the complex structure. We can argue that

$$c_k(V) = (-1)^k c_k(\bar{V}).$$

We check that a complexified bundle $E \otimes_{\mathbb{R}} \mathbb{C}$ is isomorphic to its own complex conjugate [MS74, 15.1]. For even rank Chern classes this tells us no new information, but for odd rank Chern classes they must all be 2-torsion. $\square$

Remark 1.9.5. This leads to perhaps some competing notation about how we want to index the Pontryagin classes. We could just define $p_k(E)$ to be $c_k(E \otimes_{\mathbb{R}} \mathbb{C})$, and if we did, many of the nice properties of Chern classes transfer immediately to nice properties of Pontryagin classes (for instance a Whitney sum formula would hold *integrally* instead of modulo two). We adopt the classical and more commonly seen convention that $p_k(E) := (-1)^k c_{2k}(E \otimes_{\mathbb{R}} \mathbb{C})$, in which case we have to reduce modulo two to transfer properties of Chern classes to properties of Pontryagin classes.

Proposition 1.9.6 (Properties of Pontryagin classes). Let $E \rightarrow X$ be any real vector bundle. Then

1. $p_0(E) = 1$
2. $p_i(E) = 0$ if $i > \text{rank}(E)/2$
3. For any continuous $f: Y \rightarrow X$ we have that $p_k(f^*E) = f^*p_k(E)$ for any k .
4. For two bundles $E_1, E_2 \rightarrow X$, we have that

$$p(E_1 \oplus E_2) \equiv p(E_1) \cdot p(E_2) \pmod{2}.$$

Terminology 1.9.7. When people discuss the “Chern numbers of a manifold” or the “Pontryagin numbers of a manifold,” they mean the tangent bundle of the manifold. So $c(M)$ always means $c(TM)$ and $p(M)$ always means $p(TM)$.

An important exercise is the following:

Exercise 1.9.8. Let $E \rightarrow X$ be a complex vector bundle. Then we can consider it as a real vector bundle (forgetting the complex structure) and complexify it. Show that there is a canonical isomorphism

$$E \otimes_{\mathbb{R}} \mathbb{C} \cong E \oplus \bar{E},$$

where $\bar{E}$ is the complexified vector bundle. (Hint: first do this for vector spaces, then extend fiberwise).

An immediate corollary of this exercise is the following:

Corollary 1.9.9. If $E \rightarrow X$ is a complex vector bundle, then its Pontryagin classes are related to its Chern classes via the relation:

$$\sum_{i \geq 0} (-1)^i p_i = \left(\sum_{j \geq 0} (-1)^j c_j \right) \left(\sum_{k \geq 0} c_k \right).$$

Proof. Via [Exercise 1.9.8](#) we have an isomorphism $E \otimes_{\mathbb{R}} \mathbb{C} \cong E \oplus \bar{E}$. Using the Whitney sum formula we get

$$c(E \otimes_{\mathbb{R}} \mathbb{C}) = c(E)c(\bar{E}).$$

The formula follows from this, together with the observation that $c_j(\bar{E}) = (-1)^j c_j(E)$. $\square$

Theorem 1.9.10 ([MS74, 15.6], [Hir78, 4.10.2]). If $h \in H^2(\mathbb{CP}^n, \mathbb{Z})$ denotes the hyperplane class, then

$$p(T\mathbb{CP}^n) = (1 + h^2)^{n+1}.$$

Proof. By [Proposition 1.8.19](#) we have that $c(T\mathbb{CP}^n) = (1 + h)^{n+1}$. Using the formula from [Corollary 1.9.9](#), we get that

$$\begin{aligned} \sum_{i \geq 0} (-1)^i p_i(T\mathbb{CP}^n) &= c(T\mathbb{CP}^n)c(\overline{T\mathbb{CP}^n}) \\ &= (1 + h)^{n+1}(1 - h)^{n+1} = (1 - h^2)^{n+1}. \end{aligned}$$

From this the result follows. $\square$

Example 1.9.11. We have that

$$p(T\mathbb{P}^5) = 1 + 6h^2 + 15h^4 \in H^*(\mathbb{CP}^5, \mathbb{Z}),$$

for instance.

Exercise 1.9.12. For any $n \geq 0$, prove that $p(TS^n) = 1$. That is, all Pontryagin numbers of a sphere are zero.

1.10 Cobordism

An important topic in algebraic topology is *cobordism*. Roughly speaking, two manifolds are *cobordant* if they jointly bound a manifold of one higher dimension. More precisely:

Definition 1.10.1. Let M and N be compact differentiable manifolds of real dimension n , without boundary

1. we say that M and N are *cobordant* if there exists a compact differentiable $(n + 1)$ manifold W with boundary, together with a diffeomorphism $\partial W \cong M \amalg N$
2. if M and N are oriented, we say that M and N are *oriented cobordant* if there exists an oriented differentiable $(n + 1)$ manifold W and an oriented diffeomorphism $\partial W \cong M \amalg -N$, where $-N$ denotes N with the reversed orientation.

Exercise 1.10.2. Show that cobordism is an equivalence relation.

Definition 1.10.3.

1. Denote by Ω_n^O the free abelian group on compact n -dimensional differentiable manifolds modulo the relation generated by cobordism. By convention the zero element is the empty manifold.
2. Denote by Ω_n^{SO} the free abelian group on *oriented* compact differentiable n -dimensional manifolds modulo the relation generated by oriented cobordism.

Proposition 1.10.4. Under the cartesian product of manifolds, we get graded rings

$$\begin{aligned}\Omega_*^O &:= \oplus_{n \geq 0} \Omega_n^O \\ \Omega_*^{\text{SO}} &:= \oplus_{n \geq 0} \Omega_n^{\text{SO}}.\end{aligned}$$

These are called the *cobordism ring* and the *oriented cobordism ring*, respectively. We'll predominantly be dealing with the latter.

Remark 1.10.5. There is also a notion of *complex cobordism*, defined on stably complex manifolds. This gives rise to the *complex cobordism ring* Ω_*^U , which we'll talk about later.

Example 1.10.6. Any sphere S^n represents the zero class in Ω_*^{SO} since $S^n = \partial B^{n+1}$.

Proposition 1.10.7. We have that $[\mathbb{CP}^{2k+1}] = 0$ in $\Omega_{4k+2}^{\text{SO}}$.

Proof. We leverage a fibration

$$\mathbb{CP}^1 \hookrightarrow \mathbb{CP}^{2k+1} \rightarrow \mathbb{HP}^k$$

induced by the projection

$$\begin{aligned}\mathbb{C}^{2k+2} - 0 &\rightarrow \mathbb{H}^{k+1} - 0 \\ (z_1, \dots, z_{2k+2}) &\mapsto (z_1 + z_2 j, \dots, z_{2k+1} + z_{2k+2} j).\end{aligned}$$

The fiber has structure group $\text{SO}(3)$ under the homeomorphism $\mathbb{CP}^1 \cong S^2$, hence we can think about it as the sphere bundle of an oriented rank three vector bundle. Extending to the disk bundle, we obtain a fiber sequence

$$B^3 \hookrightarrow M \rightarrow \mathbb{HP}^k,$$

where $\partial M \cong \mathbb{CP}^{2k+1}$. Hence $\mathbb{CP}^{2k+1}$ bounds. □

Theorem 1.10.8 (Thom). We have that the rational oriented cobordism ring is polynomial on the even complex projective spaces:

$$\Omega_*^{\text{SO}} \otimes \mathbb{Q} \cong \mathbb{Q}[\mathbb{CP}^2, \mathbb{CP}^4, \mathbb{CP}^6, \dots].$$

Proof. For a sketch, see [Fre12, Lecture 12] or [MS74, §16]. □

Theorem 1.10.9 ([Fre12, 2.24]). All torsion in Ω_*^{SO} is 2-torsion, and is finite in each degree.

Theorem 1.10.10 ([Hir78, 7.2.2], Thom). If $i \not\equiv 0 \pmod{4}$, then Ω_i^{SO} is finite. We have that Ω^{4k} has free part $\mathbb{Z}^{\pi(k)}$ and a finite torsion part.

Example 1.10.11. $\Omega_0^{\text{SO}} \cong \mathbb{Z}$, as a compact oriented 0-manifold is a finite set of signed points and the sum of the signs is a bordism invariant.

Example 1.10.12. $\Omega_1^{\text{SO}} \cong 0$, since every compact 1-manifold is a disjoint union of circles, and all such manifolds bound.

Example 1.10.13. $\Omega_2^{\text{SO}} \cong 0$, as every compact oriented 2-manifold bounds. Can you find an example of one that doesn't bound if we dropped the word oriented?

Example 1.10.14. $\Omega_3 \cong 0$, this is a non-trivial result of Rokhlin.

Example 1.10.15. $\Omega_4 \cong \mathbb{Z}$, generated by $\mathbb{CP}^2$.

Example 1.10.16. $\Omega_5 \cong \mathbb{Z}/2$, generated by Y^5 , which is a smooth bidegree $(1, 1)$ hypersurface in $\mathbb{CP}^2 \times \mathbb{CP}^4$.

Remark 1.10.17. The odd degree generators were constructed by Dold. For any two integers $m, n \geq 1$, consider the C_2 -action on $S^m \times \mathbb{CP}^n$ given by the antipodal map on the left and the complex conjugation action on the right. Quotienting by this action we get a manifold we call $P(m, n)$ [Dol56]. This gives a fiber bundle over $\mathbb{RP}^m$ with fiber $\mathbb{CP}^n$ and structure group C_2 . For suitable m, n , these can give generators of Ω_*^{SO} in odd degrees (see [Dol56, Satz 3]).

Example 1.10.18. $\Omega_6 \cong 0$

Example 1.10.19. $\Omega_7 \cong 0$.

Example 1.10.20. $\Omega_8 \cong \mathbb{Z} \oplus \mathbb{Z}$, generated by $\mathbb{CP}^4$ and $\mathbb{CP}^2 \times \mathbb{CP}^2$

1.10.1 Pontryagin numbers

Proposition 1.10.21 ([MS74, 14.2]). Let $E \rightarrow X$ be any real vector bundle. Then we have that

$$p(E \oplus \epsilon_{\mathbb{R}}^r) = p(E)$$

where $\epsilon_{\mathbb{R}}^r$ is a trivial rank r real vector bundle.

Proof. Complexification commutes with Whitney sums, so by multiplicativity of total Chern classes we get

$$c(E \otimes_{\mathbb{R}} \mathbb{C} \oplus \epsilon_{\mathbb{C}}^r) = c(E \otimes_{\mathbb{R}} \mathbb{C})c(\epsilon_{\mathbb{C}}^r) = c(E \otimes_{\mathbb{R}} \mathbb{C}),$$

so the total Pontryagin classes agree. $\square$

If M is some smooth compact oriented manifold, we can look at the Pontryagin classes of its tangent bundle, and we get classes $p_i(TM) \in H^{4i}(M, \mathbb{Z})$. Since M is oriented, the top-dimensional cohomology integrates to $\mathbb{Z}$, hence if M has dimension $4n$, then $p_n(TM)$ can be considered as an actual integer.

Example 1.10.22. Consider $\mathbb{CP}^4$, which is real 8-dimensional. There are two natural ways to extract integers out of its Pontryagin classes, namely

$$\begin{aligned} p_2(T\mathbb{CP}^4) &= 10 \in \mathbb{Z} \cong H^8(\mathbb{CP}^4, \mathbb{Z}) \\ p_1(T\mathbb{CP}^4)p_1(T\mathbb{CP}^4) &= 25 \in \mathbb{Z} \cong H^8(\mathbb{CP}^4, \mathbb{Z}). \end{aligned}$$

Conversely, we could consider some other real compact oriented 8-manifold, like $M = \mathbb{RP}^3 \times \mathbb{RP}^5$, and we could run the analogous computation and we'd get zero. So in some cases these Pontryagin numbers can tell the difference between two compact oriented $4n$ -manifolds.

Definition 1.10.23. Let M^{4n} be a compact oriented $4n$ -manifold, and let $I = (i_1, \dots, i_r) \vdash n$ be a partition of n . We define the I th Pontryagin number of M to be the integer

$$p_I(M) = \int_M p_{i_1}(TM) \cdots p_{i_r}(TM) \in \mathbb{Z}.$$

If $\dim M \not\equiv 0 \pmod{4}$, we just define all the Pontryagin numbers to be zero.

Here's a fun proposition:

Proposition 1.10.24 ([MS74, pp. 184-185]). Let M^{4n} be a compact oriented manifold. If any Pontryagin number $p_I(M)$ is nonzero, then M has no orientation-reversing diffeomorphisms.

Proof. If we reverse the orientation on M , all the Pontryagin classes are unchanged, but the fundamental class on M picks up a minus sign. So if M admits an orientation-reversing diffeomorphism, then $p_I(M) = -p_I(M)$, so all Pontryagin classes are zero. $\square$

We have a key property of Pontryagin classes coming from when a manifold is a boundary.

Lemma 1.10.25. Let W be an oriented $(d+1)$ -dimensional manifold with boundary, and let $M = \partial W$ be its d -dimensional compact oriented boundary. Then $p_I(M) = 0$ for all I .

Proof. If $d \not\equiv 0 \pmod{4}$, then all Pontryagin numbers are zero by definition and we're done, so let's assume $d = 4n$. Letting $j: M \hookrightarrow W$ be the inclusion of the boundary, we get a splitting of the restricted tangent bundle

$$j^*TW \cong TM \oplus \epsilon_{\mathbb{R}}^1,$$

where $\epsilon_{\mathbb{R}}^1$ is a trivial real line bundle. This comes from picking a Euclidean metric on TW , wherein we see a unique outward-pointing normal vector for the tangent bundle on M (pointing from the boundary *into* W), c.f. [MS74, 4.9].

By stability (Proposition 1.10.21) and naturality of Pontryagin classes, we now see that

$$p(TM) = p(TM \oplus \epsilon_{\mathbb{R}}^1) = p(j^*TW) = j^*p(TW),$$

so the Pontryagin classes of the tangent bundle on M are pulled back from the Pontryagin classes of the tangent bundle on W .

Finally, we see that

$$\int_M p_I(TM) = \int_{[M]} j^* p_I(TW).$$

Let $\alpha \in H_{\text{dR}}^{4k}(W, \mathbb{Z})$ be a closed $4n$ -form representing the cohomology class $p_I(TW)$. Then in the above we have that

$$\int_M j^* \alpha = \int_{\partial W} \alpha = \int_W d\alpha = 0$$

by Stokes' theorem. Therefore $p_I(M) = \int_M p_I(TM)$ is zero for all I . $\square$

This has fun applications, for instance:

Example 1.10.26. We have that $p_n(\mathbb{CP}^{2n}) = \binom{2n+1}{n}$, hence $\mathbb{CP}^{2n}$ is not a boundary of a smooth oriented $(4n+1)$ -manifold.

Example 1.10.27. All the Pontryagin numbers of S^{4n} are zero.

1.11 Genera

Lemma 1.11.1. If M and N are compact oriented manifolds such that $[M] = [N]$ in Ω_{4n}^{SO} , then all the Pontryagin numbers of M and N agree.

Proof. By [Lemma 1.10.25](#), if M is null-cobordant, then its Pontryagin numbers vanish. As mentioned before, we have that $p_I(M) = -p_I(-M)$, where $-M$ means M with its orientation reversed. We have that $[M] = [N]$ if and only if $[M] - [N]$ is null-cobordant, meaning if $M \amalg -N$ bounds. If it does, then all the Pontryagin numbers of M and $-N$ are zero, but the Pontryagin numbers of a disjoint union of manifolds are just a sum of the Pontryagin numbers of the individual manifolds. Similarly, reversing the orientation on a manifold flips the sign on its fundamental class, hence induces a factor of (-1) to the Pontryagin numbers. All of this says that if $[M] = [N]$ then

$$0 = p_I(M \amalg -N) = p_I(M) - p_I(N).$$

□

Another way to state what we just proved is the following: if I is a partition of n , then

$$p_I: \Omega_{4n}^{\text{SO}} \rightarrow \mathbb{Z}$$

is an abelian group homomorphism.

Suppose we picked a partition I_n of n for every n . Then we would get a map

$$\bigoplus_{n \geq 0} p_{I_n}: \Omega_*^{\text{SO}} \rightarrow \mathbb{Z}, \tag{1.11.2}$$

which would be an abelian group homomorphism. A very natural question to ask is the following:

Question 1.11.3. Can we pick partitions I_n for which the additive homomorphism in [Equation \(1.11.2\)](#) is multiplicative (i.e. a ring homomorphism)?

Suppose we could, then let $F = \bigoplus_{n \geq 0} p_{I_n}$. For $n = 1$, we only have one partition, so on 4-manifolds, F just inputs a 4-manifold and outputs its first Pontryagin class. For $n = 2$, there are two possibilities, namely p_2 and $p_{1,1}$.

Let's see what $F(\mathbb{CP}^2 \times \mathbb{CP}^2)$ could be. If it were multiplicative, it would be equal to $F(\mathbb{CP}^2) \cdot F(\mathbb{CP}^2)$. Since $p_1(\mathbb{CP}^2) = 3$, we have that $F(\mathbb{CP}^2 \times \mathbb{CP}^2) = 9$. We can check that

$$\begin{aligned} p_2(\mathbb{CP}^2 \times \mathbb{CP}^2) &= 9 \\ p_{1,1}(\mathbb{CP}^2 \times \mathbb{CP}^2) &= 18, \end{aligned}$$

so we have to have that F sends 8-manifolds to p_2 . Similarly we can compute

$$\begin{aligned} p_3(\mathbb{CP}^2 \times \mathbb{CP}^2 \times \mathbb{CP}^2) &= 27 \\ p_{2,1}(\mathbb{CP}^2 \times \mathbb{CP}^2 \times \mathbb{CP}^2) &= 81 \\ p_{1,1,1}(\mathbb{CP}^2 \times \mathbb{CP}^2 \times \mathbb{CP}^2) &= 162. \end{aligned}$$

So we would want F to send 12-manifolds to p_3 . Pattern-matching, we might ask F to send a $4n$ -manifold to p_n . It's not at all obvious that this is multiplicative, but it will turn out to be, and we can prove it soon. However just by looking at where $(\mathbb{CP}^2)^{\times n}$ needs to go, we can check that the only way to make this work is sending $4n$ -manifolds to p_n . We get an answer to the question above then, and the answer is yes but there is only one way to do it and it's kind of trivial.

If we picked F to send 4-manifolds to p_1 , another way to get the correct formula for $\mathbb{CP}^2 \times \mathbb{CP}^2$ to have multiplicativity is to define F to send an 8-manifold to $p_1^2 - p_2$. So if we pick a linear combination of Pontryagin numbers (or, the integration of a *polynomial* in the Pontryagin classes), we could try to ask for multiplicativity that way. So we could ask a modified question:

Question 1.11.4. For some ring R , can we cook up a ring homomorphism

$$\Omega_*^{\text{SO}} \rightarrow R$$

where, on each n , the map is an R -linear combination of the Pontryagin numbers for $4n$ -manifolds?

Let's make this more precise. For each n , we only need the Pontryagin numbers $p_1, \dots, p_n$ to define the value on Ω_n^{SO} . Denote by K_n our n th polynomial in the Pontryagin classes, that is,

$$K_n(p_1, \dots, p_n): \Omega_n^{\text{SO}} \rightarrow R.$$

We need these to satisfy two key properties:

1. K_n is homogeneous of degree n if we think of each p_i as being of degree i
2. the K_n 's are *multiplicative*, that is, if M^{4m} and N^{4n} are two compact oriented manifolds, we want

$$K_{m+n}(p_1(M \times N), \dots, p_n(M \times N)) = \sum_{j=0}^{m+n} K_j(p_1(M), \dots, p_j(M)) \cdot K_{m+n-j}(p_1(N), \dots, p_{m+n-j}(N)).$$

We can make the following definition:

Definition 1.11.5. Let R be a commutative ring, and consider the ring $R[p_1, p_2, \dots]$ in infinitely many variables p_i , with p_i of weight $4i$. We define an m -sequence (multiplicative sequence) to be a sequence of elements $K_r \in R[p_1, p_2, \dots]$ for $r \geq 0$ satisfying the following properties:

1. $K_0 = 1$
2. K_r lies in the subring $R[p_1, \dots, p_r]$
3. for any indeterminates a_i, b_i, c_i , if

$$(1 + a_1 z + a_2 z^2 + \dots)(1 + b_1 z + b_2 z^2 + \dots) = 1 + c_1 z + c_2 z^2 + \dots,$$

then

$$(1 + K_1(a_1)z + K_2(a_1, a_2)z^2 + \dots)(1 + K_1(b_1)z + K_2(b_1, b_2)z^2 + \dots) = 1 + K_1(c_1)z + K_2(c_1, c_2)z^2 + \dots$$

The point of the definition is the following observation:

Proposition 1.11.6. If an abelian group homomorphism

$$\Omega_*^{\text{SO}} \rightarrow R$$

is given by a polynomial $K_r(p_1, \dots, p_r)$ on Ω_r^{SO} , then the homomorphism is multiplicative if and only if the K_r 's are m -sequences.

Proof. Use the latter property, together with $a_i = p_i(M)$, $b_i = p_i(N)$, and $c_i = p_i(M \times N)$. □

So we now have a way to answer this question about forming multiplicative maps out of Ω_*^{SO} out of Pontryagin classes — we just need to figure out which m -sequences exist. It will be very helpful to be able to divide by integers in order to make this all work nicely, so we want the target ring to have this property (i.e. we want it to be an algebra over $\mathbb{Q}$). A commutative ring homomorphism from any ring A to a $\mathbb{Q}$ -algebra B is the same, by adjunction, as a $\mathbb{Q}$ -algebra map $A \otimes \mathbb{Q} \rightarrow B$, which is the same as a ring homomorphism between them. So we can make the following definition:

Definition 1.11.7. Given an integral domain $R \in \text{CAlg}_{\mathbb{Q}}$, we define a *genus* to be a unital ring homomorphism

$$\phi: \Omega_*^{\text{SO}} \rightarrow R.$$

(Note this is the same data as a ring homomorphism $\Omega_*^{\text{SO}} \otimes \mathbb{Q} \rightarrow R$).

We now know, roughly, that we can build a genus out of an m -sequence, and we will see shortly that the entire data of an m -sequence can be compressed into the data of a single power series over R . Magically, *every* genus comes from an m -sequence in the Pontryagin classes!

Theorem 1.11.8. For any R as in [Definition 1.11.7](#), there are bijections

$$\{\text{genera valued in } R\} \leftrightarrow \{m\text{-sequences in } R\} \leftrightarrow 1 + R[[x^2]] \quad (1.11.9)$$

between genera in R and even power series over R beginning with 1.

We'll prove this over the course of a few steps.

Lemma 1.11.10. There is a bijection between m -sequences and even power series with constant term 1.

Proof. If $Q(x) \in 1 + R[[x^2]]$ is an even power series with constant term 1, we can take some indeterminates c_i all of weight 2, and we can define the power series

$$Q(x_1) \cdots Q(x_n) \in R[[x_1, \dots, x_n]]^{\Sigma_n}.$$

This is symmetric in the x_i 's so we use the isomorphism

$$R[[x_1, \dots, x_n]]^{\Sigma_n} \cong R[[p_1, p_2, \dots]]$$

where the p_i 's are the elementary symmetric polynomials in the x_i^2 's. We can therefore write

$$Q(x_1) \cdots Q(x_n) = 1 + K_1(p_1) + K_2(p_1, p_2) + \dots + K_n(p_1, \dots, p_n) + K_{n+1}(p_1, \dots, p_n, 0),$$

where in the above expression, p_i has weight $4i$, as does K_i . We claim the K_r 's don't depend on n so long as $n \geq r$ (this is a fundamental result in symmetric function theory, see for instance [\[Mac95, I.2\]](#)). We call $\{K_r\}$ the *multiplicative sequence* or *m-sequence* associated to the genus. We observe that $Q(x)$ can be recovered uniquely from the K_r 's by

$$Q(x) := 1 + K_1(1)x^2 + \sum_{i \geq 2} K_i(1, 0, \dots, 0)x^{2i}. \quad (1.11.11)$$

(in Hirzebruch this latter sum is written $K(1+x)$). This establishes the bijection between even power series and m -sequences. $\square$

We can now construct a genus from an even power series:

Construction 1.11.12. Let $R \in \mathbf{CAlg}_{\mathbb{Q}}$ be an integral domain. Given an even power series $Q \in 1 + R[[x^2]]$, we define

$$\phi_Q: \Omega_*^{\mathrm{SO}} \rightarrow R$$

by the formula

$$\phi_Q([M]) = \int_M K_n(p_1, \dots, p_n),$$

where K_n is the m -sequence attached to Q in the bijection above.

Proposition 1.11.13. The assignment

$$\begin{aligned} 1 + R[[x^2]] &\rightarrow \{\text{genera valued in } R\} \\ Q &\mapsto \phi_Q \end{aligned} \tag{1.11.14}$$

is well-defined [HBJ94, p. 14].

Proof. We must check the resulting ϕ_Q is indeed a bordism invariant, but this follows directly from [Lemma 1.11.1](#), since Pontryagin numbers of cobordant manifolds are equal. Therefore ϕ_Q is well-defined as a function.

Additivity comes from additivity of fundamental classes in disjoint unions, from which it is clear that ϕ_Q as defined is an abelian group homomorphism. Finally we check it is a unital ring homomorphism, i.e. that it is multiplicative, but this is an immediate consequence of multiplicativity of the K_n 's. $\square$

We now define a power series out of a genus.

Construction 1.11.15. Given a genus ϕ , we define a formal power series over R by

$$g'(y) = \sum_{n=0}^{\infty} \phi([\mathbb{CP}^n]) y^n = \sum_{n=0}^{\infty} \phi([\mathbb{CP}^{2n}]) y^{2n}.$$

Integrating (with constant of integration equal to zero), we get a power series $g(y)$, and we let $f(x) = g^{-1}(x)$ denote its compositional inverse.²⁰ We then define

$$Q(x) := \frac{x}{f(x)}.$$

Proposition 1.11.16. The Q constructed out of ϕ above is well-defined, that is, it is even and starts with constant term 1.

Proof. To see it is even we have to check that f is odd, which is equivalent to g being odd, which is equivalent to g' being even, which it indeed is. To check that the constant term of Q is 1, we have to check that $f(x) = x + O(x^2)$. It is true that $f(x) = ux + O(x^2)$ since f admits a compositional inverse, and we know that $g(y) = y + O(y^2)$, hence $u = 1$. $\square$

Proposition 1.11.17. The construction of ϕ from Q is injective.

²⁰Such an inverse exists since $g(y)$ is of the form $y + a_2 y^2 + a_3 y^3 + \dots$

Proof. Since a genus is unital mapping out of a $\mathbb{Q}$ -algebra, it is determined on what it does on generators of the $\mathbb{Q}$ -algebra. In particular by [Theorem 1.10.8](#), it is determined by where it sends $[\mathbb{CP}^n]$ for each n . However in the construction above, it is clear that if ϕ and ϕ' agree on each $[\mathbb{CP}^n]$ then their resulting normalized even power series will be the same. $\square$

The last thing in order to prove [Theorem 1.11.8](#) is the following:

Proposition 1.11.18. The composite

$$\{R\text{-genera}\} \rightarrow 1 + R[[x^2]] \rightarrow \{R\text{-genera}\}$$

is the identity.

Proof. Begin with ϕ , let Q be the associated power series, and let ϕ_Q be the resulting genus. We need to check $\phi = \phi_Q$, and as above it suffices to check they agree on complex projective spaces (of even complex dimension $2n$). Let $n \geq 1$ and fix $x \in H^2(\mathbb{CP}^{2n}; \mathbb{Z})$ to be the hyperplane class. Then we have that the total Pontryagin class of complex projective space is

$$p(\mathbb{CP}^{2n}) = (1 + x^2)^{2n+1}.$$

We have that $\phi([\mathbb{CP}^{2n}])$ is $\int_{\mathbb{CP}^{2n}} K_{4n}(p_1, \dots, p_n)$. By multiplicativity of the m -series K_* , we get that

$$K_{4n}(p_1, \dots, p_n)([\mathbb{CP}^{2n}]) = K_1(x^2)^{2n+1},$$

where we recall $x^2 = p_1(\mathbb{CP}^{2n})$. As $Q(x) = K_1(p_1)$ by definition, we get the following important observation: $\phi([\mathbb{CP}^{2n}])$ is the coefficient of z^{2n} in $Q(z)^{2n+1}$, where Q is built from ϕ .

We want to check that this agrees with $\phi_Q([\mathbb{CP}^{2n}])$, that is, that this is the coefficient on y^{2n} in $g'(y)$. As above let $f = g^{-1}$ so that $Q(x) = x/f(x)$.

By the Cauchy residue theorem,²¹ we have that

$$\phi([\mathbb{CP}^{2n}]) = \text{res}_0 \left(\frac{Q(x)^{2n+1}}{x^{2n+1}} \right) dx = \text{res}_0 \left(\frac{1}{f(x)^{2n+1}} \right) dx.$$

If we let $y = f(x)$ and $g(y) = x$, then $dx = g'(y)dy$, so we get that

$$\text{res}_0 \left(\frac{1}{f(x)^{2n+1}} \right) dx = \text{res}_0 \left(\frac{1}{y^{2n+1}} \right) g'(y)dy,$$

which is the coefficient on y^{2n} in $g'(y)$, i.e. $\phi_Q([\mathbb{CP}^{2n}])$.²² $\square$

Terminology 1.11.19. In [Theorem 1.11.8](#), we call $Q(x)$ the *characteristic series* associated to a genus.

²¹Or we might call this the formal residue theorem or something, since we are not in $\mathbb{C}$ but in any characteristic zero ring.

²²There's a tiny subtlety here when we change coordinates from x to y , in that we have to check that if we were integrating a small circle around the origin in the y -plane, after applying f it didn't become a more complicated path, e.g. a path that wraps *twice* around the origin in the x -plane. We don't encounter this issue when we do u -subs in single variable calculus, but it becomes something we have to stress about when doing contour integrals. In our particular case this is a non-issue since $f(x) = x + O(x^2)$ has winding number 1.

Example 1.11.20. We had this basic m -sequence, where the n th term $K_n(p_1, \dots, p_n) = p_n$ was just the n th Pontryagin class. Earlier we had mentioned this might give an easy example of a genus. Observing that $K_n(1, 0, \dots, 0) = 0$ for all $n > 1$, by Equation (1.11.11) the even power series associated to this m -series is very simple, it is $Q(x) = 1 + x^2$. We now want to solve for the associated genus. We have that

$$Q(x) = 1 + x^2 = \frac{x}{f(x)},$$

and so $f(x) = \frac{x}{1+x^2}$.²³ We ultimately want the compositional inverse g for this, so if we set

$$y = \frac{x}{1+x^2},$$

then $(1+x^2)y = x$, i.e. $yx^2 - x + y = 0$. By the quadratic formula, we solve for x as

$$x = \frac{1 \pm \sqrt{1-4y^2}}{2y}$$

This gives two solutions in the Laurent ring, but we want the one which is actually a power series. Using the Taylor series for $\sqrt{1-t}$ we can compute that $\sqrt{1-4y^2} = 1 - 2y^2 - 2y^4 + \dots$, from which we see

$$\begin{aligned} \frac{1 + \sqrt{1-4y^2}}{2y} &= \frac{1}{y} - y - y^3 - \dots \\ \frac{1 - \sqrt{1-4y^2}}{2y} &= y + y^3 + 2y^5 + \dots \end{aligned}$$

so the latter one is the one to pick. This tells us that

$$g(y) = \frac{1 - \sqrt{1-4y^2}}{2y},$$

and the derivative of this will give us the values of our genus on even-dimensional complex projective spaces.

It doesn't look fun to derive $g(y)$, so instead let's recall something from combinatorics, namely if C_n denotes the n th Catalan number, then the generating series $c(y) = \sum_{n \geq 0} C_n y^n$ is given by

$$c(y) = \frac{1 - \sqrt{1-4y}}{2y}.$$

So we have that

$$g(y) = yc(y^2) = \sum_{n \geq 0} C_n y^{2n+1}.$$

Deriving, we get that

$$g'(y) = \sum_{n \geq 0} (2n+1) C_n y^{2n},$$

²³Since $\frac{1}{1+x^2} = \sum_{j=0}^{\infty} (-1)^j x^{2j}$, we get that $f(x) = \sum_{j=0}^{\infty} (-1)^j x^{2j+1}$.

so the genus ϕ determined by our m -series must satisfy

$$\phi([\mathbb{CP}^{2n}]) = (2n+1)C_n.$$

The Catalan numbers are, of course, given by $C_n = \frac{1}{2n+1} \binom{2n+1}{n}$, so we recover that

$$\phi([\mathbb{CP}^{2n}]) = \binom{2n+1}{n} = \int_{\mathbb{CP}^{2n}} p_n(T\mathbb{CP}^{2n})$$

as expected!

1.11.1 The L -genus

Above we saw a very basic example of a genus, namely the one determined by $Q(x) = 1 + x^2$, which has a simple m -sequence and a simple characteristic series. We could go in the reverse direction, namely we could try to build a genus where the values of $\phi([\mathbb{CP}^{2n}])$ are simple and easy to compute. When we do this we are led to the following definition, which is a fundamental object in index theory:

Definition 1.11.21. We define the L -genus

$$\phi_L: \Omega_*^{\text{SO}} \otimes \mathbb{Q} \rightarrow \mathbb{Q}$$

to be the $\mathbb{Q}$ -valued genus defined by the even characteristic series

$$Q(x) = \frac{x}{\tanh(x)}.$$

This is also equal to

$$Q(x) = 1 + \sum_{k=1}^{\infty} (-1)^{k-1} \frac{2^{2k}}{(2k)!} B_k x^{2k},$$

where the B_k are the *Bernoulli numbers*.

The following result tells us, roughly, that this is the simplest possible genus we could construct.

Proposition 1.11.22. The L -genus ϕ_L is uniquely determined by the property that $\phi_L([\mathbb{CP}^{2n}]) = 1$ for all n .

Proof. Uniqueness is immediate from the proof of [Theorem 1.11.8](#); all genera are uniquely determined by where they send complex projective spaces of even dimension. So now fix some n . We get that $Q(x) = x/\tanh(x)$, so $f(x) = \tanh(x)$. As $f^{-1}(y) = g(y)$ with $y = f(x)$, we have by the inverse function theorem that

$$g'(y) = (f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} = \frac{1}{1 - \tanh(x)^2} = \frac{1}{1 - y^2}.$$

Therefore

$$g'(y) = 1 + y^2 + y^4 + \dots = \sum_{n \geq 0} \phi_L([\mathbb{CP}^{2n}]) y^{2n}.$$

□

Alternate proof. We can just directly compute $\phi_L([\mathbb{CP}^{2n}])$ by applying the Cauchy residue theorem to ϕ_L . This is how the proof goes in [Hir78, 1.5.1]. $\square$

Remark 1.11.23. This gives us an indication as to how we might have discovered the L -genus. If we wanted to cook up a genus taking values 1 on each complex projective space, we can start with $g'(y)$ as above, and integrate to get

$$g(y) = y + \frac{y^3}{3} + \frac{y^5}{5} + \dots$$

If we were exceptionally skilled at calculus, we might recognize this as the Taylor series for $\operatorname{arctanh}(y)$, the inverse of hyperbolic tangent.

Fun fact: Arc trigonometric functions are so called because the radian measurement of an angle is, by definition, the distance of the “arc” you traverse when walking along the unit circle around that angle. So while trig functions can be thought to input distances of arcs, the inverse trig functions can be thought to *output* distances of arcs, hence the names *arcsine*, *arccosine*, *arctangent*, etc. In hyperbolic geometry there is no such relationship between angle and distance along a circle, which leads us to question the utility of the prefix “arc” when discussing inverse hyperbolic trig functions. To end the debate definitively, the ISO (International Organization for Standardization) deemed in 2009 that we should instead drop the c and be using the terms *arsinh* and *arcosh* and *artanh* for inverse hyperbolic trig functions. At least that’s what Wikipedia told me. When I went to the ISO website, the relevant sections were hidden behind a paywall, so we will be using *arctanh* out of spite.

Exercise 1.11.24. Verify that the multiplicative series attached to the L -genus satisfies

$$\begin{aligned} L_1 &= \frac{1}{3}p_1 \\ L_2 &= \frac{1}{45}(7p_2 - p_1^2) \\ L_3 &= \frac{1}{3^3 \cdot 5 \cdot 7}(62p_3 - 13p_1p_2 + 2p_1^3) \\ L_4 &= \frac{1}{3^4 5^2 7}(381p_4 - 71p_1p_3 - 19p_2^2 + 22p_1^2p_2 - 3p_1^4) \end{aligned}$$

Exercise 1.11.25. Show that

$$\phi_L([\mathbb{HP}^k]) = \begin{cases} 0 & k \text{ odd} \\ 1 & k \text{ even.} \end{cases}$$

1.12 Hirzebruch’s signature theorem

In this section we prove Hirzebruch’s signature theorem, and explore some applications.

1.12.1 Linear algebra background

Theorem 1.12.1 (Spectral theorem). Let $M \in \text{Mat}_{n \times n}(\mathbb{R})$ be a symmetric matrix with real entries. Then there exists an orthogonal matrix $P \in O(n)$ so that

$$P^T M P = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$$

where the λ_i 's are the eigenvalues of M .

Corollary 1.12.2. A real symmetric matrix has all real eigenvalues.

So given a real matrix, a natural invariant to look at is how many eigenvalues are positive, how many are negative, and how many are zero. We will consider matrices with nonzero determinant, so that none of the eigenvalues are zero, they are all either positive or negative.

Theorem 1.12.3 (Sylvester's law of inertia). Let $M \in \text{Mat}_{n \times n}(\mathbb{R})$ be a symmetric matrix, and let $P \in \text{GL}_n(\mathbb{R})$ be any invertible matrix. Then the number of real/negative/zero eigenvalues of M is the same as the number of real/negative/zero eigenvalues of $P^T M P$.

This is helpful for us, since if we have a symmetric bilinear form on a vector space, we have to *choose* a basis in which to represent it. Sylvester's law of inertia tells us that this data about how many eigenvalues are positive/negative/zero is independent of this choice. The main source of such forms come from the cup product on even-dimensional manifolds. We define the following invariant:

Definition 1.12.4. Let V be a finite-dimensional real vector space, and let

$$Q: V \times V \rightarrow \mathbb{R}$$

be a symmetric bilinear form. We define its *signature* to be

$$\text{sig}(Q) := \# \{ \text{positive eigenvalues of } Q \} - \# \{ \text{negative eigenvalues of } Q \}.$$

Example 1.12.5. Let V be 2-dimensional, pick any two linearly independent vectors v_1 and v_2 inside V , and use them to define a function

$$\text{XOR}: \{v_1, v_2\} \times \{v_1, v_2\} \rightarrow \{0, 1\}$$

which returns 1 if the vectors plugged in are the same, and returns 0 if they are different. Since v_1 and v_2 span V , we can extend this to a bilinear map

$$Q: V \times V \rightarrow \mathbb{R}.$$

We claim that $\text{sig}(Q) = 0$. To see this, we represent Q in the basis coming from v_1 and v_2 , and we get the XOR matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. This has characteristic polynomial $\lambda^2 - 1$, hence eigenvalues $+1$ and -1 . This has signature zero.

Proposition 1.12.6 ([Fre12, 11.30]). Let $Q: V \times V \rightarrow \mathbb{R}$ be a nondegenerate bilinear form, and suppose $W \subseteq V$ satisfies $\dim W = \frac{1}{2} \dim V$, and suppose that Q is identically zero when restricted to W . Then $\text{sig}(Q) = 0$.

Proof. For any nonzero $w_1 \in W$, nondegeneracy of Q implies there exists a $v_1 \in V$ so that $Q(w_1, v_1) = 1$. By shifting v_1 by some scalar multiple of w_1 , we can arrange so that $Q(v_1, v_1) = 0$. Thus in the basis $\{v_1, w_1\}$ we have that Q is of the form $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. We can then take an orthogonal complement and iterate this construction. $\square$

1.12.2 The signature of $4n$ -manifolds

Proposition 1.12.7. Let M be a $2n$ -dimensional compact oriented manifold. Then the *intersection form* is defined to be

$$Q_M: H^n(M, \mathbb{R}) \times H^n(M, \mathbb{R}) \rightarrow \mathbb{R}$$

$$(\alpha, \beta) \mapsto \int_M \alpha \cup \beta.$$

This is a bilinear form, and if $n = 2k$ (that is, M has dimension $4k$), then it is symmetric.

Proof. Bilinearity follows from the cup product being an abelian group homomorphism, and integration being a map of real vector spaces. To see symmetry, we use this property that if $x \in H^a(M, \mathbb{R})$ and $y \in H^b(M, \mathbb{R})$ then $x \cup y = (-1)^{ab} y \cup x$ [Hat01, 3.11]. In our case when $a = b = n$, if n is even then symmetry is immediate. $\square$

Definition 1.12.8. Let M be a compact oriented $4n$ -dimensional manifold. Then we define the *signature* of M , denoted $\tau(M)$ (this is the notation in [Hir78]), to be the signature of its intersection form Q_M .

Example 1.12.9. Let T^4 denote the 4-torus. Then $\tau(T^4) = 0$.

Proof. Write

$$T^4 = S^1 \times S^1 \times S^1 \times S^1,$$

and let's give it coordinates $(\theta_1, \theta_2, \theta_3, \theta_4)$, where $0 \leq \theta_i < 2\pi$. Letting all the coordinates except one vary, we obtain a 3-dimensional subspace (in fact a 3-torus), which we could denote by Σ_i . For instance:

$$\Sigma_2 := \{(\theta_1, \theta_2, \theta_3, \theta_4) \in T^4 \mid \theta_2 = 0\}.$$

We have that $[\Sigma_i] \in H_3(T^4, \mathbb{R}) \cong H^1(T^4, \mathbb{R})$ give a basis of the first cohomology as $i = 1, 2, 3, 4$, therefore $H^1(T^4, \mathbb{R}) \cong \mathbb{R}^4$.

By the Künneth isomorphism, we obtain an isomorphism of graded rings

$$H^*(T^4, \mathbb{R}) \cong \wedge^* H^1(T^4, \mathbb{R}).$$

In particular, if we let $\sigma_{ij} \in H^2(T^4, \mathbb{R})$ denote the Poincaré dual to $[\Sigma_i \cap \Sigma_j]$, we get a basis for $H^2(T^4, \mathbb{R}) \cong \mathbb{R}^6$ given by

$$\sigma_{12}, \sigma_{23}, \sigma_{14}, \sigma_{23}, \sigma_{24}, \sigma_{34}.$$

We can compute the intersection form using the wedge product on cohomology, together with the knowledge that $\sigma_{12} \wedge \sigma_{34}$ is a preferred generator for $H^4(T^4, \mathbb{Z})$ (it endows T^4 with its orientation).²⁴

In this basis, we can write out the intersection form (row $\wedge$ column) as

	σ_{12}	σ_{13}	σ_{14}	σ_{23}	σ_{24}	σ_{34}
σ_{12}	0	0	0	0	0	1
σ_{13}	0	0	0	0	-1	0
σ_{14}	0	0	0	1	0	0
σ_{23}	0	0	1	0	0	0
σ_{24}	0	-1	0	0	0	0
σ_{34}	1	0	0	0	0	0

²⁴When we take the graded tensor product of $H^*(S^1, \mathbb{Z})$ with itself, we get $\mathbb{Z}[x, y]/(x^2, y^2, xy + yx) \cong \wedge^* \mathbb{Z}\{x, y\}$.

This has characteristic polynomial $(\lambda - 1)^3(\lambda + 1)^3$, so its eigenvalues are $+1$ with multiplicity 3 and -1 with multiplicity 3, thus signature zero. $\square$

Example 1.12.10. We have that $\tau(S^{4n}) = 0$.

Proof. This is just because $H^{2n}(S^{4n}, \mathbb{R}) = 0$. $\square$

Example 1.12.11. We have that $\tau(\mathbb{CP}^{2k}) = 1$.

Proof. We have that $H^{2k}(\mathbb{CP}^{2k}, \mathbb{R}) \cong \mathbb{R}$, generated by a copy of $\mathbb{CP}^k \subseteq \mathbb{CP}^{2k}$. One way to represent this homology class by a submanifold is to take k generically chosen linear forms in the homogeneous coordinates of $\mathbb{CP}^{2k}$ and intersect them. This homology class has self-intersection one (since the intersection of $2k$ generic linear forms gives a single point), hence the matrix representing the signature form is the 1×1 matrix (1) , which has signature 1. $\square$

Example 1.12.12. If X is a smooth complex cubic surface, we have that $H^2(X, \mathbb{Z}) \cong \mathbb{Z}^{\oplus 7}$, given by the hyperplane class disjoint union the E_6 lattice. Its signature is then

$$\tau(X) = 1 - 6 = -5.$$

Proposition 1.12.13 ([Hir78, 8.2.1(III)]). Let M be a $4k$ -dimensional compact oriented manifold. If M bounds then $\tau(M) = 0$.

Proof. Following **Proposition 1.12.6**, our strategy is to find a half-dimensional subspace of $H^{2k}(M, \mathbb{Z})$ on which the intersection form is identically zero. Let $j: M \cong \partial W \hookrightarrow W$ be the oriented embedding, and let $j^*: H^{2k}(W, \mathbb{R}) \rightarrow H^{2k}(M, \mathbb{R})$ be the induced map on cohomology. The goal is to argue that the intersection form vanishes identically on $\text{im}(j^*)$. Suppose we have $\alpha, \beta \in H^{2k}(W, \mathbb{R})$. Then if Q denotes the intersection form on M , we have that

$$Q(j^*\alpha, j^*\beta) = \int_M j^*\alpha \cup j^*\beta = \int_M j^*(\alpha \cup \beta) = 0.$$

This is true by the same idea as in **Lemma 1.10.25**: we can pick a closed $4k$ -form in $H_{\text{dR}}^{4k}(W, \mathbb{R})$ representing $\alpha \cup \beta$, then use Stokes' theorem.

Thus Q vanishes identically on $\text{im}(H^{2k}(W, \mathbb{R}) \xrightarrow{j^*} H^{2k}(M, \mathbb{R}))$. We now want to argue that $\dim \text{im}(j^*) = \frac{1}{2} \dim H^{4k}(M, \mathbb{R})$. This will follow from a diagram chase – consider relative Poincaré duality for the pair (W, M) , giving the commutative diagram

$$\begin{array}{ccccc} H^{2k}(W, \mathbb{R}) & \xrightarrow{j^*} & H^{2k}(M, \mathbb{R}) & \longrightarrow & H^{2k+1}(W, M, \mathbb{R}) \\ \downarrow \cong & & \downarrow \cong & & \downarrow \cong \\ H_{2k+1}(W, M, \mathbb{R}) & \longrightarrow & H_{2k}(M, \mathbb{R}) & \xrightarrow{j_*} & H_{2k}(W, \mathbb{R}) \end{array}$$

For some element $x \in H^{2k}(M, \mathbb{R})$, by diagram chasing we can see that $x \in \text{im}(j)$ if and only if $i(x) \in \ker(j_*)$. Hence the image of j^* is canonically isomorphic to the kernel of j_* along the Poincaré duality isomorphism i . Thus

$$\dim H^{2k}(M, \mathbb{R}) = \dim \text{im}(j^*) + \dim \ker(j_*) = 2 \dim \text{im}(j^*).$$

Therefore we are done by **Proposition 1.12.6**. $\square$

Theorem 1.12.14 (Signature is a genus). The signature defines a unital ring homomorphism

$$\tau: \Omega_*^{\text{SO}} \otimes \mathbb{Q} \rightarrow \mathbb{Z},$$

that is, it is a genus.

Proof. Following [Hir78, 8.2.1] it suffices to show τ is a ring homomorphism and respects the oriented cobordism relations. It is clear that τ is additive and that it preserves additive inverses since reversing orientation will negate the signature.

The first actual thing to prove is that τ is multiplicative. If M is $4k$ -dimensional and $M \cong V \times W$, then by the Künneth isomorphism we have

$$H^{2k}(M; \mathbb{R}) \cong \bigoplus_{s=0}^{2k} H^s(V; \mathbb{R}) \otimes_{\mathbb{R}} H^{2k-s}(W; \mathbb{R}).$$

We can pick a clever basis in which the relation $\tau(M) = \tau(V)\tau(W)$ is clear (see [Hir78, p. 85]).

Finally we have to argue if V bounds then $\tau(V) = 0$, which we did in [Proposition 1.12.13](#). $\square$

Theorem 1.12.15 (Hirzebruch signature theorem). The signature *is* the L -genus.

Proof. This follows immediately by [Example 1.12.11](#), as we have previously argued the L -genus is the unique genus on oriented bordism taking the value on 1 on each complex projective space of even complex dimension. $\square$

This is a massive result in topology — it states that the signature of a *smooth closed oriented* manifold can be determined completely via its Pontryagin numbers. Let's see an example:

Example 1.12.16. For any smooth closed oriented 4-manifold M , we have that

$$\tau(M) = \left\langle \frac{p_1(M)}{3}, [M] \right\rangle.$$

Moreover we know this is *integral*! This says that p_1 of any 4-manifold is always a multiple of 3.

Example 1.12.17. For any smooth closed oriented 8-manifold M , we have that

$$\tau(M) = \frac{7p_2 - p_1^2}{45},$$

therefore its Pontryagin numbers satisfy

$$p_1^2 - 7p_2 \equiv 0 \pmod{45}.$$

For example, consider $\mathbb{CP}^4$ as an 8-manifold. By [Theorem 1.9.10](#) it has $p_1 = 5$ and $p_2 = 10$, so we get $5^2 - 7 \cdot 10$ is indeed zero modulo 45.

Example 1.12.18 (An exotic 7-sphere). Suppose we had a compact smooth 8-manifold W whose p_1 and signature τ we understood. We could solve for p_2 as

$$p_2(W) = \frac{45\tau(W) + p_1(W)^2}{7}.$$

Since p_2 is an integer, we know that the numerator on the right hand side is zero modulo 7. That is, for any smooth closed oriented 8-manifold W , we have that

$$p_1(W)^2 + 45\tau(W) \equiv 2p_1(W)^2 - \tau(W) \equiv 0 \pmod{7}. \quad (1.12.19)$$

Now, given some closed smooth oriented 7-manifold M , we know that it is the boundary of some 8-manifold by Thom's computation that $\Omega_7^{\text{SO}} = 0$ (Example 1.10.19). However this manifold W has boundary (it is not closed), hence it doesn't fall in the scope of the signature theorem, and there is no expectation that its p_1, p_2, τ must satisfy Equation (1.12.19). A very natural idea, then, is to take a manifold M^7 and find a W bounding it, then attach to it the residue of $2p_1(W)^2 - \tau(W)$ modulo 7. Viewing this as a function from smooth closed oriented 7-manifolds to $\mathbb{Z}/7$, this won't be well-defined, since M can be the boundary of many 8-manifolds, and there's again no expectation that these different 8-manifolds have the same residue modulo 7 for this combination of their p_1 and signature. However under some mild assumptions on the cohomology of M , this will turn out to be well-defined! Milnor proves in [Mil56, Theorem 1] that the following function is well-defined:

$$\lambda: \left\{ \begin{array}{l} \text{closed oriented smooth} \\ \text{7-manifolds } M^7 \text{ with} \\ H^3(M) = H^4(M) = 0 \end{array} \right\} \rightarrow \mathbb{Z}/7$$

$$M^7 = \partial W^8 \mapsto 2p_1(W^8)^2 - \tau(W^8).$$

As an example, we can consider S^7 , which is the boundary of the ball B^8 . Both p_1 and τ of the ball are zero, hence $\lambda(S^7) \equiv 0 \pmod{7}$.

What would it mean to find a manifold M^7 which is homeomorphic to S^7 but which satisfies $\lambda(M^7) \not\equiv 0 \pmod{7}$? It would mean that the bounding W^8 does not satisfy the desired congruence modulo 7, in particular the boundary of W^8 cannot be diffeomorphic to S^7 . However the boundary of W^8 is M^7 , which we said was homeomorphic to S^7 . Hence the construction of such a manifold would be an exotic 7-sphere. Milnor builds these as certain D^4 -bundles over S^4 , which are classified by the clutching construction $\pi_3(\text{SO}(4))$. For details see [Mil56].

Example 1.12.20 ([Hir71, §3]). Let $f: \mathbb{C}^7 \rightarrow \mathbb{C}$ be the function

$$f(z_1, \dots, z_7) = z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2 + z_6^3 + z_7^5,$$

and let $\epsilon > 0$ be a small positive number. Consider the manifold

$$M^{12} = \{z = (z_1, \dots, z_7) \in \mathbb{C}^7 \mid f(z_1, \dots, z_7) = \epsilon, \|z\| \leq 1\}.$$

This is a smooth 12-dimensional manifold with boundary. A computation of Brieskorn says that the boundary is homeomorphic to S^{11} . If we collapse by the boundary, we get a closed 12-dimensional manifold $N = M/\partial M$ with signature -8 .

We furthermore have that N is 5-connected, so its first five reduced homology groups vanish, and by universal coefficients, we see that $H^4(N) = 0$ as well, in particular this implies $p_1(N) = 0$. By the signature theorem we get

$$-8 = \tau(N) = \int_N L_3 = \frac{62p_3}{945}.$$

We check that $\gcd(62, 945) = 1$, and since we are conflating p_3 with an integer along the isomorphism $H^{12}(N, \mathbb{Z}) \cong \mathbb{Z}$, we see there are no integral solutions to the above.

Therefore N does not admit any smooth structure, and therefore ∂M^{11} could not be diffeomorphic to an 11-sphere, only homeomorphic.

The other rather shocking fact that follows immediately from the Hirzebruch signature theorem is:

Corollary 1.12.21. The signature of a manifold depends only on the oriented homotopy type of the manifold.

1.13 Complex genera

1.13.1 Complex cobordism

Recall Ω_*^{SO} classified smooth oriented closed manifolds up to smooth oriented cobordism. Thom assembled a space MSO whose homotopy groups came with a graded ring isomorphism $\pi_*\text{MSO} \cong \Omega_*^{\text{SO}}$ [Tho54] (see Joaquim’s appendix for more about this). This construction (now called the *Thom spectrum* construction) was very general, and in [Mil60], Milnor observed that it could be applied to the unitary group, from which he built a space MU . A natural question to ask is whether $\pi_*\text{MU}$ admits an interpretation in terms of cobordism, and he mentions that it does, and this is proven in the upcoming Part II of the paper [Mil60]. This never was written, however Thom wrote an account of these results in [Tho60, §5].²⁵

The answer is that yes, MU_* admits an interpretation in terms of cobordism, but it’s not quite as simple as “complex manifolds up to cobordism.” There are a few key issues in trying to set this up:

- ▷ *the dimension issue*: unlike oriented or real bordism, we can’t take a complex manifold to exhibit a cobordism between two complex manifolds. This is simply due to dimension reasons: if M and N are complex manifolds of complex dimension n , they have real dimension $2n$, so any cobordism $W = \partial M \amalg \partial N$ will have real dimension $(2n + 1)$, hence won’t be complex
- ▷ *the inverse issue*: for real oriented manifolds, we could just swap the orientation, forming $\bar{M}$ out of M . Complex manifolds have a canonical orientation coming from the complex structure, and if we try to form the analogous $\bar{M}$, it won’t be a complex manifold anymore.

All of this tells us that we should allow ourselves to leave the world of complex manifolds, or at least to weaken the structure we’re asking for a bit. We’ll see a posteriori that we actually didn’t lose much by doing this.

Remark 1.13.1. We want to see quickly how $\text{GL}_n(\mathbb{C})$ lives inside $\text{GL}_{2n}(\mathbb{R})$ as a subgroup. When $n = 1$, we can pick a real basis $\{1, i\}$ for $\mathbb{C} \cong \mathbb{R}^2$. In that basis, a complex number $a + bi$ acts via multiplication on some $c + di$ via $ad - bc + (bc + ad)i$. So what is the 2×2 matrix representing this? We’re solving for

$$\begin{pmatrix} - & - \\ - & - \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ad - bc \\ bc + ad \end{pmatrix}.$$

It’s clear that the matrix must be $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. This gives us a group homomorphism (exercise: check it’s multiplicative)

$$\begin{aligned} \text{GL}_1(\mathbb{C}) &\rightarrow \text{GL}_2(\mathbb{R}) \\ a + bi &\mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \end{aligned}$$

²⁵Be aware, if you read Thom’s work, that *variété* really means manifold, although in other French math it means variety, and it is very context-dependent. See [here](#) for more.

More generally, we get a map

$$\mathrm{GL}_n(\mathbb{C}) \rightarrow \mathrm{GL}_{2n}(\mathbb{R})$$

$$\begin{pmatrix} a_{11} + b_{11}i & \cdots & a_{1n} + b_{1n}i \\ \vdots & \ddots & \vdots \\ a_{n1} + b_{n1}i & \cdots & a_{nn} + b_{nn}i \end{pmatrix} \mapsto \left(\begin{array}{cc|cc} a_{11} & -b_{11} & \cdots & a_{1n} & -b_{1n} \\ b_{11} & a_{11} & & b_{1n} & a_{1n} \\ \hline & \vdots & \ddots & & \vdots \\ a_{n1} & -b_{n1} & \cdots & a_{nn} & -b_{nn} \\ b_{n1} & a_{n1} & & b_{nn} & a_{nn} \end{array} \right). \quad (1.13.2)$$

There is a C_2 -action on $\mathrm{GL}_{2n}(\mathbb{R})$ sending each 2×2 block to -1 times its transpose, and we can see that a matrix lies in the image of the group homomorphism above if and only if it is fixed under this action. This is a baby example of Galois descent.

Definition 1.13.3. Let M be a real $2n$ -manifold, and let $M \rightarrow \mathrm{BGL}_{2n}(\mathbb{R})$ be the map classifying its tangent bundle. We say that M admits an *almost complex structure* if we have a lift

$$\begin{array}{ccc} & & \mathrm{BGL}_n(\mathbb{C}) \\ & \nearrow \text{dashed} & \downarrow \\ M & \longrightarrow & \mathrm{BGL}_{2n}(\mathbb{R}), \end{array}$$

where the map $\mathrm{BGL}_n(\mathbb{C}) \rightarrow \mathrm{BGL}_{2n}(\mathbb{R})$ is induced by Equation (1.13.2). Phrased differently, the tangent bundle of M lies in the image of the map

$$\check{H}^1(M, \mathrm{GL}_n(\mathbb{C})) \rightarrow \check{H}^1(M, \mathrm{GL}_{2n}(\mathbb{R})).$$

Remark 1.13.4. An almost complex structure on TM is a vector bundle homomorphism $J: TM \rightarrow TM$ which has the property that $J^2 = -1$, that is, if we compose it twice it will act as scaling via -1 . Roughly speaking, J is intended to mimic “multiplication by i ” but it is only doing it on the fibers. To become an actual complex structure (that is, for M to genuinely be a complex manifold), we need the existence of coordinates on which J acts as multiplication by i . This need not exist.

Still, an almost complex manifold must be of even dimension (exercise). So we instead work with an even weaker notion.

Definition 1.13.5. We say that a manifold M is *stably almost complex* if $TM \oplus \epsilon_{\mathbb{R}}^k$ admits an almost complex structure for some $k \geq 0$.

Unlike almost complex manifolds, we can have odd-dimensional *stably* almost complex manifolds.

Example 1.13.6 ([GGK02, D.2]). We have that S^5 admits a stably almost complex structure, since $TS^5 \oplus \epsilon_{\mathbb{R}}^1 \cong \epsilon_{\mathbb{R}}^6 \cong \epsilon_{\mathbb{C}}^3$, however it clearly has no almost complex structure.

With that in mind, we can define complex bordism.

Definition 1.13.7. Let M and N be two compact n -manifolds stably almost complex bundles. We say they are *complex cobordant* if there exists a stably almost complex manifold W whose stably almost complex structure restricts to that of M and N . We let Ω_n^U denote the abelian group on such manifolds up to complex cobordism, and $\Omega_*^U := \bigoplus_{n \geq 0} \Omega_n^U$ the associated graded ring.

Theorem 1.13.8 (Milnor, Thom). There is a ring isomorphism

$$\pi_* \mathrm{MU} \cong \Omega_*^U.$$

One of the beautiful theorems of 20th century topology is Lazard's theorem:

Theorem 1.13.9 (Lazard's theorem). We have an isomorphism

$$\Omega_*^U \cong \mathbb{Z}[x_1, x_2, \dots]$$

where each x_i has degree $2i$.²⁶

Theorem 1.13.10. We have that

$$\Omega_*^U \otimes \mathbb{Q} \cong \mathbb{Q}[\mathbb{CP}^1, \mathbb{CP}^2, \mathbb{CP}^3, \dots].$$

Corollary 1.13.11. We see that $\Omega_{2n+1}^U = 0$, and that $\Omega_{2n}^U \cong \mathbb{Z}^{\pi(n)}$, where π is the partition function.

1.13.2 How far is complex cobordism away from complex manifolds?

A beautiful theorem that can be found in Stong's book is the following:

Theorem 1.13.12 ([Sto68, p. 130]). Let $n > 0$ and $x \in \Omega_n^U$. Then there exists a (not necessarily connected) complex manifold M so that $[M] = x$. In fact, M can be taken to be a smooth complex algebraic variety.

Thus, after the fact, we can think about Ω_*^U as complex manifolds up to an equivalence relation, and there is no need to consider stably almost complex manifolds until we define the equivalence relation.

In general, exhibiting what kinds of varieties can appear as representatives of complex cobordism classes is a very interesting line of inquiry. An example result in this direction is the following:

Theorem 1.13.13 ([BR98]). Any $x \in \Omega_{2n}^U$ can be represented by a toric manifold.

1.13.3 Chern numbers

Similarly to how we showed Pontryagin numbers were oriented cobordism invariants, we want to see that Chern numbers give complex cobordism invariants. Recall we can define Chern classes for any complex vector bundle, in particular one coming from an almost complex structure. Since Chern classes are designed to be stable invariants (they are insensitive to adding on a trivial complex vector bundle), it makes sense to define their Chern numbers.

Notation 1.13.14. Let M^{2n} be a compact $2n$ -dimensional stably almost complex manifold, which is necessarily oriented.²⁷ Let $I = (i_1, \dots, i_k)$ be a partition of n . Then we define the I th Chern number of M to be

$$c_I(M) = \int_M c_{i_1}(TM) \cdots c_{i_k}(TM) \in H^{2n}(M, \mathbb{Z}) \cong \mathbb{Z}.$$

Theorem 1.13.15 (Milnor-Novikov, [Mil62, Theorem 1"]). A closed stably almost complex manifold is the boundary of a stably complex manifold if and only if all its Chern numbers are zero.

This is our primary step to studying complex genera.

²⁶We'll say more about these x_i soon.

²⁷Every stably almost complex manifold is oriented, coming from the orientation on $TM \oplus \epsilon_{\mathbb{R}}^k$.

1.13.4 Complex genera

Definition 1.13.16. Let $R \in \text{CAlg}_{\mathbb{Q}}$ be an integral domain which is a commutative unital algebra over $\mathbb{Q}$. We define a *complex genus* to be a ring homomorphism

$$\Omega_*^U \rightarrow R.$$

Just as in the case of genera (for oriented cobordism), we can set up a correspondence to formal power series, now not necessarily assumed to be even. Now the Chern classes will play the role that the Pontryagin classes played for ordinary genera.

Theorem 1.13.17. For any unital integral domain $R \in \text{CAlg}_{\mathbb{Q}}$, there is a bijection between complex genera and formal power series with constant term 1:

$$\{\text{complex genera valued in } R\} \leftrightarrow 1 + xR[[x]].$$

A complex genus ϕ is uniquely determined by the values of $\phi([\mathbb{CP}^n])$ for all $n \geq 1$.

Proof. Completely analogous to [Theorem 1.11.8](#). □

A natural question to ask, then, is whether there is an analogous genus to the L -genus for complex genera? The answer is yes – it is the so-called *Todd genus*.

Definition 1.13.18. We define the *Todd genus*

$$\text{Td}: \Omega_*^U \rightarrow \mathbb{Q}$$

to be the unique genus satisfying $\text{Td}([\mathbb{CP}^n]) = 1$ for all $n \geq 1$.

Notation 1.13.19. We will use $\text{td}(M) \in H^*(M, \mathbb{Q})$ to denote the total Todd class, and we will denote by $\text{Td}(M) = \int_M \text{td}(TM) \in \mathbb{Q}$ the associated Todd genus.

Theorem 1.13.20 ([\[Hir78, 10.3.1\]](#)). The Todd genus is given by the characteristic series

$$Q_{\text{Td}}(x) = \frac{x}{1 - e^{-x}}.$$

Proof. We just have to check the value of $\phi_{\text{Td}}([\mathbb{CP}^n])$. By the Cauchy residue theorem, this will be the coefficient on x^n in the expression $(Q(x))^{n+1}$. We start to compute this as

$$\begin{aligned} \phi_{\text{Td}}([\mathbb{CP}^n]) &= \text{res}_0 \left(\frac{Q(x)^{n+1}}{x^{n+1}} dx \right) \\ &= \text{res}_0 \left(\frac{1}{(1 - e^{-x})^{n+1}} dx \right) \\ &= \text{res}_0 \left(\frac{e^{x(n+1)}}{(e^x - 1)^{n+1}} dx \right). \end{aligned}$$

We now want to do a u -substitution. Let $z = e^x$, so $dz = e^x dx$, and note that the location of the residue changes after this u -sub (even though its winding number doesn't). So we get

$$\phi_{\text{Td}}([\mathbb{CP}^n]) = \text{res}_1 \left(\frac{z^n dz}{(z - 1)^{n+1}} \right) = 1.$$

□

1.13.5 Computing the Todd genus

To compute the Todd genus of a manifold, we need to know its corresponding m -sequence. Let M^n be a stably almost complex n -manifold, and let $\xi_1, \dots, \xi_n$ denote its Chern roots, so that c_i is the i th elementary polynomial in them. We can then expand $\prod_{i=1}^n Q(\xi_i)$ and group terms of similar degrees to compute the m -sequence. We'll refer to the i th part of the m -sequence as $\text{td}_i(M)$, so that $\text{td}(M) = \text{td}_0(M) + \text{td}_1(M) + \dots$, and we have that $\text{Td}(M^n) = \int_M \text{td}(M) = \int_M \text{td}_n(M)$. It'll be helpful to write down the first few terms of the characteristic series for the Todd genus:

$$\frac{x}{1-e^{-x}} = 1 + \frac{x}{2} + \frac{x^2}{12} - \frac{x^4}{720} + O(x^6).$$

First, it is immediate that $\text{td}_0(M) = 1$ for all manifolds M , since $\frac{x}{1-e^{-x}}$ has constant term 1. If M is complex 1-dimensional, then $\text{td}_1 = c_1/2$. The first interesting computation is td_2 :

$$\begin{aligned} Q_{\text{Td}}(\xi_1)Q_{\text{Td}}(\xi_2) &= \left(1 + \frac{\xi_1}{2} + \frac{\xi_1^2}{12} + \dots\right) \left(1 + \frac{\xi_2}{2} + \frac{\xi_2^2}{12} + \dots\right) \\ &= 1 + \frac{c_1}{2} + \frac{\xi_1^2 + \xi_2^2}{12} + \frac{\xi_1 \xi_2}{4} + \dots \end{aligned}$$

just looking at the degree 2 terms here we get

$$\frac{1}{12} (\xi_1^2 + \xi_2^2 + 3\xi_1 \xi_2) = \frac{1}{12} ((\xi_1 + \xi_2)^2 + \xi_1 \xi_2) = \frac{c_1^2 + c_2}{12}.$$

Continuing on, we get

$$\begin{aligned} \text{td}_0 &= 1 \\ \text{td}_1 &= \frac{1}{2}c_1 \\ \text{td}_2 &= \frac{1}{12}(c_1^2 + c_2) \\ \text{td}_3 &= \frac{1}{24}c_1 c_2 \\ \text{td}_4 &= \frac{1}{720}(-c_4 + c_1 c_3 + 3c_2^2 + 4c_1^2 c_2 - c_1^4) \\ \text{td}_5 &= \frac{1}{1440}(-c_1 c_4 + c_1^2 c_3 + 3c_1 c_2^2 - c_1^3 c_2). \end{aligned}$$

Example 1.13.21. Let $\text{Gr}_{\mathbb{C}}(2, 4)$ denote the complex Grassmannian of 2-planes in 4-space. This is a complex 4-manifold, and we can compute its Todd class either by hand or using `Schubert2`, say `[GSS+]`:

```
loadPackage "Schubert2";
integral todd flagBundle({2,4})
```

```
o2 = 1
```

This tells us that $\text{Td}(\text{Gr}_{\mathbb{C}}(2, 4)) = 1$. In fact the Todd classes of all Grassmannians are 1.

Example 1.13.22. Let C be an elliptic curve, then $\text{Td}(C) = 0$.

Proof. As an abelian variety, its tangent space is trivial, hence $c_1(TC) = 0$, and therefore

$$\mathrm{Td}(C) = \int_C \frac{c_1(C)}{2} = 0.$$

□

Example 1.13.23. Any K3 surface X has $\mathrm{Td}(X) = 2$. As an example, we can consider a quartic hypersurface in $\mathbb{P}^3$ which is a model for a K3 surface, and we compute:

```
loadPackage "Schubert2";
P3 = abstractProjectiveSpace 3;
X = sectionZeroLocus OO_P3(4);
integral todd X
```

```
o4 = 2
```

1.13.6 Properties of the Todd genus

Definition 1.13.24. Let X be a topological space, and let $E \rightarrow X$ be a complex vector bundle. Then we define the *Todd class* $\mathrm{td}(E) \in H^*(X, \mathbb{Q})$ to be the class determined by the m -sequence associated to the Todd genus, evaluated at the Chern classes of E . That is:

$$\mathrm{td}(E) = 1 + \frac{c_1(E)}{2} + \frac{c_1(E)^2 + c_2(E)}{12} + \frac{c_1(E)c_2(E)}{24} + \dots \in H^*(X, \mathbb{Z}).$$

Example 1.13.25. Over a smooth manifold M with (stably almost) complex structure, we have that $\mathrm{td}(M) = \mathrm{td}(TM)$.

Notation 1.13.26. When M is an oriented n -manifold, we define $\mathrm{Td}(E) = \int_M \mathrm{td}(E) \in \mathbb{Q}$.

Proposition 1.13.27. For any two complex vector bundles $E_1, E_2 \rightarrow x$, we have an equality $\mathrm{td}(E_1 \oplus E_2) = \mathrm{td}(E_1)\mathrm{td}(E_2)$ in $H^*(X, \mathbb{Z})$.

Proof. This is an immediate consequence of the Todd genus arising via an m -sequence. □

Proposition 1.13.28. For any complex vector bundle $E \rightarrow X$ and any map $f: Y \rightarrow X$, we have that $f^*\mathrm{td}(E) = \mathrm{td}(f^*E)$.

Proof. Immediate by naturality of Chern classes. □

With this new notation in mind, we can prove the following:

Proposition 1.13.29. Let X be a compact (stably almost) complex manifold, and without loss of generality assume it is connected. Let $E \rightarrow X$ be a complex vector bundle of rank $d + 1$, and let $\mathbb{P}(E) \rightarrow X$ denote its projectivization. Then they have equal Todd genera:

$$\mathrm{Td}(\mathbb{P}(E)) = \mathrm{Td}(X).$$

Proof. Let $\pi: \mathbb{P}(E) \rightarrow X$ denote the projection. We have a short exact sequence of vector bundles

$$0 \rightarrow T\pi \rightarrow T\mathbb{P}(E) \rightarrow \pi^*TX \rightarrow 0,$$

where $T\pi$ is the *vertical tangent bundle*. By multiplicativity of Todd classes ([Proposition 1.13.27](#)) we can express the Todd genus of $T\mathbb{P}(E)$ in terms of the Todd classes of the other two bundles in the sequence:

$$\mathrm{Td}(\mathbb{P}(E)) = \int_{\mathbb{P}(E)} \mathrm{td}(T\mathbb{P}(E)) = \int_{\mathbb{P}(E)} \mathrm{td}(T\pi) \mathrm{td}(\pi^*TX).$$

Using naturality [Proposition 1.13.28](#), we can rewrite this as

$$= \int_{\mathbb{P}(E)} \mathrm{td}(T\pi) \pi^* \mathrm{td}(TX).$$

Integration is, simply, the Gysin pushforward down to a point. Since the map from $\mathbb{P}(E) \rightarrow *$ can be factored through X , we have that the above is equal to

$$= \int_X \pi_* (\mathrm{td}(T\pi) \pi^* \mathrm{td}(TX)),$$

which by the projection formula for Gysin integration ([Proposition 1.8.20](#)) is then equal to

$$= \int_X \mathrm{td}(TX) \cdot \pi_* \mathrm{td}(T\pi).$$

Just by looking at dimensions, we have that $\pi_* \mathrm{td}(T\pi) \in H^0(X, \mathbb{Q}) \cong \mathbb{Q}$, so it is just some scalar. Putting everything together we have computed that

$$\mathrm{Td}(\mathbb{P}(E)) = (\pi_* \mathrm{td}(T\pi)) \cdot \mathrm{Td}(X).$$

So we want to verify this scalar is equal to 1. To compute this, we introduce some notation: pick any basepoint $*$ $\xrightarrow{j}$ X , and consider the pullback square

$$\begin{array}{ccc} \mathbb{CP}^d & \xhookrightarrow{i} & \mathbb{P}(E) \\ \downarrow p & \lrcorner & \downarrow \pi \\ * & \xrightarrow{j} & X. \end{array}$$

Since $j^*: H^0(X) \rightarrow H^0(*)$ is an isomorphism, it suffices to compute $j^* \pi_* \mathrm{td}(T\pi)$. However by base change for Gysin maps ([Proposition 1.8.20](#)), we get that

$$j^* \pi_* \mathrm{td}(T\pi) = p_* i^* \mathrm{td}(T\pi) = p_* \mathrm{td}(i^* T\pi).$$

Observe that $i^* T\pi$ is the restriction of the vertical tangent to a point in the fiber. This is exactly the tangent space *of* the fiber. So altogether we get

$$j^* \pi_* \mathrm{td}(T\pi) = p_* \mathrm{td}(T\mathbb{CP}^d) = \int_{\mathbb{CP}^d} \mathrm{td}(\mathbb{CP}^d) = 1.$$

Therefore $\pi_* \mathrm{td}(T\pi) = 1 \in H^0(X, \mathbb{Q})$ and we are done. $\square$

Corollary 1.13.30. For X any complex manifold of dimension d , we let X^Δ denote the total flag space of its tangent bundle. Then $\mathrm{Td}(X) = \mathrm{Td}(X^\Delta)$.

Proof sketch. Use that the flag space is built as an iterated projective bundle, together with the fact that the Todd genus comes from an m -sequence, and induct with base case [Proposition 1.13.29](#). $\square$

1.13.7 Looking ahead to HRR

Just as the L -genus agrees with the index of a manifold, we will see the Todd genus agrees with a different invariant of an algebraic manifold, namely its so-called *arithmetic genus*, which is defined in terms of sheaf cohomology. This is essentially the content of *Hirzebruch-Riemann-Roch*.

1.14 The holomorphic Euler characteristic

Definition 1.14.1. Let X be any space and $F \in \text{Ab}(X)$ any sheaf. We want to define the *sheaf-theoretic Euler characteristic* $\chi(X, F)$ to be

$$\chi(X, F) = \sum_{i=0}^{\infty} \dim H^i(X, F),$$

We have to be super clear here though about:

- ▷ what we mean by dimension
- ▷ when this thing is actually convergent and hence well-defined.

For example if we could argue that $H^i(X, F)$ is a finitely generated abelian group for each i and that $H^i(X, F) = 0$ for $i \gg 0$, then $\chi(X, F)$ would be well-defined. We'll be interested in the particular case where F is a sheaf on a complex manifold. Our ultimate goal is to show the following big theorem:

for a compact complex manifold X , the integer $\chi(X, \mathcal{O}_X)$ is well-defined

We will call this thing the *holomorphic Euler characteristic*. It used to be known as the arithmetic genus but this is a bit overloaded:

Warning 1.14.2. What we are calling the holomorphic Euler characteristic, Hirzebruch calls the *arithmetic genus*. This is now overloaded notation (see for instance the discussion in [Aut, 0BY6]) — Hartshorne for instance defines the arithmetic genus of a complex variety of dimension d as $p_a(X) = (-1)^d (\chi(X, \mathcal{O}_X) - 1)$ [Har77, p. 230].

Our goal is the following:

1. define a nice collection of sheaves which have finite-dimensional $H^p(X, F)$ for all p (these are called *coherent $\mathcal{O}_X$ -modules*)
2. show that $\mathcal{O}_X$ is among this collection (Oka coherence)
3. show that $H^p(X, \mathcal{O}_X)$ vanishes for large p (this uses Hodge theory)

1.14.1 Coherent $\mathcal{O}_X$ -modules

Definition 1.14.3. Let X be a compact complex manifold, and let $F \rightarrow X$ be a sheaf of abelian groups. We say that F is a *sheaf of $\mathcal{O}_X$ -modules* if, for every $U \in \text{Open}(X)$, the abelian group $F(U)$ is a module over the ring of holomorphic functions $\mathcal{O}_X(U)$, and these module structures are compatible with restriction, in the sense that if $V \subseteq U$ and $r \in \mathcal{O}_X(U)$ and $s \in F(U)$, then $\text{res}_V^U(rs) = \text{res}_V^U(r)\text{res}_V^U(s)$.

Remark 1.14.4. If $F \in \text{Mod}_{\mathcal{O}_X}$, then F is a sheaf of complex vector spaces. In particular $H^q(X, F)$ is a complex vector space for each q , so it has a well-defined dimension.

Example 1.14.5. If $E \rightarrow X$ is a holomorphic vector bundle, then its sheaf of holomorphic sections $\text{Sec}_{\text{hol}}(-, E)$ is an $\mathcal{O}_X$ -module.

We identify a particularly nice subcategory of $\text{Mod}_{\mathcal{O}_X}$ consisting of the so-called *coherent* sheaves. We'll see in a second why they are so important.

Definition 1.14.6. Let $F \in \text{Mod}_{\mathcal{O}_X}$. We say that F is *coherent* if

1. F is *finitely type*: meaning for every $x \in X$ there exists some open neighborhood $U \ni x$ and a surjection

$$\mathcal{O}_{X|U}^{\oplus N} \twoheadrightarrow F|_U,$$

2. for every $U \subseteq X$, and for every morphism

$$\phi: \mathcal{O}_{X|U}^{\oplus n} \rightarrow \mathcal{F}|_U,$$

the kernel $\ker(\phi)$ is finite type on U .

Theorem 1.14.7 (Cartan-Serre finiteness). Let X be a compact complex manifold and F a coherent sheaf of $\mathcal{O}_X$ -modules. Then $H^q(X, F)$ is a finite-dimensional complex vector space for all $q \geq 0$.

Proof. The proof is difficult – see [GR04, VI.4] for a writeup of it. $\square$

Theorem 1.14.8 (Oka coherence). Let X be a complex manifold. Then $\mathcal{O}_X$ is a coherent $\mathcal{O}_X$ -module.

Sketch. Since we have local holomorphic charts, we can reduce to the case of $\mathcal{O}_{\mathbb{C}^n}$. In this setting, it is clear that $\mathcal{O}_{\mathbb{C}^n}$ is finite type as an $\mathcal{O}_{\mathbb{C}^n}$ -module (since it is generated everywhere locally by 1), the hard part is arguing that any $n \times 1$ matrix locally has finite kernel.

Given some neighborhood U of a point (WLOG the origin) $0 \in \mathbb{C}^n$, and some map

$$\phi: \mathcal{O}_{\mathbb{C}^n|U}^{\oplus n} \rightarrow \mathcal{O}_{\mathbb{C}^n|U},$$

we have that $\ker(\phi)$ is a submodule of $\mathcal{O}_{\mathbb{C}^n|U}^{\oplus n}$, which is finitely generated as an $\mathcal{O}_{\mathbb{C}^n|U}$ -module. So we'd like to conclude by using the fact that submodules of finitely generated modules are finitely generated, but we recall this fact only holds over *Noetherian* rings. Thus in order to prove the theorem, it would suffice to argue that $\mathcal{O}_{\mathbb{C}^n, 0}$ is Noetherian. We do this inductively on $n = \dim \mathbb{C}^n$ by using the Weierstrass preparation theorem (in fact this is one of the key applications of Weierstrass preparation). $\square$

Example 1.14.9. For any holomorphic vector bundle E of finite rank, its sheaf of holomorphic sections is a coherent $\mathcal{O}_X$ -module.

So in summary $\dim_{\mathbb{C}} H^p(X, \mathcal{O}_X) < \infty$ for every $p \geq 0$. If we can show these vanish for large p we are in business in terms of defining a holomorphic Euler characteristic. To do this we need a little Hodge theory.

1.14.2 The Dolbeault resolution

We want to show that, for a compact complex manifold X , we have that $H^p(X, \mathcal{O}_X) = 0$ for $p \gg 0$. The strategy is to produce an acyclic resolution of the sheaf $\mathcal{O}_X$ which has only finitely many nonzero terms. Such a resolution comes from the *Dolbeault sheaves*.

Let's consider the Poincaré lemma – we resolved $\mathbb{R}$ in terms of smooth differential forms. Maybe we can do a similar thing here, resolving the sheaf of holomorphic functions on X by some kind of differential forms.

The first subtlety we bump into is the following:

- ▷ for a smooth manifold X , its tangent bundle TX admits a smooth structure
- ▷ for a complex manifold X , its tangent bundle TX does *not* admit a natural holomorphic structure!

That is, if X is complex of dimension d , the transition maps to $GL_d(\mathbb{C})$ are smooth, but cannot be assumed to be holomorphic. Unlike the passage from continuous to smooth where we can “smooth out” transition functions, there is no natural way to deform these coherently to make them holomorphic. Unless X is very nice (e.g. a Stein manifold) we shouldn't expect to be able to endow smooth complex vector bundles on X with a holomorphic structure.

There is, however, a natural way to build a holomorphic vector bundle out of the tangent bundle.

Definition 1.14.10. Let X be a complex manifold. The almost complex structure $j: TX \rightarrow TX$ satisfies $j^2 = -\text{id}$ but doesn't split. However if we complexify TX , this does split. We define

$$TM \otimes_{\mathbb{R}} \mathbb{C} = T^{1,0}M \oplus T^{0,1}M$$

to be the splitting corresponding to the eigenvalues i and $-i$, respectively. We call $T^{1,0}M$ the *holomorphic tangent bundle* and $T^{0,1}M$ the *antiholomorphic tangent bundle* of M .

Remark 1.14.11. On X we can take a chart $z_1, \dots, z_n$, and by writing $z_j = x_j + iy_j$, we get a preferred basis $\frac{d}{dx_1}, \frac{d}{dy_1}, \dots, \frac{d}{dx_n}, \frac{d}{dy_n}$ of TM over the chart. This induces distinguished bases $\frac{d}{dz_1}, \dots, \frac{d}{dz_n}$ of $T^{1,0}M$, and $\frac{d}{d\bar{z}_j}$ of $T^{0,1}M$ satisfying:

$$\begin{aligned} \frac{d}{dz_j} &= \frac{1}{2} \left(\frac{d}{dx_j} - i \frac{d}{dy_j} \right) \\ \frac{d}{d\bar{z}_j} &= \frac{1}{2} \left(\frac{d}{dx_j} + i \frac{d}{dy_j} \right). \end{aligned}$$

We pick things this way so that on the dual space $T^{1,0*}M$ we have distinguished sections dz^j , and on $T^{0,1*}M$ we have distinguished sections $d\bar{z}^j$

$$\begin{aligned} dz^j &= dx^j + i dy^j \\ d\bar{z}^j &= dx^j - i dy^j. \end{aligned}$$

Notation 1.14.12. Let $\Omega^{1,0}$ be the sheaf of *smooth* sections of $T^{1,0*}M$ and let $\Omega^{0,1}$ be the sheaf of smooth sections of $T^{0,1*}M$. We can verify that $\Omega^{1,0}(U)$ can be interpreted as the space of complex-valued differential 1-forms on U consisting only of dz^i 's, and $\Omega^{0,1}$ is the space of complex-valued differential 1-forms on U consisting only of $d\bar{z}^i$'s.

For any $p, q \geq 0$ we denote by

$$\Omega^{p,q} := \bigwedge^p \Omega^{1,0} \wedge \bigwedge^q \Omega^{0,1}.$$

we refer to sections of $\Omega^{p,q}$ as (p, q) -forms.

Note 1.14.13. These sheaves above can also be defined in terms of a decomposition of $\wedge^n(T^*M \otimes_{\mathbb{R}} \mathbb{C})$ into summands over each (p, q) with $p + q = n$. When $n = 0$, the 0th wedge power of a complex line bundle is just the trivial complex line bundle. From this perspective it's easy to see that $\Omega^{0,0}$ is exactly the sheaf $C^\infty(-, \mathbb{C})$ of smooth complex-valued functions.

Remark 1.14.14. We have that $\Omega^{0,0} = C^\infty(-, \mathbb{C})$ is the sheaf of smooth complex-valued functions.

So what is the differential of a (p, q) -form? Given some

$$\omega = f(z_1, \dots, z_n) dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q},$$

it still makes sense to talk about df since f is smooth, so we define

$$d\omega := (df) \wedge dz_{i_1} \wedge \dots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \dots \wedge d\bar{z}_{j_q}.$$

There's an issue, that df breaks up into stuff involving dz_j 's and $d\bar{z}_j$'s, so $d\omega$ will be a $(p, q + 1)$ -form plus a $(p + 1, q)$ -form. Thus we could really think about this differential as the sum of *two* differentials:

Proposition 1.14.15. The differential

$$\bigoplus_{p+q=n} \Omega^{p,q} \xrightarrow{d} \bigoplus_{p+q=n+1} \Omega^{p,q}$$

breaks up naturally into the sum of two differentials

$$\begin{aligned} \partial: \Omega^{p,q} &\rightarrow \Omega^{p+1,q} \\ \bar{\partial}: \Omega^{p,q+1} &\rightarrow \Omega^{p,q+1}. \end{aligned}$$

These are defined as you might expect – $\partial f = \sum \frac{df}{dz_i} dz_i$ and $\bar{\partial} f = \sum \frac{df}{d\bar{z}_i} d\bar{z}_i$.

Exercise 1.14.16. Show that $\partial \circ \partial = 0$ and $\bar{\partial} \circ \bar{\partial} = 0$.

The core property is now the following:

Theorem 1.14.17 (Dolbeault resolution). If X is a complex manifold of dimension d , we have an acyclic resolution

$$0 \rightarrow \mathcal{O}_X \rightarrow \Omega^{0,0} \xrightarrow{\bar{\partial}} \Omega^{0,1} \xrightarrow{\bar{\partial}} \Omega^{0,2} \rightarrow \dots \rightarrow \Omega^{0,d} \rightarrow 0$$

Proof. Since we have seen that $\bar{\partial}\partial = 0$ it is clear that most of this sequence is exact. There are clearly no $(0, d + 1)$ -forms or higher since the dimension of X is only d , so this resolution is finite. To see exactness at $\Omega^{0,0}$, we observe that $\Omega^{0,0} \cong C^\infty(-, \mathbb{C})$, and a complex-valued function is holomorphic *precisely* when $\frac{d}{d\bar{z}_i}$ of it is zero for all i . In other words, when $\bar{\partial}$ of it vanishes.

The last thing to check is that the $\Omega^{0,q}$'s are acyclic. We show instead that they are fine – this is because X is a smooth manifold and therefore admits smooth partitions of unity, so we can rescale any $(0, q)$ -form by some function in our partition of unity and it preserves smoothness. $\square$

Corollary 1.14.18. If X is a complex manifold of dimension d we have that $H^q(X, \mathcal{O}_X) = 0$ for $q > d$.

Therefore the holomorphic Euler characteristic $\chi(X, \mathcal{O}_X)$ is well-defined!

1.14.3 Properties of the holomorphic Euler characteristic

Exercise 1.14.19. If

$$F_1 \rightarrow F_2 \rightarrow F_3$$

is a short exact sequence of coherent sheaves on X , show that

$$\chi(X, F_2) = \chi(X, F_1) + \chi(X, F_3).$$

Theorem 1.14.20 ([Hir78, 20.2.1]). Let X be a complex (algebraic?) manifold, and let X^Δ be the total flag space. Then

$$\chi(X, \mathcal{O}_X) = \chi(X^\Delta, \mathcal{O}_{X^\Delta}).$$

Proof. Since the Euler characteristic will be, by definition, the alternating sum of the cohomology groups, we can use that dimension is additive on short exact sequences to reduce to checking that

$$\dim_{\mathbb{C}} H^0(X, \Omega^{0,p}) = \dim_{\mathbb{C}} H^0(X^\Delta, \Omega^{0,p})$$

for all $p \geq 0$. Let $\pi: X^\Delta \rightarrow X$ denote the pullback, then π^*TX decomposes into complex analytic line bundles $\xi_1, \dots, \xi_n$. Since these bundles are complex analytic, we have that $c_1(\xi_j)$ is represented by a $(1,1)$ -form. Since π is complex analytic, we have that π^* preserves (p,q) forms. It is a theorem of Borel that $H^*(X^\Delta, \mathbb{C})$ is generated by the injective image of $\pi^*H^*(X, \mathbb{C})$, together with all the $c_1(\xi_j)$'s. In particular all the $(0,p)$ -forms in $H^*(X^\Delta, \mathbb{C})$ are pulled back from $H^*(X, \mathbb{C})$. Hence $\dim_{\mathbb{C}} H^0(X, \Omega^{0,p}) = \dim_{\mathbb{C}} H^0(X^\Delta, \Omega^{0,p})$. $\square$

1.15 Hirzebruch-Riemann-Roch

In a 1953 letter to Kodaira and Spencer, Serre conjectured that, for X a complex manifold and $W \rightarrow X$ a holomorphic vector bundle, the holomorphic Euler characteristic $\chi(X, W)$ was expressible in terms of the Chern classes of W . The general statement is now known as Riemann-Roch, or Hirzebruch-Riemann-Roch:

Theorem 1.15.1. Let X be a compact complex manifold. Then we have that

$$\text{Td}(X) = \chi(X, \mathcal{O}_X).$$

That is, its Todd genus and holomorphic Euler characteristic agree.

Proof sketch. Hirzebruch's original proof states it in the case where X is an *algebraic* manifold.²⁸ Let X^Δ denote the total flag space over X . We know that $\text{Td}(X) = \text{Td}(X^\Delta)$ by [Corollary 1.13.30](#), and that $\chi(X, \mathcal{O}_X) = \chi(X^\Delta, \mathcal{O}_{X^\Delta})$ by [Theorem 1.14.20](#), and moreover by a result of Borel we know that if X is algebraic then X^Δ will be as well. So it suffices to reduce to the case where X is a split complex algebraic manifold.

In this context, we can start taking hyperplane sections of X (this is where we use our assumption that X is algebraic), and we can use this to define the so-called virtual Euler characteristic and Todd genera $\chi_y(L_1, \dots, L_k)$ and $T_y(L_1, \dots, L_k)$, which take in a sequence of line bundles and output

²⁸Meaning a manifold with a holomorphic embedding into some $\mathbb{CP}^N$.

a polynomial in a single indeterminate y . Since X is split, we have that $TX \cong L_1 \oplus \dots \oplus L_n$ for some line bundles L_i , so we use this to derive the relations

$$\begin{aligned} (1+y)^m \chi(X) &= \sum_{l=0}^m y^l \sum_{i_1 < \dots < i_l} \chi_y(L_{i_1}, \dots, L_{i_l}) \\ (1+y)^m T(X) &= \sum_{l=0}^m y^l \sum_{i_1 < \dots < i_l} T_y(L_{i_1}, \dots, L_{i_l}). \end{aligned}$$

Thus we just need to show that $\chi_y = T_y$ for some $y \neq -1$. When $y = 1$, this reduces to a particular extension of the Hirzebruch signature theorem! $\square$

Remark 1.15.2. Hirzebruch uses that X is Kähler and algebraic throughout, and these assumptions can be dropped but not with his original proof strategy. They are removed when this result is placed in the larger context of index theory and is reproven using the Atiyah-Singer index theorem.

An immediate corollary is:

Corollary 1.15.3. The Todd genus of any compact complex manifold is an integer. Therefore for any such X :

1. If $\dim_{\mathbb{C}} X = 1$ then $\int_X c_1(TX) \equiv 0 \pmod{2}$. In particular its Euler characteristic $\chi(X)$ is even.
2. If $\dim_{\mathbb{C}}(X) = 2$ then $\int_X c_1(TX)^2 + c_2(TX) \equiv 0 \pmod{12}$
3. If $\dim_{\mathbb{C}}(X) = 3$ then $\int_X c_1(TX)c_2(TX) \equiv 0 \pmod{24}$
4. and so on...

Example 1.15.4 (Noether's formula). If X is a 2D complex surface, then HRR tells us

$$\chi(X, \mathcal{O}_X) = \frac{1}{12} (c_1(TX)^2 + c_2(TX)).$$

Note that $c_2(TX) = \chi(X)$ is the Euler characteristic. Similarly $-c_1(TX) = K_X$ is sometimes called the canonical class. So we can rewrite the above as

$$\chi(X, \mathcal{O}_X) = \frac{1}{12} (K_X^2 + \chi(X)).$$

This is often known as *Noether's formula*, and it is a core result in complex surface theory.

Two more corollaries which are worth highlighting:

Corollary 1.15.5. The holomorphic Euler characteristic depends only on the homotopy type of the complex manifold.

Corollary 1.15.6. The holomorphic Euler characteristic is a complex bordism invariant.

1.15.1 A mild generalization

More generally, we have not just $\chi(X, \mathcal{O}_X)$ but $\chi(X, E)$ for any holomorphic vector bundle E . Should this be equal to something involving the Todd genus? The answer is yes, and it takes very little extra effort to state.

Definition 1.15.7. Let $E \rightarrow X$ be a complex vector bundle on an arbitrary space, and let $\xi_1, \dots, \xi_n$ denote its Chern roots. We define the *Chern character* of E to be

$$\text{ch}(E) = e^{\xi_1} + \dots + e^{\xi_n}.$$

Example 1.15.8. Since $\mathcal{O}_X$ is the sheaf of holomorphic sections of the trivial complex line bundle, all its higher Chern classes are 1, so $\text{ch}(\mathcal{O}_X) = 1$.

Theorem 1.15.9 (HRR with coefficients). Let X be a compact complex manifold and $E \rightarrow X$ a holomorphic line bundle. Then we have that

$$\chi(X, E) = \int_X \text{ch}(E) \text{td}(TX).$$

As an example, we recover Riemann-Roch as classically stated.

Example 1.15.10. Let X be a Riemann surface, and $L \rightarrow X$ a line bundle. On the left hand side, the cohomology vanishes above the dimension of X , so we get

$$\chi(X, L) = h^0(X, L) - h^1(X, L).$$

On the right hand side, in $H^*(X, \mathbb{Q})$, all terms above degree two vanish, so we get

$$\begin{aligned} \text{ch}(L) &= 1 + c_1(L) \\ \text{td}(TX) &= 1 + \frac{c_1(TX)}{2}. \end{aligned}$$

We multiply and integrate:

$$\int_X \text{ch}(E) \text{td}(X) = \int_X (1 + c_1(L))(1 + \frac{c_1(TX)}{2}) = \int_X c_1(L) + \int_X \frac{c_1(TX)}{2} = \deg(L) + \frac{1}{2}\chi(X).$$

Hence HRR says

$$h^0(X, L) - h^1(X, L) = \deg(L) + \frac{1}{2}\chi(X).$$

Example 1.15.11. If X is 2-dimensional and $L \rightarrow X$ is a line bundle, we get

$$\chi(X, L) = \frac{c_1(L)^2}{2} + \frac{c_1(L)c_1(TX)}{2} + \frac{c_1(TX)^2 + c_2(TX)}{12}.$$

We can recognize the rightmost term as the holomorphic Euler characteristic of X by HRR, so we can rearrange all this to say

$$\chi(X, L) - \chi(X, \mathcal{O}_X) = \frac{1}{2} (c_1(L)^2 + c_1(L)c_1(TX)).$$

If we call $D = c_1(L)$ and $K_X = -c_1(TX)$ the divisor and canonical class, we rewrite this as

$$\chi(X, L) - \chi(X, \mathcal{O}_X) = \frac{1}{2} (D \cdot D - D \cdot K_X).$$

A last application that I really enjoy is, when we understand the base space fairly well, we can derive constraints on Chern classes of holomorphic vector bundles.

Example 1.15.12. Let $E \rightarrow \mathbb{CP}^3$ be any holomorphic vector bundle. Then $c_1(E)c_2(E) + c_3(E)$ is even.

Proof. The Chern character of E is similar, but we need to add on one more term. If we compute correctly, we'll get

$$\begin{aligned} \text{ch}(E) &= \text{rank}(E) + c_1(E) + \frac{c_1(E)^2 - 2c_2(E)}{2} + \frac{c_1(E)^3 - 3c_1(E)c_2(E) + 3c_3(E)}{6} \\ \text{td}(T\mathbb{CP}^3) &= 1 + 2h + \frac{22}{12}h^2 + h^3. \end{aligned}$$

Putting everything together we get

$$\chi(\mathbb{CP}^3, E) = \int_{\mathbb{CP}^3} \text{rank}(E)h^3 + \frac{11}{6}c_1(E)h^2 + (c_1(E) - 2c_2(E))h + \frac{c_1(E)^3 - 3c_1(E)c_2(E) + 3c_3(E)}{6}.$$

Throwing out everything that isn't obviously an integer, we are left with

$$\int_{\mathbb{CP}^3} \frac{11c_1(E)h^2 + c_1(E)^3 - 3c_1(E)c_2(E)}{6} \in \mathbb{Z}.$$

Therefore (by conflating cohomology classes in different degrees by multiplying by appropriate powers of the hyperplane class), we have that

$$11c_1(E) + c_1(E)^3 - 3c_1(E)c_2(E) + 3c_3(E) \equiv 0 \pmod{6}.$$

We can rearrange this to

$$c_1(E)^3 - c_1(E) + 3c_1(E)c_2(E) + 3c_3(E) \equiv 0 \pmod{6}.$$

But $x^3 - x \equiv 0$ in $\mathbb{Z}/6$ always, so we are left with $3(c_1(E)c_2(E) + c_3(E)) \equiv 0 \pmod{6}$, or:

$$c_1(E)c_2(E) + c_3(E) \equiv 0 \pmod{2}.$$

□

Appendix A

Appendices

Contents

A.1 Category theory	107
A.2 Point-set topology	109
A.3 Normalized versus unnormalized cochains	113

A.1 Category theory

If you are unfamiliar with colimits I really recommend reading Chapter 3 of [Rie16].

A.1.1 Filtered colimits

Definition A.1.1. A category is *filtered* if every finite diagram admits a cocone. Dually a category $\mathcal{C}$ is *cofiltered* if every finite diagram admits a cone.

Example A.1.2. The category $\text{Open}(X)$ is cofiltered.

Proof. We don't have to worry about morphisms commuting, so it suffices to see, for finitely many $U_1, \dots, U_n \in \text{Open}(X)$, there is a U mapping to them. Clearly we can take $U = U_1 \cap \dots \cap U_n$. $\square$

Example A.1.3. For $x \in X$, the category of open subspaces of X containing x is cofiltered.

Example A.1.4. For a space X , the category $\text{Cov}(X)$ of open covers under refinement (defined in Definition 1.2.15) is cofiltered. This is precisely the assertion that two covers admit a common refinement, and therefore by induction finitely many covers admit a common refinement.

The important thing about filtered categories is that colimits computed over them are very well behaved.

Proposition A.1.5. Filtered colimits valued in Set or Ab or any other reasonable concrete category (see SE2143601 for details for instance) commute with finite limits.

Corollary A.1.6. Filtered colimits (valued in Ab , let's say) preserve monomorphisms.

Proof. The property of a morphism $f: x \rightarrow y$ being a monomorphism is equivalent to the statement that the diagram

$$\begin{array}{ccc} x & \xrightarrow{\text{id}} & x \\ \text{id} \downarrow & \lrcorner & \downarrow f \\ x & \xrightarrow{f} & y \end{array}$$

is a pullback. This is a finite limit, hence commutes with filtered colimits. $\square$

An interesting question in the setting of abelian categories is the interaction between exactness and filtered colimits. Indeed exactness of filtered colimits is a consequence of Grothendieck's axiom AB5 for abelian categories [Gro57, 1.8.1]. We will leverage the following corollary a handful of times:

Corollary A.1.7. Let $\mathcal{A}$ be an abelian category, and let $\text{Fun}^{\text{ex}}(\mathcal{A}, \text{Ab})$ be the full subcategory of exact functors from $\mathcal{A}$ to abelian groups. Then $\text{Fun}^{\text{ex}}(\mathcal{A}, \text{Ab})$ is preserved under filtered colimits.

Proof. Filtered colimits in this category are computed levelwise, hence they commute with finite limits. Being right exact is a colimit, hence commutes with (filtered) colimits in any context. Being left exact is a pullback condition, hence a finite limit, and therefore commutes with filtered colimits. $\square$

As a remark, this is not true if the colimits are not filtered, as illustrated by the following example:

Example A.1.8. Let $F = \text{id}_{\text{Ab}}: \text{Ab} \rightarrow \text{Ab}$ be the (clearly exact) identity functor on the category of abelian groups, and consider the (non-filtered!) equalizer diagram

$$0, 2: F \rightrightarrows F,$$

where 0 is the natural transformation given levelwise by the zero homomorphism, and 2 is multiplication by 2. Then we have that the colimit of this diagram is computed by the pointwise coequalizer of 0 and 2. In particular if $C: \text{Ab} \rightarrow \text{Ab}$ is the colimit of the above diagram, then

$$C(A) = A/2A$$

for every abelian group A . But this is not an exact functor $\text{Ab} \rightarrow \text{Ab}$.

A.1.2 Cofinality

Definition A.1.9. Let J be a category, and $I \subseteq J$ a subcategory. We say $I \subset J$ is *cofinal* if, for any functor $F: J \rightarrow \mathcal{C}$, the induced map on colimits

$$\text{colim}_I F \rightarrow \text{colim}_J F$$

is an isomorphism in $\mathcal{C}$. We say $I \subseteq J$ is *final* if $I^{\text{op}} \subseteq J^{\text{op}}$ is cofinal.

The key reductive step of the Poincaré lemma relies on a recognition of cofinality (really finality, but the category gets “opped” when we take a contravariant functor out of it):

Proposition A.1.10. Let M be an n -dimensional manifold and let $x \in M$ be an arbitrary point. Then the category of open sets of M containing x admits a final system given by those open sets $U \subseteq M$ so that $U \ni x$ and U is homeomorphic to $\mathbb{R}^n$.

Proof. Fix a chart (U_0, ϕ) around x . For any $V \ni x$ open, we can intersect it with the chart to get $V \cap U_0$. In the image $\phi(V \cap U_0)$ we can find some open ball containing $\phi(x)$, and the preimage of this in M is both homeomorphic to $\mathbb{R}^n$ and contained in V . $\square$

The following result is a categorical rephrasing of an important result in differential topology. Before we can phrase it we need a definition.

Definition A.1.11. Let $\mathcal{U}$ be an open cover of X . We say that it is a *good cover* if all the $U \in \mathcal{U}$ and all the nonempty intersections of finitely many elements in $\mathcal{U}$ are contractible spaces.

Proposition A.1.12. Let M be a Riemannian manifold, and denote by $\text{Cov}^{\text{good}}(M)$ the category of good open covers of M . Then $\text{Cov}^{\text{good}}(M) \subseteq \text{Cov}(M)$ is a final subcategory.

Proof. The content of this is showing every cover refines to a good one. Fix a cover $\mathcal{U}$ of M , and now pick any point $x \in M$, and any $U \ni x$. Since every point on a Riemannian manifold admits a positive convexity radius, we can obtain a geodesically convex neighborhood around each point, and by potentially shrinking the radius we may assume it is contained in U . This is clearly convex, and it yields a good cover since the intersection of geodesically convex spaces is geodesically convex, hence contractible. $\square$

In particular, any colimiting construction we want to pursue over covers of a Riemannian manifold can be reduced to good open covers.

A wild fun fact is that the poset of open sets and refinement for a good cover can be used to recover the homotopy type of the manifold (or space) you started out with! This is the content of the *nerve lemma*.

A.2 Point-set topology

Definition A.2.1. Let $\{U_i\}_{i \in I}$ be an open cover of a space X . We say it is

- ▷ *point finite* if each $x \in X$ is contained in only finitely many U_i
- ▷ *locally finite* if each $x \in X$ has an open neighborhood $V \ni x$ so that V meets only finitely many U_i .

Clearly a locally finite open cover is point finite, but the converse need not hold (see [Mun00, p. 244] for an explicit example).

We want to rephrase compactness in terms of *refining* the cover, not just throwing out elements of the cover. To that end, it's nicer to work with a different definition. We prove a quick proposition:

Proposition A.2.2. Let X be a topological space. Then it is compact if, for every $\mathcal{U} \in \text{Cov}(X)$, there exists a refinement $\mathcal{V} \leq \mathcal{U}$ for which $\mathcal{V}$ is finite (has finitely many elements).

Proof. If X is compact (in the ordinary sense), then we can just pick $\mathcal{V}$ to be the finite subcollection of $\mathcal{U}$ guaranteed by compactness. For the reverse direction, suppose we have an open cover $\mathcal{U} = \{U_i\}_{i \in I}$, and a finite refinement $\mathcal{V} = \{V_1, \dots, V_n\}$. Then by the definition of refinement, there is some function $f: \{1, 2, \dots, n\} \rightarrow I$ so that $V_i \subseteq U_{f(i)}$ for each $1 \leq i \leq n$. Then it is clear that $\{U_{f(1)}, \dots, U_{f(n)}\}$ is a finite open cover of X , which is a subcover of our original cover $\mathcal{U}$. $\square$

As Munkres says, this definition is “awkward... but it suggests a way to generalize.”

Definition A.2.3. Let X be a space. We say it is...

$$\begin{cases} \text{compact} \\ \text{paracompact} \\ \text{point finite/metacompact} \end{cases} \quad \text{if every open cover } \mathcal{U} \text{ admits a refinement } \mathcal{V} \text{ which is } \begin{cases} \text{finite} \\ \text{locally finite} \\ \text{point finite} \end{cases}$$

It is clear that we have the following implications:

$$\text{compact} \Rightarrow \text{paracompact} \Rightarrow \text{point finite}.$$

The converse implications don't hold.

Example A.2.4. Euclidean space $\mathbb{R}^n$ is paracompact but not compact.

Example A.2.5 (Point-finite spaces need not be paracompact). The first example was, I believe, [Bin51, Example B]. Another good example is [SS78, Example 89].

Remark A.2.6. Hirzebruch, Bourbaki, and many other authors assume the spaces receiving the definitions in Definition A.2.3 are Hausdorff. We will try to be clear when the Hausdorff condition is needed.

Theorem A.2.7 (Dieudonné, [Hir78, 2.8.2]). Every locally compact¹ Hausdorff space with a countable basis is paracompact.

A.2.1 Normal spaces

The idea of a *normal* space can be thought of (loosely) as a strengthening of the Hausdorff condition. On T_1 -spaces (spaces in which singletons are closed) this is a literal strengthening of the Hausdorff condition – it says that not only can *points* be separated by open neighborhoods, actually any disjoint *closed subspaces* can be separated by open neighborhoods.

Definition A.2.8. A space is *normal* if any two disjoint closed subsets can be separated by disjoint open neighborhoods. Explicitly if $Z_1, Z_2 \subseteq X$ are each closed and $Z_1 \cap Z_2 = \emptyset$, then there exist open subspaces $U_1 \supseteq Z_1$ and $U_2 \supseteq Z_2$ so that $U_1 \cap U_2 = \emptyset$.

Theorem A.2.9 (Dieudonné, [Hir78, 2.8.1]). Every paracompact space Hausdorff is *normal*.

In particular since our manifolds are Hausdorff with countable bases, they are in particular normal.

Lemma A.2.10 (Urysohn's lemma). A space X is normal if and only if, for every two disjoint closed subsets $A, B \subseteq X$, there exists a continuous function

$$f: X \rightarrow [0, 1]$$

so that $f(A) = 0$ and $f(B) = 1$.

The following crucial result implies that, over a normal space, locally finite covers can always be “shrunk.” It is a crucial ingredient for the existence of partitions of unity.

¹Locally compact means for every $x \in X$ there exists some compact subspace $K \subseteq X$ and an open neighborhood $U \ni x$ so that $U \subseteq K$.

Theorem A.2.11 (Shrinking theorem, [Hir78, 2.8.3]). Let X be normal, and $\{U_i\}_{i \in I}$ a point finite open cover. Then X admits an open cover $\{V_i\}_{i \in I}$, indexed over the same set, so that $\overline{V_i} \subseteq U_i$.

Definition A.2.12. Let $\{U_i\}_{i \in I}$ be an open cover of a space X . We say that a system of continuous functions $\{\phi_i: X \rightarrow \mathbb{R}\}_{i \in I}$ is a *partition of unity subordinate to the cover* if:

1. $\phi_i(x) \geq 0$ for all $x \in X$
2. $\text{supp}(\phi_i) \subseteq U_i$
3. Each point has an open neighborhood meeting $\text{supp}(\phi_i)$ for only finitely many $i \in I$
4. We have that

$$\sum_{i \in I} \phi_i(x) = 1$$

for all $x \in X$. Note that sum is defined because of the previous point.

Definition A.2.13. We say an open cover of a space is *numerable* if it admits a subordinate partition of unity. We say a fiber bundle is *numerable* if the base admits a numerable cover over which the total space is trivialized [Dol63, 7.1].

Theorem A.2.14. A space is paracompact if and only if it is Hausdorff and every open cover admits a partition of unity.

Proof. For a proof see [Hir78, pp. 30–31]. □

The following result is originally due to Stone? An elegant reproof is due to Mary Ellen Rudin in the 60's. We can find the following in [Mun00, 41.4], for instance.

Theorem A.2.15. Every metric space is paracompact.

Another crucial example is that CW complexes are paracompact. Miyazaki, Dugundji, Bourgin all separately proved in 1952 that simplicial complexes were paracompact. Miyazaki shortly thereafter extended this result to all CW complexes

Theorem A.2.16 ([Miy52]). Every CW complex is paracompact.

Sketch. There's a nice outline on the nLab – we can induct on the skeleton, leveraging that disks and spheres are paracompact, and that paracompact Hausdorff spaces are closed under coproducts and pushouts along closed embeddings. □

A.2.2 Properties of open covers and neighborhoods

We say a *property* $\mathbf{P}$ is any boolean-valued assignment on objects of a category.

Notation A.2.17. We let

- ▷ $\text{Open}(X)$ be the category of open sets in X , and $\text{Open}^{\mathbf{P}}(X)$ the subcategory of those opens which satisfy $\mathbf{P}$
- ▷ $\text{Open}_x(X)$ the category of open sets in X containing a fixed point $x \in X$, and $\text{Open}_x^{\mathbf{P}}(X)$ the subcategory of those opens containing x and satisfying $\mathbf{P}$

- ▷ $\text{Cov}(X)$ the category of open covers of X and $\text{Cov}^{\mathbf{P}}(X)$ the category of covers comprised of elements satisfying $\mathbf{P}$. Explicitly:

$$\text{Cov}^{\mathbf{P}}(X) = \{\mathcal{U} \in \text{Cov}(X) : U \in \mathcal{U} \Rightarrow U \in \text{Open}^{\mathbf{P}}(X)\}.$$

Terminology A.2.18. A *local basis* (or *neighborhood basis*) of a space X at a point x is any final subcategory of $\text{Open}_x(X)$.

A core idea in the theory of open sets is the following comparison between local bases of opens satisfying a condition and covers built out of opens satisfying that same condition.

Proposition A.2.19. Let $\mathbf{P}$ be a property of the open subspaces in $\text{Open}(X)$, and assume that X is a T1 space. Then the following are equivalent:

1. for every $x \in X$, the inclusion $\text{Open}_x^{\mathbf{P}}(X) \rightarrow \text{Open}_x(X)$ is final
2. the inclusion $\text{Cov}^{\mathbf{P}}(X) \rightarrow \text{Cov}(X)$ is final

Proof. For the forward direction, pick an open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of X . For each $x \in X$, it is contained in $U_i \in \mathcal{U}$ for some i . By property (1) we can find some V_x containing x for which $V_x \subseteq U_i$ and V_x satisfies $\mathbf{P}$. Then $\mathcal{V} = \{V_x\}_{x \in X}$ is a cover in $\text{Cov}^{\mathbf{P}}(X)$ which refines $\mathcal{U}$.

Conversely if $x \in X$ and we have an open neighborhood $U \ni x$. We can complete this to a cover $\mathcal{U} = \{U, X - \{x\}\}$ since X is a T1 space.² Then by hypothesis there is a refinement $\mathcal{V} \leq \mathcal{U}$ which is in $\text{Cov}^{\mathbf{P}}(X)$. In particular $x \in V \in \mathcal{V}$ for some V , and since $V \not\subseteq X - \{x\}$, we must have that $V \subseteq U$. $\square$

Definition A.2.20. Let X be a T1 space and let $\mathbf{P}$ be the property that a space is connected. Then we say X is *locally connected* if either of the conditions in [Proposition A.2.19](#) hold.

Remark A.2.21. The condition of being an n -manifold can be rephrased in this language. If X is second countable and Hausdorff, and $\mathbf{P}$ is the property of “being homeomorphic to $\mathbb{R}^n$,” then we say X is an n -manifold if it satisfies the conditions of [Proposition A.2.19](#). This is essentially our discussion in [Proposition A.1.10](#).

Definition A.2.22. Let X be a T1 space and $\mathbf{P}$ the property of being contractible (homotopy equivalent to a one-point space). We say X is *locally contractible* if either of the conditions in [Proposition A.2.19](#) hold.

What Spanier calls *locally contractible* is something different. We, following Sella, will call that condition *semi-locally contractible*.

Definition A.2.23. Consider the property $\mathbf{P}$ which is true on an open subset $U \subseteq X$ if and only if the inclusion map $U \hookrightarrow X$ is homotopic to a constant map. We say a space X is *semi-locally contractible* if, for every $U \subseteq X$, the poset $\text{Cov}^{\mathbf{P}}(U)$ is nonempty.

Remark A.2.24. On nLab (at the time of writing), their definition of semi-locally contractible is that X admits a basis of open sets whose inclusion maps are homotopic to a constant map. This is different than the definition above.

Example A.2.25. Let X be the *infinite earring*. Then the cone on X , defined as $CX = X \times [0, 1]/X \times \{1\}$ is contractible but not semi-locally contractible.

²More generally, we start with $U \ni x$ and complete it to a cover for which U is the only open set in the cover containing our point. I think this can only be done for T1 spaces.

A.3 Normalized versus unnormalized cochains

In this section of the appendix we cop to a lie by omission (or by obfuscation?). Our definition of the Čech cochains complex for a cover $\mathcal{U} = \{U_i\}_{i \in I}$ was

$$\check{C}^q(\mathcal{U}, F) = \prod_{i_0, \dots, i_q \in I} F(U_{i_0} \cap \dots \cap U_{i_q})$$

However this notation is a bit ambiguous — are we allowing the $i_0, i_1, \dots, i_q$ to be the same? If so we’re allowing the $(q+1)$ -fold intersection of the same open set - is this what we want? In examples we did, it seems like we weren’t assuming this. We said “ah there are no 3fold overlaps so there are no Čech 2-cochains” but should we really have been allowing for these degenerate intersections where the same open set in our cover is repeated?

The answer is that *it doesn’t matter*, they both compute the same cohomology. We’re going to sketch how one proves this. We call the one where we don’t allow for repeated indices the *normalized* Čech cochains, and the one where we allow for repeated indices the *unnormalized* Čech cochains (definitions to follow).

With that being said, some computations are *much easier* in one setting or the other. When we did explicit computations like [Example 1.2.12](#) and [Example 1.2.19](#) it was beneficial to use normalized cochains, since we could conclude there are no higher cochains since there are no higher overlaps. However in the proofs of [Proposition 1.2.50](#) and [Theorem 1.3.1](#) it was more beneficial to work with unnormalized cochains since we could construct explicit chain homotopies without worrying about repeated indices.

In this section we’ll make the comparison between normalized and unnormalized cochains explicit. We need to define some local (not used outside this section) notation in order to streamline everything:

Definition A.3.1. Let X be a space, $\mathcal{U} = \{U_i\}_{i \in I}$ some open cover, and F any presheaf on X . Dust off your nearby copy of the axiom of choice and use it to fix a total ordering $\leq$ on the set I . We define

▷ the *normalized Čech cochains* to be

$$\check{C}_{\text{nor}}^q(\mathcal{U}, F) := \prod_{i_0 < i_1 < \dots < i_q} F(U_{i_0} \cap \dots \cap U_{i_q}).$$

▷ the *unnormalized Čech cochains* to be

$$\check{C}_{\text{unnor}}^q(\mathcal{U}, F) := \prod_{i_0 \leq i_1 \leq \dots \leq i_q} F(U_{i_0} \cap \dots \cap U_{i_q}).$$

In each cases, the differentials are defined the same way, however in the unnormalized cochains, differentials can be built out of identities (when we’re omitting an index that was duplicated) while in the normalized cochains this won’t occur.

The core result we want to prove is the following:

Theorem A.3.2. There is a natural map of chain complexes

$$\check{C}_{\text{nor}}^\bullet(\mathcal{U}, F) \rightarrow \check{C}_{\text{unnor}}^\bullet(\mathcal{U}, F)$$

and it is a quasi-isomorphism (meaning it induces an isomorphism on cohomology).

Proof sketch. It is clear that normalized cochains sit inside unnormalized cochains as a subcomplex, since at each level q we have a subgroup inclusion $\check{C}_{\text{nor}}^q(\mathcal{U}, F) \subseteq \check{C}_{\text{unnor}}^q(\mathcal{U}, F)$. In fact we will show that this inclusion splits. We say that a cochain $c \in \check{C}_{\text{unnor}}^q(\mathcal{U}, F)$ is *normalized* if it lies in the image of this natural map, or equivalently, it vanishes on any tuple with repeated indices (i.e. one which isn't strict).

For $0 \leq k \leq q-1$, we define an endomorphism of unnormalized cochains, essentially detecting how sensitive a cochain is to doubling up an index in the k th slot. Explicitly:

$$P_k: \check{C}_{\text{unnor}}^q(\mathcal{U}, F) \rightarrow \check{C}_{\text{unnor}}^q(\mathcal{U}, F)$$

$$c(i_0 \leq \dots \leq i_q) \mapsto c(i_0 \leq \dots \leq i_q) - c(i_0 \leq \dots \leq i_k \leq i_k \leq i_{k+2} \leq \dots \leq i_q).$$

That is, we output the difference of c and c on the tuple where we have a repeated overlap at the k th spot. Note that if c was normalized, then we have that P_k acts as the identity for each k .

We now define

$$\mathcal{P}_q := P_{q-1} \circ P_{q-2} \circ \dots \circ P_0: \check{C}_{\text{unnor}}^q(\mathcal{U}, F) \rightarrow \check{C}_{\text{unnor}}^q(\mathcal{U}, F).$$

We have a few claims about these maps $\mathcal{P}_q$:

1. $\mathcal{P}_q$ takes values in *normalized* cochains
2. $\mathcal{P}_q \circ \mathcal{P}_q = \mathcal{P}_q$, that is, it is idempotent.
3. $\{\mathcal{P}_q\}_{q \geq 0}$ is a chain map (it commutes with the differentials)

To sketch the first point – we can see that a cochain c is normalized if and only $P_k(c) = c$ for all k . Hence it suffices to verify that $P_k \circ \mathcal{P}_q = \mathcal{P}_q$ for any $0 \leq k \leq q-1$, which is immediate by definition. Once we show this, idempotent-ness is immediate. Checking it is a chain map is just some painful index-bashing, but trust me that it's true.

Hence with all our claims in hand, we get a splitting of chain complexes

$$\begin{array}{ccc} \check{C}_{\text{nor}}^\bullet(\mathcal{U}, F) & \hookrightarrow & \check{C}_{\text{unnor}}^\bullet(\mathcal{U}, F) \\ & \searrow & \downarrow \mathcal{P}_\bullet \\ & & \check{C}_{\text{nor}}^\bullet(\mathcal{U}, F). \end{array}$$

In particular the inclusion splits (it splits levelwise in fact). The remaining stuff is called the *degenerate* cochains complex. That is, our inclusion and projection exhibit a splitting of abelian groups

$$\check{C}_{\text{unnor}}^q = \check{C}_{\text{nor}}^q \oplus \check{D}^q.$$

Here $\check{D}^q$ is our notation for what is left over – it consists of those cochains which are supported only on $(q+1)$ -tuples which are *not* strictly increasing (they have a repeated entry).

The last remaining thing to check is that $\check{D}^\bullet$ has no cohomology. To do this we exhibit an explicit chain homotopy. For $0 \leq k \leq q-1$, we define

$$s^k: \check{D}^q \rightarrow \check{D}^{q-1}$$

by the formula $(s^k c)(i_0, \dots, i_{q-1}) = c(i_0, \dots, i_k, i_k, i_{k+1}, \dots, i_q)$. We then take h to be the map

$$h = \sum_{k=0}^{q-1} (-1)^k s^k: \check{D}^q \rightarrow \check{D}^{q-1}.$$

We can check that $dh + hd = \text{id}$ on $\check{D}^\bullet$, hence h exhibits a null-homotopy of the identity map on the degenerate cochains. $\square$

And obviously there's one last sanity check we should do, which is that the definitions above don't depend on the ordering we fixed on I . We skip this argument, and state the big corollary:

Corollary A.3.3. We can compute Čech cohomology with either normalized or unnormalized cochains and the answers we get in either case will be canonically isomorphic.

Remark A.3.4. This entire story and proof is *easier and generalizable* if we are willing to do a bit more category theory and work with cosimplicial objects. Indeed this is how it's normally proved. Let Δ denote the category with countably many objects $[0], [1], [2], \dots$, where the object $[n] \in \Delta$ is intended to denote the ordered set with $n+1$ elements $[n] = \{0, 1, \dots, n\}$. The morphisms $[m] \rightarrow [n]$ in Δ are weakly order-preserving maps between these sets. We say that a *cosimplicial object* in a category $\mathcal{C}$ is a functor

$$\Delta \rightarrow \mathcal{C}.$$

There are special morphisms in Δ which generate all the other morphisms. These are called *degeneracy* and *face* maps:

- ▷ *face maps*: for any $0 \leq j \leq n$ we have maps $\delta^j: [n-1] \rightarrow [n]$ uniquely defined as the injective order-preserving map which doesn't contain $j \in [n]$ in its image
- ▷ *degeneracy maps*: for any $0 \leq j \leq n$ we have maps $\sigma^j: [n+1] \rightarrow [n]$ uniquely defined as the surjective order-preserving map so that $\sigma^j(j) = \sigma^j(j+1) = j \in [n]$.

These generate all other morphisms in the sense that every morphism can be written as a composite of them [Aut, 0166]. They satisfy some natural identities, called the *simplicial identities*, see for instance [Aut, 0167].

Our Čech cochains are very naturally a cosimplicial object! Explicitly, they form a functor

$$\begin{aligned} \check{C}_{\text{unnor}}^{(-)}(\mathcal{U}, F): \Delta &\rightarrow \text{Ab} \\ [q] &\mapsto \prod_{i_{(-)} \in \text{Hom}([q], (I, \leq))} F(U_{i_0} \cap \dots \cap U_{i_q}). \end{aligned}$$

Here the product is over weakly order-preserving functions $[q] \rightarrow I$. We could define the normalized cochains by only taking a product over strictly order-preserving maps, if we like.

The Čech cochains have a cosimplicial structure since we have all this extra available stuff we can do with them – we can repeat or skip indices. Moreover the differentials on the cochain complex arise very naturally from the cosimplicial structure. This is a standard thing: any time we have a cosimplicial abelian group we can change it into a cochain complex by taking the alternating sums of the degeneracy maps. If we unwind our definition of the Čech cochains differential in Equation (1.2.8) we can see that it is exactly built that way.

Once we have all this in mind, the proof above can be rewritten in cosimplicial language:

- ▷ given any cosimplicial abelian group $A^\bullet$ (not just the unnormalized cochains, but *any* one) there is a natural “normalization” of it, defined by taking things that lie in the kernels of all the degeneracy maps:

$$NA^q := \bigcap_{k=0}^{q-1} \ker(A^q \xrightarrow{s^k} A^{q-1}).$$

- ▷ in this language, there is again a natural inclusion $NA^q \subseteq A^q$, and it splits levelwise by a similar projection map:

$$P_k := (\text{id} - d^k s^{k-1})$$

$$\mathcal{P}_q := (\text{id} - d^q s^{q-1}) \circ \cdots \circ (\text{id} - d^1 s^0).$$

- ▷ from this perspective, our three claims (that $\mathcal{P}_q$ is normalized, i.e. that it satisfies $P_k \circ \mathcal{P}_q = \mathcal{P}_q$ for any k , that $\mathcal{P}_q$ is idempotent, and that $\mathcal{P}_q$ is a chain map) follow immediately from the identities of cosimplicial maps
- ▷ the splitting off of a degenerate subcomplex and exhibiting that it has no cohomology is also a standard result in cosimplicial algebra.

The story for simplicial abelian groups can be found in [Mac67, VIII.6] or [GJ99, III.2], and the cosimplicial story is dual.

Appendix B

Student appendices

Contents

B.1	Thom's computation of MSO_* (Joaquim Bocrésion)	118
B.2	Connecting Elliptic Integrals, Curves, and Functions through Elliptic Genera (Grayson Kemplin)	126
B.3	The Hodge-de Rham spectral sequence (Brian Siew)	139
B.4	The Calabi-Eckmann Manifolds and their $\bar{\partial}$-Cohomology (Carl Scandellius)	150
B.5	Leray Spectral Sequence (David Ettel)	159

B.1 Thom's computation of MSO_* (Joaquim Bocrés)

B.1.1 Overview

Thom's seminal 1954 paper included several novel notions in topology at its time of publication. Among them included the first definition of transversality in the literature, his question of realization of homology classes by submanifolds, and the following theorem and subject of this exposition:

Theorem B.1.1 ([Hir78, Theorem 7.2.2] ; cf. [Tho54, Theorem 4.15]). The groups Ω^n are finite for $i \not\equiv 0 \pmod{4}$. In the case that $i \equiv 0 \pmod{4}$, we have $\Omega^{4k} = \mathbb{Z}^{\pi(k)} \oplus G$, for π the partition function and G a finite abelian group.

We will work towards a modern-dress contextualization and proof of this theorem à la Thom. In doing so, we will develop some of the required homotopy theory and (co)bordism theory.

B.1.2 Thom Spaces and Spectra

We will construct Thom spaces following Chapter 2 of [Tho54]. We will then generalize to the more modern notion of the Thom spectrum.

Fix $G \subset O(k)$ some closed subgroup. The universal bundle $\pi : EG \rightarrow BG$ classifies all (iso classes of) fiber bundles with structure group G via pullback along (homotopy classes of) maps $X \rightarrow BG$. We can specialize to the case of S^{k-1} -bundles via taking the associated bundle $\pi' : E'G = EG \times_G S^{k-1} \rightarrow BG$. With this data we can construct the *mapping cylinder* of π' ,

$$AG = \frac{E'G \times I \amalg BG}{(x, 1) \sim \pi'(x)}.$$

This cylinder has boundary $E'G = E'G \times \{0\}$. If we identify all points on this boundary, we obtain the *Thom space* $M(G) = AG/E'G = AG/E'G \times \{0\}$. Alternatively, by removing the boundary of AG , we can construct the *open mapping cylinder* $A^\circ G = AG - (E'G \times \{0\})$. If we take the one-point compactification $(A^\circ G)^+$ of this open mapping cylinder, this also gives $M(G)$. This space has a canonical basepoint, the point at ∞ .

The following lemma is the last bit of setup we need, and will prove very useful. We do not give a proof.

Lemma B.1.2 (The Thom Isomorphism; cf. [Tho54, p.29]). for $r > 0$ and EG is orientable, there is an isomorphism

$$H^{r-k}(BG) \cong H^r(M(G)).$$

Henceforth we will specialize to the case $M(SO(k))$ i.e. $G = SO(k)$, which is clearly a closed subgroup.

Notation B.1.3. After Thom, we denote by $\widehat{G}_k$ the oriented infinite Grassmanian, i.e. the space of k -planes in $\mathbb{R}^\infty$ together with an orientation. This is clearly a double cover of our usual Grassmanian $\text{Gr}_{\mathbb{R}}(k, \infty)$, with the projection map corresponding to forgetting the orientation of a k -plane.

This notation is of course useful since we know $BSO(k) = \widehat{G}_k$.

In the case of $M(SO(k))$, we have $E'SO(k) = \{(p, v) | p \in \widehat{G}_k, v \in p, \|v\| = 1\}$, where the norm is the standard Euclidean norm (we are already implicitly working in coordinates with $\widehat{G}_k$ being the space of oriented k -planes in $\mathbb{R}^\infty = \mathbb{R} \times \mathbb{R} \times \dots$). Now, taking $ASO(k)$, note that the fibers of $E'SO(k)$ over a point (i.e. an oriented k -plane) in $BSO(k)$ are spheres, and when we take the product with I and identify one end with the base point in BG , we get a bundle $AG \rightarrow BG$ naturally

having closed k -disks as its fibers. From here, when we remove the boundary $E'SO(k) \times \{0\}$, we turn these fibers into open k -disks. Though adding the point at ∞ ruins this elegant bundle structure, we do obtain a natural basepoint.

Definition B.1.4 (Eilenberg-MacLane Space). Given any abelian group G and $n \geq 1$ (or G any group and $n = 1$), there exists a space (in fact, a CW complex) $K(G, n)$, called an *Eilenberg-MacLane space*, with the property that $\pi_i(K(G, n)) = G$ for $i = n$ and $\pi_i(K(G, n)) = 0$ for all other $i \neq 0, n$.

It is not obvious that Eilenberg-MacLane spaces exist, though it turns out that the spaces $K(G, n)$ exist for all abelian G with $n \geq 1$ or any G with $n = 1$. Since Eilenberg-MacLane spaces are CW complexes, they are naturally based spaces.

B.1.3 Thom Spectra and Stable Homotopy

We develop the necessary homotopy theoretic tools for a modern-dress treatment of Thom's ideas. We follow Chapter 22 of [May] and Chapter 2 of [Mal26]

Definition B.1.5 (Spectrum). A spectrum is a sequence of based spaces $E = \{E_n\}$ together with a collection of bonding maps

$$\Sigma E_n \rightarrow E_{n+1}.$$

We will be slightly abusive and denote by E the data $E = \{E_n\}$ along with the bonding maps. More modern authors might instead define this as a prespectrum or sequential spectrum, though we will not need any of these more specific notions and use this definition of a spectrum.

The main benefit of spectra is that they allow us to define the *stable homotopy groups*. By the functoriality of suspension, we get maps

$$\begin{aligned} \pi_i(E_n) = [S^i, E_n] &\rightarrow [\Sigma S^i, \Sigma E_n] = [S^{i+1}, \Sigma E_n] = \pi_{i+1}(\Sigma E_n) \\ [f] &\mapsto [\Sigma f]. \end{aligned}$$

Via composition with the bonding maps we obtain a map

$$\pi_i(E_n) \rightarrow \pi_{i+1}(\Sigma E_n) \rightarrow \pi_{i+1}(E_{n+1}),$$

and this allows us to make the following definition:

Definition B.1.6 (Stable homotopy groups). Given a spectrum E , we define the stable homotopy groups $\pi_i^s(E)$ by

$$\pi_i^s(E) = \operatorname{colim}_n (\pi_{i+n}(E_n)),$$

where the diagram in question is

$$\cdots \rightarrow \pi_{i+n-1}(E_{n-1}) \rightarrow \pi_{i+n}(E_n) \rightarrow \pi_{i+n+1}(E_{n+1}) \rightarrow \cdots$$

Theorem B.1.7 (Freudenthal's Suspension Theorem; cf. [Tho54, p.36]). If X is an n -connected CW complex, then the natural suspension homomorphism $\pi_i(X) \rightarrow \pi_{i+1}(\Sigma X)$ is an isomorphism for $i \leq 2n$ and a surjection for $i = 2n + 1$

Theorem B.1.8 ("Thom's Suspension theorem"). For $n \leq k-3$, the homotopy groups $\pi_{n+k}(M(SO(k)))$ do not depend on k .

Proof. Consider the inclusion $SO(k-1) \hookrightarrow SO(k)$ given by $M \mapsto M \oplus (1) = \begin{pmatrix} M & \\ & 1 \end{pmatrix}$. The image of this inclusion is the stabilizer of $e_k \in \mathbb{R}^k$, i.e. where e_k is the k -th standard basis vector. On classifying spaces this induces a map $Bi : \widehat{G}_{k-1} \rightarrow \widehat{G}_k$, given by $W \mapsto W \oplus \mathbb{R}e_k$ (i.e. we take a $(k-1)$ -plane and orthogonally direct sum the e_k -direction). Denote by I an open interval. Following Thom, we denote by $A^\circ SO(k-1) \otimes I$ the fiberwise join of the bundles $A^\circ SO(k-1) \rightarrow \widehat{G}_{k-1}$ and $I \rightarrow *$, where the latter is just the interval sitting over a point. For our purposes, this just means that we take the product of each fiber in $A^\circ SO(k-1) \rightarrow \widehat{G}_{k-1}$ with I . Since the fibers of $A^\circ SO(k-1) \rightarrow \widehat{G}_{k-1}$ are open $(k-1)$ -disks, this produces an open k -disk bundle over $\widehat{G}_{k-1}$. With this, we can produce the following obvious map lifting Bi :

$$f : A^\circ SO(k-1) \otimes I \rightarrow A^\circ SO(k),$$

and this map intertwines with Bi via the bundle projections. Now, let $X = (A^\circ SO(k-1) \otimes I)^+$ be the one-point compactification, with compactifying point x . We can extend f by $x \mapsto \infty \in M(SO(k))$ to produce a continuous map

$$g : X \rightarrow M(SO(k)).$$

Our goal now is to understand how g acts on cohomology. Note that $A^\circ SO(k-1) \otimes I$ is the associated open disk bundle of the rank- k oriented bundle $\widehat{E}_{k-1}(\mathbb{R}^\infty) \oplus \epsilon^1$ over $\widehat{G}_{k-1}$, where $\widehat{E}_k(\mathbb{R}^\infty)$ is denoting the tautological bundle over the oriented Grassmannian $\widehat{G}_k$; ϵ^1 is a copy of the trivial rank 1 vector bundle; and the fiber over $W \in \widehat{G}_{k-1}$ is accordingly $W \times I \cong W \oplus \mathbb{R}e_k$, an open k -disk. From this it follows that its one-point compactification X is obtained from this induced bundle by the same mapping-cylinder/one-point-compactification construction used to define $M(SO(k))$, and $g : X \rightarrow M(SO(k))$ is the map induced by Bi . Since $(Bi)^* \widehat{E}_k(\mathbb{R}^\infty) \cong \widehat{E}_{k-1}(\mathbb{R}^\infty) \oplus \epsilon^1$ canonically, the bundle map $\widehat{E}_{k-1}(\mathbb{R}^\infty) \oplus \epsilon^1 \rightarrow \widehat{E}_k(\mathbb{R}^\infty)$ covering Bi induces on Thom spaces exactly our g . Naturality of the Thom isomorphism for bundle maps then yields the commutative square

$$\begin{array}{ccc} \widetilde{H}^{k+i}(M(SO(k))) & \xrightarrow{g^*} & \widetilde{H}^{k+i}(X) \\ \cong \downarrow & & \downarrow \cong \\ H^i(\widehat{G}_k) & \xrightarrow{(Bi)^*} & H^i(\widehat{G}_{k-1}) \end{array}$$

with vertical Thom isomorphisms, so it suffices to analyze $(Bi)^*$. The Gysin sequence of the oriented $(k-1)$ -sphere bundle $S^{k-1} \cong SO(k)/SO(k-1) \rightarrow \widehat{G}_{k-1} \xrightarrow{Bi} \widehat{G}_k$ looks like

$$\cdots \rightarrow H^{i-k}(\widehat{G}_k) \xrightarrow{\cup e} H^i(\widehat{G}_k) \xrightarrow{(Bi)^*} H^i(\widehat{G}_{k-1}) \rightarrow H^{i-k+1}(\widehat{G}_k) \rightarrow \cdots,$$

where $e \in H^k(\widehat{G}_k)$ is the Euler class of $\widehat{E}_k(\mathbb{R}^\infty)$. For $i \leq k-2$ both flanking terms $H^{i-k}(\widehat{G}_k)$ and $H^{i-k+1}(\widehat{G}_k)$ are zero, so $(Bi)^*$ is an isomorphism; for $i = k-1$ the left flank is zero and the right is $H^0(\widehat{G}_k) = \mathbb{Z}$, so $(Bi)^*$ is an injection. Using the commutative square, we can see that g^* is an isomorphism on $\widetilde{H}^{k+i}$ for $i \leq k-2$ and an injection for $i = k-1$.

Next, note that X and $M(SO(k))$ are both $(k-1)$ -connected. For simple connectivity when $k \geq 2$, we can apply van Kampen to $M(SO(k)) = A^\circ SO(k) \cup V$, where V is a contractible open neighbourhood of ∞ meeting $A^\circ SO(k)$ along an open sphere-bundle tube. Since $A^\circ SO(k)$ deformation retracts onto the simply connected $\widehat{G}_k = BSO(k)$ and V is contractible, we get $\pi_1(M(SO(k))) = 0$, and likewise for X . Combined with $\widetilde{H}^{<k} = 0$ from the Thom isomorphism, Hurewicz forces $\pi_{<k} = 0$. Moreover $H^*(\widehat{G}_k)$ is finitely generated in each degree (see the argument in the proof of Theorem 0.1), and via the Thom isomorphism so are $\widetilde{H}^*(M(SO(k)))$ and $\widetilde{H}^*(X)$.

The universal coefficient theorem then dualizes the cohomological conclusion to a homological one: $g_* : H_{k+i}(X) \rightarrow H_{k+i}(M(SO(k)))$ is an isomorphism for $i \leq k-2$ and a surjection for $i = k-1$. Since g is a map of simply connected spaces, a theorem of Whitehead (see [Tho54, Theorem II.6]) shows that for $i \leq k-2$ we have

$$g_* : \pi_{k+i}(X) \xrightarrow{\sim} \pi_{k+i}(M(SO(k))).$$

Lastly, we identify X with a suspension of $M(SO(k-1))$. Following Thom's notation, let $T(k-1)$ be the suspension of $M(SO(k-1))$ with poles p, p' (i.e. the two ends $\{0\} \times M(SO(k-1))$ and $\{1\} \times M(SO(k-1))$ which are identified to points in the suspension). If we denote by $a = \infty \in M(SO(k-1))$, we can consider the arc $[pap'] \subset T(k-1)$ given explicitly by $\{a\} \times [0, 1] \subset T(k-1)$ (heuristically, this is the meridian arc through a between the two pinched endpoints in the suspension). We can then let $h : T(k-1) \rightarrow X$ be the map collapsing the arc $[pap'] \subset T(k-1)$ to the compactifying point x . Concretely, $T(k-1) = M(SO(k-1)) \times [0, 1] / (M(SO(k-1)) \times \{0, 1\})$, and under the identification $M(SO(k-1)) = (A^\circ SO(k-1))^+$, the subspace $A^\circ SO(k-1) \times (0, 1) \subset T(k-1)$ maps homeomorphically onto $A^\circ SO(k-1) \otimes I \subset X$ (taking the interval I from before to be $I = (0, 1)$), while the complement (the collapsed sides $\{0\} \times M(SO(k-1))$ and $\{1\} \times M(SO(k-1))$) together with the meridian $\{a\} \times [0, 1]$ map to x . Thus $X \cong T(k-1)/[pap']$, and since $[pap']$ is a contractible subcomplex, h is a homotopy equivalence. In particular $X \simeq \Sigma M(SO(k-1))$.

Finally, $M(SO(k-1))$ is $(k-2)$ -connected, so by the Freudenthal suspension theorem (B.1.7) the suspension map

$$E : \pi_{k-1+i}(M(SO(k-1))) \rightarrow \pi_{k+i}(T(k-1))$$

is an isomorphism for $k-1+i \leq 2(k-2)$, i.e. for $i \leq k-3$. Assembling these,

$$\pi_{k-1+i}(M(SO(k-1))) \xrightarrow{E} \pi_{k+i}(T(k-1)) \xrightarrow{h_*} \pi_{k+i}(X) \xrightarrow{g_*} \pi_{k+i}(M(SO(k)))$$

is a chain of isomorphisms whenever $i \leq k-3$. It follows that for fixed n , $\pi_{n+k}(M(SO(k)))$ stabilizes as $k \rightarrow \infty$. $\square$

Corollary B.1.9. The sequences $M(SO(k))$ along with bonding maps from the proof above furnish the data of a spectrum, which we call the *Thom Spectrum*, and denote by MSO .

Theorem B.1.10 (Cobordism is Isomorphic to Homotopy). There exists an isomorphism of groups

$$\pi_n^s(MSO) \cong \Omega_n^{SO}.$$

Unfortunately, this proof relies a good deal on Thom's theory of realization of homology classes by submanifolds. Though it is the crucial step to translating homotopy theoretic statements about the Thom spectrum into statements about the cobordism groups and ring, I think it is best to blackbox it, as not to veer too far off course.

B.1.4 Constructing maps

The goal of this section is to construct maps out of $M(SO(k))$ into spaces chosen such that such maps induce isomorphisms on homotopy groups. We can do this with varying degrees of abstraction. Probably the most powerful, though least insightful and intuitive way, is to use Brown Representability:

Theorem B.1.11 (Brown Representability; cf. [Hat01, Theorem 4E.1]). Let G be an abelian group with $n \geq 1$. For any CW complex X we have

$$[X, K(G, n)]_* \cong \tilde{H}^n(X; G),$$

where the natural map is $[f] \mapsto f^*\iota$ for $\iota \in \tilde{H}^n(K(G, n); G) = \text{End}(G)$ the distinguished class corresponding to the identity map $G \rightarrow G$. Note that $[-, -]_*$ denotes the homotopy classes of maps of based spaces.

Remark B.1.12. We have limited interest in $\pi_n(X)$ and $K(G, n)$ for $n = 0$, so we can assume $n \geq 1$ and that all reduced (co)homology is just regular (co)homology.

Note that Brown Representability guarantees the existence of a map $X \rightarrow K(G, n)$ with the required behavior on cohomology: given some $u \in \tilde{H}^n(X; G)$, there exists a (homotopy class of) map $X \rightarrow K(G, n)$ with $f^*\iota = u$. However, Brown representability is very powerful and we can also construct maps $f : X \rightarrow K(G, n)$ with the desired property more manually.

We want to construct a map $f : X \rightarrow K(G, n)$ for X a CW-complex which represents the class $u \in H^n(X; G)$ via pulling back the fundamental class of $K(G, n)$. To do this, we dip our toes into obstruction theory. The idea behind obstruction theory is not too arcane. In our case, if (A, B) is a pair of a CW complex A and a subcomplex B , and $h : B \rightarrow C$ is a map, one might want to know when h admits extension over the skeleta of A . At each stage there is an obstruction class which potentially prevents extension of the map from $n - 1$ skeleta to n skeleta. More precisely, the obstruction to extending a map from $A^{n-1} \cup B$ to $A^n \cup B$ lives in a relative cohomology group with coefficients in $\pi_{n-1}(C)$. Amazingly, the vanishing of the obstruction class is equivalent to the existence of an extension. In our case this becomes especially simple, because

$$\pi_j(K(G, n)) = \begin{cases} G & j = n, \\ 0 & j \neq n. \end{cases}$$

Thus the only potentially nontrivial obstruction occurs when extending from the n -skeleton to the $(n + 1)$ -skeleton, and beyond that there are no further obstructions.

We now apply this to the case of constructing maps $X \rightarrow K(G, n)$. First, pick a cocycle z representing u . Then we can define $f_{n-1} : X^{n-1} \rightarrow K(G, n)$ by sending the $(n-1)$ -skeleton X^{n-1} of X to the basepoint $* \in K(G, n)$. Now, for each n -cell σ of X , the cocycle value $z(\sigma) \in G = \pi_n(K(G, n))$ determines, up to homotopy, a map $\sigma \simeq D^n \rightarrow K(G, n)$. This map must send $\partial\sigma$ to the basepoint $* \in K(G, n)$ and represents the class $z(\sigma) \in \pi_n(K(G, n)) \cong G$. This defines $f_n : X^n \rightarrow K(G, n)$.

To further extend f_n , for each $(n + 1)$ -cell τ , the previous step defines $f|_{\partial\tau}$. The obstruction to extending $f|_{\partial\tau}$ to a map defined on all of τ is $(\delta z)(\tau) \in \pi_n(K(G, n)) = G$. However, z is a cocycle, so that $(\delta z)(\tau) = 0$. This shows that the obstruction vanishes, and that f extends over all $(n + 1)$ -cells. Since $\pi_{n+j}(K(G, n)) = 0$ for all $j \geq 1$, there are no further obstructions in further extending f . From this we obtain the following fact:

Proposition B.1.13. For a CW-complex X and $u \in H^n(X; G)$ with $n \geq 1$, there exists a map $f : X \rightarrow K(G, n)$, unique up to based homotopy, such that $f^*\iota = u$, where $\iota \in H^n(K(G, n); G)$ is the fundamental class.

We now specialize to the case $G = \mathbb{Z}$. Consider the space

$$Y = K(\mathbb{Z}, k) \times K(\mathbb{Z}, k + 4)^{c(1)} \times K(\mathbb{Z}, k + 8)^{c(2)} \times \cdots \times K(\mathbb{Z}, k + 4m)^{c(m)} \times \cdots$$

where $c(m) = \dim_{\mathbb{Q}} H^{4m}(\widehat{G}_k; \mathbb{Q})$. Since maps to a product are the same as tuples of maps to the factors, we have

$$\begin{aligned} [X, Y] &\cong [X, K(\mathbb{Z}, k)] \times [X, K(\mathbb{Z}, k+4)] \times \cdots \times [X, K(\mathbb{Z}, k+4m)]^{c(m)} \times \cdots \\ &\simeq \prod_{m \geq 0} \widetilde{H}^{k+4m}(X)^{c(m)}. \end{aligned}$$

Proposition B.1.14. For $X = M(SO(k))$, this construction gives a map of based topological spaces inducing an group isomorphism on rational cohomology in all degrees.

Proof. Via the Thom isomorphism we have $\widetilde{H}^{k+4m}(M(SO(k))) \cong H^{4m}(\widehat{G}_k)$. Now, note that we have

$$\begin{aligned} H^*(\widehat{G}_{2\ell}; \mathbb{Q}) &= \mathbb{Q}[p_1, \dots, p_{\ell-1}, e], \text{ and} \\ H^*(\widehat{G}_{2\ell+1}; \mathbb{Q}) &= \mathbb{Q}[p_1, \dots, p_{\ell}], \end{aligned}$$

where $p_i \in H^{4i}(\widehat{G}_{2\ell}; \mathbb{Q}) = H^{4i}(\widehat{G}_{2\ell}) \otimes_{\mathbb{Z}} \mathbb{Q}$ is the i th Pontryagin class, and $e \in H^{2\ell}(\widehat{G}_k; \mathbb{Q})$ is the Euler class. Note that in particular we only have nontrivial cohomology in even degrees, since the Pontryagin and Euler classes live in even degrees.

Note that the Euler class has degree k , and some nontrivial monomial $ep_{i_1} \cdots p_{i_r}$ containing e has degree $k + 4(i_1 + \cdots + i_r)$. Thus, for this monomial to land in $H^{4m}(\widehat{G}_k; \mathbb{Q})$, we need

$$4 \mid (k + 4(i_1 + \cdots + i_r)) \implies 4 \mid k,$$

and $m = k/4 + i_1 + \cdots + i_r$. This means that the $H^{4m}(\widehat{G}_k; \mathbb{Q})$ can only contain multiples of the Euler class if $4 \mid k$ and $m \geq k/4$.

We can now abuse this and Thom's suspension theorem. By the Thom isomorphism, we have $\widetilde{H}^{i+k}(M(SO(k)); \mathbb{Q}) \cong H^i(\widehat{G}_k; \mathbb{Q})$. Note that since $\pi_{n+k}(M(SO(k)))$ does not depend on k for $k > n + 3$, we can pick $k > n \geq 4m$. Under these conditions, e cannot appear in $H^{4m}(\widehat{G}_k; \mathbb{Q}) \cong H^{4m+k}(M(SO(k)); \mathbb{Q})$. In fact, the only monomials appearing in $H^{4m}(\widehat{G}_k; \mathbb{Q})$ are monomials $p_{i_1} \cdots p_{i_r}$ with $m = i_1 + \cdots + i_r$, since each p_i is a weight- $4i$ monomial. The number of such expressions $m = i_1 + \cdots + i_r$ is just $\pi(m)$, the number of partitions of m . Thus, we see that

$$c(m) = \dim_{\mathbb{Q}} H^{4m}(\widehat{G}_k; \mathbb{Q}) = \pi(m).$$

Now, we can define a map $M(SO(k)) \rightarrow K(\mathbb{Z}, k+4m)$ for all m whose pullback maps the fundamental class of $K(\mathbb{Z}, k+4m)$ to any desired class in $H^{k+4m}(M(SO(k)))$. Note that Y has $c(m) = \pi(m)$ -many factors of $K(\mathbb{Z}, k+4m)$, and that $H^{k+4m}(M(SO(k)))$ is generated over $\mathbb{Q}$ by $c(m) = \pi(m)$ -many generators. This shows that for $u_1, \dots, u_{c(m)}$ integral cohomology classes corresponding to weight- $4m$ monomials in $p_1, \dots, p_{\ell}$ upon tensoring, we have maps $f_1, \dots, f_{c(m)} : M(SO(k)) \rightarrow K(\mathbb{Z}, k+4m)$ with the property that $f_i^* \iota = u_i$. From here we obtain a map $\prod_{i=1}^{c(m)} f_i : M(SO(k)) \rightarrow K(\mathbb{Z}, k+4m)^{c(m)}$, and further taking the product over all m gives $F : M(SO(k)) \rightarrow Y$. Note that in each degree, F^* pulls back a basis of rational cohomology of Y to a basis of rational cohomology of $M(SO(k))$. $\square$

With this identification done, one would hope to be able to relate the cohomology of $M(SO(k))$ to the homotopy groups of $M(SO(k))$, from which we could extract the stable homotopy groups $\pi_*^s \text{MSO}$. Under nice circumstances, the Hurewicz theorem would prove a useful tool for this, but the standard Hurewicz theorem will not suffice for this problem. The solution ends up being a set of homological tools coming from Serre's theory of classes.

B.1.5 Detour: Serre's Class Theory

We follow Chapter 4 of [Mil21].

Definition B.1.15 (Serre Class). A Serre class $\mathcal{C}$ is a subcategory of $\mathbf{Ab}$ such that

1. $\mathcal{C}$ contains the trivial group, i.e. $0 \in \mathcal{C}$,
2. For every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, we have A and $C \in \mathcal{C}$ if and only if $B \in \mathcal{C}$.

We can also introduce the notion of a *Serre Ring*

Definition B.1.16 (Serre Ring). A Serre class $\mathcal{C}$ is a *Serre ring* if for all $A, B \in \mathcal{C}$ we also have $A \otimes_{\mathbb{Z}} B$ and $\mathrm{Tor}_1^{\mathbb{Z}}(A, B) \in \mathcal{C}$

We might also want to do algebraic topology to Serre rings. Given any A in a Serre ring $\mathcal{C}$, we can consider $K(A, 1)$. Applying homology, we have, via the standard Hurewicz theorem, $H_1(K(A, 1)) = \pi_1(K(A, 1)) = A$. With this as motivation, we make the following definition

Definition B.1.17 (Acyclic Serre Ring). We say that a Serre ring $\mathcal{C}$ is *acyclic* if, given any $A \in \mathcal{C}$, we have $H_i(K(A, 1)) \in \mathcal{C}$ for $i \geq 1$.

This may seem to come out of nowhere, but it is natural if we instead understand $H_1(K(A, 1)) \in \mathcal{C}$ to mean something like $H_1(K(A, 1)) \equiv 0 \pmod{\mathcal{C}}$. With this as motivation, we can take any abelian group and look at it “modulo $\mathcal{C}$.” We will have that $A \equiv B \pmod{\mathcal{C}}$ iff A and B differ by elements of $\mathcal{C}$, i.e. if $A \oplus C_1 \cong B \oplus C_2$ for $C_1, C_2 \in \mathcal{C}$. Equivalently, $A \equiv B \pmod{\mathcal{C}}$ iff there is a map $A \rightarrow B$ (or $B \rightarrow A$) such that the kernel and cokernel both lie in $\mathcal{C}$. In this way, acyclicity is a generalization of acyclicity from the case of singular homology to the mod $\mathcal{C}$ -context—it is defined in terms of the vanishing of higher homology. Here is a nice example of an acyclic Serre ring:

Example B.1.18 (Finite abelian groups form an acyclic Serre ring). Consider the class $\mathcal{C}$ of finite abelian groups. This is a subcategory of $\mathbf{Ab}$, and it contains the trivial group. Notably, cardinality of finite groups is multiplicative over short exact sequences, so given a short exact sequence of finite Abelian groups, we have

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0 \implies |B| = |A| \cdot |C|,$$

so A and C are in $\mathcal{C}$ iff B is in $\mathcal{C}$. This shows that $\mathcal{C}$ is a Serre class. To further see that it is a Serre ring, note that the tensor product of finite Abelian groups is finite (one way to see this is that it is that $A \otimes_{\mathbb{Z}} B$ for A and B finite $\mathbb{Z}$ -modules is generated by tensor products of generators, of which we have $|A| \cdot |B|$ -many). That $\mathrm{Tor}_1^{\mathbb{Z}}(A, B)$ is finite for A and B finite $\mathbb{Z}$ -modules follows from the classification of finite groups, since Tor_1 distributes over direct sums and $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/a, \mathbb{Z}/b) \cong \mathbb{Z}/\gcd(a, b)$ is finite. It is much more involved to show that the higher homology groups of $K(A, 1)$ are finite for A a finite Abelian group, though it turns out that this is true. One of the easier ways to see this is that the homology $H_i(K(A, 1); \mathbb{Z})$ is isomorphic to the group homology $H_i(A, \mathbb{Z})$, and for A a finite Abelian group the higher group homology is finite.

Serre classes are useful because, for sufficiently nice Serre classes, we get mod- $\mathcal{C}$ versions of our favorite theorems. These include the following:

Theorem B.1.19 (The Mod- $\mathcal{C}$ Hurewicz Theorem; cf. [Mil21, p.108]). Assume that $\mathcal{C}$ is an acyclic Serre ring. Let X be a simply connected space and let $n \geq 2$. Then $\pi_q(X) \in \mathcal{C}$ for all $q < n$ if and only if $\tilde{H}_q(X) \in \mathcal{C}$ for all $q < n$, and in that case the Hurewicz map $\pi_n(X) \rightarrow H_n(X)$ is a mod $\mathcal{C}$ isomorphism.

Theorem B.1.20 (the Mod- $\mathcal{C}$ Whitehead Theorem; cf. [Mil21, p.109]). Let $\mathcal{C}$ be an acyclic Serre ring¹, and $f : X \rightarrow Y$ a map of simply connected spaces. Fix $n \geq 2$. The following are equivalent.

- (i) $f_* : \pi_i(X) \rightarrow \pi_i(Y)$ is a mod $\mathcal{C}$ isomorphism for $i \leq n-1$ and a mod $\mathcal{C}$ epimorphism for $i = n$, and
- (ii) $f_* : H_i(X) \rightarrow H_i(Y)$ is a mod $\mathcal{C}$ isomorphism for $i \leq n-1$ and a mod $\mathcal{C}$ epimorphism for $i = n$.

By epimorphism (resp. isomorphism) mod $\mathcal{C}$, we mean a morphism whose cokernel (resp. kernel and cokernel) lie in $\mathcal{C}$.

B.1.6 Proving Theorem 0.1

With a short primer on Serre's theory of classes out of the way, we can prove Thom's main theorem. Henceforth, let $\mathcal{C}$ denote the class of all finite groups. We have shown that this is an acyclic Serre ring.

Proof of Theorem 0.1. Firstly, we have $\Omega_n^{SO} \cong \pi_n^s(MSO) = \pi_{n+k}(M(SO(k)))$ for $k > n+3$. By [Proposition B.1.14](#), we know that the map $F : M(SO(k)) \rightarrow Y$ induces an isomorphism on rational cohomology. This means that $F_* : H_i(M(SO(k)); \mathbb{Q}) \rightarrow H_i(Y; \mathbb{Q})$ is an isomorphism for all i .

We want to use this isomorphism on rational homology to get a mod $\mathcal{C}$ isomorphism on integral homology. We use the fact that $F_* : H_i(M(SO(k))) \rightarrow H_i(Y)$ is an isomorphism modulo finite groups iff $F_* : H_i(M(SO(k)); \mathbb{Q}) \rightarrow H_i(Y; \mathbb{Q})$ is an isomorphism and the groups $H_i(M(SO(k)))$ and $H_i(Y)$ are finitely generated. To see that these groups are finitely generated, note that the Mod $\mathcal{C}$ Hurewicz theorem, for $k > 1$, we have Y simply connected (since by construction $\pi_1(Y) = 0$), and $\pi_q(Y)$ are all finitely generated. We also have $H_i(M(SO(k)); \mathbb{Z})$ finitely generated, since we $\tilde{H}^{k+i}(M(SO(k))) \cong H^i(\hat{G}_k)$, and since $\hat{G}_k$ is a CW complex with finitely many cells in all degrees (it is the inverse limit of a complex of finite Grassmanians), this cohomology is finitely generated. It follows that $H_i(M(SO(k)); \mathbb{Z})$ is finitely generated by the universal coefficient theorem.

Lastly, since $M(SO(k))$ and Y are simply connected for $k \geq 2$, we can fix n and apply the mod $\mathcal{C}$ Whitehead theorem. Applying the theorem in degrees $q \leq k+n$ gives that $F_* : \pi_{k+n}(M(SO(k))) \rightarrow \pi_{k+n}(Y)$ is an isomorphism mod $\mathcal{C}$. By inducting on n it follows that F_* is an isomorphism mod $\mathcal{C}$ in all degrees. Lastly, we can read off the answer. We

$$\pi_i(Y) \equiv \begin{cases} \mathbb{Z}^{c(m)} & \text{if } i = k + 4m \text{ for some } m \geq 0 \\ 0 & \text{else} \end{cases} \pmod{\mathcal{C}}.$$

From this, note that $\pi_{k+n}(Y) = 0$ if $4 \nmid n$, meaning that $\pi_{n+k}(M(SO(k))) = \pi_n^s(MSO) \in \mathcal{C}$, i.e. $\pi_n^s(MSO) = \Omega_n^{SO}$ is finite. Otherwise, if $4 \mid n$, we have $\pi_{n+k}(M(SO(k))) = \pi_n^s(MSO) \equiv \mathbb{Z}^{c(m)} = \mathbb{Z}^{\pi(m)} \pmod{\mathcal{C}}$, i.e. $\pi_n^s(MSO) = \Omega_n^{SO}$ has free part $\mathbb{Z}^{\pi(m)}$ for π the partition function; and torsion part some finite group G . $\square$

¹This theorem actually applies even further to Serre ideals, whose definitions consist of the same conditions as a Serre ring, but we only require one of A or B to lie in $\mathcal{C}$

B.2 Connecting Elliptic Integrals, Curves, and Functions through Elliptic Genera (Grayson Kemplin)

In this appendix, we flesh out the theory of logarithms in order to define a special class of genera: elliptic genera. Our exploration is centered around the problem of bridging the mysterious relationship between elliptic integrals, elliptic curves, and elliptic functions, a collection of mathematical phenomena that may be encountered at an undergraduate level in disparate contexts (say, on courses on special functions, algebraic geometry/number theory, and complex analysis respectively) which all ostensibly were discovered when investigating the problem of studying arc lengths of curves. We use the classical problem of arc lengths (in this case of the lemniscate) to motivate elliptic integrals, and then use them to define elliptic genera, which turn out to have other useful characterizing properties. We then review how elliptic curves correspond to elliptic functions, and then define an elliptic genus out of an elliptic function, showing how these separate halves of the “elliptic” world connect.

B.2.1 Elliptic Curves Correspond to Elliptic Functions

Any reader who has experienced the slightest brush with number theory, algebraic geometry, or even cryptography is likely to have encountered the following rich object of inquiry:

Definition B.2.1. An **elliptic curve** is a smooth curve of genus 1 with a distinguished point in the projective plane.

In the case where the underlying field of the plane is not of characteristic 2 or 3, its points $[x : y] \in \mathbb{P}^2$ satisfy $y^2 = x^3 + ax + b$ for some choice of a, b , which is perhaps a more recognizable definition of an elliptic curve. A student in an introductory complex analysis class is also likely to see the following phenomena:

Definition B.2.2. A meromorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ is an **elliptic function** precisely when it is doubly periodic, that is, when there are two $\omega_1, \omega_2 \in \mathbb{C}$ such that $\omega_2/\omega_1 \notin \mathbb{R} \cup \{\infty\}$ and $f(z + a\omega_1 + b\omega_2) = f(z) \forall z \in \mathbb{C}, a, b \in \mathbb{Z}$. $L = \{a\omega_1 + b\omega_2 \mid a, b \in \mathbb{Z}\}$ is its corresponding **period lattice**; one can therefore also define an elliptic function as a meromorphic function out of the quotient $\mathbb{C}/L$, which is homeomorphic to a torus and can be characterized by the **fundamental domain** $\{a\omega_1 + b\omega_2 \mid 0 \leq a, b \leq 1\}$ (a parallelogram).

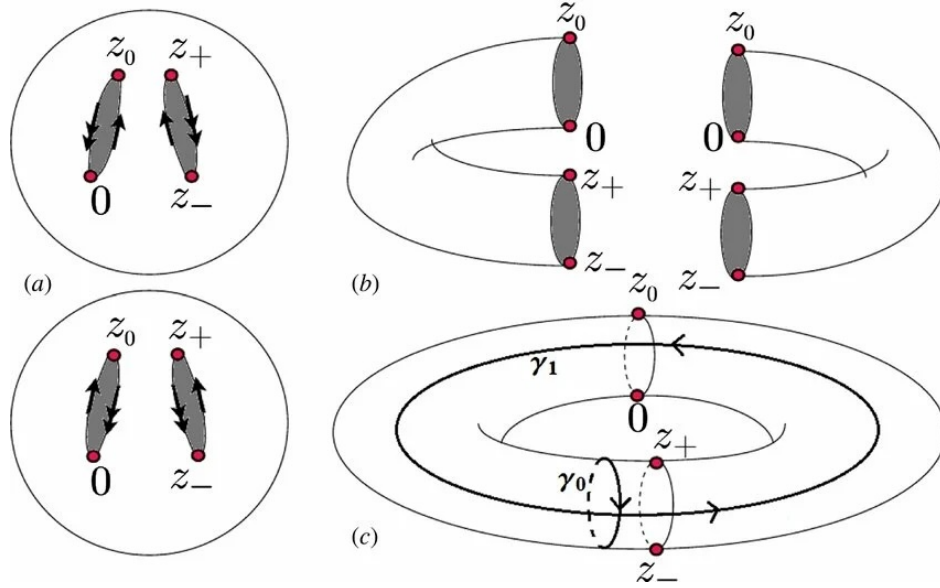
And maybe even has encountered the following definition, confusing when compared to the one above:

Definition B.2.3. In the context of complex analysis, an **elliptic curve** is a quotient $\mathbb{C}/L$ where L is some lattice as defined above, or equivalently, the embedding of a torus into the complex projective plane.

A usual part of the pedagogy for both concepts is to note that they emerged in the study of the arc length of ellipses i.e. in examining exactly the kinds of elliptic integrals we have been examining above. It is usually very unclear from a student’s perspective how these topics are related; our goal is to use the elliptic genus to exhibit this connection.

The first thing to get straight is this strange incongruency between our two separate definitions of an elliptic curve. Once we do so, it will be clear to see how elliptic functions relate to elliptic curves; they are functions on elliptic curves (as defined in the $\mathbb{C}/L$ sense). It is not so hard to see that an “algebraic” elliptic curve gives us a “complex” one when accepting general definition of an

elliptic curve as a genus 1 Riemann surface, at which point it is analytically isomorphic to a torus. A more intuitive and evocative picture comes straight out of the equational description in the case where $x^3 + ax + b = (x - e_1)(x - e_2)(x - e_3)$ has three distinct roots. In this case, we can think about $y = \pm\sqrt{x^3 + ax + b}$ as two-sheeted, where each sheet is the projectivization of the plane i.e. the Riemann sphere. For positive and negative to be well-defined on the complex plane for square roots, however, we need branch cuts, say one from e_1 to e_2 and another from e_3 to ∞ . It is not hard to see that a Riemann sphere with two cuts is homeomorphic to a cylinder $S^1 \times [0, 1]$, where each side of the boundary is one of the branch cuts. Then to form the entire elliptic curve, the positive and negative sheets are “glued together at the 0s” i.e. along the branch cuts. This corresponds to gluing the cylinders together by their ends, forming a torus. The following diagram from [GJK14] may help²:



The more subtle detail is showing this correspondence is one-to-one, that is, recovering the data of an algebraic elliptic curve from some $\mathbb{C}/L$. To do this, we construct a special function:

Definition B.2.4. Given some lattice L of complex numbers, the **Weierstrass** $\wp$ -function is a function associated to L is a function $\wp : \mathbb{C} \rightarrow \mathbb{C}$ where

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in L \setminus \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right).$$

One might wonder why that $-\frac{1}{\omega^2}$ needs to be there at all, considering without it we could write everything cleanly as $\sum_{\omega \in L} \frac{1}{(z - \omega)^2}$. The reason is to guarantee the following two analytic properties, here taken for granted:

Lemma B.2.5. The Weierstrass $\wp$ -function for any lattice L is absolutely convergent, and uniformly convergent on compact subsets of $\mathbb{C}/L$.

With this noted, we may now state:

Proposition B.2.6. The Weierstrass $\wp$ -function for any lattice L is an even elliptic function.

²If the reader has access to it, I also strongly recommend looking at the diagram in [Pen04] p. 137

Proof. Since the series is absolutely convergent and $\frac{1}{((-z)-\omega)^2} - \frac{1}{\omega^2} = \frac{1}{(z+\omega^2)} - \frac{1}{\omega^2}$ for any ω , substituting $-z$ for z merely rearranges the terms of the sum and so maintains the same value $\wp(-z) = \wp(z)$. To show that it is elliptic, we look at the derivative

$$\wp'(z) = \sum_{\omega \in L} \frac{-2}{(z - \omega)^3} :$$

It is evident that $\wp'$ is periodic on L , since any perturbation by $q \in L$ merely rearranges the terms, which does not change the sum by continued absolute convergence. Therefore, $\wp(z)$ and $\wp(z + \omega_1)$ (where L the lattice is generated by ω_1, ω_2) have the same derivative everywhere and therefore may only differ by some constant; but $\wp(\omega_1/2) = \wp(-\omega_1/2) = \wp(\omega_1/2 - \omega_1)$, forcing that constant to be 0 so $\wp(z) = \wp(z + \omega_1)$. A totally symmetric argument shows $\wp(z) = \wp(z + \omega_2)$, showing $\wp$ elliptic as desired. $\square$

The Weierstrass function encodes the data of an algebraic elliptic curve through the following result:

Proposition B.2.7. The Weierstrass $\wp$ -function for any lattice L satisfies the following differential equation:

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3,$$

where $g_2 = 60 \cdot \sum_{\omega \in L \setminus \{0\}} \frac{1}{\omega^4}$ and $g_3 = 140 \cdot \sum_{\omega \in L \setminus \{0\}} \frac{1}{\omega^6}$.

Sketch. The basic idea is to find the power series of $\wp(z)$ in terms of p . Using the power series for $\frac{1}{(z-c)^2} = \frac{1}{c^2} \cdot \frac{1}{(1-\frac{z}{c})} = \frac{1}{c^2} (\sum_{n=1}^{\infty} (z/c)^n)^2$, we get

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + \sum_{\omega \in L \setminus \{0\}} \frac{1}{\omega^2} \left(1 + \frac{z}{\omega} \dots\right)^2 - \frac{1}{\omega^2}, \\ &= \frac{1}{z^2} + \sum_{\omega \in L \setminus \{0\}} \frac{1}{\omega^2} \left(1 + \frac{2z}{\omega} + \dots\right) - \frac{1}{\omega^2} = \frac{1}{z^2} + \sum_{\omega \in L \setminus \{0\}} \left(\frac{2z}{\omega^3} + \dots\right) \end{aligned}$$

and using this derive power series for $\wp(z)^3, \wp'(z)^2$. It can be checked by looking at the terms only of degree ≤ 0 for $\wp, \wp^3, \wp'^2$ that $\wp'^2 - 4\wp^3 + g_2\wp + g_3$ only has terms of degree 2 and up. But this means it is an elliptic function with no poles, therefore bounded [bounded certainly in the fundamental parallelogram by convexity, and then across all of $\mathbb{C}$ by periodicity], and thus by Liouville's theorem a constant, thus (by the expansion having no term of degree 0) the difference equals 0 as desired. For full details see [Lan87]. $\square$

Note that this differential equation is already in the form of an algebraic elliptic curve; what this tells us is that $\{(\wp(z) : \wp'(z) : 1) \mid z \in \mathbb{C}\} \subseteq \mathbb{CP}^2$ parameterizes an elliptic curve, giving a clear way to unite the different definitions of an elliptic curve³. It can be shown that every elliptic curve is given by form $4\wp(z)^3 - g_2\wp(z) - g_3$ for some choice of L . It is clear that we can recover a specific elliptic curve from any elliptic function (namely, the one that comprises its domain), and we have defined a specific elliptic function $\wp$ for any given elliptic curve. The following impressive result finishes this correspondence:

Theorem B.2.8. Any elliptic function over a lattice L can be written of form $A(\wp(z)) + B(\wp'(z))$ for some rational functions $A(z), B(z)$.

³Indeed, this parameterization is a very valuable result when considering that nondegenerate elliptic curves cannot be rationally parameterized.

Sketch. We write any elliptic function as a sum of an odd and even elliptic function; an odd elliptic function evaluated on $\wp'$ is even (as $\text{odd} \circ \text{odd} = \text{even}$ for functions), so it suffices to show any even elliptic function is a rational function composed with $\wp$. This is done by counting the zeros and poles mod L ; see [Lan87] for the technical details. $\square$

B.2.2 Logarithms of Genera

We now turn to a slight expansion of our current general theory of genera, which will be needed to *define* our useful notion of an elliptic genus. Throughout this section, we presume the codomain of a genus $\varphi : \Omega^{SO} \rightarrow R$ is a $\mathbb{Q}$ -algebra, so it is uniquely determined by its values on $\Omega^{SO} \otimes \mathbb{Q} \cong \mathbb{Q}([\mathbb{CP}^2], [\mathbb{CP}^4], \dots)$ (since any torsion will also vanish under φ). We are already aware of how an even R -coefficient power series starting with 1, $Q(x) = 1 + \sum_{i=1}^{\infty} a_{2i}x^{2i}$, through its multiplicative sequence determines a genus φ_Q .

It may still be unclear, however, whether this correspondence is one-to-one, or equivalently, whether one can recover Q from some φ_Q . It is also not always clear how to compute a large-degree multiplicative sequence out of a power series, and therefore we may wish for a more elegant way to specify genera and determine their values. Our short-term goal for this section will then be a way to provide this correspondence, and it turns out a simple modification brings all the needed characterization.

Take our even power series $Q(x)$. Because our first term is $1 \neq 0$ (in fact any nonzero constant term is without loss of generality 1 by scaling), we may take the algebraic reciprocal $1/Q(x)$ to be a power series as well (analytically, they would agree in some radius of convergence). The formula for computing coefficients gives the first coefficient of $1/Q(x)$ is $\frac{1}{a_0} = \frac{1}{1} = 1$, so $1/Q(x)$ has leading term 1. Finally, $Q(x)$ is even, which is equivalent to $Q(x) = Q(-x)$, from which it follows $\frac{1}{Q(x)} = \frac{1}{Q(-x)}$ so $1/Q(x)$ is also an even power series with leading term 1. Then $f(x) = x/Q(x)$ is an odd power series with leading term x , so its first derivative is nonzero, which means that it can be formally inverted to some unique g (see [Niv69] for more details):

Definition B.2.9. Given φ_Q a genus defined some even power series Q , denote the compositional inverse power series of $f(x) = \frac{x}{Q(x)}$ as the **logarithm** of φ_Q , $\log_{\varphi_Q}(x)$. We correspondingly denote $f(x)$ the **exponent** of φ_Q and denote it via $\exp_{\varphi_Q}(x)$.

The logarithm of a genus, despite only ever needing the data of Q , turns out to encode a great amount of information about φ_Q :

Proposition B.2.10. Given a genus φ_Q defined by an even power series Q , we have that

$$\log_{\varphi_Q}(x) = x + \frac{\varphi_Q([\mathbb{CP}^2])}{3}x^3 + \frac{\varphi_Q([\mathbb{CP}^4])}{5}x^5 + \dots$$

Proof. Recall that the total Pontryagin class of $\mathbb{CP}^n$ is $(1 + h^2)^{2n+1}$, where $h \in H^2(\mathbb{CP}^n, \mathbb{Z})$ is a hyperplane class; clearly, breaking the Pontryagin class apart by degrees, it follows that $p_1 = h^2$ (up to a choice of scaling), so we can also write $p = (1 + p_1)^{n+1}$.

Then examining the multiplicative sequence $\{K_j\}$ defined from Q using $(1 + p_1) \cdots (1 + p_1) = 1 + p_1 + \cdots + p_n = (1 + p_1)^{n+1}$, using the property of multiplicative sequences we get

$$K(p_1, \dots, p_n) = K(p_1)^{n+1}.$$

But there only being one elementary symmetric polynomial of the x_i^2 (recall that that is what the multiplicative sequence of an even power series is defined as evaluating on) means there is only one

formal variable x_1 . So, writing the formulation of the multiplicative sequence $Q(x_1) \cdots Q(x_n)$ in the case where there is only one variable, we get $Q(x_1) = K(p_1)$. Therefore, recalling $\varphi([\mathbb{CP}^n]) = K_n(p_1, \dots, p_n)[\mathbb{CP}^n]$ and seeing that $1 + \cdots + K_n(p_1, \dots, p_n) = K(p_1, \dots, p_n) = Q(x)^{n+1}$, it follows that $\varphi([\mathbb{CP}^n])$ should correspond to the coefficient of x^n in the formal power series established by $Q(x)^{n+1}$. Here, *in the case that f converges*, complex analysis's connection to power series allows for computation via residue calculus, since via Cauchy's integral formula

$$n^{\text{th}} \text{ coefficient of } Q(x)^{n+1} = \frac{1}{n!} \left[\frac{d^n}{dx^n} Q(x)^{n+1} \right] \Big|_{x=0} = \frac{1}{2\pi i} \oint \frac{Q(x)^{n+1}}{x^{n+1}} dx$$

over some positively oriented curve around 0 with winding number 1. Using $\exp_{\varphi_Q}(x) = x/Q(x)$,

$$= \frac{1}{2\pi i} \oint \frac{1}{\exp_{\varphi_Q}(x)^{n+1}} dx.$$

We now perform a substitution $y = \exp_{\varphi_Q}(x)$, or equivalently $x = \log_{\varphi_Q}(y)$, $dx = \log'_{\varphi_Q}(y) dy$. We have to change the contour of integration, but because $\exp_{\varphi_Q}(x)$ has leading term x and the winding number is not sensitive to higher-order effects, its image is still some positively oriented curve around 0 with winding number 1. Thus, via calculus and re-applying the same correspondence used above, we get

$$= \frac{1}{2\pi i} \oint \frac{\log'_{\varphi_Q}(y)}{y^{n+1}} dy = n^{\text{th}} \text{ coefficient of } \log'_{\varphi_Q}.$$

We might here be worried that f may not converge, but it turns out we may define a *formal residue* over formal power series with all the same properties which we use here to pass from $Q(x)^{n+1}$ to $\log'(y)$, so this proof may be generalized to any case. Finally, by formal differentiation of power series

$$= (n+1) \text{ coefficient of } \log_{\varphi_Q} = \varphi_Q([\mathbb{CP}^n])$$

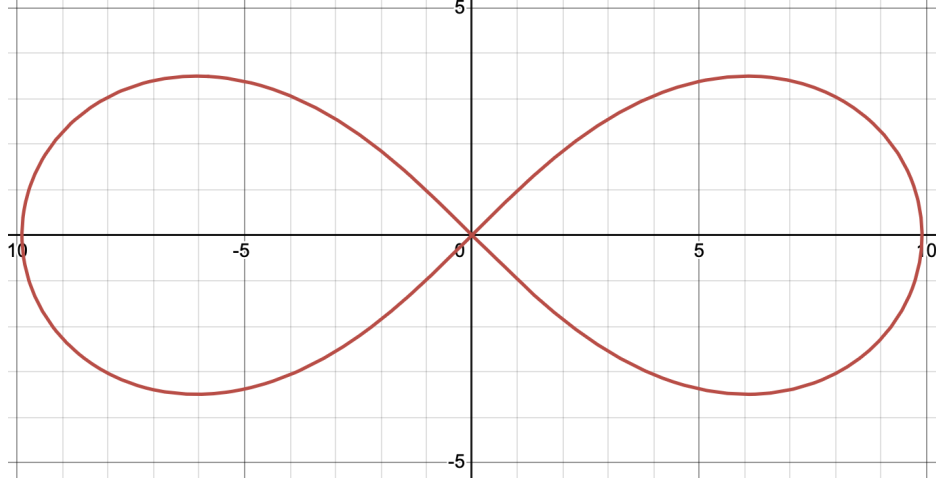
as desired. □

Therefore, given any genus φ , we can compute $\log_{\varphi}$ as a power series, compute $\exp_{\varphi}$, and compute Q out of it. This completes the one-to-one correspondence of even power series with leading term 1 and genera. One therefore is free to drop the specification of Q off of φ in general for ease of notation. It is also clear that given an odd power series $L(x)$ with leading term x , we can take $Q(x) = \frac{x}{L^{-1}(x)}$ as an even power series and define a genus of which L is the logarithm, or take $Q(x) = x/L(x)$ and make a genus where L is the exponent; therefore, we may also say there is a one-to-one correspondence between odd power series with leading term x and genera.

B.2.3 Computing the Arc Length of a Lemniscate

Our central problem of study will be to compute the various arc lengths of the *lemniscate of Bernoulli*, a planar curve determined by two foci, coordinates chosen to be $(-a, 0)$ and $(a, 0)$ such that $2a$ [$a > 0$] is the distance between the foci, containing all points such that the product of the distances to each foci equals a^2 . This curve is more familiar as the usual symbol for infinity:

[For an intuitive depiction of how the lemniscate is “drawn” from this condition, I encourage the reader to visit [this Wolfram demonstration](#)]. We already know the distance of (x, y) to $(-a, 0)$

Figure B.1: The Lemniscate of Bernoulli [$a = 7$]

is $\sqrt{(x+a)^2 + y^2}$, and likewise the distance to $(a, 0)$ is $\sqrt{(x-a)^2 + y^2}$. The product of these by definition is a^2 , and squaring on both sides gives the lemniscate as the locus of the equation

$$((x-a)^2 + y^2) \cdot ((x+a)^2 + y^2) = (x^2 + y^2 - 2ax + a^2)(x^2 + y^2 + 2ax + a^2) = a^4.$$

A classical problem for planar curves, famously simple for circles and complex for ellipses, is to find the *arc length* along the curve between any two points on it. Exploiting the symmetry of the lemniscate, we can fix one of the points to be the origin, and any point of equal distance away from the origin suffices for the other point. Thus, we seek to characterize the distance from the origin to any point on the lemniscate with distance r from the origin. First, writing $r^2 = x^2 + y^2$ to bring distance from the origin into the parameterization above, we get

$$(r^2 + a^2 - 2ax)(r^2 + a^2 + 2ax) = (r^2 + a^2)^2 - (2ax)^2 = a^4,$$

and algebra gives

$$r^4 + 2a^2r^2 - 4a^2x^2 = 0.$$

Therefore, we have a parameterization of the x -coordinate of the lemniscate $x(r)$ such that $2a^2 \cdot 2x(r)^2 = r^4 + 2a^2 \cdot r^2$ (the purpose of this odd way of writing the expression will become clear soon). We need the y -coordinate as well, and we get it by plugging in $x^2 = r^2 - y^2$:

$$r^4 + 2a^2r^2 - 4a^2(r^2 - y^2) = r^4 - 2a^2r^2 - 4a^2y^2 = 0,$$

so we have $y(r)$ such that $2a^2 \cdot 2y(r)^2 = 2a^2 \cdot r^2 - r^4$. We can now take derivatives with respect to r to get

$$2a^2 \cdot 4x(r)x'(r) = 4r^3 + 2 \cdot 2a^2 \cdot r,$$

so

$$2a^2 \cdot 2x(r)x'(r) = 2r^3 + 2a^2 \cdot r$$

and symmetrically

$$2a^2 \cdot 2y(r)y'(r) = 2a^2 \cdot r - 2r^3.$$

Using this, we can prove the arc length formula:

Proposition B.2.11. For the lemniscate of Bernoulli defined with respect to parameter $a > 0$, its arc length from the origin to a point of distance r away is

$$\sqrt{2a^2} \int_0^{r/\sqrt{2a^2}} \frac{du}{\sqrt{1-u^4}}.$$

Proof. This reduces to a computation by the above parameterization and the formula for arc length from elementary calculus. First, taking derivatives,

$$\begin{aligned} x'(r)^2 &= \frac{(2a^2 \cdot r + 2r^3)^2}{(2a^2 \cdot 2y(r))^2} = \frac{r^2(2a^2 + 2r^2)^2}{2a^2 \cdot 2(2a^2 \cdot 2y(r)^2)} = \frac{r^2(2a^2 + 2r^2)^2}{2a^2 \cdot 2(r^4 + 2a^2 \cdot r^2)} \\ &= \frac{(2a^2 + 2r^2)^2}{2a^2 \cdot 2(r^2 + 2a^2)} \end{aligned}$$

and analogously

$$y'(r)^2 = \frac{(2a^2 - 2r^2)^2}{2a^2 \cdot 2(2a^2 - r^2)}.$$

Then (and this is where keeping the $2a^2$ term separate starts to become useful)

$$\begin{aligned} x'(r)^2 + y'(r)^2 &= \frac{(2a^2 + 2r^2)^2(2a^2 - r^2) + (2a^2 - 2r^2)^2(r^2 + 2a^2)}{2a^2 \cdot 2(r^2 + 2a^2)(2a^2 - r^2)}, \\ &= \frac{16a^6}{2a^2 \cdot 2((2a^2)^2 - r^4)} = \frac{(2a^2)^2}{(2a^2)^2 - r^4} \end{aligned}$$

and thus, by the formula for arc length, the arc length of the lemniscate is

$$\int_0^r \sqrt{x'(t)^2 + y'(t)^2} dt = \int_0^r \frac{2a^2}{\sqrt{(2a^2)^2 - t^4}} dt.$$

The $2a^2$ now cleanly separates out by a u -substitution $t^2 = 2a^2 \cdot u^2$, $dt = \sqrt{2a^2} du$;

$$\begin{aligned} &= \int_0^{r/\sqrt{2a^2}} \frac{2a^2}{\sqrt{(2a^2)^2 - (2a^2)^2 u^4}} \sqrt{2a^2} du \\ &= \int_0^{r/\sqrt{2a^2}} \frac{2a^2}{2a^2 \sqrt{1-u^4}} \sqrt{2a^2} du, \\ &= \sqrt{2a^2} \int_0^{r/\sqrt{2a^2}} \frac{du}{\sqrt{1-u^4}} \end{aligned}$$

as desired. □

We will take as standard the case of $2a^2 = 1$, analyzing the cleaner integral for various choices of r

$$\int_0^r \frac{dt}{\sqrt{1-t^4}},$$

and we see from the above formulation that all other cases are obtained by scaling and a different choice of r , as would intuitively make sense for increasing the lemniscate while holding the distance from the origin the same. The above integral falls into a larger, well-known family:

B.2. CONNECTING ELLIPTIC INTEGRALS, CURVES, AND FUNCTIONS THROUGH ELLIPTIC GENERA

Definition B.2.12. An integral of form $\int_0^r \frac{dt}{\sqrt{P(t)}}$, where $P(t) = 1 - 2\delta t^2 + \epsilon t^4$ is a polynomial of degree ≤ 4 with no repeated roots, is an **elliptic integral**.

Specifically, the arc length formula above is an elliptic integral where $\delta = 0, \epsilon = -1$, and the roots $\pm 1, \pm i$ of $1 - t^4$ are distinct. We hit a major obstruction in our so-far elementary approach inasmuch as most elliptic integrals do not have solutions expressible in elementary functions. To learn anything more about them will require a change in perspective.

B.2.4 Elliptic Genera and their Properties

Focusing on the integrand $\frac{1}{\sqrt{P(t)}}$, we know $P(t)$ is a formal power series (with terms higher than 4 having coefficient 0). The square root of a formal power series can be expressed as a power series (see [Niv69]), so the entire integrand is a formal power series. Furthermore, clearly $P(t)$ is even and so $\frac{1}{\sqrt{P(-t)}} = \frac{1}{\sqrt{P(t)}}$, so the integrand is an even power series with constant term is just $\frac{1}{\sqrt{P(0)}} = \frac{1}{1} = 1$. Taking formal antidifferentiation, we get that the elliptic integral $\int_0^r \frac{dt}{\sqrt{P(t)}}$ is a formal power series of r which is odd and has leading term x . This leads to the natural definition:

Definition B.2.13. A genus $\varphi : \Omega^{\text{SO}} \rightarrow R$ is **elliptic** precisely when its logarithm is of form

$$\log_{\varphi}(x) = \int_0^x \frac{dt}{\sqrt{1 - 2\delta t^2 + \epsilon t^4}}$$

for some $\delta, \epsilon \in R$. We call the term under the radical $P_{\varphi}(t)$.

The following clean equivalent definition will be of use later, when verifying a particular kind of genus is elliptic:

Proposition B.2.14. A genus $\varphi : \Omega^{\text{SO}} \rightarrow R$ taking values in some $\mathbb{Q}$ -algebra is an elliptic genus if and only if the differential equation $(\exp'_{\varphi})^2 = P(\exp_{\varphi})$ holds for some P of form $1 - 2\delta t^2 + \epsilon t^4$.

Proof. First, suppose φ is an elliptic genus as defined above. Using the Fundamental Theory of Calculus on a formal level, it follows $\log'_{\varphi}(x) = \frac{1}{P(x)}$ for $P(t) = 1 - 2\delta t^2 + \epsilon t^4$. Then

$$\log'_{\varphi}(\exp_{\varphi}(y)) = \frac{1}{\sqrt{P(\exp_{\varphi}(y))}},$$

so via rearranging

$$P(\exp_{\varphi}(y)) = \frac{1}{(\log'_{\varphi}(\exp_{\varphi}(y)))^2}.$$

But since $\exp_{\varphi}, \log_{\varphi}$ are inverses, $\log_{\varphi}(\exp_{\varphi}(y)) = y$. Applying Chain Rule, it follows $\exp'_{\varphi}(y) \cdot \log'_{\varphi}(\exp_{\varphi}(y)) = 1$, so $\exp'_{\varphi} = \frac{1}{\log'_{\varphi}(\exp_{\varphi}(y))}$. Thus substitution gives

$$P(\exp_{\varphi}) = (\exp'_{\varphi})^2$$

as desired. Now, suppose for some genus that $(\exp'_{\varphi})^2 = P(\exp_{\varphi})$. But as we showed above $\log'_{\varphi}(x) = \frac{1}{\exp'_{\varphi}(\log_{\varphi}(x))} = \frac{1}{\sqrt{P(\exp_{\varphi}(\log_{\varphi}(x)))}} = \frac{1}{\sqrt{P(x)}}$, so via anti-differentiation $\log_{\varphi}(x) = \int \frac{1}{\sqrt{P(x)}}$ as desired. $\square$

Elliptic genera attain a host of desirable properties which we will now focus on, further justifying their existence. For example, their exponentials have a nice representation under addition:

Proposition B.2.15. For any elliptic genus φ ,

$$\exp_{\varphi}(x+y) = \frac{\exp_{\varphi}(x) \exp'_{\varphi}(y) + \exp'_{\varphi}(x) \exp_{\varphi}(y)}{1 - \epsilon \exp_{\varphi}(x)^2 \exp_{\varphi}(y)^2}.$$

Proof. We derive this from the differential equation $(\exp'_{\varphi})^2 = P(\exp_{\varphi})$. Write $f := \exp_{\varphi}$ and

$$R(x, y) := \frac{\exp_{\varphi}(x) \exp'_{\varphi}(y) + \exp'_{\varphi}(x) \exp_{\varphi}(y)}{1 - \epsilon \exp_{\varphi}(x)^2 \exp_{\varphi}(y)^2}.$$

We have the total differential $dR(x, y) = \frac{\partial R}{\partial x} dx + \frac{\partial R}{\partial y} dy$. By excruciating calculus, we get

$$\begin{aligned} (1 - \epsilon f(x)^2 f(y)^2)^2 \frac{\partial R}{\partial x} &= [f'(x) f'(y) + f''(x) f(y)] \cdot [1 - \epsilon f(x)^2 f(y)^2] \\ &\quad - [f(x) f'(y) + f'(x) f(y)] \cdot [-2\epsilon f(x) f'(x) f(y)^2], \\ &= f'(x) f'(y) - \epsilon f(x)^2 f(y)^2 f'(x) f'(y) + f(y) f''(x) - \epsilon f(x)^2 f(y)^3 f''(x) \\ &\quad + 2\epsilon f(x)^2 f(y)^2 f'(x) f'(y) + 2\epsilon f(x) f(y)^3 f'(x)^2 \end{aligned}$$

To simplify things down to the part that matters, we're only going to compute the *non-symmetric* part of this derivative, treating any symmetric part as 0 (a formal way of talking about this is considering this computation to be over the group of functions in x, y and quotienting out by the subgroup of symmetric functions:)

$$\equiv f(y) f''(x) - \epsilon f(x)^2 f(y)^3 f''(x) + 2\epsilon f(x) f(y)^3 f'(x)^2$$

The only really mysterious piece here is $f''(x)$, but we can compute this from the differential equation:

$$f'(x)^2 = 1 - 2\delta f(x)^2 + \epsilon f(x)^4,$$

so by taking derivatives

$$2f'(x) f''(x) = -4\delta f(x) f'(x) + 4\epsilon f(x)^3 f'(x),$$

from which it follows

$$f''(x) = -2\delta f(x) + 2\epsilon f(x)^3.$$

We substitute this into the above equation, still only interested in the non-symmetric part:

$$\begin{aligned} (1 - \epsilon f(x)^2 f(y)^2)^2 \frac{\partial R}{\partial x} &\equiv f(y) (-2\delta f(x) + 2\epsilon f(x)^3) - \epsilon f(x)^2 f(y)^3 (-2\delta f(x) + 2\epsilon f(x)^3) + \\ &\quad 2\epsilon f(x) f(y)^3 f'(x)^2, \\ &= -2\delta f(x) f(y) + 2\epsilon f(x)^3 f(y) + 2\delta \epsilon f(x)^3 f(y)^3 - 2\epsilon^2 f(x)^5 f(y)^3 + 2\epsilon f(x) f(y)^3 f'(x)^2, \\ &\equiv 2\epsilon f(x)^3 f(y) - 2\epsilon^2 f(x)^5 f(y)^3 + 2\epsilon f(x) f(y)^3 f'(x)^2. \end{aligned}$$

We now substitute in $f'^2 = 1 - 2\delta f^2 + \epsilon f^4$;

$$\begin{aligned} &= 2\epsilon f(x)^3 f(y) - 2\epsilon^2 f(x)^5 f(y)^3 + 2\epsilon f(x) f(y)^3 [1 - 2\delta f(x)^2 + \epsilon f(x)^4], \\ &= 2\epsilon f(x)^3 f(y) - 2\epsilon^2 f(x)^5 f(y)^3 + 2\epsilon f(x) f(y)^3 - 4\delta \epsilon f(x)^3 f(y)^3 + 2\epsilon^2 f(x)^5 f(y)^3, \end{aligned}$$

$$\begin{aligned}
 &= (2\epsilon f(x)^3 f(y) + 2\epsilon f(x) f(y)^3) - 4\delta \epsilon f(x)^3 f(y)^3, \\
 &\equiv 0.
 \end{aligned}$$

Thus $(1 - \epsilon f(x)^2 f(y)^2) \cdot \frac{\partial R}{\partial x}$ is symmetric. Since $(1 - \epsilon f(x)^2 f(y)^2)$ is never 0 in this context, we can divide by it and get that $\frac{\partial R}{\partial x}$ is symmetric and write it as $s(x, y)$. But clearly $s(x, y) = s(y, x) = \frac{\partial R}{\partial y}$ as well, so now $dR(x, y) = s(x, y)dx + s(x, y)dy = dR(x, y) \cdot d[x + y]$. Now, along the curve $x + y = c$, we get $dR(x, y) = s(x, y) \cdot 0 = 0$, from which it follows that $R(x, y)$ is constant along this curve. To determine which constant, we evaluate

$$R(0, c) = \frac{\exp_\varphi(0) \exp'_\varphi(c) + \exp'_\varphi(0) \exp_\varphi(c)}{1 - \epsilon \exp_\varphi(0)^2 \exp_\varphi(c)^2}.$$

But since $\exp_\varphi$ is an odd power series with leading term x it evaluates to 0 at 0 and its derivative evaluates to 1, so $= \frac{\exp_\varphi(c)}{1} = \exp_\varphi(x + y)$. Since this is true for any choice of c and consequently for any choice of (x, y) , we get $\exp_\varphi(x + y) = R(x, y)$ exactly as desired. $\square$

Although this formula may seem obscure, it was discovered by Euler in connection to very concrete mathematical problems. For instance, Giulio Fagnano was classically interested in the *duplication of the arc*; given some arc A of the lemniscate, he wanted to find the distance where the length of the arc from the origin was twice A , and in particular see whether that distance had a nice relation to the original distance defining A (in particular, whether it was constructible in Euclidean geometry). So say we have the arc which begins at 0 and ends at some distance r away from the origin, so the arc length is $\int_0^r \frac{1}{\sqrt{1-t^4}}$ (again, an elliptic curve for $\delta = 0, \epsilon = -1$). We can write this distance as $x = \log_\varphi(r)$, so $r = \exp_\varphi(x)$. The parameter we are interested in is

$$\exp_\varphi(2x) = \exp_\varphi(x + x) = \frac{2 \exp_\varphi(x) \exp'_\varphi(x)}{1 + \exp_\varphi(x)^4}.$$

Utilizing $\exp_\varphi'^2 = P(\exp_\varphi)$ i.e. $\exp'_\varphi = \sqrt{P(\exp_\varphi)}$ and $r = \exp_\varphi(x)$, we get

$$= \frac{2r\sqrt{1-r^4}}{1+r^4}.$$

Noting that sums, products, and square roots are all derivable in Euclidean geometry gives a positive resolution to this line of inquiry.

We conclude this sequence by directing the reader to a bevy of desirable properties and results about elliptic genera. For instance, one may still wonder at this point whether δ and ϵ have any direct meaning with respect to the genus (and in particular why we've chosen to adopt $P(t) = 1 - 2\delta t^2 + \epsilon t^4$ with an unnatural coefficient in the second term). The answer for δ can already be made apparent; taking the expansion of $[\log_\varphi(x)]' = \frac{1}{\sqrt{1-2\delta x^2+\epsilon x^4}} = 1 + \delta x^2 + \dots$ immediately gives by recalling that $\log_\varphi(x) = x + \frac{\varphi([\mathbb{CP}^2])}{3}x^3 + \dots$ that $\varphi([\mathbb{CP}^2]) = \delta$. Similarly, for $\mathbb{HP}^2$ the quaternionic projective plane one can show that $\varphi([\mathbb{HP}^2]) = \epsilon$. For this computation see [HBJ94].

We are secretly already familiar with another elliptic genus other than the case of the lemniscate:

Proposition B.2.16. The L -genus [i.e. the genus which evaluates any oriented cobordism class of manifolds to its signature] is an elliptic genus.

Proof. In the case of $\delta = \epsilon = 1$, We see that $P_\varphi(t) = 1 - 2t^2 + t^4 = (1 - t^2)^2$, so the corresponding elliptic genus φ is determined by the data

$$\log_\varphi(x) = \int_0^x \frac{dt}{1-t^2} = \tanh^{-1}(x),$$

so via compositional inverse

$$\exp_{\varphi}(x) = \tanh(x)$$

so

$$Q(x) = \frac{x}{\tanh(x)},$$

which we have previously proven is the power series determining the L -genus. $\square$

This is inspiring, since the signature has many desirable properties to take inspiration from. For instance, the signature maintains a stronger kind of multiplicative property over fiber bundles as shown by [CHS57]: under particularly nice conditions, when $E \rightarrow B$ is a principal- G bundle and G acts on a manifold M , $\text{signature}(E \times_G M) = \text{signature}(B)\text{signature}(M)$. This property has broader analogues among elliptic genera:

Theorem B.2.17. Take φ an elliptic genus, $E \rightarrow B$ a principal bundle with fiber G a compact connected Lie group, B closed and oriented manifold, and V some closed spin manifold with a smooth G -action. Then $\varphi(E \times_G V) = \varphi(B)\varphi(V)$.

This result is equivalent to the Rigidity Theorem⁴, conjectured by Witten and proved by Bott and Taubes in [BT89]. A more direct and perhaps satisfying analogue is to note that in the case of $\xi : E \rightarrow B$ a complex vector bundle of dimension k , $\mathbb{CP}(\xi)$ its projectivization fiberwise (a fiber bundle over the unitary group $U(k)$), we can use multiplicativity of signature over fiber bundles to get $\text{signature}[\mathbb{CP}(\xi)] = \text{signature}(B)\text{signature}(\mathbb{CP}^{k-1})$. In the case where k is even [$k-1$ odd], this must equal 0 since $\mathbb{CP}^{k-1}$ is a nullbordant space. Ochanine in [Och87] shows this intuition is essential to the nature of elliptic genera:

Theorem B.2.18. A genus φ is elliptic if and only if $\varphi(\mathbb{CP}[\xi]) = 0$ for all ξ even-dimensional complex vector bundles.

B.2.5 Connecting Elliptic Curves to Elliptic Genera

We now have a bridge between elliptic genera and elliptic curves, as well as between elliptic curves and elliptic functions. All that remains to complete our project of understanding the “elliptic” nature of these object is to show that a concept from one side connects to a concept from another side. The most straightforward way to do this is to show that the data of an elliptic curve can be used to create some elliptic genus. Fix some lattice $L = \{a\omega_1 + b\omega_2\}$ corresponding to an elliptic curve $\mathbb{C}/L$ and form its $\wp$ -function $\wp : \mathbb{C}/L \rightarrow \mathbb{C}$. We see that $\wp$ has exactly one pole of order two on $\mathbb{C}/L$ (by its power series), so $\wp'$ has a pole of order three. We use the following result of Liouville:

Lemma B.2.19. For any non-constant elliptic function $f : \mathbb{C}/L \rightarrow \mathbb{C}$, f has the same number of zeroes and poles (both counting multiplicity).

Sketch. Consider f'/f , which is clearly also an elliptic function defined out of $\mathbb{C}/L$. Another result of Liouville’s gives that the sum of the residues of poles of an elliptic function must be zero; but by the Argument Principle, this sum is also the number of zeroes of f minus the number of poles of f . $\square$

⁴This theorem is normally presented in much more complicated form, involving elliptic operators from partial differential equations and sequences of representations. It is not expected for the reader to know of this alternative statement.

B.2. CONNECTING ELLIPTIC INTEGRALS, CURVES, AND FUNCTIONS THROUGH ELLIPTIC GENERA

Therefore, we expect $\wp'$ to have three zeroes. Now, for any $z \in \mathbb{C}$ such that $2z \in L$, it follows from $\wp'$ being odd (as the derivative of an even function) that $\wp'(z) = \wp'(z-2z) = \wp'(-z) = -\wp'(z)$, which forces $\wp'(z) = 0$ or ∞ . There are four such points in the fundamental parallelogram: $0, \omega_1/2, \omega_2/2$, and $(\omega_1 + \omega_2)/2$, and since only 0 is a pole the rest must be exactly the three zeroes of $\wp'$. It follows that these three points cannot be poles of $\wp$, else they would be poles of $\wp'$; thus we can label

$$e_1 := \wp(\omega_1/2), e_2 := \wp(\omega_2/2), e_3 := \wp([\omega_1 + \omega_2]/2).$$

Since $0, \omega_1/2, \omega_2/2$, and $(\omega_1 + \omega_2)/2$ are exactly the zeroes of $\wp'$ and therefore also of $\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3$, it follows that we get a factorization

$$4\wp(z)^3 - g_2\wp(z) - g_3 = 4(\wp(z) - e_1)(\wp(z) - e_2)(\wp(z) - e_3)$$

which multiplies out to

$$e_1 + e_2 + e_3 = 0,$$

$$4(e_1e_2 + e_1e_3 + e_2e_3) = -g_2,$$

$$4e_1e_2e_3 = g_3.$$

Finally, remark that since $\wp$ has a pole of order 2, it takes each value twice up to multiplicity, and the points at which those values coincide at the same point (i.e. $\wp(z) - x$ has a double zero) exactly correspond to those values on which $\wp'(z) = 0$ so $x = e_1, e_2, e_3$ exactly.

Thus, the function $\wp(z) - e_1 : \mathbb{C}/L \rightarrow \mathbb{C}$ admits a double zero and so can be written as $(z - \omega_1/2)^2 h(z)$, where $h(z)$ is holomorphic and nonzero at $\omega_1/2$. Since we are working with domain $\mathbb{C}/L$, in fact, $h(z)$ is nonzero everywhere, so up to a sign we can pick a nonzero holomorphic $\sqrt{h(z)}$. This in turn gives us a choice of $\sqrt{\wp(z) - e_1} = (z - \omega_1/2)\sqrt{h(z)}$ (whose only 0 is at $\omega_1/2$); let's go on ahead and fix this sign by taking $\sqrt{\wp(z) - e_1} = \sqrt{\frac{1}{z^2} - e_1 + \dots} = \frac{1}{z} + \dots$. In turn we get

$$\frac{1}{\sqrt{\wp(z) - e_1}} = \frac{1}{(z - \omega_1/2)\sqrt{h(z)}}.$$

This function is clearly meromorphic and has one pole of order 1 at $\omega_1/2$. Bafflingly, we can immediately tell that this can't be an elliptic function on $\mathbb{C}/L$ as a result, because as we stated previously the sums of residues of poles must equal 0 and this can't happen for one pole of order 1; this is because there is still some monodromy up to a sign, which is hidden analytically but is captured in the topology of the torus $\mathbb{C}/L$ itself (this is called a quasi-elliptic function). However, we can fix this by defining a new lattice $L' = \{2a\omega_1 + b\omega_2 \mid a, b \in \mathbb{Z}\}$, on which $f(z) = \frac{1}{\sqrt{\wp(z) - e_1}}$ is an elliptic function with two poles (at $\omega_1/2$ and $3\omega_1/2$) with the property that $f(z + \omega_1) = -f(z)$ for all z (reflecting its previous quasi-elliptic nature).

Now, some magic happens. Clearly $\wp(z) - e_1$ is even since $\wp$ is even, and $\wp(z) - e_1 = \frac{1}{z^2} - e_1 + \dots = \frac{1}{z^2}(1 - e_1z^2 + \dots)$. Then we get $\sqrt{\wp(z) - e_1}$ (by our choice of sign) $= \frac{1}{z}\sqrt{1 - e_1z^2 + \dots}$ so $\frac{1}{\sqrt{\wp(z) - e_1}} = z \cdot \frac{1}{\sqrt{1 - e_1z^2 + \dots}}$. The crucial detail is that, as discussed before, the latter term can be expressed as a power series, which is clearly even since it is the radical of the even $1 - e_1z^2 + \dots$, and evaluating at $z = 0$ gives 1. Thus $\frac{1}{\sqrt{1 - e_1z^2 + \dots}}$ is an even power series with leading term 1, so $\frac{1}{\sqrt{\wp(z) - e_1}}$ is an odd power series with leading term x . We recognize that this means it can define a genus, either by setting it to be the exponential or logarithm. Looking ahead, we see that elliptic integrals must be the logarithm, so the natural thing to do is to set it equal to the exponential. The following result is as we desire:

Theorem B.2.20. Define an elliptic genus φ such that $\exp_\varphi(z) = \frac{1}{\sqrt{\wp(z) - e_1}}$ for $\wp$ over some lattice L . Then φ is an elliptic genus, where $\delta = \frac{3}{2}e_1$ and $\epsilon = (e_1 - e_2)(e_1 - e_3)$.

Proof. By a previous proposition, it suffices to show that $(\exp'_\varphi)^2 = P(\exp_\varphi)$ for $P(t) = 1 - 2\delta t^2 + \epsilon t^4$. Write $\xi = \exp_\varphi, \eta = \exp'_\varphi, x = \wp(z), y = \wp'(z)$. Then we get

$$\xi = \frac{1}{\sqrt{x - e_1}},$$

so via rearrangement

$$\begin{aligned} x &= \xi^{-2} + e_1, \\ \eta &= [(\wp(z) - e_1)^{-1/2}]' = -\frac{1}{2}(\wp(z) - e_1)^{-3/2} \cdot \wp'(z), \\ &= -\frac{y}{2}([x - e_1]^{-1/2})^3 = -\frac{y}{2}\xi^3, \end{aligned}$$

so via rearrangement

$$y = -2\eta\xi^{-3}.$$

Since we have via the differential equation of the Weierstrass function and its factorization, we have

$$y^2 = 4(x - e_1)(x - e_2)(x - e_3).$$

Substituting in gives

$$\begin{aligned} 4\eta^2\xi^{-6} &= 4(\xi^{-2} + e_1 - e_1)(\xi^{-2} + e_1 - e_2)(\xi^{-2} + e_1 - e_3), \\ &= 4\eta^{-2}(\xi^{-2} + e_1 - e_2)(\xi^{-2} + e_1 - e_3), \end{aligned}$$

and multiplying both sides by $4\xi^6$ gives

$$\eta^2 = \xi^4(\xi^{-2} + e_1 - e_2)(\xi^{-2} + e_1 - e_3).$$

Splitting $\xi^4 = \xi^2 \cdot \xi^2$ and distributing one of each to the two other terms,

$$= (1 + \xi^2[e_1 - e_2])(1 + \xi^2[e_1 - e_3]).$$

Multiplying out gives

$$= 1 + (2e_1 - e_2 - e_3)\xi^2 + (e_1 - e_2)(e_1 - e_3)\xi^4.$$

But we computed earlier that $e_1 + e_2 + e_3 = 0$, so

$$\begin{aligned} &= 1 + 3e_1\xi^2 + (e_1 - e_2)(e_1 - e_3)\xi^4 = 1 - 2\delta\xi^2 + \epsilon\xi^4, \\ &= P(\xi). \end{aligned}$$

This shows $\eta^2 = P(\xi)$ i.e. $[\exp'_\varphi]^2 = P(\exp_\varphi)$ exactly as desired. $\square$

This finally gives us a direct relationship between elliptic functions (which inform the exponent of the elliptic genus) and elliptic integrals (the logarithm of the genus). Using the fact that the logarithm is the inverse power series of the exponent, we can say, in a precise way that would have been quite difficult to see prior to the theory of genera, that the inverse function to an elliptic integral is an elliptic function.

B.3 The Hodge-de Rham spectral sequence (Brian Siew)

B.3.1 Introduction and historical context

A foundational problem in algebraic topology and complex geometry is relating the analytic properties of a complex manifold X to its underlying topological invariants, such as its de Rham cohomology $H_{\text{dR}}^k(X, \mathbb{C})$. For a complex manifold, the exterior derivative d splits into two operators, ∂ and $\bar{\partial}$, yielding the Dolbeault cohomology groups $H_{\bar{\partial}}^{p,q}(X)$.

In 1955, A. Frölicher defined a spectral sequence that relates the two. A spectral sequence is a device that iteratively computes successive approximations of cohomology. Frölicher's sequence takes the Dolbeault cohomology as its initial input (the “first page”) and converges to the de Rham cohomology [Frö55]. The purpose of this appendix is to construct this spectral sequence rigorously, prove that it degenerates at the first page for compact Kähler manifolds (following Deligne and others), and demonstrate its failure to degenerate in the non-Kähler setting via the Iwasawa manifold.

Why is this degeneration important? The degeneration of the Frölicher spectral sequence tells us that for Kähler manifolds, the analytic invariants (Dolbeault cohomology) and the topological invariants (de Rham cohomology) are not just related by a sequence of approximations, but are essentially isomorphic as vector spaces. This yields immediate and powerful restrictions on the topology of Kähler manifolds, such as the fact that their odd Betti numbers must be even, and that every holomorphic form is automatically d -closed.

B.3.2 Review of complex manifolds and spectral sequences

Complex manifolds and the Dolbeault cohomology

We begin by providing a quick review of the theory of complex manifolds and recalling some basic constructions.

Definition B.3.1. Following [Huy05, Definition 2.1.1], a *holomorphic atlas* on a differentiable manifold is an atlas of the form $\{(U_i, \varphi_i)\}$, where $\varphi_i : U_i \rightarrow \varphi_i(U_i) \subseteq \mathbb{C}^n$ is a homeomorphism, and the transition functions given by $\varphi_{ij} := \varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \rightarrow \varphi_i(U_i \cap U_j)$ are holomorphic. We then call a pair $\{(U_i, \varphi_i)\}$ a *holomorphic chart*, and call two holomorphic atlases $\{(U_i, \varphi_i)\}$, $\{(U'_j, \varphi'_j)\}$ *equivalent* if all maps $\varphi_i \circ \varphi'_j{}^{-1} : \varphi'_j(U_i \cap U'_j) \rightarrow \varphi_i(U_i \cap U'_j)$ are holomorphic.

Definition B.3.2. Following [Huy05, Definition 2.1.2], a *complex manifold* X of dimension n is a real differentiable manifold of dimension $2n$ endowed with an equivalence class of holomorphic atlases.

We recall from [Huy05, Proposition 2.6.2] that a complex manifold X must admit an *almost complex structure*, that is, some vector bundle endomorphism $I : TX \rightarrow TX$ with $I^2 = -\text{id}$, where TX is the real tangent bundle of the underlying real manifold. Complexifying the tangent bundle via $TX_{\mathbb{C}} = TX \otimes_{\mathbb{R}} \mathbb{C}$, we can extend I to $TX_{\mathbb{C}}$ via complex linearity, and since $I^2 = -1$, the eigenvalues of I on the complexified tangent bundle are exactly $\pm i$. This yields a direct sum decomposition of the complexified tangent bundle

$$TX_{\mathbb{C}} = T^{1,0}X \oplus T^{0,1}X$$

following [Huy05, Proposition 2.6.4(i)], where $T^{1,0}X$ is the eigenspace associated with i , and $T^{0,1}X$ is the eigenspace associated with $-i$. Passing to the dual, we get:

$$T^*X_{\mathbb{C}} = \Omega_X^{1,0} \oplus \Omega_X^{0,1}$$

Following the notation in [Voi02, Section 2.3], above $\Omega_X^{1,0}$ is the bundle of $(1,0)$ -covectors, and $\Omega_X^{0,1}$ is the bundle of $(0,1)$ -covectors. If we take holomorphic local coordinates $z_1, \dots, z_n$, then the bundle $\Omega_X^{1,0}$ is locally spanned by $\{dz_j = dx_j + idy_j\}_{1 \leq j \leq n}$, and $\Omega_X^{0,1}$ is locally spanned by $\{d\bar{z}_j = dx_j - idy_j\}_{1 \leq j \leq n}$. Now, we take the k^{th} exterior power on both sides, and recalling from [Lan02, Proposition XIX.1.4] how the exterior power distributes over the direct sum, this gives:

$$\bigwedge^k T^*X_{\mathbb{C}} = \bigwedge^k (\Omega_X^{1,0} \oplus \Omega_X^{0,1}) = \bigoplus_{p+q=k} \left(\bigwedge^p \Omega_X^{1,0} \otimes \bigwedge^q \Omega_X^{0,1} \right)$$

We define $\bigwedge_{\mathbb{C}}^k X := \bigwedge^k T^*X_{\mathbb{C}}$, and the bundle of (p,q) -covectors $\Omega_X^{p,q} := \bigwedge^p \Omega_X^{1,0} \otimes \bigwedge^q \Omega_X^{0,1}$. Then the above decomposition can be written as:

$$\bigwedge_{\mathbb{C}}^k X = \bigoplus_{p+q=k} \Omega_X^{p,q}$$

We further denote by $\mathcal{A}_{X,\mathbb{C}}^k$ and $\mathcal{A}_X^{p,q}$ the *sheaves of sections* of these respective bundles, and the global sections of $\mathcal{A}_X^{p,q}$, i.e. elements of $\mathcal{A}^{p,q}(X)$ are the *forms of type (p,q)* . Then following [Huy05, Corollary 2.6.8], we also have the decomposition:

$$\mathcal{A}_{X,\mathbb{C}}^k = \bigoplus_{p+q=k} \mathcal{A}_X^{p,q}$$

If we now denote by $\mathcal{A}^*(X)$ the graded space of all smooth $\mathbb{C}$ -valued differential forms on X , then we have natural projections $\Pi^k : \mathcal{A}^*(X) \rightarrow \mathcal{A}^k(X)$ and $\Pi^{p,q} : \mathcal{A}^*(X) \rightarrow \mathcal{A}^{p,q}(X)$. The exterior derivative may be extended in $\mathbb{C}$ -linear fashion to $d : \mathcal{A}_{X,\mathbb{C}}^k \rightarrow \mathcal{A}_{X,\mathbb{C}}^{k+1}$, and with the aid of the projection maps we can split it into two maps that each remind us of the usual real exterior derivative, which we denote ∂ and $\bar{\partial}$:

$$\begin{aligned} \partial &:= \Pi^{p+1,q} \circ d : \mathcal{A}_X^{p,q} \rightarrow \mathcal{A}_X^{p+1,q} \\ \bar{\partial} &:= \Pi^{p,q+1} \circ d : \mathcal{A}_X^{p,q} \rightarrow \mathcal{A}_X^{p,q+1} \end{aligned}$$

The Leibniz rule for d is inherited by ∂ and $\bar{\partial}$, that is

$$\begin{aligned} \partial(\alpha \wedge \beta) &= \partial(\alpha) \wedge \beta + (-1)^{p+q} \alpha \wedge \partial(\beta) \\ \bar{\partial}(\alpha \wedge \beta) &= \bar{\partial}(\alpha) \wedge \beta + (-1)^{p+q} \alpha \wedge \bar{\partial}(\beta) \end{aligned}$$

for any $\alpha \in \mathcal{A}^{p,q}$ and $\beta \in \mathcal{A}^{r,s}$. ∂ and $\bar{\partial}$ can be thought of as isolating the type $(p+1, q)$ forms and type $(p, q+1)$ forms respectively after passing a type (p, q) form through the exterior derivative d . Now, observe that in holomorphic local coordinates $z_1, \dots, z_n$, a general form of type (p, q) can be written as $d\alpha = \sum_{I,J} \alpha_{I,J} \wedge dz_I \wedge d\bar{z}_J$, where $|I| = p$ and $|J| = q$, and $dz_I = dz_{i_1} \wedge \dots \wedge dz_{i_p}$, and the same for J . Applying the exterior derivative gives us:

$$d\alpha = \sum_{I,J} d\alpha_{I,J} \wedge dz_I \wedge d\bar{z}_J$$

We note that $\alpha_{I,J}$ is a function, hence by definition, we have:

$$d\alpha_{I,J} = \sum_k \frac{\partial \alpha_{I,J}}{\partial z_k} dz_k + \sum_k \frac{\partial \alpha_{I,J}}{\partial \bar{z}_k} d\bar{z}_k$$

Substituting this back in, we obtain:

$$d\alpha = \sum_{I,J,k} \frac{\partial \alpha_{I,J}}{\partial z_k} dz_k \wedge dz_I \wedge d\bar{z}_J + \sum_{I,J,k} \frac{\partial \alpha_{I,J}}{\partial \bar{z}_k} d\bar{z}_k \wedge dz_I \wedge d\bar{z}_J$$

Notice that the first sum is of type $(p+1, q)$ and the second sum is of type $(p, q+1)$. So, there are no other types of forms present, and we have $d\alpha = \partial\alpha + \bar{\partial}\alpha$. Importantly, we summarize the following relations involving ∂ and $\bar{\partial}$, which are given by [Voi02, Lemma 2.29].

Lemma B.3.3. $\bar{\partial}^2 = 0$, $\partial\bar{\partial} + \bar{\partial}\partial = 0$, $\partial^2 = 0$.

Proof. These are easily derivable from $d^2 = 0$ and $d = \partial + \bar{\partial}$, for applying d^2 to some arbitrary form α of type (p, q) , we get:

$$d^2\alpha = (\partial + \bar{\partial})^2\alpha = \partial^2\alpha + (\partial\bar{\partial} + \bar{\partial}\partial)\alpha + \bar{\partial}^2\alpha = 0$$

Now, note that $\partial^2\alpha$ is a form of type $(p+2, q)$, $\bar{\partial}^2\alpha$ is a form of type $(p, q+2)$, and $(\partial\bar{\partial} + \bar{\partial}\partial)\alpha$ is a form of type $(p+1, q+1)$. As such, these must all vanish separately, giving us the desired identities. $\square$

We now recognize that fixing p and exploiting $\bar{\partial}^2 = 0$ gives us a new cochain complex

$$0 \longrightarrow \mathcal{A}^{p,0}(X) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1}(X) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,2}(X) \longrightarrow \dots$$

whence we seize the opportunity to define a new form of cohomology, following [Huy05, Definition 2.6.20].

Definition B.3.4. For a complex manifold X , the (p, q) -Dolbeault cohomology is the vector space:

$$H_{\bar{\partial}}^{p,q}(X) := H^q(\mathcal{A}^{p,\bullet}(X), \bar{\partial}) = \frac{\ker(\bar{\partial} : \mathcal{A}^{p,q}(X) \rightarrow \mathcal{A}^{p,q+1}(X))}{\operatorname{im}(\bar{\partial} : \mathcal{A}^{p,q-1}(X) \rightarrow \mathcal{A}^{p,q}(X))}$$

Spectral sequences

Before we discuss the Frölicher spectral sequence, we review the mechanism of spectral sequences of filtered complexes. We refer extensively to [Voi02, Chapters 8.3.1 and 8.3.2], but we will first build intuition for what these objects actually mean.

If one has a cochain complex, computing its cohomology can be immensely difficult. A filtration allows us to break the complex into smaller, more manageable pieces. A spectral sequence is an iterative algebraic device that computes the cohomology of the whole complex by starting with the cohomology of the small pieces and successively gluing them together to resolve ambiguities.

Definition B.3.5. Let $(K^\bullet, d)$ be a cochain complex in some abelian category. Then a *decreasingly filtered cochain complex* $(K^\bullet, d, F^\bullet)$ is the cochain complex $(K^\bullet, d)$ equipped with a decreasing family of subcomplexes

$$\dots \subseteq F^{p+1}K^\bullet \subseteq F^pK^\bullet \subseteq F^{p-1}K^\bullet \subseteq \dots \subseteq F^0K^\bullet = K^\bullet$$

such that $d(F^pK^n) \subseteq F^pK^{n+1}$.

Now, note that we have an inclusion of the complexes $F^p K^\bullet \hookrightarrow K^\bullet$, and since $F^p K^\bullet$ is a subcomplex, this commutes with the differential. We can therefore pass to cohomology and get a natural map $H^n(F^p K^\bullet) \rightarrow H^n(K^\bullet)$, from which we observe that a filtration on a complex naturally gives rise to a filtration on the cohomology, namely:

$$F^p H^n(K^\bullet) := \text{im}(H^n(F^p K^\bullet) \rightarrow H^n(K^\bullet))$$

Definition B.3.6. We define the *associated graded complex* $\text{Gr}_p^F H^n(K^\bullet)$ as successive quotients of the induced filtration on cohomology:

$$\text{Gr}_p^F H^n(K^\bullet) := F^p H^n(K^\bullet) / F^{p+1} H^n(K^\bullet)$$

With these objects, we can now begin constructing the *spectral sequence* $(E_r^{p,q}, d_r)$ associated to the filtration F . A good way to conceptualize the complex $E_r^{p,q}$, which initially appears forbidding and unwieldy, is as a book with pages indexed by $r = 0, 1, 2, \dots$. Each page consists of a two-dimensional grid of vector spaces $E_r^{p,q}$ equipped with differential arrows d_r between them, coming out of and going into each object. The directions of the arrows are dictated by the page number, and to “flip” to the next page, one takes the cohomology of the current page with respect to the arrows d_r . Let us visually define the first few pages before providing the formal proof.

- ▷ **The E_0 page:** The objects are simply the graded pieces of the original complex itself: $E_0^{p,q} = F^p K^{p+q} / F^{p+1} K^{p+q}$. Since d takes F^p to F^p , it induces a “vertical” differential $d_0 : E_0^{p,q} \rightarrow E_0^{p,q+1}$.

$$\begin{array}{ccccc} E_0^{p-1,q+1} & & E_0^{p,q+1} & & E_0^{p+1,q+1} \\ d_0 \uparrow & & d_0 \uparrow & & d_0 \uparrow \\ E_0^{p-1,q} & & E_0^{p,q} & & E_0^{p+1,q} \\ d_0 \uparrow & & d_0 \uparrow & & d_0 \uparrow \\ E_0^{p-1,q-1} & & E_0^{p,q-1} & & E_0^{p+1,q-1} \end{array}$$

- ▷ **The E_1 page:** We turn the page by taking cohomology: $E_1^{p,q} = \ker d_0 / \text{im} d_0$. The original differential d now induces “horizontal” arrows $d_1 : E_1^{p,q} \rightarrow E_1^{p+1,q}$.

$$\begin{array}{ccccc} E_1^{p-1,q+1} & \xrightarrow{d_1} & E_1^{p,q+1} & \xrightarrow{d_1} & E_1^{p+1,q+1} \\ \\ E_1^{p-1,q} & \xrightarrow{d_1} & E_1^{p,q} & \xrightarrow{d_1} & E_1^{p+1,q} \\ \\ E_1^{p-1,q-1} & \xrightarrow{d_1} & E_1^{p,q-1} & \xrightarrow{d_1} & E_1^{p+1,q-1} \end{array}$$

- ▷ **The E_2 page and beyond:** Again, we take cohomology: $E_2^{p,q} = \ker d_1 / \text{im} d_1$. The arrows now moves two steps to the right and one step down: $d_2 : E_2^{p,q} \rightarrow E_2^{p+2,q-1}$. In general, the r^{th} page differential (for a cohomological spectral sequence) goes r steps to the right and $r - 1$

steps down: $d_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$.

$$\begin{array}{ccccccc}
 E_2^{p-2,q+1} & & E_2^{p-1,q+1} & & E_2^{p,q+1} & & E_2^{p+1,q+1} \\
 & \searrow d_2 & & \searrow d_2 & & & \\
 E_2^{p-2,q} & & E_2^{p-1,q} & & E_2^{p,q} & & E_2^{p+1,q} \\
 & \searrow d_2 & & \searrow d_2 & & & \\
 E_2^{p-2,q-1} & & E_2^{p-1,q-1} & & E_2^{p,q-1} & & E_2^{p+1,q-1}
 \end{array}$$

What are we actually computing? To formally capture “taking cohomology sequentially”, we define the space of *approximate cocycles* $Z_r^{p,q}$. This represents elements in F^p that are “closed up to page r ”. Meaning, if you apply d , the result lands r levels deeper into the filtration, i.e. in F^{p+r} . Similarly, the *approximate coboundaries* $B_r^{p,q}$ are elements that come from d of something $r-1$ levels higher up.

For the purpose of the next crucial theorem, we assume our filtration is bounded, i.e. for each degree n , there exists some a such that $F^a K^n = 0$. When the filtration is bounded, the differential arrows eventually point into spaces that are zero. When all incoming and outgoing arrows to a space $E_r^{p,q}$ are zero, taking cohomology does nothing, and the space stabilizes. We denote this stable limit page $E_\infty^{p,q}$. Following [Voi02, Definition 8.26], we make the following definition:

Definition B.3.7. We say that the spectral sequence associated to the filtered cochain complex $(K^\bullet, d, F^\bullet)$ *degenerates* at E_r if, for all $k \geq r$, we have $d_k = 0$. We then have $E_r^{p,q} = E_\infty^{p,q} = \text{Gr}_p^F H^{p+q}(K^\bullet)$.

We may now state and prove the theorem of interest:

Theorem B.3.8. There exist complexes $(E_r^{p,q}, d_r)$ where $d_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$ satisfying the following conditions:

- i. $E_0^{p,q} = \text{Gr}_p^F K^{p+q} := F^p K^{p+q} / F^{p+1} K^{p+q}$, with d_0 induced by d .
- ii. The E_{r+1} page is determined by the cohomology of the E_r page. Concretely, $E_{r+1}^{p,q} = \ker(d_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}) / \text{im}(d_r : E_r^{p-r,q+r-1} \rightarrow E_r^{p,q})$.
- iii. Fix $p+q$. Then for r sufficiently large, the sequence degenerates and we have $E_r^{p,q} = E_\infty^{p,q} = \text{Gr}_p^F H^{p+q}(K^\bullet)$. We denote this convergence as $E_1^{p,q} \Rightarrow H^{p+q}(K^\bullet)$.

Proof. We begin by constructing these complexes explicitly. To do this, we must identify the elements that survive to the r -th page. We define the space of *approximate cocycles* at stage r :

$$Z_r^{p,q} := \{x \in F^p K^{p+q} \mid dx \in F^{p+r} K^{p+q+1}\}$$

Conceptually, $Z_r^{p,q}$ consists of elements living in the p -th level of the filtration whose exterior derivative is not necessarily zero, but is pushed down at least r levels deeper into the filtration. Thus, they act like cocycles “up to level r ”.

To form a cohomology group, we must quotient out by *approximate coboundaries*, which we will denote $B_r^{p,q}$. An element should be considered a boundary at stage r if it comes from the differential

of an element one step higher in the filtration from the previous stage, or if it is merely leftover from the previous stage. By shifting indices in our definition of Z , observe that:

$$\begin{aligned} Z_{r-1}^{p+1,q-1} &:= \{x \in F^{p+1}K^{p+q} \mid dx \in F^{p+r}K^{p+q+1}\} \\ Z_{r-1}^{p-r+1,q+r-2} &:= \{x \in F^{p-r+1}K^{p+q-1} \mid dx \in F^pK^{p+q}\} \end{aligned}$$

Since F is a decreasing filtration, the inclusion $Z_{r-1}^{p+1,q-1} \subseteq Z_r^{p,q}$ is obvious. Then, suppose $y \in Z_{r-1}^{p-r+1,q+r-2}$. This implies that $y \in F^{p-r+1}K^{p+q-1}$ and $dy \in F^pK^{p+q}$. Because $d^2 = 0$, we have $d^2y = 0 \in F^{p+r}K^{p+q+1}$ trivially, meaning the element dy perfectly satisfies the condition to be in $Z_r^{p,q}$. This gives us the inclusion $dZ_{r-1}^{p-r+1,q+r-2} \subseteq Z_r^{p,q}$. We can now formally define our space of approximate coboundaries:

$$B_r^{p,q} := Z_{r-1}^{p+1,q-1} + dZ_{r-1}^{p-r+1,q+r-2} \subseteq Z_r^{p,q}$$

This notation is highly suggestive of standard cohomology, and we define the objects of our spectral sequence in the usual fashion: $E_r^{p,q} := Z_r^{p,q} / B_r^{p,q}$.

We must ensure that the original differential d naturally descends to a well-defined map d_r between these quotients. First, we check that d maps the numerator of $E_r^{p,q}$ into the target numerator $E_r^{p+r,q-r+1}$. For $x \in Z_r^{p,q}$, we have $x \in F^pK^{p+q}$ and $dx \in F^{p+r}K^{p+q+1}$. Applying d again gives $d^2x = 0 \in F^{p+2r}K^{p+q+2}$ trivially, which means dx satisfies the exact condition to live in $Z_r^{p+r,q-r+1}$.

Second, we check that d maps the denominator (the boundaries $B_r^{p,q}$) into the target denominator $B_r^{p+r,q-r+1}$. For $x \in Z_{r-1}^{p+1,q-1}$, $dx \in dZ_{r-1}^{p+1,q-1}$. But looking at the target denominator, we have $B_r^{p+r,q-r+1} := Z_{r-1}^{p+r+1,q-r} + dZ_{r-1}^{p+1,q-1}$. Therefore, dx safely lands inside this space. On the other hand, if x comes from the second half of the boundary space, $x \in dZ_{r-1}^{p-r+1,q+r-2}$, then applying d yields $dx = 0$ immediately. Taken together, $d(B_r^{p,q}) \subseteq B_r^{p+r,q-r+1}$ as desired.

Since d properly sends $Z_r^{p,q}$ to $Z_r^{p+r,q-r+1}$ and $B_r^{p,q}$ to $B_r^{p+r,q-r+1}$, it descends to a map of quotients $d_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$. It naturally inherits the property $d_r^2 = 0$ from the underlying $d^2 = 0$. Now that our complexes $(E_r^{p,q}, d_r)$ are constructed, we merely need to verify the three specific conditions of the theorem.

- i. We evaluate our definitions at $r = 0$. We have $Z_0^{p,q} = F^pK^{p+q}$, and the boundary space is $B_0^{p,q} = Z_{-1}^{p+1,q-1} + dZ_{-1}^{p+1,q-2}$. Because a shift of -1 simply represents the original unfiltered space, $Z_{-1}^{p+1,q-1} = F^{p+1}K^{p+q}$ and $Z_{-1}^{p+1,q-2} = F^{p+2}K^{p+q-1}$. This implies $dZ_{-1}^{p+1,q-2} \subseteq F^{p+2}K^{p+q} \subseteq F^{p+1}K^{p+q}$. The second sum is subsumed by the first, leaving us with $B_0^{p,q} = F^{p+1}K^{p+q}$.

Taking the quotient yields $E_0^{p,q} = F^pK^{p+q} / F^{p+1}K^{p+q} =: \text{Gr}_p^F K^{p+q}$ as desired. Since the differential $Z_0^{p,q} \rightarrow Z_0^{p,q+1}$ is literally just d , the map d_0 is simply the induced differential $\bar{d} : \text{Gr}_p^F K^{p+q} \rightarrow \text{Gr}_p^F K^{p+q+1}$.

- ii. We wish to show that E_{r+1} is precisely the cohomology of E_r . Observe that if $z \in Z_{r+1}^{p,q}$, then $z \in F^pK^{p+q}$ and $dz \in F^{p+r+1}K^{p+q+1}$. Because $d^2z = 0 \in F^{p+2r}K^{p+q+2}$, we see $dz \in Z_{r-1}^{p+r+1,q-r} \subseteq B_r^{p+r,q-r+1}$. Since obviously $Z_{r+1}^{p,q} \subseteq Z_r^{p,q}$, the element z also lives in $Z_r^{p,q}$. Let $\bar{z}$ denote the class of z in $E_r^{p,q}$. Because dz lands in the boundary space B_r , it descends to 0 in the quotient, meaning $d_r \bar{z} = 0$. Thus $\bar{z} \in \ker(d_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1})$.

Now, suppose $z \in B_{r+1}^{p,q}$. By definition, we can write $z = z_1 + dz_2$ for $z_1 \in Z_r^{p+1,q-1}$ and $z_2 \in Z_r^{p-r,q+r-1}$. Since $Z_r^{p+1,q-1} \subseteq Z_{r-1}^{p+1,q-1} \subseteq B_r^{p,q}$, the z_1 term vanishes in the quotient E_r ,

giving us $\bar{z} = \overline{z_1 + dz_2} = \overline{dz_2}$. But since $z_2 \in Z_r^{p-r, q+r-1}$, the exterior derivative dz_2 is exactly what d_r produces. Thus $d_r(\bar{z}_2) = \overline{dz_2} = \bar{z}$. In other words, $\bar{z}$ is in the image of d_r . Taking the class of an element in $E_r^{p,q}$ therefore rigorously produces a well-defined map:

$$Z_{r+1}^{p,q}/B_{r+1}^{p,q} \rightarrow \frac{\ker(d_r : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1})}{\operatorname{im}(d_r : E_r^{p-r, q+r-1} \rightarrow E_r^{p,q})}$$

To prove this is a bijection, we start with surjectivity. Let $\bar{z} \in \ker d_r$, and pick some representative $z \in Z_r^{p,q}$. The assumption $d_r(\bar{z}) = 0$ means the class of dz in $E_r^{p+r, q-r+1}$ is zero, i.e., it is a boundary: $dz \in B_r^{p+r, q-r+1} = dZ_{r-1}^{p+1, q-1} + Z_{r-1}^{p+r+1, q-r}$. So there exists $w \in Z_{r-1}^{p+1, q-1}$ such that $d(z - w) \in Z_{r-1}^{p+r+1, q-r} \subseteq F^{p+r+1}K^{p+q+1}$. On the other hand, $z \in F^pK^{p+q}$ and $w \in F^{p+1}K^{p+q} \subseteq F^pK^{p+q}$, so their difference $z - w \in F^pK^{p+q}$. These two facts combined imply $z - w \in Z_{r+1}^{p,q}$. Because $w \in Z_{r-1}^{p+1, q-1} \subseteq B_r^{p,q}$, its class in the quotient is $\bar{w} = 0$, making $\overline{z - w} = \bar{z}$. Hence, we have found an element $z - w$ in $Z_{r+1}^{p,q}$ that maps to $\bar{z}$, confirming surjectivity. (For brevity, we omit the straightforward but tedious proof for injectivity.)

- iii. Fix some total degree $n = p + q$. Because our filtration is bounded, there exists some integer $\ell \geq n$ large enough such that the filtration completely trivializes around degree n : $F^\ell K^n = F^\ell K^{n-1} = F^\ell K^{n+1} = 0$. At this stage, the condition for being an “approximate” cocycle becomes exactly the condition for being a genuine cocycle of the entire complex. We have:

$$Z_{\ell+1}^{p,q} = \left\{ x \in F^p K^n \mid dx \in F^{p+\ell+1} K^{n+1} = 0 \right\} = \ker(d : F^p K^n \rightarrow F^p K^{n+1})$$

Similarly, we have:

$$Z_\ell^{p+1, q-1} = \left\{ x \in F^{p+1} K^n \mid dx \in F^{p+1+\ell} K^{n+1} = 0 \right\} = \ker(d : F^{p+1} K^n \rightarrow F^{p+1} K^{n+1})$$

Next, we analyze the dZ term in the boundary space. Since $\ell \geq n = p + q$, we must have $p - \ell \leq -q \leq 0$. Therefore, $Z_\ell^{p-\ell, q+\ell-1} = \{x \in F^{p-\ell} K^{n-1} = K^{n-1} \mid dx \in F^p K^n\}$. Applying the differential d gives us exactly $dZ_\ell^{p-\ell, q+\ell-1} = F^p K^n \cap \operatorname{im}(d)$.

We are now ready to assemble the final quotient space:

$$\begin{aligned} E_{\ell+1}^{p,q} &:= Z_{\ell+1}^{p,q}/B_{\ell+1}^{p,q} \\ &= Z_{\ell+1}^{p,q} / \left(Z_\ell^{p+1, q-1} + dZ_\ell^{p-\ell, q+\ell-1} \right) \\ &= \frac{\ker(d : F^p K^n \rightarrow F^p K^{n+1})}{\ker(d : F^{p+1} K^n \rightarrow F^{p+1} K^{n+1}) + F^p K^n \cap \operatorname{im}(d)} \end{aligned}$$

Notice that this quotient exactly isolates the cohomology of $K^\bullet$ restricted to the p -th filtration level, modulo the $(p+1)$ -th filtration level:

$$\begin{aligned} E_{\ell+1}^{p,q} &= \frac{\operatorname{im}(H^n(F^p K^\bullet) \rightarrow H^n(K^\bullet))}{\operatorname{im}(H^n(F^{p+1} K^\bullet) \rightarrow H^n(K^\bullet))} \\ &= F^p H^n(K^\bullet) / F^{p+1} H^n(K^\bullet) \\ &= \operatorname{Gr}_p^F H^{p+q}(K^\bullet) \end{aligned}$$

Since increasing r past $\ell + 1$ continues to yield zero bounding spaces, the page stabilizes to $E_\infty^{p,q} = \operatorname{Gr}_p^F H^{p+q}(K^\bullet)$ as desired.

□

B.3.3 The Hodge filtration and the Frölicher spectral sequence

With the machinery of the spectral sequence fully set up, we are now able to delve into Frölicher's construction in earnest. We set our cochain complex $K^\bullet$ to be $\mathcal{A}^\bullet(X)$, the complex of $\mathbb{C}$ -valued differential forms on a complex manifold X defined in **Section 2.1**, and we let d be the usual exterior derivative. The cohomology of this complex is then the de Rham cohomology $H_{\text{dR}}^\bullet(X)$. We can define the so-called *naive filtration* on $\mathcal{A}^\bullet(X)$, given by:

$$F^p \mathcal{A}^k(X) := \bigoplus_{r \geq p, r+s=k} \mathcal{A}^{r,s}(X)$$

Recall $d = \partial + \bar{\partial}$, and given $\mathcal{A}^{r,s}(X)$ in the decomposition of $F^p \mathcal{A}^k(X)$, we have $\partial(\mathcal{A}^{r,s}(X)) \subseteq \mathcal{A}^{r+1,s}(X) \subseteq F^p \mathcal{A}^{k+1}(X)$, $\bar{\partial}(\mathcal{A}^{r,s}(X)) \subseteq \mathcal{A}^{r,s+1}(X) \subseteq F^p \mathcal{A}^{k+1}(X)$, so $d(F^p \mathcal{A}^k(X)) \subseteq F^p \mathcal{A}^{k+1}(X)$, implying $F^\bullet$ is a valid filtration.

The naive filtration is obviously bounded, since given k we can take $p > k$, and since no r satisfies $r > k$ and $r + s = k$, the direct sum is empty and $F^p \mathcal{A}^k(X) = 0$. As such, we can start to apply **Theorem 2.7** and construct our spectral sequence $(E_r^{p,q}, d_r)$. We begin by computing the E_0 page. By definition:

$$E_0^{p,q} = F^p \mathcal{A}^{p+q}(X) / F^{p+1} \mathcal{A}^{p+q}(X) \cong \mathcal{A}^{p,q}(X)$$

Then, we determine the differential $d_0 : E_0^{p,q} \rightarrow E_0^{p,q+1}$. This is induced by the exterior derivative d . For a given $x \in \mathcal{A}^{p,q}(X)$, $dx = \partial x + \bar{\partial} x$. The term ∂x is a form of type $(p+1, q)$, and hence in $\mathcal{A}^{p+1,q}(X) \subseteq F^{p+1} \mathcal{A}^{p+q+1}(X)$. It therefore vanishes when passing to $E_0^{p,q+1}$. On the other hand, $\bar{\partial} x$ is a form of type $(p, q+1)$, and hence in $\mathcal{A}^{p,q+1}(X) \cong E_0^{p,q+1}$, so the induced differential $d_0 = \bar{\partial}$ precisely.

We are now ready to assemble the next page, the E_1 page. Recall that the E_1 page is determined by the cohomology of the E_0 page along the d_0 arrows. Therefore, we have:

$$\begin{aligned} E_1^{p,q} &= \ker \left(d_0 : E_0^{p,q} \rightarrow E_0^{p,q+1} \right) / \text{im} \left(d_0 : E_0^{p,q-1} \rightarrow E_0^{p,q} \right) \\ &= \ker \left(\bar{\partial} : \mathcal{A}^{p,q}(X) \rightarrow \mathcal{A}^{p,q+1}(X) \right) / \text{im} \left(\bar{\partial} : \mathcal{A}^{p,q-1}(X) \rightarrow \mathcal{A}^{p,q}(X) \right) \end{aligned}$$

But referring to **Definition 2.4**, this is nothing but the (p, q) -Dolbeault cohomology of X ! In other words, we have:

$$E_1^{p,q} \cong H_{\bar{\partial}}^{p,q}(X)$$

At this point, we invoke [Huy05, Corollary 2.6.21], which tells us that the Dolbeault cohomology of X computes the cohomology of the sheaf of holomorphic p -forms on X , Ω_X^p , giving us an isomorphism $H_{\bar{\partial}}^{p,q}(X) \cong H^q(X, \Omega_X^p)$, and a clean way to express the data of the E_1 page:

$$E_1^{p,q} \cong H^q(X, \Omega_X^p)$$

Finally, from **Theorem 2.7(iii)**, we have one of the key results of this Appendix:

Theorem B.3.9. The Frölicher spectral sequence converges $E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H_{\text{dR}}^{p+q}(X)$, and agrees with the Dolbeault cohomology on the E_1 page.

B.3.4 The Hodge decomposition and degeneration for compact Kähler manifolds

Recall from **Definition 2.8** that a spectral sequence degenerates at the E_r page if all arrows from that page onward are trivial, i.e. $d_k = 0$ for all $k \geq r$. This immediately causes all subsequent pages to be identical. We show that, for a particular class of complex manifold, the Frölicher spectral sequence defined in the previous section degenerates immediately. We begin with a definition from [Voi02, Chapter 3]:

Definition B.3.10. A *Kähler manifold* is a complex manifold equipped with a Hermitian metric h , whose associated Kähler form $\omega = -\Im(h)$ satisfies $d\omega = 0$ (closed).

If our manifold X is Kähler, and we let $d : \mathcal{A}^{k-1}(X) \rightarrow \mathcal{A}^k(X)$ be the exterior derivative as per usual, we define the adjoint operator $d^* : - * \circ d \circ * : \mathcal{A}^k(X) \rightarrow \mathcal{A}^{k-1}(X)$, where $*$ denotes the *Hodge star operator* (see [Huy05, Section 1.2]). Analogously, we can define $\partial^* : \mathcal{A}^{p,q}(X) \rightarrow \mathcal{A}^{p-1,q}(X)$ and $\bar{\partial}^* : \mathcal{A}^{p,q}(X) \rightarrow \mathcal{A}^{p,q-1}(X)$.

Definition B.3.11. (Laplacian) We define the operators $\Delta_d = d^*d + dd^*$, $\Delta_\partial = \partial^*\partial + \partial\partial^*$, and $\Delta_{\bar{\partial}} = \bar{\partial}^*\bar{\partial} + \bar{\partial}\bar{\partial}^*$. Observe immediately that the latter two preserve bidegree, i.e. are bihomogeneous.

These are connected by the so-called Kähler identities [Huy05, Proposition 3.1.12]; we state without proof one of the important ones since the proof is rather lengthy and analysis-heavy.

Lemma B.3.12. If X is Kähler, then $\Delta_d = 2\Delta_\partial = 2\Delta_{\bar{\partial}}$.

Corollary B.3.13. Let X be a Kähler manifold and $\alpha \in \mathcal{A}^k(X)$. α is d -harmonic $\iff \alpha$ is ∂ -harmonic $\iff \alpha$ is $\bar{\partial}$ -harmonic. Moreover, Δ_d is bidegree preserving, i.e. bihomogeneous. If α is d -harmonic, then all of its components of type (p, q) are also d -harmonic.

Proof. The first two assertions follow trivially from **Lemma 4.3**. For the last assertion, note we have $0 = \Delta_d \alpha = \sum_{p+q=k} \Delta_d \alpha^{p,q}$, and since $\Delta_d \alpha^{p,q}$ is a form of type (p, q) , they must all vanish separately. $\square$

Corollary B.3.14. Let X be a Kähler manifold, let $\mathcal{H}_d^k(X)$ be the space of d -harmonic k -forms, let $\mathcal{H}_d^{p,q}(X)$ be the space of d -harmonic forms of type (p, q) , and let $\mathcal{H}_{\bar{\partial}}^{p,q}(X)$ be the space of $\bar{\partial}$ -harmonic forms of type (p, q) . Then:

$$\mathcal{H}_d^k(X) = \bigoplus_{p+q=k} \mathcal{H}_d^{p,q}(X) = \bigoplus_{p+q=k} \mathcal{H}_{\bar{\partial}}^{p,q}(X)$$

From here, [Voi02, Theorems 5.23, 5.24] provide isomorphisms $\mathcal{H}_d^k(X) \rightarrow H_{\text{dR}}^k(X, \mathbb{C})$ and $\mathcal{H}_{\bar{\partial}}^{p,q}(X) \rightarrow H_{\bar{\partial}}^{p,q}(X)$ in the case where X is compact. In other words, every de Rham cohomology class contains a unique d -harmonic representative, and every Dolbeault cohomology class contains a unique $\bar{\partial}$ -harmonic representative. With this, we obtain the celebrated:

Theorem B.3.15. (Hodge decomposition) If X is a compact Kähler manifold, then:

$$H_{\text{dR}}^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H_{\bar{\partial}}^{p,q}(X) = \bigoplus_{p+q=k} H^q(X, \Omega_X^p)$$

Armed with the Hodge decomposition, we can now prove the following important result.

Theorem B.3.16. If X is a compact Kähler manifold, the Frölicher spectral sequence associated to X degenerates at the E_1 page.

Proof. Let $b_k := \dim H_{\text{dR}}^k(X, \mathbb{C})$ be the k^{th} Betti number, and let $h^{p,q} := \dim H^q(X, \Omega_X^p)$ be the Hodge numbers. We note that the theory of spectral sequences gives us:

$$E_{\infty}^{p,q} = F^p H_{\text{dR}}^{p+q}(X, \mathbb{C}) / F^{p+1} H_{\text{dR}}^{p+q}(X, \mathbb{C}) = \text{Gr}_p^F H_{\text{dR}}^{p+q}(X, \mathbb{C})$$

Now, we have:

$$b_k = \dim H_{\text{dR}}^k(X, \mathbb{C}) = \sum_{p=0}^k \dim \text{Gr}_p^F H_{\text{dR}}^k(X, \mathbb{C}) = \sum_{p=0}^k \dim E_{\infty}^{p,k-p} = \sum_{p+q=k} \dim E_{\infty}^{p,q}$$

Then, since $E_{r+1}^{p,q}$ is always calculated from $E_r^{p,q}$ via taking cohomology, we have $\dim E_{r+1}^{p,q} \leq \dim E_r^{p,q}$. Thus, $\dim E_{\infty}^{p,q} \leq \dim E_1^{p,q} = h^{p,q}$. But taking dimensions on both sides of the Hodge decomposition gives us $b_k = \sum_{p+q=k} h^{p,q}$. Substituting this into our inequality gives:

$$\sum_{p+q=k} h^{p,q} = b_k = \sum_{p+q=k} \dim E_{\infty}^{p,q} \leq \sum_{p+q=k} h^{p,q}$$

Since the left and right sides are equal, we must have strict equality for every p and q , i.e. $\dim E_{r+1}^{p,q} = \dim E_r^{p,q}$ for all $r \geq 1$. As noted in the proof to [Voi02, Theorem 8.28], the dimension preservation guarantees $d_r = 0$ for all $r \geq 1$. As such, the spectral sequence degenerates at the E_1 page. $\square$

Why is **Theorem 4.7** so powerful? The fact that a spectral sequence degenerates at the very first page means that $E_1^{p,q} = E_{\infty}^{p,q}$. By definition, this yields:

$$H^q(X, \Omega_X^p) \cong \text{Gr}_p^F H_{\text{dR}}^{p+q}(X, \mathbb{C})$$

We list some immediate applications of this theorem to further illustrate its importance and power.

Corollary B.3.17. Every odd Betti number of a compact Kähler manifold must be even.

Proof. The Laplacian Δ_d is a real operator, complex conjugation provides an isomorphism $\overline{H_{\bar{\partial}}^{p,q}(X)} \cong H_{\bar{\partial}}^{q,p}(X)$. Therefore, their dimensions, the Hodge numbers $h^{p,q}$, satisfy $h^{p,q} = h^{q,p}$. Consequently, the k^{th} Betti number is $b_k = \sum_{p+q=k} h^{p,q}$. If k is odd, the terms $h^{p,q}$ and $h^{q,p}$ pair up perfectly, and b_k is therefore even. $\square$

This provides an immediate topological obstruction to admitting a Kähler metric. For instance, the Hopf surface $S^1 \times S^3$ is a complex manifold but its first Betti number is $b_1 = 1$. Hence, it can never be given a Kähler metric.

Corollary B.3.18. Every holomorphic form on a compact Kähler manifold is d -closed.

Proof. Consider a globally defined holomorphic p -form $\alpha \in H^0(X, \Omega_X^p)$. In our spectral sequence, this corresponds to a class in $E_1^{p,0}$. Since the sequence degenerates at E_1 , the differential $d_1 : E_1^{p,0} \rightarrow E_1^{p+1,0}$ must be exactly zero. However, we established that d_1 is precisely induced by the ∂ operator. Therefore, $\partial\alpha = 0$. Since α is holomorphic, we already know $\bar{\partial}\alpha = 0$. Consequently, $d\alpha = (\partial + \bar{\partial})\alpha = 0$. Thus, on a compact Kähler manifold, every holomorphic form is automatically closed. $\square$

B.3.5 Failure of degeneration

The Iwasawa manifold

We now exhibit a compact complex manifold for which the Frölicher spectral sequence does *not* degenerate at the E_1 page. Here, we take reference from **Example 3.A.34** of [Huy05]. Define the complex Heisenberg group as follows:

$$G = \left\{ \begin{pmatrix} 1 & z_1 & z_3 \\ 0 & 1 & z_2 \\ 0 & 0 & 1 \end{pmatrix} \mid z_1, z_2, z_3 \in \mathbb{C} \right\}$$

If we further define $\Gamma = G \cap \mathrm{GL}(3, \mathbb{Z}[i])$, then the quotient $X := G/\Gamma$ is a compact complex manifold of complex dimension 3, called the *Iwasawa manifold*. On G , consider the left-invariant forms of type $(1, 0)$:

$$\begin{aligned} \varphi^1 &= dz_1 \\ \varphi^2 &= dz_2 \\ \varphi^3 &= dz_3 - z_1 dz_2 \end{aligned}$$

These descend to global forms of type $(1, 0)$ on X , and satisfy the following equations:

$$\begin{aligned} d\varphi^1 &= d\varphi^2 = 0 \\ d\varphi^3 &= -\varphi^1 \wedge \varphi^2 \end{aligned}$$

Since $d\varphi^j$ has no $(0, 1)$ component, we have:

$$\bar{\partial}\varphi^1 = \bar{\partial}\varphi^2 = \bar{\partial}\varphi^3 = 0$$

Thus, all three define classes in $H_{\bar{\partial}}^{1,0}(X)$. In fact, we have $h^{1,0}(X) = \dim H_{\bar{\partial}}^{1,0}(X) = 3$. Now consider the forms of type $(0, 1)$ $\bar{\varphi}^1, \bar{\varphi}^2, \bar{\varphi}^3$. We have:

$$\begin{aligned} \bar{\partial}\bar{\varphi}^1 &= \bar{\partial}\bar{\varphi}^2 = 0 \\ \bar{\partial}\bar{\varphi}^3 &= -\bar{\varphi}^1 \wedge \bar{\varphi}^2. \end{aligned}$$

As such, $\bar{\varphi}^1$ and $\bar{\varphi}^2$ define nontrivial Dolbeault classes, while $\bar{\varphi}^3$ does not. Hence, $h^{0,1}(X) = \dim H_{\bar{\partial}}^{0,1}(X) = 2$. On the other hand, the first Betti number of the Iwasawa manifold is $b_1(X) = 4$. Therefore:

$$\dim E_1^{1,0} + \dim E_1^{0,1} = h^{1,0}(X) + h^{0,1}(X) = 3 + 2 = 5 > 4 = b_1(X)$$

But if the Frölicher spectral sequence degenerated at E_1 , then by **Theorem 2.8(iii)** we would have

$$\sum_{p+q=1} \dim E_1^{p,q} = \sum_{p+q=1} \dim E_{\infty}^{p,q} = \dim H_{dR}^1(X, \mathbb{C}) = b_1(X),$$

which is impossible. Hence the Frölicher spectral sequence of the Iwasawa manifold does *not* degenerate at the E_1 page.

B.4 The Calabi-Eckmann Manifolds and their $\bar{\partial}$ -Cohomology (Carl Scandellius)

The following explores the class of compact, simply-connected complex manifolds $\mathbf{M}_{\mathbf{p},\mathbf{q}}$ discovered by Eugenio Calabi and Beno Eckmann [CE53]. We construct the spaces and then, after some Kähler geometry preliminaries, we prove that $\mathbf{M}_{\mathbf{p},\mathbf{q}}$ do not admit a Kähler metric and are thus non-algebraic. We end with a note on their $\bar{\partial}$ -cohomology.

B.4.1 Construction of $\mathbf{M}_{\mathbf{p},\mathbf{q}}$

We want to specify a complex structure on $S^{2p+1} \times S^{2q+1}$, where $p, q \in \mathbb{Z}$. We will then realise this as the total space of a principal fiber bundle over $\mathbb{CP}^{p+q}$.

Let $V_{\alpha\beta}$ be the complement of the zero set of (z_α, z'_β) , the (α, β) -th components of the coordinates on $\mathbb{C}^{p+1} \times \mathbb{C}^{q+1}$. These are open by continuity of the restrictions of coordinate projections. Any $(z, z') \in S^{p+1} \times S^{q+1}$ has some nonzero component in each product sphere. So, $\{V_{\alpha\beta}\}_{\alpha=0,\dots,p, \beta=0,\dots,q}$ cover $S^{p+1} \times S^{q+1}$.

Theorem B.4.1. Each cover element $V_{\alpha\beta}$ can be furnished with local holomorphic coordinates. In particular, let $\mathbf{T}_{(1,\tau)} \cong \mathbb{C}^2/(\mathbb{Z} \oplus \mathbb{Z}\tau)$ be the complex 2-torus for a fixed $\tau \in \mathbb{C} - \mathbb{R}$. We claim we can produce the sequence

$$V_{\alpha\beta} \xrightarrow{\Phi_{\alpha\beta}} \mathbb{C}^{p+q} \times \mathbf{T}_{(1,\tau)} \xrightarrow{\{\zeta_\ell\}_\ell} \mathbb{C}^{p+q+1}$$

where $\Phi_{\alpha\beta}$ is a homeomorphism and $\{\zeta_\ell\}_\ell$ a complex atlas. We can then pull back the complex structure on $\mathbb{C}^{p+q} \times \mathbf{T}_{(1,\tau)}$ to $V_{\alpha\beta}$.

We will take this construction in two steps. First, we construct the homeomorphism Φ . Let $(w_\alpha, w'_\beta) : V_{\alpha\beta} \rightarrow \mathbb{C}^p \times \mathbb{C}^q$ define smooth maps on each chart $V_{\alpha\beta}$ component-wise as follows:

$$w_{\alpha,j} = \frac{z_j}{z_\alpha}, \quad w'_{\beta,k} = \frac{z'_k}{z'_\beta}, \quad j = 0, \dots, p, \quad k = 0, \dots, q. \quad (\text{I})$$

Fix $\tau \in \mathbb{C} - \mathbb{R}$. We define the map $t_{\alpha\beta} : V_{\alpha\beta} \rightarrow \mathbf{T}_{(1,\tau)}$ by

$$t_{\alpha\beta} = \frac{1}{2\pi i} (\log z_\alpha + \tau \cdot \log z'_\beta) \mod (1, \tau), \quad (\text{II})$$

noting $\log$ is single-valued over the fundamental lattice domain. This is differentiable and well-defined on the equivalence classes of $\mathbf{T}_{(1,\tau)}$. In fact, we claim $\Phi_{\alpha\beta} := (w_\alpha, w'_\beta, t_{\alpha\beta})$ defines our desired homeomorphism from the lemma.

Proof 1/2. Fix α, β . Take any point in $\mathbb{C}^p \times \mathbb{C}^q \times \mathbf{T}_{(1,\tau)}$, denoted by $(p_{\alpha,j}, p'_{\beta,k}, t \mod (1, \tau))_{j \neq \alpha, k \neq \beta}$. We are free to choose $p_{\alpha,\alpha} = p'_{\beta,\beta} = 1$. Any point $(z_0, \dots, z_p, z'_0, \dots, z'_q) \in \Phi_{\alpha\beta}^{-1}(\mathbb{C}^p \times \mathbb{C}^q \times \mathbf{T}_{(1,\tau)})$ in the preimage must satisfy $\sum_0^p |z_j|^2 = \sum_0^p |z'_j|^2 = 1$, for $z_\alpha, z_\beta \neq 0$. So, using (I),

$$\left(\sum_0^p |p_{\alpha,j}|^2 \right)^{-1} = |z_\alpha|^2, \quad \left(\sum_0^p |p_{\beta,k}|^2 \right)^{-1} = |z'_\beta|^2, \quad z_\alpha, z_\beta \neq 0.$$

Similarly, using $\log z - \log |z| = i \arg z$, we re-write (II) to

$$\arg z_\alpha + \tau \arg z'_\beta = 2\pi t + i(\log |z_\alpha| + \tau \log |z'_\beta|) \mod (2\pi, 2\pi\tau).$$

Since $\tau \notin \mathbb{R}$, $\arg(\cdot) \in \mathbb{R}$, this uniquely determines $\arg z_\alpha + \tau \arg z'_\beta \pmod{2\pi}$, varying continuously with $t_{\alpha\beta}, |z_\alpha|, |z'_\beta|$. Combining these two results, z_α, z_β are determined uniquely by the initial choice of point in $\mathbb{C}^p \times \mathbb{C}^q$, and vary continuously with it. The rest of the coordinates are defined via (I) by $z_j = p_{\alpha,j} z_\alpha$, and $z'_k = p'_{\beta,k} z'_\beta$. The resulting (z, z') has $|z| = |z'| = 1$ and uniquely $\Phi_{\alpha\beta}(z, z') = (p_\alpha, p_\beta, t)$. Since (z, z') varies continuously with (p_α, p_β, t) , and we already have forward continuity, we conclude $\Phi_{\alpha\beta}$ defines a homeomorphism $V_{\alpha\beta} \xrightarrow{\cong} \mathbb{C}^{p+q} \times \mathbf{T}_{(1,\tau)}$. $\square$

Since $\mathbb{C}^{p+q} \times \mathbf{T}_{(1,\tau)}$ is a complex manifold, we can pull back its complex structure under Φ to $V_{\alpha\beta}$. It remains to show that these charts on $V_{\alpha\beta}$ are biholomorphically compatible across α, β .

Proof 2/2. On any $V_{\alpha\beta} \cap V_{\gamma\delta}$, for coordinates $(w_\alpha, w'_\beta), (w_\gamma, w'_\delta)$ we set $w_{\alpha\alpha} = w'_{\beta\beta} = w_{\gamma\gamma} = w'_{\delta\delta} = 1$, we have transition maps $V_{\alpha\beta} \rightarrow V_{\gamma\delta}$ on each subchart:

$$w_{\gamma j} = \frac{w_{\alpha j}}{w_{\alpha\gamma}}, \quad w'_{\delta, k} = \frac{w'_{\beta k}}{w'_{\beta\delta}},$$

$$t_{\gamma\delta} = t_{\alpha\beta} + \frac{1}{2\pi i} (\log w_{\alpha\gamma} + \tau \log w'_{\beta\gamma}) \pmod{(1, \tau)}, \quad (\text{III})$$

and since $w_{\alpha\gamma}, w'_{\beta\delta} \neq 0$ over $V_{\alpha\beta} \cap V_{\gamma\delta}$, we have biholomorphic transition maps. $\square$

We denote the complex manifold formed from the product space $S^{2p+1} \times S^{2q+1}$ endowed with the above complex structure by $\mathbf{M}_{\mathbf{p}, \mathbf{q}, \tau}$. The choice of τ will not be of interest to us, so we will often omit it from the subscript.

Theorem B.4.2. We can realise $\mathbf{M}_{\mathbf{p}, \mathbf{q}, \tau}$ as the total space of a principal fiber bundle over $\mathbb{CP}^{p+q}$.

Proof. Let $\pi : z \in \mathbb{C}^{p+1} \mapsto \text{span}_{\mathbb{C}}(z)$. Under restriction, $\pi : S^{2p+1} \rightarrow \mathbb{CP}^p$ homogenises affine coordinates: $z = (z_0, \dots, z_p) \mapsto [z_0 : \dots : z_p]$. This defines a surjection with great circles $\pi^{-1}\pi(z) = \{\lambda z : |\lambda| = 1\} \cong S^1$ as the fibers. That is, π gives the well-known *Hopf fibration*

$$S^1 \hookrightarrow S^{2p+1} \xrightarrow{\pi} \mathbb{CP}^p.$$

Let π' denote the same map from $\mathbb{C}^{q+1}$. We define the product map $\Pi : (z, z') \mapsto (\pi(z), \pi(z'))$ and restrict its domain to $S^{2p+1} \times S^{2q+1}$. By scaling invariance, our earlier $w_{\alpha j}, w'_{\beta k}$ coordinates descend via Π to give holomorphic affine coordinates on $\mathbb{CP}^p \times \mathbb{CP}^q$. As such, Π is the (holomorphic) fiber surjection with fibers $\Pi^{-1}\Pi(w_{\alpha j}, w'_{\beta k}, t_{\alpha\beta}) = (w_{\alpha j}, w'_{\beta k}, \mathbf{T}_{(1,\tau)})$. We have the fiber sequence

$$\mathbf{T}_{(1,\tau)} \hookrightarrow \mathbf{M}_{\mathbf{p}, \mathbf{q}, \tau} \xrightarrow{\Pi} \mathbb{CP}^p \times \mathbb{CP}^q,$$

for any choice of $\tau \notin \mathbb{R}$. Principality follows since the structure group acts on $\mathbf{T}_{(1,\tau)}$ by translations $\pmod{(1, \tau)}$ and, per (III), does so holomorphically in $w_{\alpha\gamma}, w'_{\beta\gamma}$ (or, more precisely, in each $w_{\alpha j}, w'_{\beta k}$). This group is exactly $\mathbf{T}_{(1,\tau)}$. So, $\mathbf{M}_{\mathbf{p}, \mathbf{q}}$ is a complex principal $\mathbf{T}_{(1,\tau)}$ -bundle. $\square$

We package the information of this principal fiber bundle into the notation $\xi_{p,q}$. Let us now examine what this space looks like for a particular case of p, q .

Example B.4.3 ($\mathbf{M}_{\mathbf{p}, 0, \tau}$). When $p = 0$, the manifold $\mathbf{M}_{\mathbf{p}, 0, \tau}$ is isomorphic to the *Hopf manifold* $\mathbf{H}_{\mathbf{p}}$ as complex manifolds (and analogously when instead $q = 0$).

To see this, we take global affine coordinates on $S^{2p+1} \times S^1$ by $(z_\alpha)_{0,\dots,p} \times (e^{2\pi i \rho})_{\rho \in \mathbb{R} \bmod 1}$. Then, on any chart $V_{\alpha 0}$ with local holomorphic coordinates $(w_{\alpha j}) \times w'_{0,k}$ defined as above, we have

$$u_j := z_j e^{2\pi i \tau \cdot \rho} = w_{\alpha j} z_\alpha e^{\tau \log e^{2\pi i \rho}} : (V_{\alpha 0})_j \rightarrow \mathbb{C}.$$

For each u_j , we can locally choose an holomorphic branch of $\log e^{2\pi i \rho}$, with branch choices differing by constant multiplicative factor $e^{2\pi i \tau \cdot N}$ for $N \in \mathbb{Z}$, since ρ is only defined $\bmod 1$. Note $|e^{2\pi i \tau}| \neq 1$ since $\tau \notin \mathbb{R}$. This locally holomorphic but globally multi-valued function holomorphic is a bona fide single-valued holomorphism on the universal covering space, since paths non-trivial in homotopy lift to trivial paths (upon which the function must thus follow a single branch). That is, we get an entire function $\tilde{u}_j : \widetilde{\mathbf{M}_{p,0}} \rightarrow \mathbb{C}$ and, collectively, $(u_j) : \widetilde{\mathbf{M}_{p,0}} \rightarrow \mathbb{C}^{p+1} - 0$, satisfying $\tilde{u}_j(\gamma^N x) = a^N \tilde{u}_j(x)$ for $\gamma \in \text{Deck}(\widetilde{\mathbf{M}_{p,0}})$. So, u_j descends to a single-valued holomorphism on the quotient

$$(u_j) : \mathbf{M}_{p,0} \cong \widetilde{\mathbf{M}_{p,0}} / \text{Deck}(\widetilde{\mathbf{M}_{p,0}}) \rightarrow (\mathbb{C}^{p+1} - 0) / \langle \cdot a \rangle = \mathbf{H}_p,$$

which is, in particular, a biholomorphism.

B.4.2 Non-algebraicity

Preliminaries: Hodge Theory for Kähler Manifolds

Definition B.4.4 (Kähler manifolds). Let M be a complex manifold. It determines, by pulling back i -action from $\mathbb{C}^n$ through the local frames, an endomorphism $J : TM \rightarrow TM$, $J^2 = -I$, its (integrable) complex structure. Separately, M carries a Riemannian structure if we associate a Riemannian metric g , a smoothly-varying inner product on the tangent bundle: $g : TM \times TM \rightarrow \mathbb{R}$. This Riemannian metric, on M associated with its complex structure, is a Hermitian metric (and thus associates with M a Hermitian structure) if it is compatible with the complex structure:

$$g(Ju, Jv) = Jg(u, v), \quad \forall u, v \in T_p(M), \forall p \in M.$$

This Hermitian metric has an associated (non-degenerate) 2-form $\omega(u, v) := g(Ju, v)$. If $d\omega = 0$ then ω is a symplectic form. In particular, we call it the Kähler form on M and it associates with M a Kähler structure (M, J, g, ω) , collectively a Kähler manifold.

We closely follow Lee's exposition [Lee24] on the central Hodge theory in what follows.

Theorem B.4.5 (Hodge Theorem for Riemannian Manifolds). For M an oriented,⁵ compact Riemannian manifold. Then, the composite map

$$\mathcal{H}^k(M) \hookrightarrow \mathcal{Z}(M) \rightarrow H_{dR}^k(M; \mathbb{C}),$$

for $\mathcal{Z}(M)$ the closed k -forms, is an isomorphism of vector spaces.

Proof. Let $\alpha \in \mathcal{H}^k(M)$. Then $d^* \alpha = 0$ so $\langle \alpha, d\gamma \rangle = \langle d^* \alpha, \gamma \rangle = 0$. Let $\kappa = \alpha + d\sigma$ for $(k-1)$ -form σ be another representative of the same cohomology class. Then

$$\|\kappa\|^2 = \|\alpha\|^2 + \underbrace{2\Re\langle \alpha, d\sigma \rangle}_{=0} + \|d\sigma\|^2 \geq 0$$

⁵Since we are only working with complex manifolds, we always have a canonical orientation.

with equality if and only if $d\sigma = 0$.⁶ This gives injectivity.

Take $\alpha \in \mathcal{A}^k(M) : d\alpha = 0$. If $k = 0$, then $d^*\alpha = 0$ so α is harmonic, being closed and co-closed. If $k \geq 1$, then, since $\ker \Delta_d = \ker \Delta_{d^*} \subset \ker d$, by properties of the adjoint, $\langle d^*\alpha, \ker \Delta_d \rangle = \langle \alpha, d \ker \Delta_d \rangle = 0$.

So there is a $\beta \in \mathcal{A}^{k-1} : \Delta_d \beta = d^*\alpha$. Expanding Δ_d and re-arranging: $dd^*\beta = d^*(\alpha - d\beta) \in \text{im } d^*$. But $d^2 = 0$, so the left-hand side is in $\ker d$. Since $\text{im } d^* \perp \ker d$, we must have $d^*(\alpha - d\beta) = 0$. Then $\alpha - d\beta$ is co-closed as well as closed by assumption. So $\alpha - d\beta \in \mathcal{H}^k$ is a harmonic representative of $[\alpha] \in H_{dR}^k(M; \mathbb{C})$. $\square$

Theorem B.4.6 (Hodge Decomposition and Duality for Kähler Manifolds). Let M be a Kähler n -manifold. Then:

1. For $k = 0, \dots, 2n$ there is the *Hodge Decomposition*

$$\mathcal{H}^k(M) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}(M),$$

where $\mathcal{H}^{p,q}(M)$ are the harmonic (p, q) -forms on M . In particular,

$$H_{dR}^k(M; \mathbb{C}) \cong \bigoplus_{p+q=k} H^{p,q}(M).$$

This is independent of our choice of Kähler metric on M . We define $h^{p,q} = \dim_{\mathbb{C}} H^{p,q}(M)$ as the (p, q) *Hodge number* of M . Then, the Betti and Hodge numbers are related by

$$b_k = h^{k,0} + h^{k-1,1} + \dots + h^{1,k-1} + h^{0,k}.$$

2. Conjugation defines a conjugate-linear bijection $\mathcal{H}^{p,q}(M) \leftrightarrow \mathcal{H}^{q,p}(M)$ which we call *Hodge Duality*. In particular,

$$h^{p,q} = h^{q,p}$$

We will need some lemmata for the proof.

Definition B.4.7 (Lefschetz Operator). For M a Kähler manifold with Kähler form ω , we define the *Lefschetz Operator*:

$$L_\omega : \mathcal{A}^{p,q} \rightarrow \mathcal{A}^{p+1,q+1}, \quad L_\omega \eta = \omega \wedge \eta.$$

Definition B.4.8 (Interior Product). Axiomatically, let $\iota_X : \Omega^p \rightarrow \Omega^{p-1}$ be the anti-derivation characterised by

$$(\iota_X \omega)(X_1, \dots, X_{p-1}) = \omega(X, X_1, \dots, X_{p-1})$$

for vector fields X_i . As an anti-derivation on the exterior algebra of differential forms, it obeys a form of Leibniz's Rule:

$$\iota_X(\alpha \wedge \gamma) = (\iota_X \alpha) \wedge \gamma + (-1)^{\deg \alpha} \alpha \wedge (\iota_X \gamma).$$

In the following, we use the operator commutator $[P, Q] = PQ - QP$.

Lemma B.4.9 (The Kähler Identities). On a Kähler manifold, $\bar{\partial}^* = -i[L_\omega^*, \partial]$.

⁶Equivalently, α is the unique minimiser of the Hodge norm $\|\alpha\| = \int_M \alpha \wedge * \alpha$ within its cohomology class. Here $*$: $T_p^{(1)} M \rightarrow T_p^{(1)} M$ is the real-linear *Hodge *-operator* satisfying $*^2 = -1$ and locally given by $*dx = dy, *dy = -dx$.

Proof. It suffices to check on Euclidean space, with metric components⁷ $\bar{g}_{k\bar{l}} = \frac{1}{2}\delta_{kl}$. The associated Kähler form is $\omega = \frac{i}{2} \sum_j dz^j \wedge d\bar{z}^j$. Then, for $\eta \in \mathcal{A}^\bullet$,

$$\begin{aligned} [\bar{\partial}^*, L]\eta &= \bar{\partial}^*(\omega \wedge \eta) - \omega \wedge (\bar{\partial}^* \eta) \\ &= -2\iota_{\partial_{\bar{z}^j}} \partial_j \left(\bar{g}_{k\bar{l}} dz^k \wedge d\bar{z}^l \wedge \eta \right) + \bar{g}_{k\bar{l}} dz^k \wedge d\bar{z}^l \wedge (2\iota_{\partial/\partial \bar{z}^j} \partial_j \eta) \\ &= -2\iota_{\partial_{\bar{z}^j}} \left(\bar{g}_{k\bar{l}} dz^k \wedge d\bar{z}^l \wedge \partial_j \eta \right) + \bar{g}_{k\bar{l}} dz^k \wedge d\bar{z}^l \wedge (2\iota_{\partial_{\bar{z}^j}} \partial_j \eta) \\ &= idz^j \wedge \partial_j \eta \\ &= i\partial \eta. \end{aligned}$$

Since $[L^*, \partial] = \overline{[\bar{\partial}^*, L]^*}$, we get our result. $\square$

Lemma B.4.10. On a Kähler manifold, the Hodge Laplacian satisfies $\Delta_d := dd^* + d^*d = \Delta_\partial + \Delta_{\bar{\partial}}$.

Proof. Using Lemma B.4.9 and $\partial^2 = 0$, we get identities

$$\bar{\partial}^* \partial + \partial \bar{\partial}^* = -i[L_\omega^*, \partial] \partial - i\partial[L_\omega^*, \partial] = 0 = \partial^* \bar{\partial} + \bar{\partial} \partial^*$$

with the last equality following by conjugation and the forward reasoning. Then,

$$\begin{aligned} \Delta_d &= (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) + (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) \\ &= \partial\partial^* + \partial^*\partial + \underbrace{\partial\bar{\partial}^* + \bar{\partial}^*\partial + \bar{\partial}\partial^* + \partial^*\bar{\partial}}_{=0} + \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial} \\ &= \Delta_\partial + \Delta_{\bar{\partial}} \end{aligned}$$

$\square$

Lemma B.4.11. On a Kähler manifold, $\Delta_d = 2\Delta_\partial = 2\Delta_{\bar{\partial}}$.

Proof. Using Lemma B.4.9 for the first line and its conjugate for the second, we have

$$\begin{aligned} \Delta_\partial &= \partial\partial^* + \partial^*\partial = i\partial[L_\omega^*, \bar{\partial}] + i[L_\omega^*, \bar{\partial}]\partial = i\partial L_\omega^* \bar{\partial} - i\partial \bar{\partial} L_\omega^* + iL_\omega^* \bar{\partial} \partial - i\bar{\partial} L_\omega^* \partial \\ \Delta_{\bar{\partial}} &= \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial} = -i\partial[L_\omega^*, \partial] - i[L_\omega^*, \partial]\bar{\partial} = i\partial L_\omega^* \bar{\partial} + i\bar{\partial} \partial L_\omega^* - iL_\omega^* \bar{\partial} \bar{\partial} - i\bar{\partial} L_\omega^* \partial \end{aligned}$$

and equality follows from $\partial\bar{\partial} = -\bar{\partial}\partial$, which itself follows from $d^2 = 0$. Apply Lemma B.4.10. $\square$

We can now prove the Hodge decomposition and duality:

Proof. Let $\alpha \in \mathcal{H}^k(M)$. Consider the decomposition $\alpha = \sum_{p+q=k} \alpha^{p,q}$ where $\alpha^{p,q} \in \mathcal{A}^{p,q}$. Note

$$\begin{aligned} \bar{\partial} : \mathcal{A}^{p,q} &\rightarrow \mathcal{A}^{p,q+1} \\ \bar{\partial}^* : \mathcal{A}^{p,q} &\rightarrow \mathcal{A}^{p,q-1}. \end{aligned}$$

So, using Lemma B.4.11, $\Delta_d : \mathcal{A}^{p,q} \rightarrow \mathcal{A}^{p,q}$ preserves type. Thus all $\Delta_d \alpha^{p,q} = 0$, so $\alpha^{p,q} \in \mathcal{H}^{p,q}$. Since no (p, q) -form is also (p', q') , we get the desired decomposition. The Hodge theorem then provides our desired decomposition in cohomology.

For Hodge duality, our lemma guarantees that $\mathcal{H}_{\bar{\partial}}^{p,q} = \mathcal{H}_\partial^{p,q}$. Since conjugation sends $\mathcal{H}_{\bar{\partial}}^{p,q}$ to $\mathcal{H}_\partial^{q,p}$, we have our result. $\square$

⁷That is, the g_{ij} defining the Riemannian metric on M via a basis of the cotangent bundle: $g = \sum_{i,j} g_{ij} dx^i \otimes dx^j$.

Theorem B.4.12 (Betti Numbers of compact Kähler manifolds). For compact $(M, h) : \dim_{\mathbb{C}} M = n$ Kähler manifold, all even Betti numbers are positive, and all odd Betti numbers are even.

Proof. For the statement about odd Betti numbers: the Hodge decomposition gives

$$b_k = (h^{2m+1,0} + h^{2m,1} + \dots + h^{m+1,1}) + (h^{m,m+1} + \dots + h^{1,2m} + h^{0,2m+1}).$$

These parenthetic terms are equal, by Hodge duality, and we conclude b_k is even.

For the statement about even Betti numbers: let ω be the fundamental 2-form associated with the Hermitian metric h on M . Then $\wedge^k \omega$ defines a non-zero (real De Rham) cohomology class $[\wedge^k \omega] \in H_{dR}^{2k}(M, \mathbb{R})$ for $k = 1, \dots, n$. In particular, $b_{2k} = \dim H_{dR}^{2k} > 0$ by the De Rham theorem for smooth manifolds. $\square$

In fact, as conjectured by Kodaira and proven, in the case of elliptic surfaces, by Miyaoka [Miy74] and, in the remaining case of K3 surfaces, by Siu [Siu83], we can strengthen this result when $\dim_{\mathbb{C}} M = 2$. Any compact complex surface admits a Kähler metric if and only if its first Betti number is even.

$\mathbf{M}_{p,q}$ non-Kähler

Theorem B.4.13. $\mathbf{M}_{p,q}$ does not admit a Kähler structure.

Proof of Theorem 1. We can immediately see the Betti numbers from (as topological spaces) $\mathbf{M}_{p,q} \cong S^{2p+1} \times S^{2q+1}$:

$$\begin{aligned} b_k &= \dim H_k(S^{2p+1} \times S^{2q+1}) \\ &= \dim \bigoplus_{i+j=k} H_i(S^{2p+1}) \otimes H_j(S^{2q+1}) \\ &= \begin{cases} 1 & k = 0 \text{ or } 2p+1 \text{ or } 2q+1 \text{ or } 2p+2q+2 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

provided $p \neq q$, and $\dim H_{2p+1}(S^{2p+1} \times S^{2p+1}) = 2$. So, by Theorem B.4.12, $\mathbf{M}_{p,q}$ is non-Kähler. $\square$

$\mathbf{M}_{p,q}$ non-algebraic

Lemma B.4.14. Complex projective space $\mathbb{CP}^n$ can be endowed with the *Fubini-Study* metric. Axiomatically, this is the unique metric invariant under $SU(n+1)$ action on $\mathbb{CP}^n$. The associated symplectic form is closed, so this endows $\mathbb{CP}^n$ with a Kähler structure.

Theorem B.4.15. A non-Kähler manifold is non-projective.

Proof. Any projective manifold M inherits the Fubini-Study Kähler metric on $\mathbb{CP}^n$ by restriction. $\square$

As the upshot of our previous discussion on $\mathbf{M}_{p,q}$ being non-Kähler, they are in particular non-algebraic.

B.4.3 $H_{\bar{\partial}}^{(r,s)}(\mathbf{M}_{\mathbf{p},\mathbf{q}})$ Groups

We now characterise the $\bar{\partial}$ -cohomology $\mathbf{M}_{\mathbf{p},\mathbf{q}}$. We start with some basic definitions. Some knowledge of spectral sequences for fiber bundles will be assumed. Sufficient background can be found in the other appendices or in Ch. 3, Sec. 5 of Griffiths and Harris' exposition [GH78]. We closely follow Borel's exposition in Appendix II to Hirzebruch's book [Hir78], to which we owe the main theorems presented below and their proofs.

Definition B.4.16 ($\bar{\partial}$ -cohomology groups $H_{\bar{\partial}}^{p,q}(M)$). Let M be complex n -manifold. Since $\bar{\partial}^2 = 0$, we have the cochain complex (p -th Dolbeault complex)

$$0 \rightarrow \mathcal{A}^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1} \rightarrow \dots \rightarrow \mathcal{A}^{p,n} \rightarrow 0$$

whereupon the corresponding Dolbeault cohomology groups can be defined

$$H_{\bar{\partial}}^{p,q}(M) = \frac{\ker(\bar{\partial} : \mathcal{A}^{p,q} \rightarrow \mathcal{A}^{p,q+1})}{\operatorname{im}(\bar{\partial} : \mathcal{A}^{p,q-1} \rightarrow \mathcal{A}^{p,q})}.$$

As usual, we can show Dolbeault cohomology defines a functor and that its cohomology of a complex manifold is invariant under biholomorphism.

Definition B.4.17 (Base and fiber bundle degree). We refer to the integer $f_{[V]} = [F] \in H^2(V, \mathbb{Z})$ as the *degree* of the fiber bundle F , for an algebraic manifold V . The analogous case is used for the base space, denoted $b_{[V]}$.

Definition B.4.18 (The complex vector bundle $\mathbf{H}^{p,q}(F)$). Let $\xi = (E, B, F, \pi)$ be a complex analytic bundle, with F compact connected, and with transition maps $f_{\alpha\beta} = U_{\alpha} \cap U_{\beta} \rightarrow G$ into the structure group, with $\{U_{\alpha}\}_{\alpha \in A}$ a cover of B . We let $\mathbf{H}^{p,q}(F)$ denote the bundle with total space obtained via

$$\bigcup_{b \in B} H^{p,q}(F_b), \quad F_b = \pi^{-1}(b), b \in B,$$

and with transition functions $f_{\alpha\beta}^0 = f_{\alpha\beta} \circ \varphi^0$, where φ^0 is a representation of G in $\mathbf{GL}(H^{p,q}(F))$.

Theorem B.4.19. For our $\xi_{p,q}$, and more generally, for holomorphic fiber bundles (E, B, F, π) with spaces all connected and the fiber compact, and with structure group G of ξ acting trivially on $H_{\bar{\partial}}(F)$, there exists a spectral sequence (E_r, d_r) satisfying:

1. The group $E_r = \bigoplus_{f_{[V]}, b_{[V]}, (p,q)} E_r^{p,q, f_{[V]}, b_{[V]}}$ is 4-graded. Notably, $E_r^{p,q,s,t} = 0$ if $p+q = s+t$ or any index is negative. We have the associated differential complex:

$$\dots \rightarrow E_r^{p,q,s,t} \xrightarrow{d_r} E_r^{p,q+1,s+t-r+1} \rightarrow \dots$$

2. If $p+q = s+t$, then

$$E_2^{p,q,s,t} \cong \bigoplus_{i \geq 0} H^{i,s-i}(B, W \otimes \mathbf{H}^{p-i,q-s+1}(F)),$$

where W is a complex vector bundle on B .

3. Let $\hat{W} = \pi^*W$. Then, $E_R \implies H_{\bar{\partial}}(E, \hat{W})$. For all $p, q \geq 0$,

$$\mathrm{Gr}H^{p,q}(E, \hat{W}) = \bigoplus_{s+t=p+q} E_{\infty}^{p,q,s,t}$$

for a suitable filtration of $H^{p,q}(E, \hat{W})$.

4. If $W = 1$, then (E_r, d_r) consists of differential anti-commutative algebras, and the isomorphism in the last display is compatible with the products.

A proof for Theorem B.4.19 can be found in Borel's exposition in Appendix II of Hirzebruch's book [Hir78] but is otherwise beyond the scope of this paper.

We can now turn to characterising the $\bar{\partial}$ -cohomology of $\mathbf{M}_{\mathbf{p},\mathbf{q}}$. We denote the connected component of the group of biholomorphic automorphisms on X a compact, complex manifold containing its identity by $\mathrm{Aut}^0(X)$. This is the group consisting of $g \in \mathrm{Aut}(X)$ for which there is a continuous family of complex automorphisms $f_{t \in [0,1]} \in \mathrm{Aut}$ with $f_0 = I, f_t = g$.

Lemma B.4.20. $\mathrm{Aut}^0(\mathbf{M}_{\mathbf{p},\mathbf{q}})$ acts trivially on $H_{\bar{\partial}}(\mathbf{M}_{\mathbf{p},\mathbf{q}})$.

Proof. Note ξ meets the conditions for Theorem B.4.19, so we know $E_2 = H_{\bar{\partial}}(\mathbb{CP}^{p+q}) \otimes H_{\bar{\partial}}(\mathbf{T})$ for the associated spectral sequence. Since $\mathrm{Aut}^0(\mathbf{M}_{p,q})$ is an automorphism group of ξ , it operates on the spectral sequence. For compact Kähler manifolds, the Hodge decomposition shows that their Aut^0 component act trivially on $\bar{\partial}$ -cohomology. So, the $\mathrm{Aut}^0(\mathbf{M}_{p,q})$ action on $H_{\bar{\partial}}(\mathbf{M}_{p,q})$ and $H_{\bar{\partial}}(\mathbf{T})$ is trivial, and thus also on their tensor product $E_2 = H_{\bar{\partial}}(\mathbb{CP}^{p+q}) \otimes H_{\bar{\partial}}(\mathbf{T})$. Since the automorphism commutes with the differentials of the spectral sequence, its action on the later pages is also trivial. In particular, the action on $E_{\infty} = \mathrm{Gr}H_{\bar{\partial}}(\mathbf{M}_{p,q})$ is trivial. In fact, any compact subgroup of $\mathrm{Aut}^0(\mathbf{M}_{p,q})$ acts trivially on $H_{\bar{\partial}}(\mathbf{M}_{p,q})$ and it follows that all $\mathrm{Aut}^0(\mathbf{M}_{p,q})$ acts trivially. $\square$

Lemma B.4.21 (Hartog's Extension Theorem). For a holomorphic function F on several complex variables, an isolated singularity is always removable.⁸

Proof. This proof is Hörmander's [Hor73]. We extend F by $\varphi f - \psi$ for smooth φ, ψ , where φ vanishes wherever f is undefined. This extension is holomorphic if we restrict our choice of ψ such that $\bar{\partial}\psi = \bar{\partial}(\varphi f) = f\bar{\partial}\varphi + \varphi\bar{\partial}f = f\bar{\partial}\varphi$. We choose $\varphi \equiv 1$ everywhere away from a compact subset of the support of F , so $\bar{\partial}\varphi = 0$ there. Then, since $\bar{\partial}(f\bar{\partial}\varphi) = 0$, the Poincaré lemma for compactly supported differential forms tell us that there exists a compactly supported function ψ satisfying our first differential equality.

Since ψ has compact support, it vanishes in some open connected component containing ∞ . On this component, $\varphi f - \psi = f$. So, our extension thus agrees with F on an open set and thus (by rigidity of holomorphic functions) globally where both are defined. $\square$

Lemma B.4.22. $H^{1,0}(\mathbf{M}_{0,\mathbf{q}}) = 0$, for $q \geq 1$.

Proof. Recall from Example 1 that $\mathbf{M}_{0,q} \cong \mathbf{H}_q$ can be identified with the Hopf manifold constructed as $(\mathbb{C}^{q+1} - 0)/\langle \gamma : z \mapsto cz, c \neq 1 \rangle$. For $\omega \in \Omega^*(\mathbf{M}_{0,q})$, pulling back under the quotient, $\pi^*\omega = g_1 dz_1 + \dots + g_{q+1} dz_{q+1}$ for $g_i \in \mathcal{O}(\mathbb{C}^{q+1} - 0)$. Since $\pi^*\omega$ is invariant under γ , we can iterate γ to get $g_i(c^n z) = c^{-n} g_i(z)$. Then either $|c| < 1$ and $|c^{-n}| \rightarrow \infty$ as $c^n z \rightarrow 0$, or we get the same blowup using invariance under γ^{-1} .

In particular, g_i is unbounded near the origin unless $g_i \equiv 0$. But Hartog's extension theorem would require g to extend holomorphically over 0 and thus be bounded in some neighbourhood of 0. $\square$

⁸In fact, F can fail to be defined on a compact set, as is the usual statement of Hartog's, but we will not need this.

Theorem B.4.23. For $p \leq q$,

$$H_{\bar{\partial}}(\mathbf{M}_{\mathbf{p},\mathbf{q}}) \cong \mathbb{C}[x_{1,1}]/(x_{1,1}^{p+1}) \otimes \Lambda(x_{q+1,q}, x_{0,1}).$$

Proof. Suppose first $p = 0$. In the spectral sequence of $\xi_{0,q} = (\mathbf{M}_{\mathbf{0},\mathbf{q}}, \mathbf{P}^q, \mathbf{T}, \pi)$, we have directly from Theorem B.4.19 (taking the trivial vector bundle over $\mathbf{P}^q$) that

$$\begin{aligned} E_2^{0,q,s,t} &\cong \bigoplus_{i \geq 0} H^{i,s-i}(\mathbb{CP}^q, 1 \otimes \mathbf{H}^{-i,q-s+i}(\mathbf{T})) \\ E_2 &\cong H^*(\mathbb{CP}^q) \otimes H^*(\mathbf{T}) \cong \mathbb{C}[x_{1,1}]/(x_{1,1}^{q+1}) \otimes \Lambda(x_{q+1,q}, x_{0,1}), \end{aligned}$$

for non-zero representatives $x_{1,1} \in H_{\bar{\partial}}^{1,1}(\mathbb{CP}^q)$ and $x_{0,1} \in H_{\bar{\partial}}^{0,1}(\mathbf{T})$ and where $\Lambda(x_{q+1,q}, x_{0,1})$ is the free exterior algebra over $\mathbb{C}$ with the given generators. The element $x_{0,1}$ generates $E_2^{0,1,0,1} \cong H^{0,0}(\mathbb{CP}^q) \otimes H^{0,1}(\mathbf{T}) \cong \mathbb{C}^{\otimes 2} \cong \mathbb{C}$ and d_2 sends it into $E_2^{0,2,2,0} = 0$. All higher d_r act similarly on $x_{0,1}$, sending it into $E^{\dots, 2-r, < 0} = 0$. So, $d_2(x_{0,1}) = 0$, so $x_{0,1} \in E_{\infty}$ is a *permanent cocycle*. The same cannot be true of $x_{1,0}$, or it would generate a non-trivial class in cohomology $[x_{1,0}] \in H^{1,0}(\mathbf{M}_{\mathbf{0},\mathbf{q}})$, contrary to Lemma 7. So, we can assume that $d_2(x_{1,0}) = x_{1,1}$. We can then compute E_3 as follows:

Every element of E_2 is a $\mathbb{C}$ -linear combination of $(x_{1,1}^k \otimes 1), (x_{1,1}^k \otimes x_{1,0}), (x_{1,1}^k \otimes x_{0,1}), (x_{1,1}^k \otimes (x_{1,0} \wedge x_{0,1}))$, where $0 \leq k \leq q$. These d_2 sends, respectively, to $(0), (x_{1,1}^{k+1} \otimes 1), (0), (x_{1,1}^{k+1} \otimes x_{0,1})$. Since $x_{1,1}^{q+1} = 0$ also $d_2(x_{1,1}^q \otimes x_{1,0}) = 0 = d_2(x_{1,1}^q \otimes (x_{1,0} \wedge x_{0,1}))$. So,

$$E_3 \cong \Lambda(y_{q+1,q}, x_{0,1}).$$

where the first generator denotes the E_3 class coming from $x_{1,1}^q \otimes x_{1,0}$. Because of their fiber degree, $d_r y_{q+1,q} = d_r x_{0,1} = 0, r \geq 3$. So, $E_{r+1} = \ker d_r / \text{im } d_r = E_r$ and, inductively, $E_3 \cong E_{\infty} \cong H_{\bar{\partial}}(\mathbf{M}_{\mathbf{0},\mathbf{q}})$, also multiplicatively by Theorem B.4.19.4. This gives us our result for $p = 0$, since

$$H_{\bar{\partial}}(\mathbf{M}_{\mathbf{0},\mathbf{q}}) \cong \Lambda(x_{q+1,q}, x_{0,1}) \cong \mathbb{C}[x_{1,1}]/(x_{1,1}) \otimes \Lambda(y_{q+1,q}, x_{0,1}).$$

Now, let $0 < p \leq q$ and take the fibering $\xi = (\mathbf{M}_{\mathbf{p},\mathbf{q}}, \mathbb{CP}^p, \mathbf{M}_{\mathbf{0},\mathbf{q}}, \pi)$. The structure group is a one-dimensional complex torus. It sits inside $\text{Aut}(\mathbf{M}_{\mathbf{p},\mathbf{q}})$ and, in particular, since it contains the identity and is connected, sits inside $\text{Aut}^0(\mathbf{M}_{\mathbf{p},\mathbf{q}})$. So, Lemma 5 applies and the structure group acts trivially on the cohomology $H_{\bar{\partial}}(\mathbf{M}_{\mathbf{0},\mathbf{q}})$. After noting also that $\mathbf{M}_{\mathbf{p},\mathbf{q}}, \mathbb{CP}^p, \mathbf{M}_{\mathbf{0},\mathbf{q}}$ are all connected and $\mathbf{M}_{\mathbf{0},\mathbf{q}}$ is compact, we can again use Theorem B.4.19 to get the associated spectral sequence with second page given by

$$E_2 = \mathbb{C}[x_{1,1}]/(x_{1,1}^{p+1}) \otimes \Lambda(x_{q+1,q}, x_{0,1}).$$

By the same reasoning as earlier along with the $\bar{\partial}$ -type constraints imposed by $p \leq q$, both $x_{0,1}$ and $x_{q+1,q}$ are permanent cocycles. So, for $r \geq 2$ the spectral sequence differential is the zero map. Thus, inducting as in the last case, $E_2 \cong E_{\infty} \cong H_{\bar{\partial}}(\mathbf{M}_{\mathbf{p},\mathbf{q}})$ additively. Compatibility with products follows since the generators of $H_{\bar{\partial}}(\mathbf{M}_{\mathbf{p},\mathbf{q}})$ have the same relations $x_{1,1}^{p+1} = 0$ and $x_{p+1,p}^2 = x_{0,1}^2 = 0$ as the E_2 generators. So, we conclude

$$H_{\bar{\partial}}(\mathbf{M}_{\mathbf{p},\mathbf{q}}) \cong \mathbb{C}[x_{1,1}]/(x_{1,1}^{p+1}) \otimes \Lambda(x_{q+1,q}, x_{0,1}),$$

for $0 < p \leq q$, per our claim, and thus we have finished our characterisation of $H_{\bar{\partial}}(\mathbf{M}_{\mathbf{p},\mathbf{q}})$ for $p \leq q$. $\square$

B.5 Leray Spectral Sequence (David Ettel)

In this appendix we will give an introduction to the Leray spectral sequence. We will specialize to the so-called Leray-Serre spectral sequence and prove Leray-Hirsch theorem as well as more general statement along the way. Finally, we will apply the Leray-Hirsch theorem to prove that if $E \rightarrow X$ is a vector bundle and $\mathbb{P}(E)$ is its projectivization, then $H^*(\mathbb{P}(E), \mathbb{Z})$ is free over $H^*(X, \mathbb{Z})$. Before we dive into the mathematical theory, it is worthwhile to look at how and when it was first developed as its history is quite inspirational.

B.5.1 History

The Leray spectral sequence is named after its inventor Jean Leray, a French mathematician of the twentieth century (*1906 - †1998). Jean Leray's most decisive mathematical turn came during his imprisonment in Oflag XVIIA in Austria from 1940 to 1945. Trained mainly in analysis and fluid mechanics, he deliberately shifted toward algebraic topology while in captivity, partly to avoid being used for technical work by the German war effort. Nota bene: Leray spectral sequences are thus a memorial of civilian resistance against the Nazi regime. In the prison camp he helped organize an improvised university and developed a new, deeply local approach to topology.

It was in this prison setting that the ideas behind sheaves, sheaf cohomology, and eventually the Leray spectral sequence first emerged. Leray wanted tools that could study very general spaces without relying so heavily on simplicial methods. This led him to organize cohomological information in successive layers. After the war, these wartime ideas were announced in 1946 and later reshaped by Cartan, Koszul, Borel, and Serre into the modern form of the Leray spectral sequence [Mil00].

B.5.2 Spectral Sequences

Why We Care About Spectral Sequences

A spectral sequence is a computational device for cohomology. It applies when the object we want to understand, such as the sheaf cohomology $H^\bullet(X, \mathcal{F})$, is difficult to compute directly but comes equipped with a filtration whose associated graded pieces are more tractable. Each page $E_r^{p,q}$ is then an approximation to the answer, and each differential d_r is a correction term. Successive pages refine the approximation until, at the E_∞ page, what remains is the associated graded of a filtration on $H^\bullet(X, \mathcal{F})$. For us, the Leray spectral sequence will be the central tool for relating cohomology upstairs and downstairs along a map $f : X \rightarrow Y$, and the filtration it produces on $H^k(X, \mathcal{F})$ is the object of interest.

An Intuitive Introduction

For this introduction, which is mostly based on [Vak25], we will specialize to abelian groups, but all of the constructions can be applied to modules over any ring R or even more generally to any abelian category. For a rigorous and formal introduction to spectral sequence we refer you to Brian Siev's section of the appendix.

Definition B.5.1 (Double Complexes). A *double complex* is best described by the following picture.

$$\begin{array}{ccccccc}
& \vdots & & \vdots & & \vdots & \\
& d_{\uparrow}^{0,2} \uparrow & & d_{\uparrow}^{1,2} \uparrow & & d_{\uparrow}^{2,2} \uparrow & \\
E^{0,2} & \xrightarrow{d_{\rightarrow}^{0,2}} & E^{1,2} & \xrightarrow{d_{\rightarrow}^{1,2}} & E^{2,2} & \xrightarrow{d_{\rightarrow}^{2,2}} & \dots \\
& d_{\uparrow}^{0,1} \uparrow & & d_{\uparrow}^{1,1} \uparrow & & d_{\uparrow}^{2,1} \uparrow & \\
E^{0,1} & \xrightarrow{d_{\rightarrow}^{0,1}} & E^{1,1} & \xrightarrow{d_{\rightarrow}^{1,1}} & E^{2,1} & \xrightarrow{d_{\rightarrow}^{2,1}} & \dots \\
& d_{\uparrow}^{0,0} \uparrow & & d_{\uparrow}^{1,0} \uparrow & & d_{\uparrow}^{2,0} \uparrow & \\
E^{0,0} & \xrightarrow{d_{\rightarrow}^{0,0}} & E^{1,0} & \xrightarrow{d_{\rightarrow}^{1,0}} & E^{2,0} & \xrightarrow{d_{\rightarrow}^{2,0}} & \dots
\end{array}$$

In words it is a collection of abelian groups $E^{p,q}$ indexed by $p, q \in \mathbb{Z}$ and families of "upward" morphisms

$$d_{\uparrow}^{p,q} : E^{p,q} \rightarrow E^{p,q+1}$$

and "rightward" morphisms

$$d_{\rightarrow}^{p,q} : E^{p,q} \rightarrow E^{p+1,q}$$

between them. As we might have guessed already from the figure above, we use the first coordinate p to index the column and the second coordinate q to index the row of the $E^{p,q}$ and their respective morphisms $d_{\uparrow}^{p,q}$ and $d_{\rightarrow}^{p,q}$. To remember which is which, think of them as coordinates in the x, y -plane. When talking about the morphisms, we generally omit the indices and just write $d_{\uparrow}$ or $d_{\rightarrow}$. This abuse of notation makes the following conditions for the morphisms of a double complex easier to state. We require that

$$d_{\uparrow}^2 = 0 \quad \text{and} \quad d_{\rightarrow}^2 = 0$$

and that one of the following two holds

$$d_{\rightarrow} d_{\uparrow} = d_{\uparrow} d_{\rightarrow} \quad \text{or} \quad d_{\rightarrow} d_{\uparrow} = -d_{\uparrow} d_{\rightarrow},$$

i.e. that they either all commute or all anticommute. For the sake of simplicity, we will from now on assume to always be in the anticommutative case, the commutative case can be handled analogously.

Remark B.5.2. In some contexts we might be about variations where we instead want to use downward or leftward morphisms instead. In some scenarios we also specifically require certain $E^{p,q}$ to be zero. In our context we will assume that all $E^{p,q}$ with $p < 0$ or $q < 0$ are zero, which is hinted at by the figure above, but this is not a property that we required in the formal definition. This case is usually denoted as the *first quadrant double complex*.

Suppose now that we are given such a double complex. Then we can construct a "single" complex out of it.

Definition B.5.3 (Total Complex). We define the *total complex* $E^{\bullet}$ corresponding to a double complex as the collection of

$$E^k := \bigoplus_{p \in \mathbb{Z}} E^{p, k-p}$$

connected by morphisms

$$d := d_{\rightarrow} + d_{\uparrow}.$$

You might recall that in the context of homological algebra we usually require that the maps of a complex satisfy $d^2 = 0$ and indeed

$$d^2 = (d_{\rightarrow} + d_{\uparrow})^2 = d_{\rightarrow}^2 + (d_{\rightarrow} d_{\uparrow} + d_{\uparrow} d_{\rightarrow}) + d_{\uparrow}^2 = 0.$$

Example B.5.4. It is worth trying to write out what this total complex looks like in our example B.5.1 where we assume that $E^{p,q} = 0$ for $p < 0$ or $q < 0$. In the scenario of such a *first quadrant double complex* we have

$$\begin{aligned}
 E^0 &= \dots \oplus E^{-1,0-(-1)} \oplus E^{0,0-0} \oplus E^{1,0-1} \oplus \dots \\
 &= \dots \oplus E^{-1,1} \oplus E^{0,0} \oplus E^{1,-1} \oplus \dots \\
 &= \dots \oplus \{0\} \oplus E^{0,0} \oplus \{0\} \oplus \dots \\
 &\cong E^{0,0} \\
 E^1 &\cong E^{0,1} \oplus E^{1,0} \\
 E^2 &\cong E^{0,2} \oplus E^{1,1} \oplus E^{2,0}
 \end{aligned}$$

i.e. they correspond to the diagonals of slope minus one.

The first thing any good AlgeBro wants to do when seeing a complex is to take its cohomology. And really, we will use the cohomology of the total complex to associated a "hypercohomology" to the double complex.

Definition B.5.5 (Cohomology of the Double Complex). We define the *cohomology of a double complex* to be the cohomology of its associated total complex.

Exercise B.5.6. Draw up a double complex constructed out of your favorite abelian groups, derive its total complex and finally its cohomology.

With the language of a double complex in hand we are now able to give what Vakil calls an "approximate definition" of spectral sequences.

Definition B.5.7 (Spectral Sequence (approximately)). We define a *spectral sequence* (with *rightward orientation*⁹) to be a sequence of tables or pages

$$\rightarrow E_0, \rightarrow E_1, \rightarrow E_2, \rightarrow E_3, \dots$$

where each page is a collection of elements $E_r^{p,q}$ and where $E_0^{p,q} = E^{p,q}$. They have differential maps

$$\rightarrow d_r^{p,q} : \rightarrow E_r^{p,q} \longrightarrow \rightarrow E_r^{p-r+1, q+r} \quad \text{satisfying} \quad \rightarrow d_r^{p,q} \circ \rightarrow d_r^{p+r-1, q-r} = 0$$

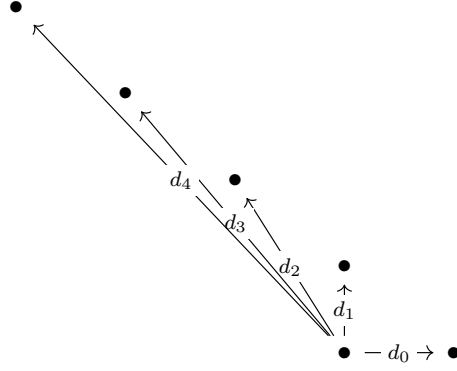
between them for $r \in \mathbb{Z}^{\geq 0}$ such that the cohomology $\ker \rightarrow d_r^{p,q} / \text{im } \rightarrow d_r^{p+r-1, q-r}$ of $\rightarrow d_r$ at $\rightarrow E_r^{p,q}$ satisfies the isomorphism

$$\ker \rightarrow d_r^{p,q} / \text{im } \rightarrow d_r^{p+r-1, q-r} \cong \rightarrow E_{r+1}^{p,q}. \quad (\text{B.5.8})$$

Once the orientation of the spectral sequence is clear, the left subscript $\rightarrow$ is often ignored which is exactly what we will be doing in this introduction from now on.

At first sight the formula for the differential maps $d_r^{p,q}$ might seem quite confusing. But, once again, a simple image should be enough to help us remember their definition.

⁹The orientation refers to the fact that the 0-th differential is the rightward one from earlier: $d_0 = d_{\rightarrow}$.



While you should remember this visual, do not forget that the morphisms in it apply to different pages! Another way to remember it is to notice that the difference of the addition to the x and y components always stems from the difference of the indices of the grade 1 components of $E^\bullet$.

To actually define the $E_r^{\bullet, \bullet}$ and $d_r^{\bullet, \bullet}$, we have to describe them in terms of the $E^{\bullet, \bullet}$. For this intuitive introduction we will only describe them for $r = 0, 1, 2$ and hope the reader believes in the continuation for higher r .

Before we continue, note the following immediate result.

Lemma B.5.9. For all $r > i$ the r -th page $E_r^{p,q}$ is a subquotient¹⁰ of the i -th page $E_i^{p,q}$.

In particular, this tells us that

$$E^{p,q} \cong 0 \quad \Rightarrow \quad E_r^{p,q} \cong 0 \quad \forall r > 0$$

as $E^{p,q} \cong E_0^{p,q}$.

Proof. This follows immediately from the isomorphisms in B.5.8. □

A direct consequence of this lemma is that in a first quadrant double complex for fixed p, q and sufficiently large r , we get the super-short exact sequences

$$\begin{array}{ccc} 0 & \xleftarrow{d_r^{p,q}} & E_r^{p,q} \\ & & \xleftarrow{d_r^{p+r-1, q-r}} 0 \end{array}$$

since $E_r^{p+r-1, q-r}$ and $E_r^{p-r+1, q+r}$ are both zero as $E^{p+r-1, q-r}$ and $E^{p-r+1, q+r}$ are zero for sufficiently large r by definition of the first quadrant double complex B.5.2.

Now recall that the isomorphism B.5.8 tells us that

$$\ker d_r^{p,q} / \operatorname{im} d_r^{p+r-1, q-r} \cong E_{r+1}^{p,q}.$$

In combination with the super-short exact sequence we just discovered, this means that for a first quadrant double complex we have canonical isomorphisms

$$E_r^{p,q} \cong E_{r+1}^{p,q} \cong E_{r+2}^{p,q} \cong \dots$$

for sufficiently large r . This allows us to define the *infinity page*.

¹⁰By a *subquotient* of an abelian group G we mean a quotient H/K of subgroups $K \subseteq H \subseteq G$. Note that for abelian groups all subgroups are abelian and normal. Hence all subquotients are again (abelian) groups.

Definition B.5.10 (Infinity Page). The *infinity page* E_∞ is defined by

$$E_\infty^{p,q} := \lim_{r \rightarrow \infty} E_r^{p,q}.$$

Remark B.5.11. Obviously, similar arguments apply in other scenarios where the double complex has similar zero configurations, for example when all but finitely many rows are zero.

Now let us discuss the first few pages of the spectral sequence corresponding to a first quadrant double complex and present them visually.

The 0th page. Recall that we defined the differential d_0 on $E_0^{\bullet,\bullet} \cong E^{\bullet,\bullet}$ to be $d_\rightarrow$ from the double complex. Thus, on the 0-th page each row is a complex and looks like

$$\begin{array}{ccccccc} \xrightarrow{d_0} & E_0^{p-1,q+1} & \xrightarrow{d_0} & E_0^{p,q+1} & \xrightarrow{d_0} & E_0^{p+1,q+1} & \xrightarrow{d_0} \\ \\ \xrightarrow{d_0} & E_0^{p-1,q} & \xrightarrow{d_0} & E_0^{p,q} & \xrightarrow{d_0} & E_0^{p+1,q} & \xrightarrow{d_0} \\ \\ \xrightarrow{d_0} & E_0^{p-1,q-1} & \xrightarrow{d_0} & E_0^{p,q-1} & \xrightarrow{d_0} & E_0^{p+1,q-1} & \xrightarrow{d_0} \end{array}$$

The 1st page. Recall that the entries on E_1 are defined by the row cohomologies of E_0 :

$$E_1^{p,q} \cong \ker d_0^{p,q} / \operatorname{im} d_0^{p-1,q}$$

and that the differential maps are now "upwards"

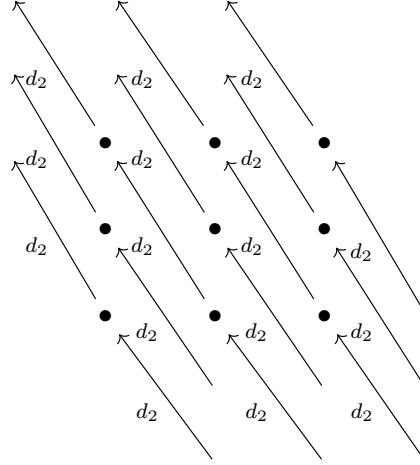
$$d_1^{p,q} : E_1^{p,q} \rightarrow E_1^{p,q+1}$$

and are just the induced maps on cohomologies stemming from $d_\uparrow$ of the first quadrant double complex. We can check that these morphisms turn the columns into complexes. Schematically, the first page thus looks like this¹¹:

$$\begin{array}{ccccc} & & d_1 \uparrow & & d_1 \uparrow & & d_1 \uparrow \\ & & \bullet & & \bullet & & \bullet \\ & & d_1 \uparrow & & d_1 \uparrow & & d_1 \uparrow \\ & & \bullet & & \bullet & & \bullet \\ & & d_1 \uparrow & & d_1 \uparrow & & d_1 \uparrow \\ & & \bullet & & \bullet & & \bullet \\ & & d_1 \uparrow & & d_1 \uparrow & & d_1 \uparrow \end{array}$$

The 2nd page. The entries of E_2 are defined by the cohomologies of the columns in E_1 . Now there are natural morphisms pointing to the entry one to the left and two up whose square is 0 and the second page looks schematically like

¹¹For the sake of readability, we refrain from actually writing out the cohomology groups.



Now the attentive reader might have discovered a pattern in the schematic structure of the pages.

We end this introduction with a little theorem.

Theorem B.5.12. There is a filtration

$$E_{\infty}^{0,q} \xrightarrow{E_{\infty}^{1,q-1}} ? \xrightarrow{E_{\infty}^{2,q-2}} \dots \xrightarrow{E_{\infty}^{q,0}} H^k(E^{\bullet}).$$

Here the notation $A \xrightarrow{Q} B$ means that $B/A \cong Q$.

As a consequence of this theorem we can say that a spectral sequence $E_{\bullet}^{\bullet}$ converges to $H^{\bullet}(E^{\bullet})$ or that a page $E_r^{\bullet}$ abuts to $H^{\bullet}(E^{\bullet})$.

Sometimes the filtration allows us to fully determine $H^k(E^{\bullet})$ as we will later see in the context of the Leray spectral sequence.

Example B.5.13 (Leray Spectral Sequence). Let $f : X \rightarrow Y$ be a continuous map of paracompact Hausdorff spaces and let $\mathcal{F}$ be a sheaf of abelian groups on X . Choose an injective resolution $\mathcal{F} \rightarrow \mathcal{I}^{\bullet}$ in the category of sheaves of abelian groups on X , and let $\mathcal{U}$ be an open cover of Y . Applying the pushforward f_* and then the Čech complex with respect to $\mathcal{U}$ produces a double complex

$$K^{p,q} = C^p(\mathcal{U}, f_*\mathcal{I}^q),$$

with horizontal differential the Čech coboundary and vertical differential induced by $\mathcal{I}^{\bullet}$. The Leray spectral sequence is the spectral sequence of this double complex, filtered by rows. Its E_0 page is $K^{p,q}$ equipped only with the horizontal differential.

$$\begin{array}{ccccccc} & & \vdots & & \vdots & & \vdots \\ C^0(\mathcal{U}, f_*\mathcal{I}^2) & \xrightarrow{d_0} & C^1(\mathcal{U}, f_*\mathcal{I}^2) & \xrightarrow{d_0} & C^2(\mathcal{U}, f_*\mathcal{I}^2) & \rightarrow & \dots \\ C^0(\mathcal{U}, f_*\mathcal{I}^1) & \xrightarrow{d_0} & C^1(\mathcal{U}, f_*\mathcal{I}^1) & \xrightarrow{d_0} & C^2(\mathcal{U}, f_*\mathcal{I}^1) & \rightarrow & \dots \\ C^0(\mathcal{U}, f_*\mathcal{I}^0) & \xrightarrow{d_0} & C^1(\mathcal{U}, f_*\mathcal{I}^0) & \xrightarrow{d_0} & C^2(\mathcal{U}, f_*\mathcal{I}^0) & \rightarrow & \dots \end{array}$$

Taking cohomology of the rows produces E_1 .

$$\begin{array}{ccccccc}
& \vdots & & \vdots & & \vdots & \\
& \uparrow & & \uparrow & & \uparrow & \\
H_C^0(Y, f_* \mathcal{I}^2) & & H_C^1(Y, f_* \mathcal{I}^2) & & H_C^2(Y, f_* \mathcal{I}^2) & & \cdots \\
d_1 \uparrow & & d_1 \uparrow & & d_1 \uparrow & & \\
H_C^0(Y, f_* \mathcal{I}^1) & & H_C^1(Y, f_* \mathcal{I}^1) & & H_C^2(Y, f_* \mathcal{I}^1) & & \cdots \\
d_1 \uparrow & & d_1 \uparrow & & d_1 \uparrow & & \\
H_C^0(Y, f_* \mathcal{I}^0) & & H_C^1(Y, f_* \mathcal{I}^0) & & H_C^2(Y, f_* \mathcal{I}^0) & & \cdots
\end{array}$$

Taking cohomology of the columns then yields E_2 :

$$\begin{array}{ccccccc}
& \vdots & & \vdots & & \vdots & \\
& \swarrow d_2 & & \swarrow d_2 & & \swarrow d_2 & \\
H^0(Y, R^3 f_* \mathcal{F}) & & H^1(Y, R^3 f_* \mathcal{F}) & & H^2(Y, R^3 f_* \mathcal{F}) & & \cdots \\
& \swarrow d_2 & & \swarrow d_2 & & \swarrow d_2 & \\
H^0(Y, R^2 f_* \mathcal{F}) & & H^1(Y, R^2 f_* \mathcal{F}) & & H^2(Y, R^2 f_* \mathcal{F}) & & \cdots \\
& \swarrow d_2 & & \swarrow d_2 & & \swarrow d_2 & \\
H^0(Y, R^1 f_* \mathcal{F}) & & H^1(Y, R^1 f_* \mathcal{F}) & & H^2(Y, R^1 f_* \mathcal{F}) & & \cdots \\
& \swarrow d_2 & & \swarrow d_2 & & \swarrow d_2 & \\
H^0(Y, R^0 f_* \mathcal{F}) & & H^1(Y, R^0 f_* \mathcal{F}) & & H^2(Y, R^0 f_* \mathcal{F}) & & \cdots
\end{array}$$

B.5.3 Existence of the Leray Spectral Sequence

Before stating the existence theorem for the Leray spectral sequence, we first introduce the notion of a system of local coefficients on X . This section is based on [McC00].

Definition B.5.14 (System of Local Coefficients on X). Let X be a topological space. A *system of local coefficients on X* is a locally constant sheaf on X .

Such a local system is called *simple* if the action of $\pi_1(X)$ on its fibers is trivial.

This notion is not abstract nonsense: every fibration $F \hookrightarrow E \xrightarrow{\pi} X$ produces one canonically. A loop γ in X based at x lifts, via the homotopy lifting property, to a homotopy self-equivalence of the fiber $\pi^{-1}(x) \simeq F$, and therefore induces an automorphism of $H^q(F; R)$. The resulting monodromy representation $\pi_1(X, x) \rightarrow \text{Aut}(H^q(F; R))$ is precisely the data of a local coefficient system $\mathcal{H}^q(F; R)$ on X , and it is this system that appears in the E_2 -page of the Leray–Serre spectral sequence. Calling it *simple* means the monodromy action is trivial, so that $H^q(F; R)$ sits over X as the constant sheaf.

Theorem B.5.15 (Leray–Serre Spectral Sequence). Let R be a commutative ring with unit. Suppose

$$F \hookrightarrow E \xrightarrow{\pi} X$$

is a fibration, where F is connected and X is path-connected. Then there exists a first-quadrant spectral sequence of algebras $\{E_r^{*,*}, d_r\}$ converging to

$$H^*(E; R)$$

as an algebra, with

$$E_2^{p,q} \cong H^p(X; H^q(F; R)),$$

where $H^q(F; R)$ is regarded as a local coefficient system on X induced by the fibration.

Proof. See [McC00]. □

B.5.4 Leray-Hirsch Theorem

Now that we know what a Leray spectral sequence is and that it exists, we can use it to prove the famous Leray-Hirsch Theorem which has many useful applications. One of them we will explore in the next chapter. In this chapter we will prove the Leray-Hirsch Theorem by constructing it as a corollary of a slightly more powerful result that we will prove first. But once again, we will have to start with some definitions before we can state the results.

Definition B.5.16 (Space of Finite Type, [McC00]). We say that a topological space X is of *finite type* if $\dim_k H^n$ is finite for all n .

Example B.5.17. Another common definition of *finite type* is that X has the homotopy type of CW complex. While we choose not to use this definition here, CW complexes still provide a good illustration of what finite type means in practice.

The next condition captures the key cohomological behavior of the fiber that makes the Leray-Hirsch argument work. Roughly speaking, it says that every cohomology class on the fiber already comes from a class on the total space.

Definition B.5.18 (Totally Nonhomologous to Zero [McC00]). Let $F \xrightarrow{\iota} E \rightarrow X$ be a fibration. We say that F is *totally nonhomologous to zero in E with respect to the ring R* if the induced homomorphism of cohomologies

$$\iota : H^*(E; R) \rightarrow H^*(F; R)$$

is surjective.

It turns out that this property is, under sufficiently nice conditions on the underlying fibration, enough to make the spectral sequence collapse at the second table.

Theorem B.5.19. [McC00] Let $F \xrightarrow{\iota} E \xrightarrow{\pi} X$ be a fibration with F connected, X path connected and of finite type, and for which the system of local coefficients on X is simple.

F is totally nonhomologous to zero in E with respect to k if and only if the spectral sequence, $E_r(X, E, F)$, collapses at the E_2 -term.

Proof. First note that we can decompose ι^* as

$$\iota^* : H^q(E; k) \twoheadrightarrow E_\infty^{0,q} = E_{q+1}^{0,1} \hookrightarrow \dots \hookrightarrow E_3^{0,q} \hookrightarrow E_2^{0,q} = H^q(F; k).$$

For a derivation of this expression consult theorem 5.9 in [McC00].

Suppose the spectral sequence collapses at the E_2 term. This means

$$E_2 = E_3 = E_4 = \dots = E_\infty.$$

Then all of the inclusions are actually equalities and in particular surjections and consequently the map ι^* is onto as well. Therefore F is totally nonhomologous to zero in E with respect to k if the spectral sequence collapses at the E_2 -term.

Conversely, suppose that F is totally nonhomologous to zero in E with respect to k . Then ι^* is surjective and all of the inclusions must be onto as well, forcing them to be equalities. Now by definition, we have

$$E_{r+1}^{0,q} = \ker (d_r : E_r^{0,q} \rightarrow E_r^{r,q-r+1}).$$

Thus the equality

$$E_{r+1}^{0,q} = E_r^{0,q}$$

forces

$$d_r \mid_{E_r^{0,q}} \equiv 0.$$

Since k is a field, $V := H^q(F; k)$ is a k -vector space, say with basis $\{e_\alpha\}_{\alpha \in I}$. Because the local system $\mathcal{H}^q(F; k)$ is simple, it is the constant sheaf with stalk V , so

$$\mathcal{H}^q(F; k) \cong \underline{V} \cong \bigoplus_{\alpha \in I} \underline{k}.$$

Cohomology commutes with direct sums of coefficients, giving

$$H^p(X; \mathcal{H}^q(F; k)) \cong \bigoplus_{\alpha \in I} H^p(X; k) \cong H^p(X; k) \otimes_k V = H^p(X; k) \otimes_k H^q(F; k).$$

Therefore, we also have that

$$E_2^{p,q} \cong H^p(X; k) \otimes_k H^q(F; k) \tag{B.5.20}$$

This means in particular that

$$\begin{aligned} E_2^{p,0} &\cong H^p(B; k) \otimes_k H^0(F; k) \\ &\cong H^p(B; k) \otimes_k k \\ &\cong H^p(B; k) \end{aligned} \tag{B.5.21}$$

and

$$\begin{aligned} E_2^{0,q} &\cong H^0(B; k) \otimes_k H^q(F; k) \\ &\cong k \otimes_k H^q(F; k) \\ &\cong H^q(F; k). \end{aligned} \tag{B.5.22}$$

Plugging [B.5.21](#) and [\(B.5.22\)](#) back into [\(B.5.20\)](#) we then get that

$$E_2^{p,q} \cong E_2^{p,0} \otimes_k E_2^{0,q}. \tag{B.5.23}$$

Now note that $d_2 : E_2^{p,0} \rightarrow E_2^{p+2,-1}$ is the zero map since $E_2^{p+2,-1}$ lies outside of the first quadrant:

$$d_2 \mid_{E_2^{p,0}} \equiv 0.$$

Now finally note that d_2 is a derivation and for $x \in E_2^{p,0}$, $y \in E_2^{0,q}$ we can write

$$d_2(xy) = d_2(x) y + (-1)^{|x|} x d_2(y).$$

But we just proved that $d_2(x) = 0$ as well as $d_2(y) = 0$. Therefore, $d_2(xy) = 0$. Together with [\(B.5.23\)](#) this implies that $d_2 \equiv 0$ and consequently

$$E_3 = E_2.$$

We can continue the same argument for d_3 and so on to establish that the spectral sequence collapses at E_2 if we assume that F is totally nonhomologous to zero in E with respect to a field k . This concludes the proof. □

This theorem allows us to now state the famous Leray-Hirsch Theorem as a simple corollary.

Corollary B.5.24 (Leray-Hirsch [McC00]). Let $F \xrightarrow{i} E \xrightarrow{\pi} X$ be a fibration with F connected, X path connected and of finite type, and for which the system of local coefficients on X is simple.

If F is totally nonhomologous to zero in E with respect to a field k , then

$$H^*(E; k) \cong H^*(X; k) \otimes_k H^*(F; k)$$

as vector spaces.

Proof. Theorem B.5.19 tells us that F is totally nonhomologous to zero in E with respect to a field k if and only if the spectral sequence collapses at the E_2 term. This tells us in particular that

$$H^p(X; k) \otimes_k H^q(F; k) \cong E_2^{p,q} = E_\infty^{p,q}.$$

But the latter is just

$$F^p H^{p+q}(E; k) / F^{p+1} H^{p+q}(E; k)$$

where the $F^p H^n$ come from the filtration

$$H^n(E; k) = F^0 H^n \supseteq F^1 H^n \supseteq F^2 H^n \supseteq \dots$$

This means that the graded pieces of $H^*(E; k)$ are the same as these of $H^*(X, k) \otimes_k H^*(F, k)$. Since we are over a field this gives us an isomorphism of vector spaces. □

The significance of the Leray-Hirsch theorem is that it gives a powerful criterion under which the cohomology of a fibration splits in the simplest possible way. In general, the cohomology of the total space E can be much more complicated than the cohomology of the base and the fiber taken separately, since nontrivial differentials and extension phenomena may occur. The Leray-Hirsch theorem identifies a situation in which this complexity disappears: if the fiber is totally nonhomologous to zero in E , then the cohomology of E is, as a vector space, exactly the tensor product of the cohomology of the base and the cohomology of the fiber. For this reason, the theorem is both conceptually important, as it explains when a fibration behaves cohomologically like a product, and practically useful, since it reduces questions about $H^*(E; k)$ to separate computations on X and F .

B.5.5 Application

Finally, now that all the Leray machinery is set up, we can move on to the main result of this appendix section. To begin, we will have to make a simple definition.

Definition B.5.25 (Projectivization of a Vector Bundle). Let $E \rightarrow X$ be a rank r vector bundle. Then we define the projectivization $\mathbb{P}[E]$ of E to be the fiber bundle defined by projectivizing the fibers of E .

Example B.5.26. Consider the real 2-sphere S^2 and its tangent bundle $TS^2 \rightarrow S^2$, a rank-2 real vector bundle. The projectivization $\mathbb{P}(TS^2) \rightarrow S^2$ assigns to each point $p \in S^2$ the space of unoriented tangent directions at p , i.e. the set of lines through the origin in the tangent plane $T_p S^2$. Each fiber is therefore $\mathbb{RP}^1 \cong S^1$, so $\mathbb{P}(TS^2) \rightarrow S^2$ is a circle bundle over the 2-sphere.

Theorem B.5.27. Let $E \rightarrow X$ be a vector bundle over a finite type path connected space X for which the system of local coefficients on X is simple. Let $\mathbb{P}[E]$ denote its projectivization. Then $H^*(\mathbb{P}(E); \mathbb{Z})$ is free over $H^*(X; \mathbb{Z})$.

Proof. This is a direct consequence of Leray-Hirsch. We have the fibration

$$\mathbb{P}^{r-1} \hookrightarrow \mathbb{P}[E] \rightarrow X$$

The fiber $\mathbb{P}^{r-1}$ is famously connected. It is also nonhomologous¹² to zero in $\mathbb{P}[E]$ over $\mathbb{Z}$.

Therefore, Leray-Hirsch specializes to

$$H^*(\mathbb{P}[E]; \mathbb{Z}) \cong H^*(X; \mathbb{Z}) \otimes_{\mathbb{Z}} H^*(\mathbb{P}^{r-1}; \mathbb{Z}) \quad (\text{B.5.28})$$

By Hatcher [Hat01] theorem 3.19

$$H^*(\mathbb{P}^{n-1}; \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^n) \cong \bigoplus_{i=0}^{n-1} \alpha^i \mathbb{Z} \cong \bigoplus_{i=0}^{n-1} \mathbb{Z}$$

Note that we here assume that the vector bundle is complex, in case of a real vector bundle the isomorphism is dependent on the q of H^q . Plugging this into equation (B.5.28) we get that

$$\begin{aligned} H^*(\mathbb{P}[E]; \mathbb{Z}) &\cong H^*(X; \mathbb{Z}) \otimes_{\mathbb{Z}} \bigoplus_{i=0}^{n-1} \mathbb{Z} \\ &\cong \bigoplus_{i=0}^{n-1} (H^*(X; \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}) \\ &\cong \bigoplus_{i=0}^{n-1} H^*(X; \mathbb{Z}) \end{aligned}$$

where the last isomorphism comes from the fact that $\mathbb{Z}$ is an identity object for $\otimes_{\mathbb{Z}}$. Now the last expression is literally the free group over $H^*(X; \mathbb{Z})$ of rank n . This concludes the proof and everyone can go home with a smile on their faces. $\square$

Why should we care about this particular result? It is relevant because it tells us that the cohomology of the projectivization $\mathbb{P}[E]$ is governed in a remarkably simple way by the cohomology of the underlying base space X . Since the fiber of $\mathbb{P}[E] \rightarrow X$ is a projective space, whose cohomology is well understood, the theorem says that no unexpected cohomology classes appear in passing from the fiber and the base to the total space. In this sense, $\mathbb{P}[E]$ behaves cohomologically as well as one could hope: its cohomology is built freely from that of X together with the standard classes coming from the fiber.

This is useful both conceptually and computationally. On the computational side, freeness over $H^*(X; \mathbb{Z})$ is the key step toward the projective bundle formula, which gives an explicit description of $H^*(\mathbb{P}[E]; \mathbb{Z})$ in terms of $H^*(X; \mathbb{Z})$ and the Chern classes of E . On the conceptual side, this theorem provides one of the most important applications of the Leray-Hirsch theorem and serves as a model for many other arguments in the theory of characteristic classes.

¹²The projective bundle $\mathbb{P}(E)$ carries a tautological line bundle $\mathcal{O}_{\mathbb{P}(E)}(-1)$, whose fiber over a point $\ell \in \mathbb{P}(E_x)$ is the line $\ell \subset E_x$ itself. Let $\tilde{\alpha} := c_1(\mathcal{O}_{\mathbb{P}(E)}(1)) \in H^2(\mathbb{P}(E); \mathbb{Z})$ be the first Chern class of its dual. Since Chern classes commute with pullback and $\mathcal{O}_{\mathbb{P}(E)}(1)$ restricts to $\mathcal{O}_{\mathbb{P}^{r-1}}(1)$ on each fiber, $\iota^* \tilde{\alpha} = \alpha$ is the hyperplane class. The classes $1, \iota^* \tilde{\alpha}, \dots, (\iota^* \tilde{\alpha})^{r-1}$ therefore generate $H^*(\mathbb{P}^{r-1}; \mathbb{Z}) = \mathbb{Z}[\alpha]/(\alpha^r)$, so ι^* is surjective. See [Hat01].

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